



MICROBIOLOGY CULTURE MEDIA MANUAL

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ACETAMIDE BROTH

Cat: 1211

For the confirmation of *Pseudomonas aeruginosa* in bottled water

Formula in grams per liter

Acetamide	10,00	Sodium Chloride	5,00
Dipotassium Phosphate	1,39	Monopotassium Phosphate	0,73
Phenol Red	0,012		

Final pH 7,0 ± 0,2 at 25°C

Preparation

Dissolve 17,2 grams of the medium in one litre of distilled water. If needed, heat gently to dissolve completely. Sterilize by filtration. **DO NOT AUTOCLAVE**. Aseptically dispense into sterile test tubes.

Uses

In this medium the acetamide is the sole source of carbon, whose utilization by many bacteria indicates deamination which is shown by a color change from orange-red to purple-red. It is adopted by the CeNAN, for confirmation of *Pseudomonas aeruginosa* (presence). Inoculate with one or two loopfuls from a tube of presumptive medium (Asparagine Broth) and incubate at 37°C for 48 hours.

A positive reaction turns the medium to an intense purple-red. *P. aeruginosa* is confirmed by a positive asparagine test and a positive acetamide test.

Bibliography

Kelly, N.M., C.T. Keans (1.983) Acetamide Broth for Isolation of *Pseudomonas aeruginosa* from patients with cystic fibrosis. J. Clin. Microbiol 17:159-163.
CeNAN (1.982) Técnicas para el Examen microbiológico de Alimentos y Bebidas. Madrid.

Microbiological Test

Microorganisms	Growth	Change to purple red
<i>Escherichia coli</i> ATCC 25922	Inhibited	-
<i>Proteus mirabilis</i> ATCC 29906	Inhibited	-
<i>Pseudomonas aeruginosa</i> ATCC 9027	Satisfactory	+
<i>Pseudomonas aeruginosa</i> ATCC 25668	Satisfactory	+

AMIES TRANSPORT MEDIUM WITH CHARCOAL

Cat : 1535

For transport and maintenance of microbiological samples

Formula in grams per liter

Activated Charcoal.....	10,00	Sodium Chloride.....	3,00
Disodium Phosphate.....	1,10	Sodium Thioglycollate.....	1,00
Potassium Chloride.....	0,20	Monopotassium Phosphate.....	0,20
Calcium Chloride.....	0,10	Magnesium Chloride.....	0,10
Agar N° 2.....	7,50		

Final pH 7,3 ± 0,2 at 25°C

Preparation

Suspend 23 grams of the medium in one litre of distilled water. Mix well. Heat agitating frequently and boil for one minute or until completely dissolved. Distribute in tubes and sterilize at 121°C (15 lbs. sp.) for 15 minutes. Maintain an homogeneous mixture of the charcoal throughout the medium by inverting the tubes as they cool.

Uses

Transport media are formulated to maintain the viability of microorganisms without significant increase in growth. Amies developed his formula (1967) with charcoal upon proving that *N. gonorrhoeae* increased its survival rate when charcoal swabs were used. The charcoal swab

method was not optimal as the collection of the specimen sometimes removed the charcoal. Amies solved this problem by incorporating charcoal into the formulation, that neutralizes fatty acids that are toxic to microorganisms. Is recommended for throat, vaginal, and wound samples.

Bibliography

Amies C.R. (1,967) "A Modified Formula for the Preparation of Stuart's Transport Medium". Can. J. Public Health 58: 296-300.

Microbiological Test

Microorganisms	Growth
<i>Neisseria gonorrhoeae</i> ATCC 19424	Satisfactory
<i>Brucella abortus</i> ATCC 4315	Satisfactory
<i>Streptococcus pneumoniae</i> ATCC 6303	Satisfactory
<i>Shigella flexneri</i> ATCC 12022	Satisfactory
<i>Salmonella typhi</i> ATCC 6539	Satisfactory

AMIES TRANSPORT MEDIUM W/O CHARCOAL

Cat. 1530

For transport and maintenance of microbiological samples

Formula in grams per liter

Sodium Chloride	3,00	Disodium Phosphate	1,10
Sodium Thioglycollate	1,00	Potassium Chloride	0,20
Monopotassium Phosphate	0,20	Calcium Chloride	0,10
Magnesium Chloride	0,10	Agar N° 2	7,50

Final pH 7,3 ± 0,2 at 25°C

Preparation

Suspend 13 grams of the medium in one litre of distilled water. Mix well. Heat agitating frequently and boil for one minute or until completely dissolved. Distribute in tubes and sterilize at 121°C (15 lbs. sp.) for 15 minutes.

Uses

Transport media are chemically defined, semisolid, non-nutritive, phosphate buffered media that provide a reduced environment. In this medium an inorganic phosphate buffer has substituted the glycerophosphate buffer (as in modified Stuart Transport Medium). The metabolism of glycerophosphate by some coliforms and other Gram-negative *bacilli* allowed massive

overgrowth of these organisms on the swabs and fecal samples. The NaCl concentration (0,3%) is ideal for the preservation of *N. gonorrhoeae*.

Use a sterile cotton swab for the collection of the specimens and insert into the base of the medium tube. Cut off any excess swab to allow a proper cap closure.

Bibliography

Amies C.R. (1,967) "A Modified Formula for the Preparation of Stuart's Transport Medium". Can. J. Public Health 58: 296-300.

Microbiological Test

Microorganisms	Growth
<i>Neisseria gonorrhoeae</i> ATCC 19424	Satisfactory
<i>Brucella abortus</i> ATCC 4315	Satisfactory
<i>Streptococcus pneumoniae</i> ATCC 6303	Satisfactory
<i>Shigella flexneri</i> ATCC 12022	Satisfactory
<i>Salmonella typhi</i> ATCC 6539	Satisfactory

ANAEROBIC AGAR

Cat. 1000

For the cultivation of anaerobes, specially of *Clostridium* species

Formula in grams per liter

Casein Peptone.....	17,50	Soy Peptone.....	2,50
Sodium Chloride.....	2,50	L-Cystine.....	0,40
Dextrose.....	10,00	Sodium Thioglycollate.....	2,00
Sodium Sulfoxyl Formaldehyde.....	1,00	Methylene Blue.....	0,002
Bacteriological Agar.....	15,00		

Final pH 7,2 ± 0,2 at 25°C

Preparation

Suspend 51 grams of the medium in one litre of distilled water. Soak for 10-15 minutes. Mix well and heat with agitation. Boil for one minute or until the medium is completely dissolved. Sterilize in the autoclave at 121°C (15 lbs. sp.) for 15 minutes. The medium can be incubate in anaerobes jar or with Brewer lids for anaerobiosis.

Uses

Three reducing agents generate an strong and stable descent of the oxidation-reduction potential, thus securing good anaerobic conditions. Methylene blue acts as the redox indicator.

The seeding of the sample (clinical or food) can be performed by surface inoculation or by emptying. That is, by inoculating and mixing the product to study with the medium, melted and cooled to 45-50°C. Normally the sample should never be heated to destroy the vegetative forms of the anaerobe, as the anaerobes non sporeformers will be also destroyed. Nevertheless, sometimes it would be useful to heat the sample when sporeformers such as *Clostridium* are sought, except *C. Perfringens*, which rarely forms spores. When heating is indicated, warm the sample suspended in a liquid diluent (peptone water, buffering phosphate solution, etc.) for 10 minutes between 70°C-80°C.

The plates of Anaerobic Agar can also be incubated in a normal atmosphere covering the surface of the plates with a Brewer lid. In this case, it is important to leave about 1,5 cm on the outer edge of the plate uninoculated. With care place the Brewer lid on the plate to obtain a hermetic seal. The central part of the lid should not touch the surface of the plate but form a chamber of 2-5 mm.

When growth is observed, open the plate and pick the desired colonies. Incubate longer if necessary. If the medium has not been prepared shortly above the surface. before its use, it is necessary to heat and remelt it to expel the dissolved oxygen.

If for some reason the sample can not be streaked on the Anaerobic Agar plate, place the sample in Thioglycollate Medium without Indicator previously heated and cooled. Incubate until the next day and seed the Anaerobic Agar plate. Thioglycollate Medium without Indicator is an excellent enrichment broth and frequently this method gives better results than direct seeding.

Bibliography

Brewer, J.H. 1.942 A new Petri dish and technique for use in the cultivation of anaerobes and microaerophiles Science 95:587.
Marshall, R.T. (ed.) 1.992, Standard methods for the microbiological examination of dairy products, 16 Th ed. American Public Health Association. Washington D.C

Microbiological Test

Microorganisms	Growth
<i>Clostridium butyricum</i> ATCC 9690	Good
<i>Clostridium perfringens</i> ATCC 12919	Good
<i>Clostridium sporogenes</i> ATCC 11437	Good

ANTIBIOTIC MEDIUM N° 1 (SEED AGAR) Cat. 1520

Medium prepared according to the formulation specified by the Food and Drug Administration of the U.S.A.
Pharmacopoeia

Formula in grams per liter

<i>Gelatin Peptone</i>	6,00	<i>Casein Peptone</i>	4,00
<i>Yeast Extract</i>	3,00	<i>Beef Extract</i>	1,50
<i>Dextrose</i>	1,00	<i>Bacteriological Agar</i>	15,00

Final pH 6,6 ± 0,2 at 25°C

Preparation

Suspend 30,5 grams of the medium in one litre of distilled water. Mix well. Heat with frequent agitation. And boil for one minute. Distribute into appropriate containers and sterilize at 121°C (15 lbs. sp.) for 15 minutes.

Uses

1. ASSAY PLATES

Seed Agar is used as an inoculum substrate. It is melted and cooled to 48°C and inoculated according to the specific antibiotic in test. Use 2 ml of the liquid culture to inoculate 100 ml of the Seed Agar. Agitate the mixture gently to produce an homogeneous distribution and pour 4 ml on each plate of solidified Base Agar (21 ml).

It is very important that the seed layer is evenly distributed over the entire surface of the Base Agar. Once the seed layer is solid you can place cylinders for the adequate solutions, normal and antibiotic tests. The standard and the problem are added as described before. This method is used for testing the potency of bacitracin and penicillin preparations.

Seed Agar is used for the basic layer as well as the seed layer for the assay of chloramphenicol in plates. With a higher pH, the medium is used for the assay of erythromycin, carbomycin and neomycin. This formula is available in dehydrated form under the name Neomycin Test Agar (Antibiotic Medium N° 11).

2. PREPARATION OF TEST CULTURES

Seed Agar is the chosen medium to prepare the test cultures used in some methods of plate assays. For example, in the assay broth for chloramphenicol, chlortetracycline, erythromycin and penicillin potency tests. It is also used to prepare spore suspensions of *Bacillus subtilis* for the assay of streptomycin.

3. ENUMERATION OF MICROORGANISMS

Seed Agar can be used to determine the number of microorganisms in many antibiotic preparations.

4. DETERMINATION OF ANTIBIOTICS IN MILK

The milk used to manufacture fermented products is tested for inhibitory substances, such as residual antibiotics in the treatment of mastitis, which can interfere with the normal activity of the initial culture. Disk diffusion methods are utilized to detect the presence of residual antibiotics.

Bibliography

Grove and Randall. Assay Methods of Antibiotics, Medical Encyclopedia Inc. New York 1955. United States Pharmacopoeial Convention. 1.955. The United States, pharmacopoeia, 23rd Ed. Biological Tests and Assays, p. 1690-1696. The United States Pharmacopoeial Convention, Rockville, Md.

Microbiological Test

Microorganisms	Growth	Inhibition zones
<i>Staphylococcus aureus</i> ATCC 6538P	Satisfactory	Cephalotine, Cloramphenicol, Penicilline
<i>Micrococcus luteus</i> ATCC 9341	Satisfactory	Cephalotine, Cloramphenicol, Penicilline
<i>Staphylococcus epidermidis</i> ATCC 12228	----	----
<i>Bacillus subtilis</i> ATCC 6633	Satisfactory	----
<i>Bacillus cereus</i> ATCC 11778	Satisfactory	----

ANTIBIOTIC MEDIUM N° 2 (BASE AGAR) Cat. 1002

Standard medium used for the preparation of the basal layer in the Antibiotics Microbiological assay

Formula in grams per liter

Gelatin Peptone 6,00	Yeast Extract..... 3,00
Beef Extract..... 1,50	Bacteriological Agar..... 15,00

Final pH 6,6 ± 0,2 at 25°C

Preparation

Suspend 25,5 grams of medium in one litre of distilled water. Heat with frequent agitation for one minute. Sterilize at 121°C (15 lbs. sp.) for 15 minutes. Cool at 45-50°C and pour into sterile Petri dishes.

Uses

Base Agar is an standard medium used to prepare the base layer in the microbiological assay of antibiotics.

This medium is prepared in accordance with the Food and Drug Administration (FDA) and USP guidelines. It is used to prepare the base layer in the microbiological assay of antibiotics such as bacitracin, chloramphenicol and penicillin. The sample can be tested by two methods- dilution and diffusion in an agar plate.

The diffusion method is the most common and can be performed using various techniques; cylinders, punched-hole or paper disc tests.

To perform the antibiotic test the Base Agar should be prepared on the same day as the test.

For the cylinder method, pour 21 ml. of medium into a Petri dish (20x100 mm.) and cover to avoid dehydration.

Once the medium has solidified, add 4 ml. of the seed layer inoculated with the standardized culture for the particular antibiotic to be tested. Be sure to obtain an even and level distribution of this layer. The layer is allowed to solidify and the cylinders are placed on the surface. The dilutions of the antibiotic will be added to these cylinders.

The plate is incubated for 24 hours at 35-37°C. The zones of inhibition are observed, measured and compared with the calibration curve determined by adding known amounts of the same antibiotic under the same experimental conditions.

Bibliography

Grove and Randall. Assay Methods of Antibiotics, Medical Encyclopedia Inc. New York 1955. United States Pharmacopoeial Convention. 1.955. The United States, pharmacopoeia, 23rd Ed. Biological Tests and Assays, p. 1690-1696. The United States Pharmacopoeial Convention, Rockville, Md.

Microbiological Test

Microorganisms	Growth
<i>Staphylococcus aureus</i> ATCC 6538-P	Good
<i>Micrococcus luteus</i> ATCC 10240	Good
<i>Staphylococcus epidermidis</i> ATCC 12228	Good

ANTIBIOTIC MEDIUM N° 3

Cat. 1534

To evaluate the antibiotic activity

Formula in grams per liter

<i>Gelatin Peptone</i>	5,00	<i>Dipotassium Phosphate</i>	3,68
<i>Sodium Chloride</i>	3,50	<i>Yeast Extract</i>	1,50
<i>Beef Extract</i>	1,50	<i>Monopotassium Phosphate</i>	1,32
<i>Dextrose</i>	1,00		

Final pH 7,0 ± 0,2 at 25°C

Preparation

Suspend 17,5 grams of the medium in one litre of distilled water. Mix well. Soak for 10-15 minutes. Heat, with frequent agitation and boil for one minute until completely dissolved.

Distribute into appropriate containers and sterilize at 121°C (15 lbs. sp.) for 15 minutes.

Uses

This liquid medium is prepared according to the formula specified by the Food and Drug Administration (FDA) and the United States Pharmacopoeia (USP).

Antibiotic Medium N° 3 can be used with the following microbiological methods for antibiotic assays:

1. Cylinder method in plates.
2. Serial dilution method.
3. Turbidimetric method.

In the cylinder method in plates, Antibiotic Medium N° 3 is used to resuspend the inoculum in the potency assay for penicillin, erythromycin, neomycin, chlortetracycline and chloramphenicol.

The serial dilution method is used for penicillin assay. Lastly, this medium can also be used in the turbidimetric determination of the potency of bacitracin, streptomycin and terramycin. The turbidimetric method is based on the inhibition of growth of a microbial culture in a fluid medium containing a uniform solution of an antibiotic. Use of this method is appropriate only when test samples are clear.

Bibliography

Grove and Randall. Assay Methods of Antibiotics, Medical Encyclopedia Inc. New York 1955. United States Pharmacopoeial Convention. 1.955. The United States, pharmacopoeia, 23rd Ed. Biological Tests and Assays, p. 1690-1696. The United States Pharmacopoeial Convention, Rockville, Md.

Microbiological Test

Microorganisms	Growth	Inhibition zones
<i>Staphylococcus aureus</i> ATCC 6538P	Satisfactory	Kanamicine, Tetraciline
<i>Micrococcus luteus</i> ATCC 9341	Satisfactory	
<i>Klebsiella pneumoniae</i> ATCC 10031	Satisfactory	Streptomycin

ANTIBIOTIC MEDIUM N° 5 (FOR STREPTOMYCINE ASSAYS) Cat. 1524

Used in the potency assay of streptomycin with yeast extract

Formula in grams per liter

Gelatin Peptone 6,00	Yeast Extract 3,00
Beef Extract 1,50	Bacteriological Agar 15,00

Final pH 7,9 ± 0,2 at 25°C

Preparation

Suspend 25,5 grams of medium in one litre of distilled water. Mix well. Heat with frequent agitation and boil for one minute until completely dissolved. Distribute into appropriate containers and sterilize at 121°C (15 lbs. sp.) for 15 minutes.

Uses

This agar can be used in the cylinder plate method for the assay of streptomycin, generally with *Bacillus subtilis* as the test organism. This method is based on the diffusion of an antibiotic solution from a cylinder placed on the surface of an inoculated agar medium. The diameter of a zone of inhibition after incubation depends, in part, on the concentration or activity of the antibiotic. This method is used in the assay of commercial preparations of antibiotics, as well as in the quantitative determination of

antibiotics in body fluids, animal feeds and other materials. Plates are prepared and incubated following the guidelines of the FDA and the USP. It is the same formula as Base Agar but with an elevated pH to be compatible with streptomycin.

Bibliography

Grove and Randall. Assay Methods of Antibiotics, Medical Encyclopedia Inc. New York 1955. United States Pharmacopocical Convention. 1.955. The United States, pharmacopoeia, 23rd Ed. Biological Tests and Assays, p. 1690-1696. The United States Pharmacopocical Convention, Rockville, Md.

Microbiological Test

Microorganisms	Growth	Inhibition zones
<i>Bacillus subtilis</i> ATCC 6633	Good	Gentamicyn, Streptomycin

ANTIBIOTIC MEDIUM N° 8 (BASE AGAR WITH LOW pH) Cat. 1004

Used for plate assay of antibiotics such as tetracycline

Formula in grams per liter

<i>Gelatin Peptone</i>	6,00	<i>Yeast Extract</i>	3,00
<i>Beef Extract</i>	1,50	<i>Bacteriological Agar</i>	15,00

Final pH 5,7 ± 0,1 at 25°C

This medium has the same formula as Antibiotic Medium N° 2 (Base Agar) with the difference that the pH of the final medium has been adjusted to 5,7.

Preparation

Suspend 25.5 grams of medium in one litre of distilled water. Mix well. Heat with frequent agitation and boil for one minute. Sterilize at 121°C (15 lbs. sp.) for 15 minutes and cool at 45°-50°C and dispense into sterile Petri dishes. The activity (potency) of an antibiotic can be demonstrated under suitable conditions by its inhibitory effect on microorganisms. Reduction in antimicrobial activity may reveal changes not demonstrated by chemical methods.

Uses

Base Agar with low pH is used to prepare the basal layer for the assay of tetracycline's and other antibiotics.

Prepare the inoculum for assay by washing growth from a fresh 24-48 hours agar slant, issuing sterile distilled water or saline water.

Bibliography

Grove and Randall. Assay Methods of Antibiotics, Medical Encyclopedia Inc. New York 1955. United States Pharmacopocial Convention. 1.955. The United States, pharmacopoeia, 23rd Ed. Biological Tests and Assays, p. 1690-1696. The United States Pharmacopocial Convention, Rockville, Md.

Microbiological Test

Microorganisms	Growth	Dilutions assay in series
<i>Bacillus cereus</i> ATCC 11778	Good	Tetracycline
<i>Staphylococcus aureus</i> ATCC 6538	Good	Tetracycline, Chlortetracycline

ANTIBIOTIC MEDIUM N° 11 (NEOMYCIN ASSAY AGAR) Cat. 1528

To analyse the neomycin content as per FDA and U.S.A. Pharmacopoeia

Formula in grams per liter

<i>Gelatin Peptone</i>	6,00	<i>Casein Peptone</i>	4,00
<i>Yeast Extract</i>	3,00	<i>Beef Extract</i>	1,50
<i>Dextrose</i>	1,00	<i>Bacteriological Agar</i>	15,00

Final pH 7,9 ± 0,2 at 25°C

Preparation

Suspend 30,5 grams of the medium in one litre of distilled water. Mix well. Heat with frequent agitation and boil for one minute. Distribute into appropriate containers and sterilize at 121°C (15 lbs. sp.) for 15 minutes.

Uses

Medium specially prepared to analyze the neomycin content in pharmaceutical preparations as per FDA and the U.S.A Pharmacopoeia. It can also be used to test other antibiotics, including erythromycin and carbomycin. Neomycin Assay Agar is used in the cylinder plate method for the assay of neomycin. It has the same formula as Seed Agar (casein peptone agar from the USA Pharmacopoeia) but with an higher pH, while the seed agar is slightly acid.

This agar can be used in plates as either the base or seed layer as well as to prepare the *S. aureus* PCJ 209-P inoculum. It can also be used to prepare the *Klebsiella pneumoniae* PCL 602 or ATCC 10031 inoculum which is used in the turbidimetric assay for neomycin. The inoculum for the erythromycin assay is *S. lutea* 9314.

Bibliography

United States Pharmacopoeial Convention. 1995. The United States pharmacopoeia, 23rd ed. Biological Tests and Assays, p. 1960-1696. The United States Pharmacopoeial Convention, Rockville, M.D.
Federal Register. 1992. Tests and methods of assay of Antibiotics and Antibiotic-Containing Frugs. Fed. Regist. 21:436.100-436-106.

Microbiological Test

Microorganisms	Growth	Null inhibition
<i>Micrococcus luteus</i> ATCC 9431	Good	Ampicillin, Erythromycin
<i>Staphylococcus aureus</i> ATCC 6538	Good	Kanamycin, Neomycin
<i>Staphylococcus epidermis</i> ATCC 12228	Good	Oleandomycin, Paramycin

ASPARAGINE BROTH

Cat. 1207

For the presumptive identification and enumeration (MPN) of *Pseudomonas aeruginosa*

Formula in grams per liter

Monopotassium Phosphate	10,00	Asparagine	2,00
Dipotassium Phosphate	1,00	Magnesium Sulfate	0,50

Final pH 7,0 ± 0,2 at 25°C

Preparation

Suspend 13,5 grams of the medium in one litre of distilled water with 8 ml. of glycerol. Heat agitating until completely dissolved. Dispense and sterilize at 121°C (15 lbs. sp.) for 15 minutes.

To obtain a double strength broth, dissolve 27 grams of the medium and add 16 ml. of glycerol.

Uses

This medium is an excellent enrichment broth for *P. aeruginosa* because the formula contains a strictly mineral base with asparagine as the sole source of carbon.

Asparagine Broth is recommended for enumeration by the MPN method with 5 tubes/series inoculating 10 ml., 1 ml. and 0,1 ml.

All tubes are incubated at 37°C for 48 hours.

The appearance of growth with or without pigmentation is considered a presumptive test for the presence of *P. aeruginosa* and counts are determined using the MPN tubes. Confirmation is made by subculturing a loopful from each turbid tube into Acetamide Broth.

Bibliography

APHA. Standard Methods for Examination of Water and waste water, 14th ea. 1975.

Microbiological Test

Microorganisms	Growth
<i>Pseudomonas aeruginosa</i> ATCC 27853	Good
<i>Pseudomonas aeruginosa</i> ATCC 10145	Good

AZIDE BLOOD AGAR BASE

Cat. 1113

For the isolation of streptococci and staphylococci. When added 5% of sheep blood, it allows the research of hemolytic reactions.

Formula in grams per liter

Peptone mixture.....	10,00	Sodium Chloride	5,00
Beef Extract.....	3,00	Sodium Azide.....	0,20
Bacteriological Agar.....	15,00		

Final pH 7,2 ± 0,2 at 25°C

Preparation

Suspend 33.2 grams of the medium in one litre of distilled water. Mix well. Heat with frequent agitation and boil for one minute until complete dissolution. Dispense, in appropriate containers and sterilize at 121°C (15 lbs. sp.) for 15 minutes. Cool to 45°C and aseptically add 5% of sterile defibrinated sheep blood. Mix well and pour into Petri dishes.

Uses

Sodium Azide has proved to have a bacteriostatic effect on Gram negative bacteria, thus, this medium is used for the isolation of streptococci and staphylococci in clinical specimens, water, foods, etc. 0.01% Sodium Azide in blood agar was reported to prevent the swarming of *Proteus* and allows the selective isolation from mixed bacterial populations. Gram-negative organisms are

inhibited by sodium azide it can be supplemented with 5% of sheep blood allows the investigation of hemolytic reactions of fastidious pathogens..

Bibliography

Edwards, S. J. 1933 The diagnosis of *Streptococcus mastitis* by cultura methods. J. Comp. Pathol. Ther. 46:211.
Lichstein, H. C., and M.L. Snyder. 1941. The inhibition of the spreading growth of *Proteus* and other bacteria to permit the isolation of associated streptococci. J. Bacteriol. 42:653.

R:22 Toxic when swallowed
S:45 In case of accident or uneasiness, seek medical advise immediately. Show the label if possible

Microbiological Test

Microorganisms	Growth	Hemolytic Test
<i>Neisseria meningitidis</i> ATCC 13090	Good	----
<i>Staphylococcus faecalis</i> ATCC 19433	Good	Alfa/gamma
<i>Staphylococcus epidermidis</i> ATCC 12228	Good	----
<i>Streptococcus pneumoniae</i> ATCC 6303	Good	Alfa
<i>Streptococcus pyogenes</i> ATCC 19615	Good	Beta
<i>Escherichia coli</i> ATCC 25922	----	----

BACILLUS CEREUS SELECTIVE AGAR BASE

Cat. 1124

 For the enumeration and isolation of *Bacillus Cereus* in food, according to MOSSEL

Formula in grams per liter

<i>Meat peptone</i>	10,00	<i>Sodium chloride</i>	10,00
<i>D-Mannitol</i>	10,00	<i>Beef extract</i>	1,00
<i>Phenol red</i>	0,025	<i>Bacteriological agar</i>	12,00

Final pH 7,1 ± 0,2 at 25°C
Preparation

Suspend 43 grams of the medium in 900 ml. of distilled water. Heat agitating frequently until complete dissolution. Sterilize in the autoclave at 121°C for 15 minutes. Cool to 45-50°C and add 100 ml. of an sterile egg yolk emulsion and, if desired, 0.01 to 0.1 gr. of Polymixin in sterile dissolution, per litre of medium.

Uses

This medium was been adapted to meet the needs of *Bacillus cereus*, and was proposed by Mossel et al. (1967) for the enumeration, detection and isolation of *Bacillus cereus* in food.

Bacillus cereus is negative-mannitol. The mannitol content allows the separation of the accompanying mannitol-positive flora, which are characterized by a yellow color.

Bacillus cereus is resistant to certain concentrations of Polymixin, which inhibits the accompanying flora.

Bacillus cereus forms lecithinase. The indissoluble degradation products of the lecithin of egg yolk accumulate around the cereus colonies, forming a white precipitate. Inoculated plates should be incubated for 18-40 hours at 32°C, the colonies of *Bacillus cereus* will appear red and surrounded by a ring of precipitation.

Bibliography

Donovan, K.O.: A Selective Medium for *Bacillus Cereus* in Milk, J. appl. Bact., 21; 100:103 (1958)
 Mossel. D.A.A. Koopman, M.J. a Jongerius, E.: Enumeration of *Bacillus Cereus* in Foods. Appl. Microbiol., 15; 650:653 (1967)

Microbiological Test

Microorganisms	Growth	Colony colour	Precipitation
<i>Bacillus cereus</i> ATCC 1178	Acceptable	Red	+
<i>Bacillus subtilis</i> ATCC 6051	Acceptable	Yellow	-
<i>Proteus mirabilis</i> ATCC 29906	Inhibited	Colourless	-
<i>Staphylococcus aureus</i> ATCC 6538	Inhibited	Yellow	+

BAIRD PARKER AGAR BASE (EUROPEAN PHARMACOPOEIA)

Cat. 1100

Used for the selective isolation of coagulase-positive staphylococci

Formula in grams per liter

Glycine.....	12,00	Casein Pancreatic Digest	10,00
Sodium Pyruvate.....	10,00	Beef Extract.....	5,00
Lithium Chloride	5,00	Yeast Extract.....	1,00
Bacteriological Agar	20,00		

Final pH 6,8 ± 0,2 at 25°C

Preparation

Suspend 63 grams of the medium in one litre of distilled water. Mix well. Heat with frequent agitation and boil for one minute until complete dissolution. Sterilize in autoclave at 121°C (15 lbs. sp.) for 15 minutes. Cool to 45°- 50° C and add 10 ml. of a 1% potassium tellurite solution and 50 ml. of a egg yolk emulsion. Homogenize gently and pour into Petri dishes.

Refrigerate in sealed containers or in tubes or bottles with screw caps. The base, can be kept for long periods of time, and can be melted as needed.

Uses

This medium is widely used and is included in many Standard Methods Procedures for testing goods, dairy products, etc. The prepared plates of the complete medium should be used within 24 hours. The plates should be dry before inoculation (the drying can be made by incubating at 35-37°C for approximately 10 minutes before use).

Baird Parker Agar Base is used for the selective and selective isolation and enumeration of coagulase positive staphylococci. Contains Lithium Chloride and Potassium

Tellurite to inhibit the accompanying flora and Glycine and Pyruvate to facilitate the staphylococci growth. Prepare the sample in an adequate solution, dilute it and place from 0.1 ml. to 1.0 ml. of the appropriate dilution in the plates. Spread the inoculum over the entire surface. Incubate at 35-37°C for 24-36 hours. Typical *S. aureus* colonies are black, shiny, convex and surrounded by a clear zone of approximately 2-5 mm in diameter.

Some other microorganisms, which occasionally grow on this medium, are micrococci which form small dark or black colonies, yeasts which form white colonies and some species of *Bacillus* which form dark brown matte colonies.

Bibliography

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Sharp, Neave and Reider. J. App. Bact. 28:390, 1962. Baird-Parker and Devenport J. App. Bact. 28:390, 1965. Tardio and Bact. J. AOAC. 54:728, 1971.

Microbiological Test

Microorganisms	Growth	Colony colour	Lecitinase Transparence around the colonies
<i>Bacillus subtilis</i> ATCC 6633	Slight-null	Brown	-
<i>Escherichia coli</i> ATCC 25922	null	----	-
<i>Staphylococcus epidermidis</i> ATCC 12228	Slight-good	Black	-
<i>Staphylococcus aureus</i> ATCC 6538	Good	Black	+
<i>Staphylococcus aureus</i> ATCC 25923	Good	Black	+
<i>Proteus mirabilis</i> ATCC 25933	Good	Brown	-

B.C.P. AGAR

Cat. 1051

Lactose Agar with Bromcresol Purple used for the isolation of coliforms

Formula in grams per liter

<i>Polipectone</i>	5,00	<i>Beef Extract</i>	3,00
<i>Lactose</i>	10,00	<i>Bromcresol Purple</i>	0,025
<i>Bacteriological Agar</i>	10,00		

Final pH 6,8 ± 0,2 at 25°C

Preparation

Suspend 28 grams of the medium in one litre of distilled water. Mix carefully. Heat with frequent agitation and boil for one minutes. Distribute into appropriate containers and sterilize in the autoclave at 121°C (15 lbs. sp.) for 15 minutes.

Uses

It is a non inhibitor medium used for the isolation of enterobacteria. It allows to differentiate species in base of lactose fermentation. When lactose is fermented it produces acid that changes the color of the medium from purple to yellow. Blue colonies are lactose-negative and yellow colonies are lactose-positive. Reading must be made after 18-24 hours as longer incubation times may cause the diffusion of the acid in the medium and result in an error.

Appearance of lactose-positive cultures.

Klebsiella.....mucoid

E. coli.....mucoid

Slow lactose-fermenting (lactose +) *E. coli* types can present a bluish color on the periphery of the colony after 18 hours of incubation.

Bibliography

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 Lennette, E.H., Ballows, A., Hausler, W.J.Jr., and Shadomy, H.J. Manual of Clinical Microbiology. 4th ed. 1985 Washington D.C.: American society for Microbiology.
 Mac Faddin, Jean F., Media for Isolation-Cultivation-Identification-Maintenance of Medical Bacteria Vol.1 1985 Baltimore, MD. Williams & Wilkins.

Microbiological Test

Microorganisms	Growth	Change to purple red
<i>Escherichia coli</i> ATCC 25922	Satisfactory	Yellow
<i>Klebsiella pneumoniae</i> ATCC 13883	Satisfactory	Yellow
<i>Salmonella typhimurium</i> ATCC 14028	Satisfactory	Blue
<i>Shigella sonnei</i> ATCC 25931	Satisfactory	Blue

BIGGY AGAR

Cat. 1006

For the isolation and identification of *Candida* spp.

Formula in grams per liter

Dextrose	10,00	Glycine	10,00
Bismuth Ammonium Citrate.....	5,00	Sodium Sulfite.....	3,00
Yeast Extract.....	1,00	Bacteriological Agar.....	16,00

Final pH 6,8 ± 0,2 at 25°C

Preparation

Suspend 45 grams of the medium in one litre of distilled water. Mix well and heat with frequent agitation. Boil for no more than one minute. Cool to 45-50°C, swirl the medium gently and pour into sterile Petri dishes with 20 ml per dish. **Do not autoclave.**

Uses

Biggy Agar is an abbreviation for Bismuth Glucose Glycine Yeast Agar. Is used to isolate *C. albicans* and *C. tropicalis*, and to differentiate the species according to the Nickerson method:

Candidiasis is the most frequently encountered opportunistic fungal infection. *Candida* species can be present in clinical specimens as a result of environmental contamination, colonization or actual disease process.

C. albicans Smooth brown-black colonies with a thin mycelial border and no color diffusion into the surrounding medium.

C. tropicalis Discrete smooth dark brown colonies with a prominent black center and thin mycelial border and a color diffusion into the medium after 3 days incubation.

C. krusei Wrinkled, flat colonies brown to black in color with a yellow color diffusion.

C. parakrusei Medium-sized, flat, normally wrinkled colonies reddish-brown in color with a big yellow mycelia border.

C. stellatoidea Medium-sized flat colonies dark brown in color with only slight mycelia.

Freshly poured plates should only be used. Inoculation onto slanted surfaces is not generally satisfactory.

Bibliography

Nickerson, W.J. 1953. Reduction of inorganic substances by yeasts. I. Extracellular reduction of sulfite by species of *Candida*. J. Infect. Dis. 93:43. Warren, N.G., and K.C. Hazen. 1955 *Candida*, *Cryptococcus* and other yeasts of medical importance, p. 723-737. IN P.R. Murray, E.J. Baron, M.A. Pfaller, F.C. Tenover and R.H. Tenover (ed.), Manual of clinical microbiology, 6th ed. American Society for Microbiology, Washington D.C.

Microbiological Test

Microorganisms	Growth	Change to purple red
<i>Candida albicans</i> ATCC 10231	Satisfactory	Brown to black
<i>Candida pseudotropicalis</i>	Satisfactory	Brown to red
<i>Escherichia coli</i> ATCC 25922	Inhibited	--
<i>Staphylococcus aureus</i> ATCC 25923	Inhibited	--

BILE ESCULIN AGAR

Cat. 1031

For the isolation and presumptive identification of Group D streptococci

Formula in grams per liter

<i>Ox Bile</i>	40,00	<i>Peptone Bacteriological</i>	5,00
<i>Beef Extract</i>	3,00	<i>Esculin</i>	1,00
<i>Ferric Citrate</i>	0,50	<i>Bacteriological Agar</i>	15,00

Final pH 6,6 ± 0,2 at 25°C
Preparation

Suspend 64 grams of the medium in one litre of distilled water. Mix well. Heat with frequent agitation and boil until completely dissolved. Dispense into appropriate and sterilize at 121°C (15 lbs. sp.) for 15 minutes. Overheating can cause darkening of the medium. If tubes are used, allow to solidify in a slanted position.

Uses

Group D streptococci grow well on this differential medium because the ox bile in the formula does not inhibit them while the other Gram-positive bacteria are inhibited.

On the other hand, the hydrolysis of esculin to esculetin in this bile medium (differential test for enterococci) is shown by the dark brown colour of the medium. Tolerance to bile and the ability to hydrolyze esculin that reacts with the ferric citrate constitutes a reliable presumptive test for the identification of Group *D streptococci*.

The brown color (positive reaction) around the colonies appears after 18-24 hours of incubation at a temperature of 35-37°C.

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 Swan, A. 1954. The use of bile-esculin medium and of Moxed's technique of Lancefield grouping in the identification of enterococci (group D streptococci). J. Clin Pathol 7:160
 Facklam, R.R. and M.D. Moody 1970 Presumptive identification of group D streptococci, The bile esculin test. Appl. Microbiol 20:245.
 Farmer J.J. III 1995 Enterobacteriaceae P.R. Murray, E.J. Baron, M.A. Pfaller, F.C. Tenover and R.H. Tenover (eds) Manual of clinical microbiology, 6th ed. American Society for Microbiology, Washington, D.C.

Microbiological Test

Microorganisms	Growth	Change to purple red
<i>Streptococcus faecalis</i> ATCC 11700	Satisfactory	+
<i>Streptococcus faecalis</i> ATCC 19433	Satisfactory	+
<i>Streptococcus faecium</i> ATCC 8043	Satisfactory	+
<i>Streptococcus pyogenes</i> ATCC 12344	Null	-
<i>Streptococcus pneumoniae</i> ATCC 6301	Null	-
<i>Staphylococcus aureus</i> ATCC 25923	Satisfactory	+(light)
<i>Escherichia coli</i> ATCC 25922	Light	-

BILE ESCULIN AZIDE AGAR

Cat. 1005

Selective medium for the isolation and presumptive identification of Group D streptococci

Formula in grams per liter

<i>Tryptone</i>	17,00	<i>Ox Bile</i>	10,00
<i>Beef Extract</i>	5,00	<i>Sodium Chloride</i>	5,00
<i>Proteose Peptone n° 3</i>	3,00	<i>Esculin</i>	1,00
<i>Ferric Ammonium Citrate</i>	0,50	<i>Sodium Azide</i>	0,150
<i>Bacteriological Agar</i>	15,00		

Final pH 7,0 ± 0,2 at 25°C

Preparation

Suspend 56,6 grams of the medium in one litre of distilled water. Mix well. Heat with frequent agitation and boil until totally dissolved. Dispense in appropriate containers and sterilize at 121°C (15 lbs. sp.) for 15 minutes. Overheating can cause darkening of the medium. If tubes are used, allow to solidify in a slanted position.

Uses

The same as the uses of Bile Esculin Agar except that by adding the sodium azide the medium becomes selective, inhibiting the Gram-negative bacteria. Bile Esculin Azide Agar is a modification of Bile Esculin Agar by adding sodium azide and reducing the concentration of bile. The resulting medium is more selective but still provides for rapid growth and efficient recovery of group D streptococci. The ability to hydrolyze esculin in the presence of bile is a characteristic of enterococci and group D streptococci.

R:22 Toxic when swallowed

S:45 In case of accident or uneasiness, seek medical advice immediately. Show the label if possible

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Swan, A. 1954. The use of bile-esculin medium and of Maxted's technique of Lancefield grouping in the identification of enterococci (group D streptococci) J. Clin. Pathol 7:160.
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Microbiological Test

Microorganisms	Growth	Esculin
<i>Streptococcus faecalis</i> ATCC 11700	Good	+
<i>Streptococcus faecium</i> ATCC 8043	Good	+
<i>Streptococcus pyogenes</i> ATCC 12344	Null	-
<i>Escherichia coli</i> ATCC 25922	Null	-

BISMUTH SULFITE AGAR (WILSON BLAIR) Cat. 1011

Highly selective medium for the isolation of *Salmonella typhi* as well as other enteric bacilli from faeces, water and diverse foods.

Formula in grams per liter

<i>Bacteriological peptone</i>	10,00	<i>Bismuth Sulfite Indicator</i>	8,00
<i>Beef Extract</i>	5,00	<i>Dextrose</i>	5,00
<i>Dissodium Phosphate</i>	4,00	<i>Ferrous Sulphate</i>	0,30
<i>Brilliant Green</i>	0,025	<i>Bacteriological Agar</i>	20,00

Final pH 7,7 ± 0,2 at 25°C

Preparation

Suspend 52 grams of the medium in one litre of distilled water. Mix well. Heat with frequent agitation and boil for one minute. Cool the medium to 45°C (very important) pour into Petri plates without stopping the agitation. **DO NOT AUTOCLAVE.**

Uses

As this a very strong inhibitor medium, it is recommended to inoculate also some other selective media less inhibitors, as Levine EMB Agar, MacConkey Agar, XLD Agar, Hektoen Enteric Agar, etc. Generally, Bismuth Sulfite Agar is inoculated by streaking the surface to obtain isolated colonies but the pour plate inoculation method can be also utilized, mixing perfectly and allowing the plate to solidify. All plates are incubated 24-48 hours at 35-37°C.

The solidified plates should have a uniform, opaque, cream to pale green appearance. If kept in refrigeration, the medium will slowly oxidize, once it turns to a definite green color it should be discarded. It is recommended to keep the plates refrigerated for 4 days before use to reduce inhibition and thus be able to isolate *Salmonella typhimurium*.

In the presence of H₂S, *salmonellas* reduce the iron salts and bismuth to iron sulfate, which produces a black colony, and to metallic bismuth that precipitates in the culture medium forming a bright sheen but less darker than the colony it surrounds. The intensity of the black colony as well as the metallic sheen can be increased by leaving the plates at room temperatures for 2-3 hours in the light.

Colonies of coliforms, *Shigella* (which generally do not grow) and *Proteus* are green, brown or black but does not blacken the medium. Plates should be incubated at 35-37°C for 48 hours.

Bibliography

1. Wilson, W.J., and E.M. Blair 1.926 A combination of Bismuth and Sodium Sulfite affording an enrichment and selective medium for the typhoid-paratyphoid groups of bacteria. J. Pathol. Bactend 29:310.
- United States Pharmacopoeial Convention 1.995. The United States Pharmacopoeia 23rd ed.

Microbiological Test

Microorganisms	Growth	Colony colour
<i>Enterobacter aerogenes</i> ATCC 13048	Null -Scarce	Brown-Green
<i>Escherichia coli</i> ATCC 25922	Null -Scarce	Brown-Green
<i>Salmonella enteritidis</i> ATCC 13076	Satisfactory	Bright metallic black
<i>Salmonella typhi</i> ATCC 19430	Satisfactory	Bright metallic black
<i>Shigella flexneri</i> ATCC 12022	Null -Scarce	Brown
<i>Streptococcus faecalis</i> ATCC 29212	-----	-----

BLOOD AGAR BASE

Cat. 1108

Suitable for the isolation and cultivation of fastidious microorganisms.

Formula in grams per liter

Heart Infusion.....	10,00	Meat Peptone.....	10,00
Sodium Chloride.....	5,00	Bacteriological Agar.....	15,00

Final pH 7,3 ± 0,2 at 25°C

Preparation

Suspend 40 grams of the medium in one litre of distilled water. Leave to stand for 5 minutes and mix well until a uniform suspension is obtained. Heat with gentle agitation and boil for one minute. Sterilize to 121°C (15 lbs. sp.) for 15 minutes. Cool to 45-50°C, and add 5 -10% sterile defibrinated blood, homogenize and pour into Petri plates.

Uses

For the isolation, cultivation and detection of hemolytic reaction of fastidious microorganisms. Blood Agar Base is suitable to isolate and cultivate a wide range of microorganisms with difficult growth. Upon adding blood, it can be utilized for determining hemolytic reactions. Once the medium has been melted and cooled to 45 °C you can add 5-10% of defibrinated sterile sheep blood, in this case you can recuperate Haemophilus. Be careful to avoid bubble formation when adding the blood to the cooled medium and rotate the flask or bottle slowly to

create a homogeneous solution. The medium can then be poured into dishes and solidified.

You can also inoculate the empty Petri dish with a small amount of specimen material and then pour the medium at 50°C, swirl the plate gently to homogenize the inoculum.

In some laboratories the medium is prepared in screw-capped tubes which can be inoculated at 45°C and then poured into sterile Petri dishes.

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Tharshis and Frish AM. J. Clin. Path. 21:101, 1951.

Microbiological Test

Microorganisms	Growth	Transparency
<i>Neisseria meningitidis</i> ATCC 13090	Good	----
<i>Staphylococcus aureus</i> ATCC 25923	Good	Beta
<i>Staphylococcus epidermidis</i> ATCC 12228	Good	----
<i>Streptococcus pneumoniae</i> ATCC 6303	Good	Alfa
<i>Streptococcus pyogenes</i> ATCC 19615	Good	Beta

NALIDIXIC ACID BLOOD AGAR BASE

Cat. 1128

For the differentiation of the hemolytic activity of *Streptococcus* and *Listeria monocytogenes*

Formula in grams per liter

Heart Infusion	10,00	Meat Peptone	10,00
Sodium Chloride	5,00	Bacteriological Agar	15,00
Nalidixic Acid	0,04		

Final pH 7,3 ± 0,2 at 25°C

Preparation

Suspend 40 grams of the medium in one liter of distilled water. Leave to stand for 5 minutes and mix well until a uniform suspension is obtained. Heat with frequent agitation and boil for one minute. Sterilize to 121° C (15 lbs.sp) for 15 minutes. Cool to 45-50°C and add 5-10 % sterile defibrinated blood, homogenize and pour into Petri plates.

Uses

The addition of nalidixic acid inhibits the accompanying flora and renders Blood Agar base a selective medium for *Streptococcus* and *Staphylococcus*, and making it also possible to differentiate *Listeria monocytogenes*. Be careful to avoid bubble formation when adding the blood. Pour into Petri dishes. The medium can be inoculated or seeded and incubated at 37°C for 18-24 hours.

Streptococcal colonies will have 2 to 3mm of diameter; colourless or smooth round white and will produce α haemolysis (*Streptococcus pneumoniae*) β (*Streptococcus pyogenes*) or negative (*Streptococcus bovis*).

Listeria colonies will be little than 1-2mm of diameter Colourless or smooth white and with weak β Haemolysis

Bibliography

Cruikshank, R. (1972) Medical Microbiology. 11th Edition. Livingstone. London.

Microbiological Test

Microorganisms	Growth	Haemolysis
<i>Staphylococcus aureus</i> ATCC 25923	Partially inhibited	Beta
<i>Staphylococcus epidermidis</i> ATCC 12228	Good	-
<i>Streptococcus pneumoniae</i> ATCC 6303	Good	Alfa
<i>Streptococcus pyogenes</i> ATCC 19615	Good	Beta
<i>Escherichia coli</i> ATCC 25922	Inhibited	-

BORDET GENGOU AGAR BASE

Cat. 1107

For the detection and isolation of *Bordetella pertussis* and *Bordetella parapertussis*

Formula in grams per liter

Proteose Peptone	10,00	Sodium Chloride	5,50
Potato Infusion	4,50	Bacteriological Agar	16,00

Final pH 6,7 ± 0,2 at 25°C

Preparation

Suspend 36 grams of the medium in one litre of distilled water with 10 ml. of glycerol. Leave to stand for 5 minutes and mix well until a uniform suspension is obtained. Heat with gentle agitation and boil for one minute. Sterilize to 121°C (15 lbs. sp.) for 15 minutes. Cool to 45-50 °C and add 10% sterile defibrinated blood, homogenize and pour into Petri plates.

The medium can be made selective by aseptically adding 0,25 units of penicillin/ml.

Uses

Pour 30-40 ml. of the complete medium into each Petri dish and keep them in a humid environment. Plates should have a bright cherry red colour. Use two plates per patient, one plate with penicillin and another without it. Once inoculated by using a swab or platinum loop incubate the plates at 37°C for 48-72 hours in a humid environment.

Colonies of *B. pertussis*, after 48-72 hours, are almost transparent, with an unclear edge, smooth, slightly elevated and shiny, less than 1 mm. in diameter.

Colonies of *B. parapertussis* grow faster and at 48 hours are well developed with a similar appearance to *B. pertussis* giving a blackish-green tint to the medium. Colonies of Gram-positive cocci usually are opaque and darker.

All suspect colonies should be identified by serological methods.

Bibliography

Bordet, J. y Gengou, O. Ann.Inst. Pasteur 20, 731-741 American Public Health Association (1963) "Diagnostic Procedures and Reagents" 4th Ed. APHA Inc., New York p. 150, 294-5.

Microbiological Test

Microorganisms	Growth
<i>Bordetella bronchiseptica</i> ATCC 4617	Satisfactory
<i>Bordetella pertussis</i> ATCC 8467	Satisfactory
<i>Bordetella parapertussis</i>	Satisfactory

BRAIN HEART INFUSION AGAR

Cat. 1048

Recommended for the development of fastidious microorganisms

Formula in grams per liter

<i>Peptone mixture</i>	10,00	<i>Beef Heart Infusion</i>	10,00
<i>Calf Brain Infusion</i>	7,50	<i>Sodium Chloride</i>	5,00
<i>Dipotassium Phosphate</i>	2,50	<i>Dextrose</i>	2,00
<i>Bacteriological Agar</i>	15,0		

Final pH 7,4 ± 0,2 at 25°C

Preparation

Suspend 52 grams of the medium in one litre of distilled water. Mix well. Heat with frequent agitation and boil for one minute. Dispense and sterilize at 121°C (15 lbs. of pressure) for 15 minutes. Before using the medium swirl gently to distribute the possible precipitate. To prepare a selective medium for fungi, the sterilized and melted medium should be cooled to 50°C, before adding the appropriate antibiotics.

Occasionally a small amount of sediment may appear which should be resuspended with a gentle swirl before dispensing.

Uses

For the cultivation of fastidious microorganisms. Brain Heart Infusion Agar (BHIA) is a solid medium rich in nutrients, suitable for the cultivation of several fastidious strains of bacteria, fungi, and yeasts. Brain Heart Infusion Agar is used for the cultivation of a wide variety of microorganisms such as *Streptococcus* and *Pneumococcus*.

If 10% sterile defibrinated blood is added, the medium can be used for the cultivation and isolation of *Histoplasma capsulatum*. With the addition of antibiotics the medium can be used for the isolation of fungi.

Brain Heart Infusion Agar with cycloheximide and chloramphenicol is recommended for the isolation of fungi difficult to grow such as *H. capsulatum* and *Blastomyces*.

Occasionally BHIA plates are used for general sensitivity tests. However it is not suitable to determine haemolytic reactions as this medium has a high dextrose concentration and it may give atypical readings.

Bibliography

Creitz and Pucket A.J. Clin. Path 24:1318, 1954. Golding and Davidson Modern, Hospital, 92:April 1954

Microbiological Test

Microorganisms	Growth
<i>Neisseria meningitidis</i> ATCC 13090	Good
<i>Streptococcus pneumoniae</i> ATCC 6303	Good
<i>Streptococcus pyogenes</i> ATCC 19615	Good
<i>Aspergillus niger</i> ATCC 16404	Good

BRAIN HEART INFUSION BROTH

Cat. 1400

Used for the culture of pathogenic cocci and other microorganisms

Formula in grams per liter

<i>Gelatin peptone</i>	10,00	<i>Beef Heart Infusion</i>	10,00
<i>Calf Brain Infusion</i>	7,50	<i>Sodium Chloride</i>	5,00
<i>Disodium Phosphate</i>	2,50	<i>Dextrose</i>	2,00

Final pH 7,4 ± 0,2 at 25°C

Preparation

Suspend 37 grams of the medium in one litre of distilled water. Heat slightly if needed until complete dissolution. For blood cultures add 0,5 to 1,0 grams of agar per liter of rehydrated medium. Boil for one minute. Dispense and sterilize in autoclave at 121°C (15 lbs. of pressure) for 15 to 20 minutes. For best results the medium should be used on the same day, or boiled or heated for a few minutes, then left to cool before using.

Uses

For the cultivation of fastidious germs. Brain Heart Infusion Broth is a liquid medium very rich in nutrients and especially used for the cultivation of fastidious organisms difficult to grow like *streptococci*,

pneumococci, and *meningococci*. It is very useful for blood cultures.

This medium is very versatile and supports the growth of many fastidious organisms. Adding 0,1% of agar reduces the flow of convection currents of oxygen and encourages the development of anaerobes and microaerophiles.

The liquid medium should be used the same day of the preparation. If not, heat in a boiling water bed to expel the dissolved oxygen.

Bibliography

Chapman. Trans. N.Y. Acad. Science. 9:52, 1946. Newman. J. Milk and Food Technol. 13:226, 1950.
Roseburg, Epps, and Clark. J. Infection Diseases, 74:131, 1944.
APHA Diagnostic Procedures and Reagents. 3rd Edition, 1951.

Microbiological Test

Microorganisms	Growth
<i>Neisseria meningitidis</i> ATCC 13090	Satisfactory
<i>Streptococcus pneumoniae</i> ATCC 6303	Satisfactory
<i>Streptococcus pyogenes</i> ATCC 19615	Satisfactory
<i>Brucella abortus</i> ATCC 4315	Moderate

BRILLIANT GREEN AGAR

Cat. 1078

Highly selective medium for the isolation of *Salmonella*

Formula in grams per liter

Peptone mixture	10,00	Lactose	10,00
Sucrose	10,00	Sodium chloride	5,00
Yeast extract	3,00	Phenol red	0,08
Brilliant green	0,0125	Bacteriological Agar	20,00

Final pH 6,9 ± 0,2 at 25°C

Preparation

Suspend 58 grams of the medium in one litre of distilled water. Mix well and heat with frequent agitation. Boil for one minute. Dispense and sterilize at 121°C (15 lbs. sp.) for 15 minutes. Cool the medium to 45-50°C pour into Petri plates, and if necessary, leave to dry about 2 hours with the covers partially removed.

Uses

As this medium is very inhibitor, inoculate the plates with a loop fully loaded with the material under study. At the same time inoculate other selective media that are less inhibitive such as Desoxycholate Agar, *Salmonella Shigella* Agar, XLD Agar, MacConkey Agar, EMB Agar, Hektoen Enteric Agar. When there is a suspicion that the material under study contains low concentrations of *Salmonella*, it is necessary to initially inoculate the sample in Tetrathionate Broth or Selenite Cystine Broth.

The medium, which has a coffee color at the beginning, changes to red during the incubation at 37°C. The germs which degrade the lactose are completely inhibited, and some of the not inhibited strains present green-yellow colonies, opaque and surrounded by a yellowish halo. The lactose negative microorganisms, such as *Salmonella* and *Proteus* form colonies of a pale pink color, transparent and surrounded by a brilliant red halo. Some *Proteus* form red colonies.

Bibliography

American Public Health Association. Standard Methods for the Examination of Water and Waster water, 11th Edition APHA, New York, 1960. American Public Health Association. Recommended Methods for the Microbiological Examination of Foods, APHA, Inc. New York, 1958.

Microbiological Test

Microorganisms	Growth	Colony colour
<i>Escherichia coli</i> ATCC 25922	Inhibited-moderate	Yellow-green
<i>Salmonella enteritidis</i> ATCC 13076	Good	Pink-white
<i>Staphylococcus aureus</i> ATCC 25923	Inhibited	----
<i>Salmonella typhi</i> ATCC 19430	Inhibited-moderate	Red
<i>Salmonella typhimurium</i> ATCC 14028	Good	Pink-white

BRILLIANT GREEN BILE AGAR

Cat. 1010

For the determination of the degree of contamination by coliforms in drinking water and wastewater

Formula in grams per liter

Brilliant Green.....	29,50 mcg	Gelatin Peptone.....	8,25
Lactose.....	1,90	Basic Fuchsin.....	0,077
Erioglaucine.....	0,0649	Ferric Chloride.....	0,0295
Sodium Sulfite.....	0,025	Monopotassium Phosphate.....	0,0153
Ox bile.....	0,00295	Bacteriological Agar.....	10,15

Final pH 6,9 ± 0,2 at 25°C

Preparation

Suspend 20,6 grams of the medium in one litre of distilled water. Soak for 5-10 minutes to hydrate correctly the agar. Heat with frequent agitation and boil for one minute. Sterilize in the autoclave at 121°C (15 lbs. sp.) for 15 minutes. Cool to 45-50°C and pour into Petri dishes.

Uses

Brilliant Green Bile Agar can be used to assess the degree of contamination of water samples, of diverse foods as well as in other products. For the enumeration of *coliform* bacteria you should employ sample dilutions which yield between 10-50 colonies per plate using the pour plate method. Therefore, several dilutions should be made in the melted medium, poured and once they have jellified, incubated at 35-37°C for 17-19 hours.

The coliform colonies have an intensely red center zone surrounded by a pink halo sharply outlined against the uniformly blue background of the medium.

The medium is sensitive to light, which reduces its effectiveness and changes its color from strong blue to purple or pink. The medium should be prepared immediately before use and, if necessary, stored in the dark for as little time as possible.

Bibliography

Methods for the Examination of Water and Wastewater, 10th Ed
APHA, Inc. New York, 1958.
Recommended Methods for the Microbiological Examination of
Foods, APHA, Inc. New York, 1958.

Microbiological Test

Microorganisms	Growth	Colony colour
<i>Escherichia coli</i> ATCC 25922	Good	Red
<i>Salmonella enteritidis</i> ATCC 13076	Good	Colourless
<i>Staphylococcus aureus</i> ATCC 25923	----	----
<i>Enterobacter aerogenes</i> ATCC 13048	Good	Pink

BRILLIANT GREEN BILE BROTH 2%

Cat. 1228

For the detection of coliforms of sanitary importance

Formula in grams per liter

Dehydrated Ox Bile	Lactose
Gelatin Peptone.....	Brilliant Green.....
20,0	10,0
10,0	0,0133

Final pH 7,2 ± 0,2 at 25°C

Preparation

Suspend 40 grams of the medium in one litre of distilled water. Heat with frequent agitation until complete dissolution. Dispense in volumes of 10 ml. in test tubes with gas collecting tubes (Durham) when the sample has 1 ml. or less volume. To analyze samples of 10 ml. of product, dissolve 80 grams of the medium in a liter of distilled water, distribute in the same manner. In both cases, sterilize at 121°C (15 lbs. of pressure) for 15 minutes. **DO NOT OVERHEAT.**

Uses

Brilliant Green Bile Broth 2% is a selective medium recommended by APHA for the cultivation of coliforms in drinking water, waste water, milk and dairy products, and other products of sanitary concern.

The brilliant green and the bile inhibit the development of coliforms accompanying flora, it also stops the growth of the anaerobes lactose fermenters such as *Clostridium perfringens* which could give false positive reactions at 44°C. The presence of gas after incubation for 24 to 48 hours is considered a positive test for the coli-enterobacter group. It is recommended to incubate at 32-35°C, preferably at 32°C for milk analysis.

Bibliography

Standard Methods for the Examination of Water and Sewage, 9th. Edition 195, 1946. Standard Methods for the Examination of Dairy Products, 9th. Edition 152, 1948.

Microbiological Test

Microorganisms	Growth	Gas
<i>Escherichia coli</i> ATCC 25922	Good	+
<i>Enterobacter aerogenes</i> ATCC 13048	Good	+
<i>Staphylococcus aureus</i> ATCC 25923	Inhibited	----
<i>Streptococcus faecalis</i> ATCC 19433	Inhibited	----

BRILLIANT GREEN SELENITE BROTH

Cat. 1221

Used for the selective enrichment of *Salmonella* species

Formula in grams per liter

<i>Yeast Extract</i>	5,00	<i>Gelatin Peptone</i>	5,00
<i>D-Mannitol</i>	5,00	<i>Sodium Selenite</i>	4,00
<i>Dipotassium Phosphate</i>	2,65	<i>Monopotassium Phosphate</i>	1,02
<i>Sodium Taurocholate</i>	1,00	<i>Sodium Sulfapyridine</i>	0,50
<i>Brilliant Green</i>	0,005		

Final pH 7,4 ± 0,2 at 25°C

Preparation

Suspend 24 grams of the medium in one litre of distilled water. Mix well. Heat slowly until completely dissolved. Dispense into sterile containers. DO NOT STERILIZE THE MEDIUM. Keep refrigerated at 4°C in the dark, it is not recommended to store it longer than 8 days. Once prepared use as soon as possible.

Uses

Once made the pre-enrichment in bottles, pass 10 ml to Brilliant Green Selenite Broth. Incubate at 37°C for 48 hours. After 24 hours subculture to plated media such as Brilliant Green Agar, Desoxycholate Citrate Agar (DCA)

and Hektoen Enteric Agar (HE Agar) to obtain isolated colonies. Incubate these plates at 37°C for 48 hours.

Repeat the subculture to plated selective media after 48 hours of incubation of the enrichment broth. Observe the plated media after 24 and 48 hours, keeping in mind the appearance and color of colonies on each medium.

Bibliography

International Standard. ISO 3565. (1975).
Meal and Meat Products-Detection of *Salmonella* (Reference Method). ISO 3565 (1975).

R: 22/22/23 Toxic by inhalation and swallowing
Danger of accumulative effects

S:23/45 Do not inhale vapors. In case of an accident or uneasiness visit the doctor immediately. Show the label if possible

	Brilliant Green Agar	D C A	H E AGAR
<i>Salmonella</i>	Pink to red with a red halo	Colourless to pale pink at 18 hours. As they grow larger, opaque with gray to black center as incubation time increases	Greenish-blue. Centers may or may not be black
<i>Shigella</i>	No growth	Initially colourless, then pale pink	Greenish, moist, convex

Microbiological Test

Microorganisms	Inoculum Concentration	Growth	
		6 hours	24 hours
<i>Escherichia coli</i> ATCC 25928	approx. 99%	< 30%	< 5%
<i>Salmonella typhimurium</i> ATCC 14028	approx. 1%	> 70%	> 95%

BRILLIANT GREEN TETRATHIONATE BILE BROTH (EUR. PHARM.) Cat. 1253

Medium for the selective enrichment of *Salmonella* in food, faeces.

Formula in grams per liter

<i>Calcium Carbonate</i>	20,00	<i>Potassium Tetrathionate</i>	20,00
<i>Meat peptone</i>	8,60	<i>Ox Bile</i>	8,00
<i>Sodium Chloride</i>	6,40	<i>Brilliant Green</i>	0,07

Final pH 7,0 ± 0,2 at 25°C

Preparation

Suspend 63 grams of the dehydrated medium in one litre of distilled water. Heat with a gentle frequent agitation until complete dissolution, but without boiling. Pour into adequate containers homogenizing the medium well enough to distribute the calcium carbonate. **DO NOT STERILIZE IN AUTOCLAVE.** The growth of *Proteus* is inhibited by taking the pH to 6,5 or also by adding Novobiocine at 0,4%.

Uses

This is a medium recommended by the European Pharmacopoeia. The Ox Bile inhibits the development of

Gram + Bacteria and allows the development of intestinal bacteria.

Once the medium has been inoculated incubate at 37°C. Make the investigation during the following days.

Proteus development is inhibited by lowering the pH to 6,6 or by adding Novobiocine 0,4%.

Bibliography

European Pharmacopoeia. 2002 4th. Edition.
Microbiological examination of non sterile products PS 137-140.

Microbiological Test

Microorganisms	Concentration inoculum	Growth: 6-24 hours	
<i>Escherichia coli</i> ATCC 25922	approx. 99%	< 30%	< 5%
<i>Salmonella typhimurium</i> ATCC 14028	approx. 1%	> 70%	> 95%

BRUCELLA AGAR

Cat. 1012

For the cultivation of *Brucella* in diverse clinical specimens, foods, and other materials of sanitary interest.

Formula in grams per liter

<i>Meat Peptone</i>	10,00	<i>Casein Peptone</i>	10,00
<i>Sodium Chloride</i>	5,00	<i>Yeast Extract</i>	2,00
<i>Dextrose</i>	1,00	<i>Sodium Bisulfite</i>	0,10
<i>Bacteriological Agar</i>	15,00		

Final pH 7,0 ± 0,2 at 25°C

Preparation

Suspend 43 grams of the medium in one litre of distilled water. Mix well and heat with frequent agitation. Boil for one minute or until the medium dissolves completely. Sterilize in the autoclave at 121°C (15 lbs. sp.) for 15 minutes. Pour into petri dishes. Once solidified, invert the plates to dry up excess moisture.

Uses

Rich in nutrients and growth factors, it is very suitable to grow and isolate fastidious microorganisms. It is used extensively to isolate *Brucella* from materials contaminated with other bacteria and for the production of clostridial toxins.

Successfully used to isolate *Brucella* from diverse specimens contaminated with microflora, both saprophytes and commensals, in clinical samples as well as in foods. It can also be utilized in the isolation of many anaerobes both of human and animal origin. Food samples in study (milks, creams, meats, viscera, etc.) can be inoculated directly on the plates of *Brucella* Agar, while suspensions or macerations in sterile saline solution of clinical specimens such as organs, feces, scrapings of vaginal mucous, etc., are more convenient. Inoculations should be made in duplicate - one plate incubated at the desired temperature and one plate in CO₂.

Brucella Agar can be made selective to yield higher numbers of positive isolations by adding 1,4 mg/l of ethyl violet and the following antibiotic package:

<i>Polymixin B Sulfate</i>	6000 U/l
<i>Cycloheximide (Actidione)</i>	100 mg/l
<i>Bacitracin</i>	100 mg/l

If you need to restrict swarming of *Proteus* add 2-3 g/l of bacteriological agar to the medium.

Note: For an excellent medium for anaerobes, add 5% sterile sheep blood, 5 mcg/ml. of hemin and 10 mcg/ml. of Vitamin K3 (menadione) to the basal medium, culture and incubate under anaerobic conditions. This blood enriched *Brucella* Agar will be selective if antibiotics or other inhibitory agents such as oxbile (2 g/l) are added.

Bibliography

Kzudas and Morse, J. Bact. 66:502, 1953
Rennoux G. Ann. Inst. Pasteur, 87:325, 1954
Standard Methods for the Examination of Dairy Products. 10th Ed. APHA, Inc. New York, 1960
Smith Louis Ds. The Pathogenic Anaerobic Bacteria. C. Thomas Pub., Springfield, Il, 1975.
Koneman, Allen, Dowell, and Sommers. Color Atlas and Textbook of Diagnostic Microbiology, J.B. Lippincott, Philadelphia, 1979.

Microbiological Test

Microorganisms	Growth
<i>Brucella abortus</i> ATCC 4315	Good
<i>Brucella melitensis</i> ATCC 4309	Good
<i>Brucella suis</i> ATCC 4314	Good

BRUCELLA BROTH

Cat. 1223

For the cultivation of *Brucella* from diverse materials of medical and sanitary interest.

Formula in grams per liter

<i>Meat Peptone</i>	10,00	<i>Casein Peptone</i>	10,00
<i>Sodium Chloride</i>	5,00	<i>Yeast Extract</i>	2,00
<i>Dextrose</i>	1,00	<i>Sodium Bisulfite</i>	0,10

Final pH 7,0 ± 0,2 at 25°C

Preparation

Suspend 28 grams of the medium in one litre of distilled water. Mix well. Heat with frequent agitation and boil one minute until completely dissolved. Dispense and sterilize in the autoclave at 121°C (15 lbs. sp.) for 15 minutes.

0

Uses

Brucella Broth is used to cultivate *Brucella* and other bacteria from clinical material, foods and other materials of sanitary importance

It is a medium rich in nutrients and growth factors, excellent to grow fastidious microorganisms.

It is used extensively to isolate *Brucella* from mixed flora and for clostridial toxin production. It can also be used in blood culture bottle systems.

Brucella species are level 3 pathogens and cause brucellosis a zoonotic disease with a domestic animal-reservoir. It is usually transmitted through milk, dairy products, meat and direct contact with infected animals.

Bibliography

Isenberg, H.D. (ed.) 1992. Clinical microbiology procedures handbook. American Society for Microbiology, Washington, D.C.
 Hausler, W.J. (ed.). 1976. Standard methods for the examination of dairy products, 14th ed. American Public Health Association, Washington, D.C.

Microbiological Test

Microorganisms	Growth
<i>Brucella abortus</i> ATCC 4315	Good
<i>Brucella melitensis</i> ATCC 4309	Good
<i>Brucella suis</i> ATCC 4314	Good

BRYANT- BURKEY BROTH BASE

Cat. 1247

Medium for enumeration in milk and of lactate fermenters *Clostridium* spores, dairy products particularly used for detecting *Clostridium tyrobutyricum* responsible for the "late cheese spoilage"

Formula in grams per liter

<i>Tryptone</i>	15,00	<i>Beef Extract</i>	7,50
<i>Yeast Extract</i>	5,00	<i>Sodium Acetate</i>	5,00
<i>L- Cysteine</i>	0,50	<i>Resazurin</i>	0,0025

Final pH 5,9 ± 0,2 at 25°C

Preparation

Suspend 33,0 grams of the dehydrated medium in one litre of distilled water. Add 10 ml of 50% sodium lactate. Heat with frequent agitation until complete dissolution. Dispense in tubes and sterilize at 121°C (15 lbs. sp.) for 15 minutes.

Uses

This medium is used for *Clostridium tyrobutyricum* detection, which is the bacteria that causes the "late cheese expoliage". To count the bacteria use the most probably number method (MPN), considering positive the

tubes with growth and gas production. Read results after incubation at 37°C ± 2°C for 7 days.

Bibliography

BRYANT M.P. and BURKEY L.A: 1956. The characteristics of lactate-fermenting sporeforming anaerobes from silage. J. Bact., 43-46 CERF. O. et BERGERE J.L. 1968. Numeration des spores de *Clostridium* et son application au lait et aux produits laitiers. Numeration des différents groupes de *Clostridium*. Le lait, 48, 501-519.

Microbiological Test

Microorganisms	Growth	Gas production
<i>Clostridium tyrobutyricum</i> EMD 132	Satisfactory	+
<i>Clostridium perfringens</i> ATCC 10543	Satisfactory	none
<i>Staphylococcus aureus</i> ATCC 25923	Moderate	none
<i>Pseudomonas aeruginosa</i> ATCC 27853	----	----

BUFFERED PEPTONE WATER

Cat. 1402

Recommended as a diluent for the homogenization of samples in microbiological analysis of food.

Formula in grams per liter

<i>Bacteriological Peptone</i> 10,00	<i>Disodium Phosphate</i>9,00
<i>Sodium Chloride</i> 5,00	<i>Monopotassium Phosphate</i> 1,50

Final pH 7,0 ± 0,2 at 25°C

Preparation

Dissolve 25,5 grams of the medium in one litre of distilled water. Mix well. Distribute into appropriate containers and sterilize at 121°C (15 lbs. sp.) for 15 minutes.

Uses

This medium is recommended as a diluent for the homogenization of food samples containing suspected contaminants such as *Salmonella*, etc. Changes in pH may cause damages to bacteria growth. This media

maintains a high pH due to its phosphates content. This medium complies with the recommendations of the International Standard Organization ISO (1933) and the German DIN Regulations 10181 and 10160 for the examination of milk, meat and meat products.

Bibliography

M.R. Pascual Anderson (1982) Techniques for Microbiological Analysis of Foods and Drinks, CeNAN.

Microbiological Test

Microorganisms	Growth
<i>Salmonella enteritidis</i> ATCC 13076	Satisfactory
<i>Salmonella typhi</i> ATCC 19430	Satisfactory
<i>Salmonella typhimurium</i> ATCC 14028	Satisfactory

BUFFERED PEPTONE WATER (EUROPEAN PHARMACOPOEIA) Cat. 1401

Recommended as a diluent for the homogenization of samples in
microbiological analysis of food

Formula in grams per liter

Disodium Phosphate..... 7,23	Sodium Chloride 4,30
Monopotassium Phosphate..... 3,56	Bacteriological Peptone..... 1,00

Final pH 7,0 ± 0,2 at 25°C

Preparation

Suspend 16,1 grams of the medium in one litre of distilled water. Mix well. Distribute into appropriate recipients and sterilize at 121°C (15 lbs. sp.) for 15 minutes.

Uses

A feature common to all selective media is that sublethally injured organisms are not detected, as they are relevant for the quality of foods a resuscitation must be included in examination procedures.

If the product to be examined has antimicrobial activity this must be adequately neutralized. This medium is recommended by the European Pharmacopoeia as a

diluent for the homogenization of food samples for microbiological analysis.

Before sterilizing the medium it can be added from 1 to 10 gr/l of polysorbate (20 or 80)., which will act as a surface-active agent or inactivator of antimicrobial agents.

Bibliography

European Pharmacopoeia 4 th Edition 2002. 126-138.

Microbiological Test

Microorganisms	Growth
<i>Salmonella enteritidis</i> ATCC 13076	Satisfactory
<i>Salmonella typhi</i> ATCC 19430	Satisfactory
<i>Salmonella typhimurium</i> ATCC 14028	Satisfactory
<i>Staphylococcus aureus</i> ATCC 6538P	Satisfactory

CALCIUM CASEINATE AGAR

Cat. 1069

Selective medium for recount and research of proteolytic microorganisms in foods

Formula in grams per liter

<i>Meat Peptone</i>	5,00	<i>Sodium Chloride</i>	5,00
<i>Beef Extract</i>	3,00	<i>Casein (Hammarsten)</i>	2,50
<i>Calcium Hydroxide</i>	0,15	<i>Bacteriological Agar</i>	13,50

Final pH 7,2 ± 0,2 at 25°C
Preparation

Suspend 29 grams of the medium in one litre of distilled water. Mix well. Heat and boil agitating frequently until complete dissolution. Sterilize in autoclave at 121°C (not more than 15 lbs. sp.) for 15 minutes. Pour into Petri dishes shaking the medium to mix well the resulting precipitate.

Uses

This medium contains casein, which is degraded by proteolytic organisms thus forming clear zones surrounding the colonies. The finished medium is turbid especially if 5-10 g/l of powdered milk is added. Colonies of proteolytic organisms are easily recognized by the clearing zone around them.

Inoculation can be made by streaking the surface of the plate or by the pour plate method. Incubation is usually for 2-3 days.

Count only the colonies with clearing zones. Covering the surface of the plate with 5-10% acetic acid can improve differentiation of colonies.

Bibliography

Frazier, W.C., a. RUPP, P.: Studies on the proteolytic bacteria of milk. A. medium for the direct isolation of caseolytic milk bacteria. J. Bact. 16 57-63 (1928).

Microbiological Test

Microorganisms	Growth	Transparence halo (clearing)
<i>Bacillus cereus</i> ATCC 11778	Good	+
<i>Pseudomonas aeruginosa</i> ATCC 27853	Good	+
<i>Proteus vulgaris</i> ATCC 13315	Good	-
<i>Escherichia coli</i> ATCC 25922	Good	-
<i>Enterobacter cloacae</i> ATCC 13047	Good	-

CARY BLAIR MEDIUM

Cat. 1529

Transport medium recommended for the collection and transport of clinical specimens

Formula in grams per liter

Sodium Chloride.....	5,00	Sodium Thioglycollate.....	1,50
Disodium Phosphate.....	1,10	Calcium Chloride.....	0,09
Agar N° 2.....	5,50		

Final pH 8,4 ± 0,2 at 25°C

Preparation

Suspend 13,2 grams of the medium in one litre of distilled water. Mix well. Heat with frequent agitation and boil slowly until completely dissolved. Dispense into screw-capped test tubes and place in flowing steam for 15 minutes. Allow to cool at room temperature and tighten the caps to avoid water loss.

Uses

Cary-Blair medium has a low nutrient content and a phosphate buffer system (in place of glycerophosphate) which inhibits the massive growth of strains such as *Escherichia coli* and *Klebsiella aerogenes*. These organisms possess specific dehydrogenases that break down sodium glycerophosphate.

This medium has a low oxidation/reduction potential, which assures bacterial survival for long periods of time.

Cary-Blair Medium has been described as especially good for epidemiological studies of *Vibrio parahaemolyticus* for long term survival (up to 35 days at temperatures from 22-31°C) of rectal swabs. Long recovery times have been reported for *Pasteurella pestis* as well as for *salmonellas* and *shigellas*.

Cotton swabs are used for the collection of the samples, placed in the transport medium tube to the bottom of the tube and the excess is cut off to allow for cap closure.

Bibliography

Cary, S.G. and E.B. Blair 1964. New transport medium for shipment of clinical specimens. J. Bacteriol.
Cary, S.G., M.S. Mathew, M.H. Fusillo, and C. Hasking 1965 Survival of Shigella and Salmonella in a new transport medium. Am. J. Clin. Path.

Microbiological Test

Microorganisms	Growth
<i>N. meningitis</i> ATCC 13090	Satisfactory
<i>N. gonorrhoeae</i> ATCC 19424	Satisfactory
<i>St. pneumoniae</i> ATCC 6301	Satisfactory
<i>Shigella flexneri</i> ATCC 12022	Satisfactory
<i>Bordetella pertussis</i> ATCC 9340	Satisfactory
<i>Haemophilus influenzae</i> ATCC 19418	Satisfactory

CETRIMIDE AGAR BASE

Cat. 1102

Medium for the selective isolation and identification of *Pseudomonas aeruginosa*.

Formula in grams per liter

<i>Gelatin Peptone</i>	20,00	<i>Potassium Sulfate</i>	10,00
<i>Magnesium Chloride</i>	1,40	<i>Cetrimide</i>	0,30
<i>Bacteriological Agar</i>	13,60		

Final pH 7,2 ± 0,2 at 25°C

Preparation

Suspend 45,3 grams of the medium in one litre of distilled water. Add 10 ml of glycerol. Heat agitating frequently, and boil for one minute. Dispense and sterilize and autoclave at 118 to 121°C (12-15 lbs. sp.) for 15 minutes.

Uses

Cetrimide Agar Base promotes the production of pyocyanin a water soluble pigment as well as fluorescence, under ultraviolet light, of *Pseudomonas aeruginosa*, which constitutes a presumptive identification. Cetrimide is the selective agent as it inhibits the growth of the accompanying microbial flora. Typical *P. aeruginosa*

colonies are greenish or yellowish green in color. Pyorubin-producing strains form reddish colonies. The identification is completed by performing the oxidase test. Inoculate the plates by spreading the sample and incubate aerobically up to 48 hours at 35° C.

Bibliography

King, Ward and Raney. J. Lab. and Clin. Med. 44:301, 1954.
Brown and Lowbury. J. Clin. Path. 18:752, 1965.
Lowbury. J. Clin. Path. 4:66, 1951. Lowbury and Collins. J. Clin. Path. 8:47, 1955.

Microbiological Test

Microorganisms	Growth
<i>Escherichia coli</i> ATCC 25922	Inhibited
<i>Pseudomonas aeruginosa</i> ATCC 27853	Satisfactory
<i>Staphylococcus aureus</i> ATCC 25923	Inhibited

CHAPMAN STONE AGAR

Cat. 1017

Selective and differentiation medium to isolate staphylococci in foods

Formula in grams per liter

Ammonium Sulfate.....	75,00	Sodium Chloride	55,00
Gelatin	30,00	Casein Peptone	10,00
Mannitol.....	10,00	Yeast Extract.....	2,00
Dipotassium Phosphate.....	5,00	Bacteriological Agar.....	15,00

Final pH 7,0 ± 0,2 at 25°C

Preparation

Suspend 202 grams of the medium in one litre of distilled water. Mix well. Heat with frequent agitation and boil until dissolved. Sterilize at 121°C (15 lbs. sp.) for 10 minutes. Pour into Petri dishes.

Uses

Chapman Stone Agar is used similarly to Staphylococcus N° 110 Agar but contains ammonium sulfate to detect the gelatinase activity (Stone reaction). The medium is opaque white.

The samples suspected of containing pathogenic staphylococci are inoculated heavily and incubated from 30-32°C for 48 hours.

Any pigmented colony (yellow or weakly orange) which is surrounded by a clear zone is probably a pathogenic *staphylococcus* causing poisoning by contaminated foods. It is recommended to pick the colony and emulsify in Brain Heart Infusion Broth (0,1-0,2 ml.) and perform the coagulase test.

At the same time it is convenient to add a drop of bromocresol purple to the colony site to determine mannitol fermentation: a yellow color formation is a positive reaction. The zones or clear halos around the colonies indicate degradation by the enzyme gelatinase (gelatin proteolysis).

The *staphylococcal* colonies which cause food poisoning by ingestion of the enterotoxin which they produce are yellow, yellow-gold or orange, ferment mannitol, are coagulase-positive, produce beta-hemolysis in media such as Blood Agar and are gelatinase-positive (positive Stone reaction). Pale colonies, practically lacking in color, not producing pigment, should not be considered as positives, even if they are surrounded by a clear zone (halo).

Bibliography

Chapman J. Bact. 1945, 50: 201 Recommended Methods for the Microbiological Examination of Foods APHA. Inc. New York 1958. Standards Methods for Examination of Dairy Products, 1st Ed. APHA. Inc. New York, 1960.

Microbiological Test

Microorganisms	Growth	Halo
<i>Escherichia coli</i> ATCC 25922	Inhibited	-
<i>Staphylococcus epidermidis</i> ATCC 12228	Satisfactory	+
<i>Staphylococcus aureus</i> ATCC 25923	Satisfactory	+

CHLORAMPHENICOL AGAR

Cat. 1301

Selective medium to isolate and count moulds in milk and dairy products

Formula in grams per liter

Dextrose.....	20,00	Yeast Extract	5,00
Cloramphenicol.....	0,10	Bacteriological Agar	12,00

Final pH 6,6 ± 0,2 at 25°C

Preparation

Suspend 37,1 grams of the medium in one litre of distilled water. Mix well to obtain an homogeneous suspension. Heat with frequent agitation and boil for one minute until completely dissolved. Distribute and sterilize at 121°C (15 lbs. sp.) for 15 minutes. **DO NOT OVERHEAT** as it will facilitate the hydrolysis of the components and the medium will remain soft.

Uses

This medium is recommended by International Dairy Federation (FIL-IDF), International Organization for Standardization (ISO), and DIN for isolation and

enumeration of yeasts and moulds in milk and dairy products.

The presence of Cloramphenicol inhibits most of contaminant bacteria in the medium.

Bibliography

FIL-IDF(1991) Standard 94B. Enumeration of yeast and moulds. Colony Count Technique at 25°C.
 ISO (1981) ISO/DIS 6611: Milk and Milk products: Enumeration of yeast and moulds colony counts technique at 25°C.
 DIN Standard 10186. Mikrobiologische Milch Untersuchung. Bestimmung der Anzahl von Hefen und Schimmelpilzen

Microbiological Test

Microorganisms	Growth
<i>Candida albicans</i> ATCC 2091	Satisfactory
<i>Candida tropicalis</i> ATCC 750	Satisfactory
<i>Escherichia coli</i> ATCC 25922	Inhibited
<i>Staphylococcus aureus</i> ATCC 25923	Inhibited

CLED AGAR (CYSTINE LACTOSE ELECTROLYTE DEFICIENT) Cat. 1016

For the cultivation of gram positive and gram negative urinary tract bacteria.
It inhibits the Proteus swarming

Formula in grams per liter

Lactose	10,00	Casein Peptone	4,00
Gelatin Peptone	4,00	Beef Extract	3,00
L-Cystine	0,128	Bromothymol Blue	0,02
Bacteriological Agar	15,00		

Final pH 7,3 ± 0,2 at 25°C

Preparation

Suspend 36 grams of the medium in one litre of distilled water. Soak 10-15 minutes and mix well. Heat slowly while stirring frequently boil for a minute. Sterilize in the autoclave at 121°C (15 lbs. of sp.) for 15 minutes. Pour into Petri dishes. When the medium is solidified, invert the plates to avoid excess moisture.

Uses

CLED Agar is a non selective differential plating medium for the growth and enumeration of urinary tract microorganisms. Omitting sodium chloride inhibits the Proteus swarming and supports the growth of a great majority of bacteria causing urinary tract infections and is used to differentiate and identify them. The presence of bacterial contaminants like *diphtheroids*, *lactobacilli* and other microbes indicate the degree of care taken with the handling of the urine specimen.

Urinary cultures should be performed with the first early morning sample after careful cleansing of the genital area. Do not use the first portion of the urine stream but collect the sample from the midstream. The microorganisms which cause infection in the urinary tract are generally abundant and of only one species. *E. coli* is the organism most frequently isolated. The seeding of the sample can be made by the dilution method or by

streaking on the surface of agar with a calibrated loop. Count the colonies after 18 hours of incubation at 35°C. Report the number of colonies per ml. of urine. Remember that a count of 100.000 (10)⁵/ml. or more is an indication of a significant clinical urinary tract infection.

CHARACTERISTICS OF THE COLONIES *Escherichia coli*: are large, elevated yellow, opaque, with a center slightly darker. The agar is yellow. *Enterobacter*: are similar to *E. coli*: are but mucoid and larger in size. Yellow agar. *Klebsiella*: are large, yellow or yellowish-white. Highly mucoid and elevated. It can present a light blue shade. Yellow agar. *Proteus*: are Blue, translucent with irregular edges. Slightly elevated. *Pseudomonas*: are Pale blue-green. Typical matte surface and irregular edges. "Sweet" odor. Blue-green agar *Salmonella*, *Shigella*, *Serratia*, and *Providencia*: are From blue to intense blue. *Streptococcus faecalis*: are Very small, from 0.4 mm, yellow, opaque. Yellow agar. *Staphylococcus*: are small, yellow intense colors, opaque. Yellow agar. *Corynebacteria*: are Very small, gray.

Bibliography

Bebis, T. D. J. Med. Lab. Technol, 26-38-41, 1968. Mackey, J. R. and Sandys, G.H. 1965. B.M.H. 1 1173. Mackey, J.R. and Sandys, G.H. 1966. B.M.H. 1 1173. Guttman, D. and Nailer G.R.E., 1967 B.M.J. 2 343-345.

Microbiological Test

Microorganisms	Growth	Colour of the medium
<i>Enterobacter aerogenes</i> ATCC 13048	Satisfactory	Light yellow-blue
<i>Escherichia coli</i> ATCC 25922	Satisfactory	Yellow
<i>Proteus vulgaris</i> ATCC 13315 (swarming inhibited)	Satisfactory	Blue-blue green
<i>Staphylococcus aureus</i> ATCC 13315	Satisfactory	Light yellow - ●
<i>Streptococcus faecalis</i> ATCC 19433	Satisfactory	Light yellow - ●

● = without changes

CLED AGAR WITH ANDRADE'S INDICATOR

Cat.1303

Modification of Cled Agar to increase the differentiation of the colonies

Formula in grams per liter

<i>Lactose</i>	10,00	<i>Gelatin Peptone</i>	4,00
<i>Casein Peptone</i>	4,00	<i>Beef extract</i>	3,00
<i>L-Cystine</i>	0,128	<i>Andrade's indicator</i>	0,10
<i>Bromothymol blue</i>	0,02	<i>Bacteriological Agar</i>	15,00

Final pH 7,5 ± 0,2 at 25°C
Preparation

Suspend 36,2 grams of the medium in one litre of distilled water. Mix well and heat to boiling with frequent agitation until completely dissolved. Distribute and sterilize at 121°C (15 lbs. Sp.) for 15 minutes. Mix well before pouring into Petri dishes.

Uses

The typical composition of this medium, is similar to Cled Agar, but with Andrade's indicator added, it improves colony detection, and microorganism identification.

Bibliography

Bevis T.D. (1968) J. Med. Lab. Technol.25,38-41. Furniss A.L., Lee J.V. and Donovan T.J. (1978) P.H.L.S. Monograph series, London, H.M.S.O.,11.

Microbiological Test

Microorganisms	Growth	Medium colour
<i>P. mirabilis</i> ATCC 10975	Satisfactory	Transparent greenish blue
<i>Escherichia coli</i> ATCC 25922	Satisfactory	Semitransparent brilliant rose
<i>Staph. aureus</i> ATCC 25923	Satisfactory	Golden yellow. Lactose's Fer
<i>Staph. albus</i> spp.	Satisfactory	White porcelain or slightly pink
<i>K. aerogenes</i> ATCC 13882	Satisfactory	Mucoids, greyish green
<i>E. Faecalis</i> ATCC 29212	Satisfactory	Intense orange yellow
<i>Strep. pyogenes</i> ATCC 19615	Satisfactory	Greenish grey, opacous and small

CLOSTRIDIUM PERFRINGENS AGAR BASE MEMBRANE FILTRATION METHOD

Cat.1132

Selective Medium Base for the enumeration and isolation of *Clostridium perfringens*

Formula in grams per liter

<i>Tryptose</i>	30,00	<i>Yeast extract</i>	20,00
<i>Sucrose</i>	5,00	<i>L-cysteine Hydrochloride</i>	1,00
<i>MgSO₄ · 7 H₂O</i>	0,10	<i>Bromcresol purple</i>	0,04
<i>Bacteriological Agar</i>	15,00		

Final pH 7,6 ± 0,3 at 25°C

Preparation

Suspend 71,14 grams of the medium in one litre of distilled water. Heat with frequent agitation and boil for one minute. Sterilize at 121°C (15 lbs sp) for 15 minutes.

Cool to 45-50°C and add:

D-Cycloserine: 400 mg.

Polymixin sulphate: 25 mg.

Indoxyl D-glucoside: 60 mg. (dissolved in 8 ml. of distilled water).

Phenolphthalein diphosphate: 20 ml. (0,5% sterile solution).

FeCl₃6H₂O diphosphate: 2 ml. (0,5% sterile solution).

Uses

The mCP agar method is a two-step membrane-filtration method for the detection of *Clostridium perfringens* (*C. perfringens*) in environmental waters.

The mCP method can be used for monitoring all types of waters. *C. perfringens* is present in large numbers in human and animal wastes, and its spores are resistant to wastewater-treatment practices, extremes in

temperature, and environmental stress. The method has been recommended for use for examination of chlorinated waters and untreated water containing industrial wastes lethal to non-sporeforming bacteria, sewage sludge, and situations in which the detection of remote as well as recent pollution is desirable.

Bibliography

Armon, R., and Payment, P., 1988, A modified m-CP medium for enumerating *Clostridium perfringens* from water samples: Canadian Journal of Microbiology, v.34, p.78-79.

Bisson, J.W., and Cabelli, V.J., 1979, Membrane filter enumeration method for *Clostridium perfringens*: Applied and Environmental Microbiology, v. 37, no.1, p. 55-66.

Microbiological Test

Microorganisms	Growth	Colony colour
<i>Clostridium perfringens</i>	Good	Opaque yellow or that change to pink or red after 20-30 seconds exposure to ammonium hydroxide vapors.

COLUMBIA AGAR BASE (EUROPEAN PHARMACOPOEIA) Cat. 1104

Used for the isolation and cultivation of fastidious microorganisms

Formula in grams per liter

<i>Casein pancreatic digest</i>	10,00	<i>Meat peptic digest</i>	5,00
<i>Yeast extract</i>	5,00	<i>Hear pancreatic digest</i>	3,00
<i>Sodium Chloride</i>	5,00	<i>Corn Starch</i>	1,00
<i>Bacteriological agar</i>	13,50		

Final pH 7,3 ± 0,2 at 25°C
Preparation

Suspend 42,5 grams of the medium in one litre of distilled water. Mix well. Heat with frequent agitation and boil for one minute. Distribute into appropriate containers and sterilize in autoclave at 121°C (15 lbs. sp.) for 15 minutes. The medium is generally enriched with defibrinated sterile blood, serum or some other material.

Uses

Columbia Agar Base is a highly nutritive general purpose medium for the cultivation of fastidious organisms, especially when supplemented with plain or chocolate blood. It can also be used as a selective isolation medium by adding antimicrobial agents. Columbia Agar Base is used extensively as a medium base for a variety of culture formulations in medical bacteriology. The hemolytic reactions in blood agar are genuinely defined. The majority of the common pathogenic bacteria, however, grow well without the addition of blood.

Columbia Agar Base is used satisfactorily in other formulas supplemented by enrichments and/or various inhibitory agents.

With the addition of 5-10% sterile defibrinated sheep, rabbit or human blood and, especially when the patient is receiving antibiotic treatment, addition of 1,0 ml. of VCN suspension and 1,0 ml. of *PRONADISA* Polyenrichment to 100 ml. of medium. Columbia Agar Base becomes an excellent chocolate agar, which can be used to isolate pathogenic *gonococci* and *meningococci*, as good or better than Thayer-Martin Medium.

Bibliography

Ellner, Stossel, Drakeford and Vasi. AM J. Clin. Path. 45:502-504, 1966.

Microbiological Test

Microorganisms	Growth	Hemolysis
<i>Neisseria meningitidis</i> ATCC 13090	Good	----
<i>Staphylococcus aureus</i> ATCC 25923	Good	Beta/Gamma
<i>Streptococcus pneumoniae</i> ATCC 6303	Good	Alpha
<i>Streptococcus pyogenes</i> ATCC 19615	Good	Beta

CTA MEDIUM (CYSTINE TRYPTICASEIN) Cat. 1502

For maintenance of strains and in motility and fermentation studies

Formula in grams per liter

Casein Peptone.....	20,00	Sodium Chloride	5,00
L-Cystine	0,50	Sodium Sulfite	0,50
Phenol Red.....	0,017	Bacteriological Agar.....	2,50

Final pH 7,3 ± 0,2 at 25°C

Preparation

Suspend 28,5 grams of the medium in one litre of distilled water. If desired, add 0,5 to 1,0% carbohydrate for specific fermentation tests. Homogenize and heat to boiling for one minute until completely dissolved. Distribute in screw-capped tubes and sterilize at 115-118°C (12 lbs pressure).for 15 minutes.

Cool in a vertical position. Store at room temperature. The medium can be stored for long periods of time in refrigeration if the tubes are tightly capped. The CTA Medium should be used right after preparation, or the tubes should be boiled with loose caps and cooled immediately before use.

Uses

The Cystine Trypticasein Medium is convenient for the preservation and determination of the motility of microorganisms difficult to cultivate. Adding carbohydrates to the medium makes it possible to determine the fermentation reactions of these microorganisms, e.g., pathogenic *Neisseria*.

The fastidious organisms such as *Neisseria*, *Pasteurella*, *pneumococci*, *streptococci*, *Brucella*, *Corynebacteria*, and *Vibrio* grow without adding carbon dioxide, serum, or any other enrichment substances.

Motility is easily determined in the semi-solid medium. The stabbed cultures of motile organisms display development

outside the line of inoculation. The non-motile microorganisms only along the inoculated stab line, while the surrounding agar remains clear.

CTA Medium is recommended especially for the differentiation of fastidious microorganisms by fermentation reactions.

For fermentation tests with members of *Neisseria*, inoculate only the surface of the tubes. The facultative microorganisms such as streptococci and strictly anaerobic microorganisms can be inoculated by stabbing at half the depth of the tube.

The acid reactions can be easily observed because the acid formed does not spread immediately throughout the entire tube. The majority of cultures display an alkaline change when there is no fermented carbohydrate present. CTA Medium is also convenient for the fermentation tests and classification of yeasts.

Bibliography

Vera J. Bact. 55:531, 1948. Peterson and Hartsell J. Inf. Dis. 96:75, 1975. Myers and Kashy AJPH. 51:1872, 1962. Alford, Wiese and Guntor. J. Bact. 69:516, 1955. Kroeger and Sibel. J. Bact. 58:270, 1949. Vera and Petran. Bull. Nati. Assin. Clin. Lab. 5:90, 1954. Fahlberg, Dukes and Gunthrio. J. Invest. Derma. 29:111, 1955.

Microbiological Test

Microorganisms	Growth	Motility
<i>Escherichia coli</i> ATCC 25922	Good	+
<i>Staphylococcus aureus</i> ATCC 25923	Good	-

CZAPEK DOX MODIFIED AGAR

Cat. 1015

Medium used for the cultivation of fungi and bacteria which use sodium nitrates as sole source of nitrogen.

Formula in grams per liter

Sucrose.....	30,00	Sodium Nitrate.....	2,00
Magnesium Glycerophosphate.....	0,50	Potassium Chloride.....	0,50
Potassium Sulfate.....	0,35	Ferrous Sulfate.....	0,01
Bacteriological Agar.....	12,00		

Final pH 6,8 ± 0,2 at 25°C

Preparation

Suspend 45,4 grams of the medium in one litre of distilled water. Mix well. Heat agitating frequently and boil until completely dissolved. Dispense into appropriate containers and sterilize by autoclaving at 121°C (15 lbs. sp.) for 15 minutes.

Uses

Czapek-Dox Modified Agar is a semi-synthetic medium which contains sodium nitrate as a sole source of nitrogen. It has the advantage of a chemically defined formulation, which has been modified in the original formula by the substitution of the magnesium sulfate and potassium phosphate for the magnesium glycerophosphate in this formula. The medium is utilized commonly for the cultivation of *fungi* and *chlamydospore* formation by *C. albicans*.

In general, the medium should be cooled to 45-50°C before pouring in order to avoid excess water moisture on the plates. Dispense approximately 12 ml. in a 90 mm. diameter Petri dish. Store the plates in an inverted position. Inoculate with a straight needle taking the precaution to invert the plates to protect the medium surface from airborne spores.

Times and temperatures of incubation vary considerably according to the type of fungi. As a general rule, incubate from 1-2 weeks at room temperature (approximately 25°C) for moulds and 24-48 hours for *C. albicans*.

Bibliography

Thom and Raper. Manual of Aspergilli. Williams and Wilkins Co., Baltimore, MD 1945.
Smith G. An Introduction to Industrial Mycology 5th Ed. Arnold LR London, 1960.

Microbiological Test

Microorganisms	Growth
<i>Aspergillus niger</i> ATCC 16404	Satisfactory
<i>Saccharomyces cerevisiae</i> ATCC 976	Null/light
<i>Bacillus subtilis</i> ATCC 6633	Moderate
<i>Candida albicans</i> ATCC 10231	Moderate
<i>Staphylococcus aureus</i> ATCC 25923	Inhibited

CZAPEK DOX MODIFIED BROTH

Cat. 1250

Medium used for the cultivation of fungi and bacteria which use sodium nitrates as sole source of nitrogen

Formula in grams per liter

Sucrose	30,00	Sodium Nitrate	3,00
Dipotassium Phosphate.....	1,00	Potassium Chloride.....	0,50
Magnesium sulfate.....	0,50	Ferrous Sulfate	0,01

Final pH 6,8 ± 0,2 at 25°C

Preparation

Suspend 35 grams of the medium in one litre of distilled water. Mix well and boil until complete dissolution. Dispense into appropriate containers and sterilize by autoclaving at 121°C (15 lbs. sp.) for 15 minutes.

Uses

Czapek-Dox Broth Modified is similar to Czapek-Dox Agar Modified, lacking the agar, and is used to grow bacteria and fungi which are capable of utilizing sodium nitrate as a sole source of nitrogen.

It has the advantage of a chemically defined formulation, which has been modified in the original formula by the substitution of the magnesium sulfate and potassium phosphate for the magnesium glycerophosphate in this formula. The medium is utilized commonly for the

cultivation of *fungi* and *chlamydospore* formation by *C. albicans*.

It is useful in a variety of microbiological procedures, including soil microbiology and fungi and mildew resistance tests. It will yield a moderately good growth of most saprophytic aspergilli and characteristic mycelia and conidia.

Bibliography

Thom y Raper. Manual of Aspergilli. Williams and Wilkins Co. Baltimore Md. 1945.
Smith G. An Introduction to Industrial Mycology 5th Ed Arnold LR London 1960.

Microbiological Test

Microorganisms	Growth
<i>Aspergillus niger</i> ATCC 16404	Satisfactory
<i>Saccharomyces cerevisiae</i> ATCC 976	Null/Slight
<i>Bacillus subtilis</i> ATCC 6633	Moderate
<i>Candida albicans</i> ATCC 10231	Moderate
<i>Staphylococcus aureus</i> ATCC 25923	Inhibited

DCLS AGAR (DESOXYCHOLATE, LACTOSE, SUCROSE) Cat. 1045

Selective medium for the isolation of gram negative enteric bacilli

Formula in grams per liter

<i>Sodium Citrate</i>	10,00	<i>Proteose Peptone</i>	7,00
<i>Sodium Thiosulfate</i>	5,00	<i>Lactose</i>	5,00
<i>Sucrose</i>	5,00	<i>Beef Extract</i>	3,00
<i>Sodium Desoxycholate</i>	2,50	<i>Neutral Red</i>	0,03
<i>Bacteriological Agar</i>	12,00		

Final pH 7,2 ± 0,2 at 25°C

Preparation

Suspend 49,5 grams of the medium in one litre of distilled water. Mix well. Heat to boiling until completely dissolved avoid overheating. **DO NOT AUTOCLAVE**. Cool to 45°-50°C and dispense into Petri dishes.

Uses

DCLS Agar is a selective medium for the primary isolation of *Salmonella* and *Shigella* from foods, feces and urine. It is also used to isolate *Vibrio cholerae*. It inhibits gram-positive growth and partially inhibits coliforms and *Proteus*.

It can be used with direct streaking or better, with enrichment in Selenite Broth, for example, for *salmonellas*. It is preferable to inoculate in duplicate; one heavily and the other diluted. Incubation is for 18-24 hours at 35-37°C with identification characteristics:

Red colonies: *P. vulgaris*, *coliforms*

Colourless or weakly pink colonies: *Salmonella*, *Shigella*

The presence of 2 sugars in the formulation assures the formation of red colonies of those organisms, which ferment one or both of the sugars.

The majority of *Shigella* organisms yield colourless colonies but some strains of *S. flexneri*, as well as other species of *Shigella*, grow rapidly giving colonies that are weakly pink, but are distinguished easily from *Proteus* or the coliforms. If *Salmonella* or *Shigella* is suspected, the colonies should be subcultured on other media for identification, such as Kligler Iron Agar, Motility Test Medium, or Triple Sugar Iron Agar.

Bibliography

Hajna A.A. - J. Bact. 1945. 40: 516-517

Microbiological Test

Microorganisms	Growth	Colony colour	Precipitation
<i>Bacillus cereus</i> ATCC 1178	Null		
<i>Escherichia coli</i> ATCC 25922	Null/Slight	Rose-red	-
<i>Salmonella typhimurium</i> ATCC 14028	Good	colourless	-
<i>Salmonella choleraesuis</i> ATCC 13312	Good	colourless	-
<i>Salmonella enteritidis</i> ATCC 13076	Good	colourless	-
<i>Proteus vulgaris</i> ATCC 13315	Light	colourless/rose	±
<i>Pseudomonas aeruginosa</i> ATCC 27853	Moderate	colourless	-
<i>Staphylococcus aureus</i> ATCC 25923	Null		

DESOXYCHOLATE AGAR

Cat. 1020

Used for the cultivation of Gram-negative enteric bacilli

Formula in grams per liter

Peptone Mixture.....	10,00	Lactose.....	10,00
Sodium Chloride.....	5,00	Dipotassium Phosphate.....	2,00
Sodium Citrate.....	1,00	Sodium Desoxycholate.....	1,00
Neutral Red.....	0,033	Ferric Citrate.....	1,00
Bacteriological Agar.....	16,00		

Final pH 7,3 ± 0,2 at 25°C

Preparation

Suspend 46 grams of the medium in one litre of distilled water. Soak for 10-15 minutes. Mix well. Heat with frequent agitation and boil until completely dissolved. Cool to 45-50°C and pour into Petri dishes. **DO NOT AUTOCLAVE.**

NOTE: Overheating may increase the inhibition degree.

Uses

Desoxycholate Agar is a selective and differential medium for the isolation and enumeration of coliform microorganisms in milk, dairy products, and different types of water

The desoxycholate and the citrate salts inhibit the development of the gram-positive organisms. For the determination and enumeration of coliforms in water and milk, 1 ml. of the diluted sample must be added per tube when the melted medium is at 45-50°C. If the food sample is suspected of low number of organisms, inoculate with larger volumes (1-5 ml.) of undiluted sample.

The recovery of organisms is sometimes facilitated by adding a thin layer over the inoculated and solidified agar.

The colonies of the lactose fermenters which grow under the surface of the medium are brilliant red or pink, and in general, lenticular or ellipsoid. On the other hand, the colonies on the surface are large and pink for *E. coli*, while those of *Enterobacter* are pale on the edges and pink colored in the center.

The colonies of the microorganisms which do not ferment lactose such as *Salmonella*, *Shigella*, and *Proteus* are colourless.

Bibliography

Standard Methods for the Examination of Dairy Products. 1 Ed. APHA, Inc. New York, 1960. Standard Methods for the Examination of Water and Wastewater, APHA, Inc. New York, 1960.

Microbiological Test

Microorganisms	Growth	Change to purple red
<i>Escherichia coli</i> ATCC 25922	Good	Pink with bile precipitate
<i>Salmonella typhimurium</i> ATCC 14028	Good	Colourless
<i>Staphylococcus aureus</i> ATCC 25923	Inhibited	----

DESOXYCHOLATE CITRATE AGAR

Cat. 1067

Highly selective medium for the isolation of enteric pathogens, specially *Salmonella* and *Shigella*

Formula in grams per liter

<i>Sodium Citrate</i>	20,00	<i>Peptone Proteose n° 3</i>	10,00
<i>Lactose</i>	10,00	<i>Pork Infusion</i>	9,50
<i>Sodium Desoxycholate</i>	5,00	<i>Ferric Ammonium Citrate</i>	2,00
<i>Neutral Red</i>	0,02	<i>Bacteriological Agar</i>	13,50

Final pH 7,5 ± 0,2 at 25°C

Preparation

Suspend 70 grams of the medium in one litre of distilled water. Mix well. Heat with frequent agitation and boil until completely dissolved. Do not overheat. **DO NOT AUTOCLAVE**. Cool to 45-50°C and pour into Petri dishes.

Uses

Desoxycholate Citrate Agar is a modification of desoxycholate agar and is ideal for the investigation of pathogenic *enterobacteria* in highly contaminated foods. The gram-positive organisms are totally inhibited by the sodium citrate and sodium desoxycholate. *Proteus* and coliforms are also highly inhibited.

It is recommended to heavily seed the sample on the plate.

A previous enrichment in Selenite Broth can also be used.

Salmonella typhi, *S. paratyphi* and *Shigella* types yield colourless (lactose-negative) colonies while lactose-positive organisms like *E. coli* are pink to red. This is due to the neutral red in which presence lactose fermenting bacteria form red colonies while non fermenting will appear clear, with or without black centers. Lactose-fermenting colonies mass have a desoxycholate precipitation zone around them.

Bibliography

Leifson E. 1935. New culture media based on sodium desoxycholate for the isolation of intestinal pathogens and for the enumeration of colon bacilli in milk and water. J. Pathol. Bacteriol. 40: 581-599.

Farmer III, J.J. and MT. Kelly. 1991 Enterobacteriaceae. P. 360-383. In A. Balows, W. J. Hausler, Jr., K.L. Hermann, H.D. Isenberg and H.J. Shadomy (ed.), Manual of clinical microbiology, 5th ed. American Society for Microbiology.

Microbiological Test

Microorganisms	Growth	Colony colour	H2S
<i>Klebsiella pneumoniae</i> ATCC 13883	Moderate	Red	-
<i>Escherichia coli</i> ATCC 25922	Light	precipitated red	-
<i>Salmonella enteritidis</i> ATCC 13076	Good	colourless	+
<i>Salmonella typhimurium</i> ATCC 14028	Good	colourless	+
<i>Shigella flexneri</i> ATCC 12022	Good	colourless	-
<i>Streptococcus faecalis</i> ATCC 19433	Inhibited	----	-

DESOXYCHOLATE LACTOSE AGAR

Cat. 1025

Differential and slightly selective medium used for the isolation of gram negative enteric bacilli

Formula in grams per liter

Peptone Bacteriological	10,00	Lactose	10,00
Sodium Chloride	5,00	Sodium Citrate	2,00
Sodium Desoxycholate	0,50	Neutral Red	0,03
Bacteriological Agar	15,00		

Final pH 7,1 ± 0,2 at 25°C

Preparation

Suspend 42,5 grams of the medium in one litre of distilled water. Heat till boiling to dissolve. Completely the medium. Avoid overheating. **DO NOT AUTOCLAVE.** Prepared plates may present a slight precipitate.

Uses

Desoxycholate Lactose Agar is prepared according to the recommendations of the APHA for the examination of water, milk and food products, especially for *coliforms*.

In general, it is used for the enumeration of colonies by the dilution method. This is accomplished by adding 1 ml. of the desired dilution to an empty Petri dish and pouring on the cooled (45-50°C) medium. If the product has not been diluted (e.g. pasteurized milk), it can be added directly to the melted medium and plates poured.

It is convenient to put a second layer of medium on the plate after initial solidification.

Coliform colonies are lenticular, pink or bright red. Differentiation is made on the basis of the lactose fermentation, Lactose fermenters in presence of neutral red give red colonies. While non fermenters give clear colonies.

If no second layer is applied, the colonies of *E. coli* which develop on the surface of the plate are large and pink while *E. aerogenes* are pale with a pink center.

Bibliography

Standard Methods for the Examination of Dairy Products. Eleventh Edition APHA Inc. New York 1960.
Recommended Methods for the Microbiological Examination of Foods APHA Inc. New York 1960.
American Public Health Association. 1960. Standard methods for the examination of water and wastewater, 11th ed. American Public Health Association, Washington, D.C.

Microbiological Test

Microorganisms	Growth	Colour colony	Precipitated
<i>Escherichia coli</i> ATCC 25922	Good	red	+
<i>Klebsiella pneumoniae</i> ATCC 13883	Good	red	+
<i>Enterobacter cloacae</i> ATCC 13047	Good	rose	±
<i>Salmonella typhimurium</i> ATCC 14028	Good	Colourless	-
<i>Shigella flexneri</i> ATCC 12022	Good	Colourless	-
<i>Streptococcus faecalis</i> ATCC 11700	Null/light	Colourless	-
<i>Staphylococcus aureus</i> ATCC 23923	Null		

DEXTROSE AGAR

Cat. 1021

Used for the obtaining total counts of microorganisms and for general laboratory purposes

Formula in grams per liter

<i>Polypeptone</i>	10,00	<i>Dextrose</i>	10,00
<i>Sodium chloride</i>	5,00	<i>Beef Extract</i>	3,00
<i>Bacteriological Agar</i>	15,00		

Final pH 6,9 ± 0,2 at 25°C

Preparation

Suspend 43 grams of the medium in one litre of distilled water. Mix well until a uniform suspensions is obtained. Heat with frequent agitation and boil for one minute. Dispense and sterilize at 121°C (15 lbs. sp.) for 15 minutes.

Uses

Dextrose Agar is a medium suitable to cultivate a wide variety of microorganisms with or without added blood. The high dextrose concentration yields abundant growth is less time than other media. It can also be used in the microbiological analysis of frozen products, for which it is necessary to acidify the medium with approximately 7,1 ml. of a 10% tartaric acid solution per litre of medium after it has been sterilized and cooled to 45-50°C.

Do not attempt to remelt the medium after it has been acidified because the agar will hydrolyze and not gel correctly.

It is a general use medium but is not appropriate for hemolytic studies because of the high content of dextrose.

Bibliography

Recommended Methods for the Microbiological Examination of Foods APHA Inc., New York.
COMPENDIUM OF METHODS FOR THE MICROBIOLOGICAL EXAMINATION OF FOOD. 3RD edition APHA 1992.

Microbiological Test

Microorganisms	Growth
<i>N. meningitis</i> ATCC 13090	Satisfactory
<i>N. gonorrhoeae</i> ATCC 19424	Satisfactory
<i>St. pneumoniae</i> ATCC 6303	Satisfactory
<i>St. pyogenes</i> ATCC 19615	Satisfactory
<i>Bordetella pertusis</i> ATCC 9797	Satisfactory
<i>Clostridium perfringens</i> ATCC 12919	Acceptable

DEXTROSE BROTH

Cat. 1203

Medium used for the study of glucose fermentation

Formula in grams per liter

Casein peptone 10,00 Dextrose 5,00
Sodium chloride 5,00

Final pH 7,3 ± 0,2 at 25°C

Preparation

Suspend 20 grams of the medium in one liter of distilled water. Mix well and heat slightly until completely dissolved. Dispense into tubes with Durham fermentation (gas collection) tubes. Sterilize at 118°C (12 lbs sp) for 15 minutes.

Uses

This medium is used to cultivate fastidious microorganisms as well as to detect gas formation from enteric bacilli through the glucose fermentation

Bibliography

Norton, 1932. Bacteriology of pus. J. Lab. Clin. Med.
MacFaddin J.D. 1985 Media for isolation cultivation identification maintenance of medical bacteria.
Williams & Wilking, Baltimore. MD.

Microbiological Test

Microorganisms	Growth	Gas production
<i>Shigella flexneri</i> ATCC 12022	Satisfactory	-
<i>Escherichia coli</i> ATCC 25922	Satisfactory	+

DNASE TEST AGAR (DEOXYRIBONUCLEASE) Cat. 1028

Used for the detection of desoxiribonuclease activity

Formula in grams per liter

Casein Peptone	15,00	Soy Peptone	5,00
Sodium Chloride	5,00	Deoxyribonucleic Acid	2,00
Bacteriological Agar	15,00		

Final pH 7,3 ± 0,2 at 25°C

Preparation

Suspend 42 grams of the medium in one litre of distilled water. Mix well to obtain a homogeneous suspension. Heat with frequent agitation and boil for one minute. Sterilize in an autoclave at 118-121°C (15 lbs. sp.) for 15 minutes. Cool to 45-50°C and pour into sterile Petri dishes. If desired, add 5% blood to the medium without mannitol to prepare a blood agar medium.

Uses

Make a heavy band streak (2 cm. in length) of the test organism (e.g. *Staphylococci*, *Pseudomonas*, *Serratia*, *Bacillus*, etc.) on the surface of the plate. You can simultaneously place in the same plate 4 to 5 different samples. Incubate for 18 to 24 hours at 35°C.

After good growth add a drop of 1N hydrochloric acid or a few drops of 0,1% toluidine blue solution. With some strains it is necessary to increase the concentration of HCl to 2N to obtain a good positive reaction, the appearance of a well defined bright clear halo around the bacterial streak.

In presence of diluted hydrochloric acid, the reaction with DNA in the culture medium forms a hazy precipitate. Colonies producing desoxiribonuclease appear surrounded by a zone or a clear halo which contains fractions of soluble nucleotides from the degradation of DNA, which are not precipitated by the hydrochloric acid.

Results

In the presence of hydrochloric acid DNA se positive: A clear zone surrounding the inoculum streak with the rest of

the plate remaining opaque. The positive reaction takes approximately 5 minutes to form.

DNase negative: Absence of clear halo around the inoculum streak.

In the presence of toluidine blue:

DNase positive: Appearance of a pink halo surrounding the inoculum streak. The rest of the plate remains blue.

DNase negative: Absence of the pink halo surrounding the inoculum streak.

Nevertheless, for some fastidious organisms it may be necessary to add blood. The addition of diluted hydrochloric acid forms a well defined but opaque halo with DNase positive organisms. The DNase medium with blood should not be used in the study of hemolytic reactions and should only be added in absolutely necessary cases.

Weckman and Catting (1957), Disalvo (1959) and Fusillo and Weis (1959) proved that coagulase positive staphylococci degraded the DNA by hydrolysis and are considered DNase positive.

Bibliography

Blair E.B. Emerson, J.S. and Tull, S.C. Am. J.Clin.Poth, 47:30-39, 1957. Disalvo Med. Tech. Bull. 9:191, 1958.
Weckman and Catting J. Bact. 73: 747, 1957.

Microbiological Test

Microorganisms	Growth	DNase
<i>Streptococcus pyogenes</i> ATCC 19615	Satisfactory	+
<i>Staphylococcus epidermidis</i> ATCC 12228	Satisfactory	-
<i>Staphylococcus aureus</i> ATCC 25923	Satisfactory	+
<i>Serratia marcescens</i> ATCC 8100	Satisfactory	+

E. COLI COLIFORMS CHROMOGENIC MEDIUM

Cat. 1340

Selective medium for the simultaneous detection of *E. Coli* and total coliform microorganisms in water and food samples.

Formula in grams per liter

<i>Sodium Chloride</i>	5,00	<i>Phosphate buffer</i>	4,90
<i>Bacteriological peptone</i>	3,00	<i>Sodium pyruvate</i>	1,00
<i>Tryptophane</i>	1,00	<i>Sorbitol</i>	1,00
<i>Chromogenic mixture</i>	0,36	<i>Tergitol -7</i>	0,10
<i>Bacteriological Agar</i>	10,00		

Final pH 6,8 ± 0,2 at 25°C

Preparation

Suspend 26,4 grams of the medium in one liter of distilled water. Heat with frequent agitation to boiling, and keep it boiling for 1 minute. Avoid overheating. **Do not autoclave**. Dispense in Petri dishes. Store the plates in the refrigerator, protected from light.

Uses

The interaction of medium ingredients, such as peptone, sorbitol, etc, grants a quick colony growth, including infectious coliform micro-organisms. Gram + bacteria, as well as some Gram – ones, are inhibited by Tergitol-7, which does not affect coliforme bacteria. The coliform characteristic enzyme, B-D-galactosidase, cleaves Salmon-GAL substrate, and gives a salmon to red colour to the coliforme colonies. X-glucuronide substrate, is used for B-D-glucuronidase detection, which is a *E. coli* characteristic enzyme. *E. coli* bacteria, cleaves both

substrates Salmon-Gal y X-glucuronide, giving a dark blue to violet colour to the colonies, being easily to distinguish them from other coliforme colonies, that have a salmon to red colour. As an additional part of the *E. Coli* confirmation the addition of tryptophane to the medium allows to perform the Indole test.

Bibliography

Alonso, J.L. Soriano, K., Amoros I., Ferrus, M.A. 1998 Cevartitatie determination of *E. Coli* and fecal coliforms in water using a chromogenic medium. J. Environ. Sci Health 33.

Microbiological Test

Microorganisms	Growth	Colour
<i>Escherichia coli</i> ATCC 25922	Satisfactory	Blue-dark violet
<i>Escherichia coli</i> ATCC 11775	Satisfactory	Blue-dark violet
<i>Citrobacter freundii</i> ATCC 8090	Satisfactory	Salmon
<i>Salmonella enteritidis</i> ATCC 13076	Good	Colourless
<i>Streptococcus faecalis</i> 19433	None	-----

EC MEDIUM

Cat. 1522

For the determination and enumeration of coliforms organisms in water

Formula in grams per liter

<i>Tryptose</i>	20,00	<i>Lactose</i>	5,00
<i>Sodium Chloride</i>	5,00	<i>Dipotassium Phosphate</i>	4,00
<i>Bile Salts N° 3</i>	1,90	<i>Monopotassium Phosphate</i>	1,50

Final pH 6,9 ± 0,2 at 25°C
Preparation

Suspend 37,4 grams of the medium in one liter of distilled water. Heat agitating frequently until the medium is completely dissolved. Dispense in test tubes containing gas collecting tubes (Durham) and boil for 5 minutes. **DO NOT AUTOCLAVE.**

Uses

EC is the abbreviation of *Escherichia Coli*. This medium improves the detection methods of the coliform group and *E. Coli* and it can be used to investigate drinking water, wastewater treatment systems and in general water-quality monitoring, as well as in foods.

It is required a prior enrichment in presumptive media to obtain an optimal recovery of fecal coliforms when using EC Medium.

Lactose fermentation with gas production is evidence of the presence of coliforms after incubation at 37°C for 48 hours.

If growth from positive tubes (at 37°C) is reinoculated and reincubated at 45,5°C and yields positive growth, confirmation of *E. coli* can then be made by using the appropriate biochemical tests (indol, citrate, etc.).

Formation of gas at 37°C.....Coliforms

Formation of gas at 37°C & 45,5°C*E. coli*

Bibliography

Hajna and Perry 1944 A.P.H.A.

Ray B. 1986 Impact of bacterial injury and repair in food microbiology. Its past, present and future J. Food Prot.

Microbiological Test

Microorganisms	Growth
<i>Bacillus subtilis</i> ATCC 6633	Inhibited -
<i>Enterobacter aerogenes</i> ATCC 13048	Inhibited -
<i>Escherichia coli</i> ATCC 25922	Good +
<i>Streptococcus faecalis</i> ATCC 19433	Inhibited -

ELLIKER MEDIUM

Cat. 1539

For the cultivation of streptococci and lactobacilli in dairy products

Formula in grams per liter

<i>Tryptone</i>	20,00	<i>Yeast Extract</i>	5,00
<i>Dextrose</i>	5,00	<i>Lactose</i>	5,00
<i>Sucrose</i>	5,00	<i>Sodium Chloride</i>	4,00
<i>Gelatin</i>	2,50	<i>Sodium Acetate</i>	1,50
<i>Ascorbic Acid</i>	0,50		

Final pH 6,8 ± 0,2 at 25°C

Preparation

Suspend 48,5 grams of the medium in one litre of distilled water. Mix well. Heat to boiling to dissolve the medium completely. Dispense and sterilize at 121°C (15 lbs. sp.) for 15 minutes.

Uses

The medium is recommended for the general cultivation of *streptococci* and *lactobacilli* prepared according to the formula of Elliker which has a slightly acidic pH and contains sufficient nutrients to support the sodium acetate inhibits gram negative bacteria.

Bibliography

Elliker, P.R.A. W. Anderson and G. Hannesson 1956. An agar culture medium for lactic acid streptococci and lactobacilli. J. Dairy Sci. 39:1611 Splittstoeg.
Vanderzant C. and D.F. Splittstoeg 1992. Compendium of methods for the microbiological association of food, APHA 3rd edition.

Microbiological Test

Microorganisms	Growth
<i>Lactobacillus casei</i> ATCC 7469	Satisfactory
<i>Lactobacillus lactis</i> ATCC 8000	Satisfactory
<i>Streptococcus cremoris</i>	Satisfactory

ENDO AGAR BASE **Cat. 1118**

For the determination of coliforms in waters, dairy products and food in general

Formula in grams per liter

<i>Bacteriological Peptone</i>	10,00	<i>Lactose</i>	10,00
<i>Potassium Phosphate</i>	3,50	<i>Sodium Sulfite</i>	2,50
<i>Bacteriological Agar</i>	10,00		

Final pH 7,5 ± 0,2 at 25°C

Preparation

Suspend 36 grams of the medium in one litre of distilled water. Add 4 ml. of an alcoholic solution at 10% (p/v) of basic fuchsin (ethyl alcohol at 95%). Mix well. Boil until completely dissolved. Sterilize at 121°C (15 lbs. sp.) for 15 minutes. Mix well before pouring it.

Note: Basic fuchsin is a potential carcinogen and precautions should be taken to avoid inhalation of the dye powder as well as contact with the skin.

First AID: In case of contact with eyes, rinse immediately with plenty of water and seek medical advice, also if breathing become difficult or if swallowed.

Uses

Endo Agar is used for the differentiation of lactose-positive and -negative bacteria of the intestinal tract, particularly for confirmation of presumptive tests for coliforms. Acid and

aldehyde production by lactose-fermenting organisms such as *E. coli* produce a characteristic red coloration to the colony and the area surrounding it, along with a brilliant gold metallic sheen. Non-lactose fermenters form colourless and transparent colonies.

To confirm presumptive positive *coliforms*, tubes of Endo Agar can be inoculated, incubated at 35-37°C and examined for acid and gas production.

Bibliography

Endo S. 1904 uber ein verfahren Zum Nachweiss der Typhusbacillen.
A.P.H.A. 1975 Standard methods for the examination of water and wastewater. 14th edition.

Microbiological Test

Microorganisms	Growth	Colony colour
<i>Enterobacter aerogenes</i> ATCC 13048	Satisfactory	Red
<i>Salmonella typhi</i> ATCC 6539	Satisfactory	Colourless
<i>Shigella sonnei</i> ATCC 25931	Satisfactory	Colourless
<i>Escherichia coli</i> ATCC 25922	Satisfactory	Red with metallic shine

ENDO LES AGAR BASE

Cat. 1137

A Standard Methods Medium for membrane-filter technique used for detection and enumeration of coliform micro-organisms in water

Formula in grams per liter

<i>Lactose</i>	9,40	<i>Tryptose</i>	7,50
<i>Casein Peptone</i>	3,70	<i>Meat Peptone</i>	3,70
<i>Sodium Chloride</i>	3,70	<i>Dipotassium Phosphate</i>	3,30
<i>Sodium Sulfite</i>	1,60	<i>Yeast Extract</i>	1,20
<i>Monopotassium Phosphate</i>	1,00	<i>Sodium Desoxicholate</i>	0,10
<i>Sodium Lauryl sulphate</i>	0,05	<i>Bacteriological Agar</i>	15,00

Final pH 7,2 ± 0,2 at 25°C

Preparation

Dissolve 50,25 grams of the medium in one litre of distilled water with 20 ml of ethanol 95 % (v/v). Add 0,8 g. of basic fuchsin. Mix well, heat agitating constantly till boiling and completely dissolved. Sterilize in autoclave at 121°C (15 lbs. sp.) for 15 minutes. Cool to 45-50° and pour into plates.

Uses

This medium is a modification of Endo Base Agar (Cod. 1118), for the membrane-filter technique. It uses Lauryl Sulphate Broth as previous enrichment, and thus obtaining

more growth and more brilliant colonies. It's used for enumerating coliforms in water by membrane filtration. LES stands for Lawcenc Experimental Station.
First AID: In case of contact with eyes, rinse immediately with plenty of water and seek medical advice, also if breathing become difficult or if swallowed.

Bibliography

APHA (1980) Standard Methods for the Examination of Water and Wastewater. 15th. Ed. Washington, D.C.

Microbiological Test

Microorganisms	Growth	Colony colour
<i>S.typhi</i> ATCC 6539	Satisfactory	colourless
<i>S.sonnei</i> ATCC25931	Satisfactory	colourless
<i>E.coli</i> ATCC25922	Satisfactory	Brilliant red to black
<i>E.aerogenes</i> ATCC13048	Satisfactory	Brilliant red to black

ENTEROCOCCUS CONFIRMATORY AGAR

Cat. 1018

Used to confirm the presence of enterococci in water and other sources of sanitary interest

Formula in grams per liter

Casein Peptone	5,00	Yeast Extract	5,00
Dextrose.....	5,00	Sodium Azide	0,40
Methylene Blue.....	0,01	Bacteriological Agar	15,00

Final pH 8,0 ± 0,2 at 25°C

Preparation

Suspend 30,4 grams of the medium in one litre of distilled water. Heat with frequent agitation and boil until dissolution is complete (approximately one minute). Dispense in test tubes and sterilize in an autoclave at 121°C (15 lbs. sp.) for 15 minutes. Leave in a slanted position to solidify.

Uses

This medium is used to confirm the presence of enterococci in water and other sources of sanitary interest. In order to do so aseptically add a volume of *Enterococcus* Confirmatory Broth, which has the same formulation but lacks the agar, to cover half of the slanted surface. It is preferable that the Confirmatory Broth contain 6,5% sodium chloride and 65 units of penicillin per 100 ml. Using growth from the *Enterococcus* Presumptive Broth, inoculate both the surface as well as the broth in the

Confirmatory Agar/Broth mixture tube. The tubes are incubated at 35-37°C for 12 hours and are examined to detect the presence of small pinpoint colonies. Perform a Gram stain and observe under a microscope looking for large chains of ovoid cells. Immediately perform a catalase test by adding to the tube in study 5 ml. of H₂O₂. If there is no generation of gases (negative test), this constitutes the confirmation of enterococci in the sample.

Bibliography

Winter and Sandholzer U. S. Det. Interior Fishery, Leaflet 201 Part II, Nov. 1946.
Ewing W.H. 1986. Edwards and Ewing's identification of enterobacteriaceae 4th edition.

Microbiological Test

Microorganisms	Growth
<i>Escherichia coli</i> ATCC 25922	Inhibited
<i>Streptococcus faecalis</i> ATCC 19433	Satisfactory
<i>Streptococcus faecium</i> ATCC 29212	Satisfactory

E.M.B. (EOSIN METHYLENE BLUE) AGAR

Cat. 1039

For the isolation and differentiation of coliforms from other enterobacteria of medical and sanitary interest

Formula in grams per liter

<i>Bacteriological Peptone</i>	10,00	<i>Lactose</i>	5,00
<i>Sucrose</i>	5,00	<i>Dipotassium Phosphate</i>	2,00
<i>Eosin Y</i>	0,40	<i>Methylene Blue</i>	0,065
<i>Bacteriological Agar</i>	13,50		

Final pH 7,2 ± 0,2 at 25°C

Preparation

Suspend 36 grams of the medium in one litre of distilled water. Mix well. Heat with frequent agitation and boil for one minute. Sterilize in autoclave at 121°C (15 lbs. sp.) for 15 minutes. Cool to 45-50°C. Swirl gently, avoiding the formation of bubbles and pour into Petri dishes.

Uses

It is similar to Levine EMB, used for the study of *enterobacteria*. It is widely used in medical bacteriology, in techniques recommended by the APHA and for the detection and enumeration of *coliform* microorganisms, which can contaminate foods and drinking water. Due to the lactose and sucrose, the medium can be differential in primary culture: salmonellas and shigellas which are lactose-negative can be differentiated from other lactose-negative but sucrose-positive organisms such as *Proteus vulgaris*, *Citrobacter* and *Aeromonas*.

The accompanying microflora which hinders the isolation of medically important organisms are inhibited by the dyes in the formula, especially gram-positives.

It can also be used for the rapid identification of *C. albicans* (incubated in CO₂) and sometimes to isolate *Nocardia*.

E. coli Elevated or slightly convex. 2-3 mm. in diameter, with transmitted light blue-black center with a narrow, clear

edge. Blue-green metallic sheen with reflected light. Some strains show no metallic sheen. Small tendency to confluent growth.

E. aerogenes Klebsiella Large colonies, 4-6 mm. in diameter, mucoid with a tendency to run together. Usually no metallic sheen. With transmitted light, gray-brown centers with clear edges.

Salmonella Shigella Slightly elevated, medium size 1-2 mm. in diameter. Transparent, from colourless to amber.

C. albicans Feathery, spider-like colony after 24-48 hours incubation in CO₂ at 35-37°C. Never presents a typical colonial appearance.

Coagulase-positive staphylococci Very small punctiform, colourless and inhibited.

Proteus species When there is no swarming, similar to *Salmonella* and *Shigella*. Swarming can be minimized by adding a very small amount of alpha-p-nitrophenyl-glycerol.

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American Public Health Association. Diagnostic Procedures and Reagents. 2nd Ed. APHA, Inc. New York, 1950.
American Public Health Association. Examination of Dairy Products. 10th Ed. APHA, Inc. New York, 1953.
Society of American Bacteriologists. Manual of Microbiological Methods MacGraw-Hill New York, 1957.

Microbiological Test

Microorganisms	Growth	Colony colour
<i>Enterobacter aerogenes</i> ATCC 13048	Satisfactory	pink
<i>Escherichia coli</i> ATCC 25922	Satisfactory	green with metallic shine
<i>Salmonella typhimurium</i> ATCC 14028	Satisfactory	colourless
<i>Pseudomonas aeruginosa</i> ATCC 10145	Good	colourless
<i>Staphylococcus aureus</i> ATCC 25923	Poor-nul	colourless

E.S.T.Y. BROTH

Cat. 1254

For the cultivation of lactic streptococci

Formula in grams per liter

Tryptone..... 5,00	Soy peptone5,00
Beef extract..... 5,00	Yeast extract.....2,50
Ascorbic Acid..... 0,50	Magnesium sulfate0,25
Disodium Glicerophosphate..... 19,00	

Final pH 6,9 ± 0,2 at 25°C

Preparation

Suspend 37,25 grams of the medium in 900 ml of distilled water. Mix well. Heat to boiling with frequent agitation until complete dissolution. Adjust final volume to 1000 ml. Sterilize by autoclaving at 121°C (15 lbs sp) for 15 minutes. Allow to cool to 45-50°C and add 50 ml of a sterile solution of 10% lactose.

Uses

Its utilization have been described for bacteriophages Assays.
It's recommended for initial culture maintenance which produce acids in its metabolism.

This medium has a high buffer capability due to the disodium glycerophosphate which acts as a regulator pH agent, and inhibits the growth of *Lactobacillus bulgaricus* isolating *S. thermophilus* from yogurt. The Ascorbic acid stimulates the growth of lactic streptococci.

Bibliography

Reiter B., and J.D. Oram. 1962. Nutritional studies on cheese starters. I. Vitamin and aminoacid requirements of single strain starters. J. Dairy Res.
International Dairy Federation 1981 Identification and enumeration of microorganisms in fermented mil KS.

Microbiological Test

Microorganisms	Growth
<i>Lactobacillus bulgaricus</i> ATCC 11842	Inhibited
<i>Streptococcus thermophilus</i> ATCC 14486	Satisfactory

E.S.T.Y. MEDIUM

Cat. 1555

Selective medium for the enumeration of *Streptococcus termophilus* in yogurt

Formula in grams per liter

<i>Disodium Glicerophosphate</i>	19,00	<i>Tryptone</i>	5,00
<i>Soy peptone</i>	5,00	<i>Beef extract</i>	5,00
<i>Yeast extract</i>	2,50	<i>Ascorbic Acid</i>	0,50
<i>Magnesium Sulphate</i>	0,25	<i>Bacteriological Agar</i>	11,00

Final pH 6,9 ± 0,2 at 25°C

Preparation

Suspend 48,30 grams of the medium in 900 ml of distilled water. Mix well. Heat to boiling with frequent agitation until complete dissolution. Adjust final volume to 1000 ml. Sterilize by autoclaving at 121° C (15 lbs sp) for 15 minutes. Allow to cool to 45-50°C and add 50 ml. of an sterile solution of 10% lactose.

Uses

Lactic streptococci produce acid and are difficult to grow, this medium buffers the acid from the lactose fermentation while the ascorbic acid promotes the growth of lactic streptococci. Its a recommended medium for

Streptococcus Termophilus isolation and enumeration in yogurt, due to the glycerophosphate high concentration it inhibits *Lactobacillus bulgaricus* development. It has been recommended by Milky International Federation for this use.

Bibliography

Terzaghi, B.E. and W. E. Sandine. 1975 Improved medium for lactic streptococci and their bacteriophages. Appl. Microbiol 29:807-813.

International Dairy Federation 1981. Identification and enumeration of micro-organisms in fermented milks. Joint IDF/ISO/AOAC Group E44.

Microbiological Test

Microorganisms	Growth
<i>Lactobacillus bulgaricus</i> ATCC 11842	Negative
<i>Streptococcus termophilus</i> ATCC 14486	Positive

EUGON AGAR

Cat. 1036

To obtain eugonic cultures of most microorganisms

Formula in grams per liter

Casein Peptone	15,00	Dextrose	5,50
Soy Peptone	5,00	Sodium Chloride	4,00
L-Cystine	0,70	Sodium sulfite	0,20
Bacteriological Agar	15,00		

Final pH 7,0 ± 0,2 at 25°C

Preparation

Suspend 45,4 grams of the culture medium in one litre of distilled water. Heat with frequent agitation and boil for one minute. Dispense and sterilize at 118°C (12 lbs. sp.) for 15 minutes. Cool the medium to 45-50°C and add 5-10% sterile defibrinated sheep or rabbit blood, if desired.

Uses

This medium yields a high level of growth of microorganisms (eugonic growth) even with the bacteria more difficult to cultivate, such as *Haemophilus*, *Neisseria*, *Pasteurella*, *Brucella*, *Lactobacillus*, etc. It is very useful in medical bacteriology as well as microbiology of foods. Likewise, this medium is ideal for cultivating delicate pathogenic microorganisms and for obtaining high counts of bacterial cultures in the preparation of antigens and vaccines.

In food bacteriology it is widely used to detect the presence of lactic bacilli in raw meats, smoked sausages,

etc., as well as in the bacteriological analysis of milk and other dairy products. It is employed for the enumeration of colonies in canned foods, and in general, in the detection of sanitation problems which are presented in the food industry.

The medium can be made richer by adding 1,0 ml. of Polyenrichment for every 100 ml. of medium while the addition of defibrinated blood, chocolate or not, permits the development of *Histoplasma capsulatum* and *Nocardia*. Also it is used for the analysis of clinical materials such as blood and cerebrospinal or pleural fluids which generally contain pure cultures.

Bibliography

Vera M.J. Bact. 54:14, 1947. Pelczar and Vera Milk Plant Monthly, 38-30, 1949.
Frank J. Bact. 70:269, 1955. Ramos C., Mario "Manual of Milk and Lactides". Edition of Author, Berna 12:201, Mexico 6, D.F., 1976.

Microbiological Test

Microorganisms	Growth
<i>Neisseria meningitidis</i> ATCC 13090	Good
<i>Streptococcus pneumoniae</i> ATCC 6303	Good
<i>Streptococcus pyogenes</i> ATCC 19615	Good
<i>Brucella abortus</i> ATCC 4315	Good

EVA BROTH (ETHYL VIOLET AZIDE BROTH, LITSKY) Cat. 1230

For the confirmation of enterococci and as a detector of fecal contamination in water

Formula in grams per liter

Peptone mixture.....	20,00	Glucose.....	5,00
Sodium Chloride.....	5,00	Sodium Chloride.....	5,00
Dipotassium Phosphate.....	2,70	Monopotassium Phosphate.....	2,70
Sodium Azide.....	0,40	Ethyl Violet.....	0,0008

Final pH 7,0 ± 0,2 at 25°C

Preparation

Suspend 40,8 grams of the medium in one litre of distilled water. Dissolve the medium and dispense in 10 ml. amounts into test tubes and sterilize at 121°C (15 lbs. sp.) for 15 minutes. It is recommended to use a large inoculum as the medium is very selective and it is used in the second phase of confirmation.

Uses

This medium is specific for enterococci. The sodium azide and the Ethyl Violet inhibit all gram-positive bacilli and gram-positive cocci except enterococci. EVA Broth should be used in conjunction with Rothe Broth (Glucose Broth with Azide) for the investigation of fecal *streptococci* in water and food products. It is a very selective medium for the presence of *streptococcal* organisms which are a sign of fecal contamination.

The presence of enterococci in water and other specimens indicates fecal contamination.

The tubes are inoculated with the appropriate dilutions in a series of 3 tubes for each dilution and incubated at 37°C for 48 hours. The appearance of turbidity and eventually the formation of a violet (purple) button of growth in the bottom of the tube is characteristic of fecal *streptococcal* growth.

Bibliography

Litsky W. Mallmann W.L. Fifield C.W. A.J.P.H. 1953, 43, 873-879.
Mallman and Seligman. 195 A.J.P.H. 40:286.

Microbiological Test

Microorganisms	Growth
<i>Escherichia coli</i> ATCC 25922	Inhibited
<i>Staphylococcus aureus</i> ATCC 25923	Inhibited
<i>Streptococcus pyogenes</i> ATCC 19615	Inhibited
<i>Streptococcus faecalis</i> ATCC 29212	Satisfactory
<i>Streptococcus faecalis</i> ATCC 19433	Satisfactory

EWING MALONATE BROTH MODIFIED

Cat. 1212

For the enterobacteria differentiation , specially *Salmonella* from Arizona

Formula in grams per liter

<i>Sodium Malonate</i>	3,00	<i>Ammonium Sulphate</i>	2,00
<i>Sodium Chloride</i>	2,00	<i>Yeast Extract</i>	1,00
<i>Dipotassium Phosphate</i>	0,60	<i>Monopotassium Phosphate</i>	0,40
<i>Dextrose</i>	0,25	<i>Bromthymol Blue</i>	0,025

Final pH 6,7 ± 0,2 at 25°C

Preparation

Suspend 9.3 grams of the medium in one liter of distilled water. Dispense in appropriate test tubes and volumes and sterilize in an autoclave at 121°C (15 lbs. sp.) for 15 minutes.

Uses

Ewing Malonate Broth is widely used to distinguish microorganisms that utilize malonate, such as *Enterobacter*, *Klebsiella*, and strains of Arizona, from those that are not able to utilize it, such as *Escherichia*, *Salmonella*, *Serratia*, and some others.

Inoculate the broth with the suspect culture and incubate at 35°C for 48 hours. The organisms that develop have the capacity to utilize the malonate, alkalinizing the medium and changing it to a blue color. The organisms that do not utilize malonate do not produce a color change and the medium retains the original green color.

Bibliography

Leifson, E. J. Bact. 26:329, 1933. Ewing W. H. Identification of Enterobacteriaceae, Burgess Publishing Co., Minneapolis, Minn., 1972.

Microbiological Test

Microorganisms	Growth	Medium colour
<i>Enterobacter aerogenes</i> ATCC 13048	Satisfactory	Blue
<i>Escherichia coli</i> ATCC 25922	Satisfactory	Green
<i>Klebsiella pneumoniae</i> ATCC 13833	Satisfactory	Blue
<i>Salmonella typhimurium</i> ATCC 14028	Satisfactory	Green
<i>Salmonella arizonae</i> ATCC 13314	Satisfactory	Blue

FECAL COLIFORMS AGAR BASE

Cat. 1127

Medium for membrane-filter technique at high temperature, used for detection, and enumeration of fecal coliform micro-organisms

Formula in grams per liter

Lactose	12,50	Bacteriological Peptone	10,00
Proteose Peptone n°3	5,00	Sodium Chloride	5,00
Yeast Extract	3,00	Bile Salts n°3	1,50
Aniline blue	0,10	Bacteriological Agar	15,00

Final pH 7,4 ± 0,2 at 25°C
(without Rosolic Acid)

Preparation

Suspend 52 grams of the medium in one litre of distilled water. Dissolve until complete dilution. Add 10 ml of rosolic acid at 1% in NaOH 0,2N. Mix well to obtain an homogeneous suspension. Heat with frequent agitation till boiling. Cool to 45-50°C and pour into Petri dishes.

Uses

This medium is suitable for membrane-filter technique at high temperature. This medium is used for detection, and enumeration of fecal coliform micro-organisms.

The differential indicator system (with aniline blue and rosolic acid). Gives the colonies of fecal coliforms a blue colour, while the rest of microorganisms will become grey.

Bibliography

Geldreich, Clark and Kabler, 1963, USPHS, HEW. Personal Communication.
Geldreich, Clark, Huff and Bert, 1965, Journal of American Water Works Association, 57:208.

Microbiological Test

Microorganisms	Growth	Colony colour
<i>Escherichia coli</i> ATCC 25922	Satisfactory	blue
<i>Salmonella typhimurium</i> ATCC 14028	Satisfactory	grey
<i>Shigella flexneri</i> ATCC 12022	Satisfactory	grey
<i>Streptococcus faecalis</i> ATCC 1943	Inhibited	----

FECAL COLIFORMS BROTH BASE

Cat. 1121

For the detection and enumeration of fecal coliform organisms through the membrane filter technique at high temperature

Formula in grams per liter

<i>Lactose</i>	12,50	<i>Tryptose</i>	10,00
<i>Proteose peptone n° 3</i>	5,00	<i>Sodium chloride</i>	5,00
<i>Yeast extract</i>	3,00	<i>Bile salts n° 3</i>	1,50
<i>Aniline blue</i>	0,10		

Final pH (without Rosolic Acid) 7,4 ± 0,2 at 25°C

Preparation

Suspend 37,1 grams of the medium in one litre of distilled water. Add 10 ml. of Rosolic Acid at 1% in NaOH0.2H solution and heat to boiling. Cool at room temperature and add 2 ml. of broth to each sterile absorbent pad placed in a Petri dish.

Uses

Place the membrane filter, which the sample has been filtered through, on the upper part of the saturated absorbent pad. Close the Petri dish hermetically.

Immerge the closed dishes in a water bath at 44.5°C for 24 hours. Take it out from the water bath, observe coliforms and count the colonies.

The differential indicator system (with aniline blue and rosolic acid) gives the colonies of fecal coliforms a blue colour while the rest of microorganisms will become grey.

Bibliography

Geldreich, Clark and Kabber, 1963, USPHS, HEN. Personal Communication.
 Geldreich, Clark, Huff and Bert, 1965, Journal of American water works Association, 57:208.

Microbiological Test

Microorganisms	Growth 44,5°C	Growth 35°C	Colony colour
<i>Escherichia coli</i> ATCC 25922	Good	Good	Blue
<i>Salmonella typhimurium</i> ATCC 14028	Inhibited	Good	Grey
<i>Shigella flexneri</i> ATCC 12022	Inhibited	Good	Grey
<i>Streptococcus faecalis</i> ATCC 19433	Inhibited	Inhibited	----

GC AGAR BASE

Cat. 1106

Used for the isolation and cultivation of gonococci

Formula in grams per liter

Peptone mixture.....	15,00	Sodium Chloride	5,00
Dipotassium Phosphate.....	4,00	Corn Starch.....	1,00
Monopotassium Phosphate.....	1,00	Bacteriological Agar.....	10,00

Final pH 7,2 ± 0,2 at 25°C

Preparation

Suspend 7,2 grams of the medium in 100 ml. of distilled water to make a double strength base. Mix well and leave to stand for 5 minutes. Heat with frequent agitation and boil for one minute. Autoclave at 121°C (15 lbs. sp.) for 15 minutes. Also autoclave 100 ml. of 2% solution of hemoglobin made by gradually adding water to two grams of dry hemoglobin to obtain a uniform suspension, before exposing it to the heat of the autoclave. Cool both solutions to 50°C., add the hemoglobin and the other supplements to the base as desired, and pour into plates.

Cool both flasks to 50°C and aseptically add the hemoglobin to the GC Agar Base and mix gently. Add 2,0 ml. of the Polyeenrichment supplement. Mix carefully to avoid bubbles. This completed medium is the general purpose Chocolate Agar. Adding 2,0 ml. of the antimicrobial mixture VCN the medium becomes Thayer-Martin Medium. Pour into plates or tubes with screw caps. Allow tubes to solidify with a long slant.

Uses

The specimen should be placed on the surface of the plate making sure that a heavy inoculum is contained in a relatively small area. Streaking out from this area will produce well isolated colonies. Incubate in a humid atmosphere of 5-10% CO₂ at 35°C for 24-48 hours.

The typical colonies of *N. gonorrhoeae* on Thayer-Martin Medium are grayish-white, opaque, at times shiny, finely granular in appearance, variable in size (1-2 mm.), round with entire or lobate edges and mucoid after 48 hours of incubation.

For suspect isolated colonies, perform a Gram stain and oxidase test. In carbohydrate studies utilizing CTA Medium with selected 1% sugars, *N. gonorrhoeae* ferments only glucose with acid but no gas production. *N. meningitidis* ferments both glucose and maltose with acid but no gas production. The carbohydrate tests are incubated for 1-4 days at 35°C aerobically without CO₂.

Some strains of *N. gonorrhoeae* are inhibited by the antimicrobial agents in selective formulas such as Thayer-Martin Medium so it is wise to streak a non-selective Chocolate Agar plates to culture these organisms.

Bibliography

Bailey and Scott. Diagnostic Microbiology. Fifth Edition, 1978. The C.V. Mosby Company. St. Louis, USA. Preparation of Transgrow. Sept. 15, 1971. Venereal Disease Research Lab., C.D.C. Atlanta, Ga., USA.
Thayer, J. D. Martin J. E., 1966. Improved medium selective for the cultivation of *N. gonorrhoeae* and *N. meningitidis*. Public Health Rep. 81, 559-562.

Microbiological Test

Microorganisms	Growth
<i>Haemophilus influenzae</i> ATCC 19418	Satisfactory
<i>Neisseria meningitidis</i> ATCC 13090	Satisfactory
<i>Neisseria gonorrhoeae</i> ATCC 19424	Satisfactory
<i>Streptococcus pneumoniae</i> ATCC 6303	Satisfactory
<i>Streptococcus pyogenes</i> ATCC 19615	Satisfactory

GELATIN LACTOSE MEDIUM

Cat. 1526

Recommended for the confirmation of *Clostridium perfringens*

Formula in grams per liter

<i>Gelatin</i>	120,00	<i>Tryptose</i>	15,00
<i>Lactose</i>	10,00	<i>Yeast Extract</i>	10,00
<i>Phenol Red</i>	0,05		

Final pH 7,5 ± 0,2 at 25°C

Preparation

Suspend 155 grams of the medium in one litre of distilled water. Heat agitating frequently until completely dissolved. Sterilize in autoclave at 121°C (15 lbs. sp.) for 15 minutes.

Uses

This medium is used for the confirmation of *Clostridium perfringens*. The lactose fermentation is indicated by the presence of gas bubbles as well as a colour change of the medium from red to yellow.

C. perfringens usually liquifies the gelatin after 24-44 hours.

Bibliography

APHA. 3rd Edition Compendium of methods for the microbiological examination of foods.
 Métodos Analíticos del Laboratorio del Instituto Nacional del Consumo (CICC). Alimento I Ministerio de Sanidad y Consumo 1.999.

Microbiological Test

Microorganisms	Colour change to yellow (Gas production)	Gelatinase
<i>Clostridium perfringens</i>	+	+
<i>Clostridium bifermentans</i>	-	+

GIOLITTI CANTONI BROTH

Cat. 1232

For the detection of *S. aureus* in food samples

Formula in grams per liter

Mannitol.....	20,00	Tryptone	10,00
Beef Extract.....	5,00	Yeast Extract.....	5,00
Lithium Chloride	5,00	Sodium Chloride	5,00
Sodium Pyruvate.....	3,00	Glycine	1,20

Final pH 6,9 ± 0,2 at 25°C

Preparation

Suspend 54,2 grams of the medium in one litre of distilled water. Mix well. Heat slowly until completely dissolved. Dispense in 19 ml. into test tubes and sterilize at 121°C (15 lbs. sp.) for 15 minutes. Cool and add 0,3 ml. of a sterile 3,5% potassium tellurite solution to each tube.

Uses

This medium was designed by Giolitti and Cantoni to facilitate the growth of *S. aureus* by incorporating mannitol and pyruvate in the formula, even when present in low numbers in food samples. The growth of gram-negative, lactose-negative bacilli is inhibited by the glycine and the potassium tellurite.

The medium should be inoculated immediately after sterilization and cooling when there is no dissolved air in the medium. If not used immediately, tubes should be heated before use to drive off the dissolved air.

Duplicate tubes should be inoculated with 1 ml. of each serial dilution and the caps tightened. Incubate at 37°C for 48 hours, examining the tubes each day.

The test is considered negative for *S. aureus* if no blackening of the medium is observed. If blackening is present throughout the tube or in the bottom of the tube, subculture to an isolation medium such as Baird Parker Agar and observe for positive growth of black colonies surrounded by a clearing zone.

The International Dairy Federation recommends this medium in a procedure for detecting *S. aureus* in dairy products, using it as an enrichment medium from which selective media are inoculated.

Bibliography

Giolitti, C. and Cantoni, C. (1966) "A Medium for the Isolation of Staphylococci from Foodstuffs", J. Appl. Bact. 29, 395.
International Dairy Federation. 1978 IDF Standard GOA: 1978.

Microbiological Test

Microorganisms	Growth
<i>Escherichia coli</i> ATCC 25922	Inhibited
<i>Micrococcus luteus</i> ATCC 10240	Inhibited
<i>Staphylococcus aureus</i> ATCC 6538	Satisfactory (blackish)
<i>Staphylococcus aureus</i> ATCC 25923	Satisfactory (blackish)

GLUCOSE BROTH (DEXTROSE BROTH) Cat. 1203

Medium used for the study of glucose fermentation

Formula in grams per liter

Casein Peptone 10,00 Dextrose 5,00
 Sodium Chloride 5,00

Final pH 7,3 ± 0,2 at 25°C

Preparation

Suspend 20 grams of the medium in one liter of distilled water. Mix well and heat slightly until completely dissolved. Dispense into tubes with Durham fermentation (gas collection) tubes. Sterilize at 118°C (12 lbs sp) for 15 minutes.

Uses

Glucose Broth is used primarily for the cultivation and confirmation of *streptococci* from primary isolation of the product in study.

If an suitable pH indicator is added, (Phenol red, bromothymol blue, etc.), the medium can be utilized for fermentation studies of glucose.

Bibliography

J. Dental Research, 1:205, 1919 Am. J. Clin. Path 21:884, 1951

Microbiological Test

Microorganisms	Growth	Gas production
<i>Escherichia coli</i> ATCC 25922	Satisfactory	+
<i>Shigella flexneri</i> ATCC 12022	Satisfactory	-

GLUCOSE CHLORAMPHENICOL AGAR

Cat. 1094

Selective medium for isolation and enumeration of yeast and moulds in milk and dairy products.

Formula in grams per liter

Glucose	20,00	Yeast Extract.....	5,00
Cloramphenicol	0,20	Bacteriological Agar.....	15,00

Final pH 6,6 ± 0,2 at 25°C

Preparation

Suspend 40,2 grams of the dehydrated medium in one litre of distilled water. Mix well and heat agitating frequently until completely dissolved. Pour the solution into appropriate containers and sterilize it at 121°C (15 lbs. of steam pressure) for 15 minutes.

Uses

The International Dairy Federation (FIL-IDF) recommends this medium, for the isolation and enumeration of yeast and moulds in milk and dairy products. This medium has been adopted by DIN and ISO standards.

Bibliography

FIL-IDF(1991) Standard 94B. Enumeration of yeast and moulds. Colony Count Technique at 25°C.
ISO (1981) ISO/DIS 6611: Milk and Milk products: Enumeration of yeast and moulds colony count technique at 25°C.
DIN Standard 10186. Mikrobiologische Milchuntersuchung. Bestimmung der Anzahl von Hefen und Schimmelpilzen

Microbiological Test

Microorganisms	Growth
<i>Escherichia coli</i> ATCC 25922	Inhibited
<i>Candida albicans</i> ATCC 2091	Satisfactory
<i>Staphylococcus aureus</i> ATCC 25923	Inhibited
<i>Aspergillus</i> spp.	Satisfactory
<i>Lactobacillus casei</i> ATCC 9595	Inhibited

GLUCOSE CHLORAMPHENICOL BROTH

Cat. 1258

Selective medium for the isolation and enumeration of yeast and moulds in milk and dairy products using the MPN (most probably number) method.

Formula in grams per liter

Glucose 20,00 Yeast Extract 5,00
 Chloramphenicol 0,20

Final pH 6,6 ± 0,2 at 25°C

Preparation

Suspend 25,2 grams of the medium in one litre of distilled water. Pour into appropriate containers and sterilize it at 121°C (15 lbs. of steam pressure) for 15 minutes.

Uses

International Dairy Federation (FIL-IDF) recommends this liquid medium, for the isolation and enumeration of yeast and moulds in milk and dairy products, using the most probably number (MPN) method.

Bibliography

FIL-IDF(1991) Standard 94B. Enumeration of yeast and moulds. Colony Count Technique at 25°C.
 ISO (1981) ISO/DIS 6611: Milk and Milk products: Enumeration of yeast and moulds colony count technique at 25°C.
 DIN Standard 10186. Mikrobiologische Milchuntersuchung. Bestimmung der Anzahl von Hefen und Schimmelpilzen

Microbiological Test

Microorganisms	Growth
<i>Escherichia coli</i> ATCC 25922	Inhibited
<i>Candida albicans</i> ATCC 2091	Satisfactory
<i>Staphylococcus aureus</i> ATCC 25923	Inhibited
<i>Aspergillus</i> spp.	Satisfactory
<i>Lactobacillus casei</i> ATCC 9595	Inhibited

GN ENRICHMENT BROTH

Cat. 1248

For the selective culture of Gram negative Enterobacteria, especially *Shigellas*,
from all types of research materials

Formula in grams per liter

<i>Tryptose</i>	20,00	<i>Sodium chloride</i>	5,00
<i>Sodium citrate</i>	5,00	<i>D-Mannitol</i>	2,00
<i>Dipotassium Hydrogen phosphate</i>	4,00	<i>Potassium Dihydrogen phosphate</i>	1,50
<i>Dextrose</i>	1,00	<i>Sodium desoxycholate</i>	0,50

Final pH 7,0 ± 0,2 at 25°C

Preparation

Suspend 39 grams of the medium in one liter of distilled water. Heat with frequent agitation to dissolve the medium completely. Dispense into tubes and sterilize at 121°C (15 lbs. sp.) for 15 minutes

Uses

GN stands from Gram Negative, as this medium is used for isolating and cultivating gram negative microorganisms.

The GN Enrichment Broth encourages the growth of *Salmonellas* and *Shigellas* due to its content in mannitol, as it favors growth of mannitol-fermenting *Salmonella* and *Shigella* over mannitol non fermenting species such as *Proteus*.

The gram-positive microorganisms are inhibited by the presence of citrate and desoxycholate.

If *Proteus* and *Pseudomonas aeruginosa* are present, its growth in the first hours of incubation is very scarce, it does not occur the same with *Salmonellas* and *Shigellas*.

Bibliography

Hajna, A.A. 1955. A new enrichment broth medium for gram-negative organisms of the intestinal group. Public Health Lab. 13:83-89.
MacFaddin, J.F. 1985 Media for isolation-cultivation-identification-maintenance of medical bacteria, vol 1, p. 357-359. Williams & Wilkins, Baltimore, MD.

Microbiological Test

Microorganisms	Growth
<i>Shigella flexneri</i> ATCC 12022	Satisfactory
<i>Salmonella typhimurium</i> ATCC 14028	Satisfactory
<i>Escherichia coli</i> ATCC 25922	Satisfactory
<i>Streptococcus faecalis</i> ATCC 11700	Light
<i>Bacillus cereus</i> ATCC 11778	Inhibited

HEKTOEN ENTERIC AGAR

Cat. 1030

For the isolation and differentiation of enteric pathogens such as *Salmonella*, *Shigella*, and other enterobacteria

Formula in grams per liter

<i>Meat Peptone</i>	12,00	<i>Lactose</i>	12,00
<i>Sucrose</i>	12,00	<i>Bile Salts N° 3</i>	9,00
<i>Sodium Chloride</i>	5,00	<i>Sodium Thiosulfate</i>	5,00
<i>Yeast Extract</i>	3,00	<i>Salicin</i>	2,00
<i>Ferric Ammonium Citrate</i>	1,50	<i>Acid Fuchsin</i>	0,10
<i>Bromthymol Blue</i>	0,065	<i>Bacteriological Agar</i>	14,00

Final pH 7,5 ± 0,2 at 25°C

Preparation

Suspend 76 grams of the medium in one liter of distilled water. Heat and boil until completely dissolved. **DO NOT OVERHEAT. DO NOT STERILIZE IN AUTOCLAVE.** Cool to 55°C-60° C and pour into Petri dishes.

Uses

The difference between coliforms and other enteric organisms is easily discerned on Hektoen Enteric Agar. The colonies of coliforms are salmon to orange in color, while *Salmonella* and *Shigella* are green to greenish-blue. *Proteus* is not inhibited but produces a yellowish-green colony when it grows. The colonies of *Proteus* and *Salmonella* can present a black center if they form H₂S.

The specimen is seeded by streaking directly on the surface of the medium, or is first enriched in tetrathionate broth, selenite cystine broth, or GN broth and incubated at 35°C for 18 to 24 hours. It is recommended to seed the sample at the same time on other selective media for enterobacteria because a larger number of positive cultures will be obtained. These can be, for example,

Eosin Methylene Blue Agar, MacConkey Agar, SS Agar, Brilliant Green Agar, Desoxycholate Lactose Agar, or XLD Agar. The indicator system of acidity or alkalinity is the bromothymol blue and acid fuchsin.

E. coli, while suppressed, and other organisms which utilize lactose, sucrose, and/or salicin with production of acid, give colonies whose tones vary from yellow to orange to salmon. The *salmonellas* and *shigellas* are green or bluish-green. *Salmonellas* presents colonies with clear edges and black centers, from the formation of iron sulfide resulting from H₂S production.

Bibliography

King, S. & Metzger Appl. Microbiol. 16:577, 1968. King, S. & Metzger Appl. Microbiol. 16:579, 1968.
 Isenberg, Kominos & Siegel. Appl. Microbiol. 18:656, 1969.
 Hoben, Aston & Peterson Appl. Microbiol. 26:126, 1973.
 Polloch & Dalhgren. Appl. Microbiol. 27:197, 1974. Peloxv, Lavirotte & Pons Microbia, Tomo I No. 1, 1975.
 Goo et al Appl. Microbiol. 26:288, 1973.

Microbiological Test

Microorganisms	Growth	Colony colour
<i>Enterobacter cloacae</i> ATCC 13047	Acceptable	Orange
<i>Escherichia coli</i> ATCC 25922	Acceptable	Orange
<i>Salmonella enteritidis</i> ATCC 13076	Satisfactory	Blue-greenish
<i>Salmonella Thyphimurium</i> ATCC 14028	Satisfactory	Blue-greenish
<i>Shigella flexneri</i> ATCC 12022	Satisfactory	Blue-greenish
<i>Streptococcus faecalis</i> ATCC 11700	----	----

INDOL NITRATE MEDIUM (TRYPTICASEIN NITRATE MEDIUM) Cat. 1504

For the differentiation of microorganisms on the basis of indol production and the reduction of nitrate to nitrite

Formula in grams per liter

Casein Peptone.....	20,00	Disodium Phosphate	2,00
Dextrose	1,00	Potassium Nitrate.....	1,00
Bacteriological Agar	1,00		

Final pH 7,2 ± 0,2 at 25°C

Preparation

Suspend 25 grams of the medium in one liter of distilled water. Mix well. To perform motility and gas detection tests add 2 grams of agar. Heat agitating frequently until boiling and completely dissolved. Dispense into test tubes till half their height and sterilize in autoclave at 121 °C (15 lbs. sp) for 15 minutes. If the prepared medium is semisolid allow to solidify the tubes in a vertical position.

Uses

Utilize the medium during the first 2 days after preparation. If kept longer, heat in a waterbath to boiling to regenerate the medium.

Identification of negative gram bacilli

The test for indol should be conducted at 24-48 hours incubation (or after good bacterial growth) by adding a few drops of Kovacs or Ehrlichs reagents. A positive test is the formation of a pink to red color in the reagent layer after a few minutes. Nitrate reduction tests are conducted using Gries reagent consisting of 2 solutions:

Solution A

Sulfanilic acid.....	8 g
Acetic acid 5N	1 litre

Solution B

Alpha-naphthylamine	5 g
Acetic acid 5N	1 litre

Store refrigerated (4°C). Generally both reactivities (A and B) are stable for approximately 3 months.

For investigation of nitrate reduction, use 3 separate tubes. A positive control (*Escherichia coli*), a negative control (*Acetobacter calcoaceticus*) and the third tube for the study.

Procedures

1. Inoculate heavily each tube by stabbing.
2. Incubate at 35°C for 8, 12, and 24 hours.
3. Add approximately 10 drops of the Solutions A and B, or drops of Solution A plus 5 drops of Solution B.
4. The formation of a red color in 1-2 minutes indicates reduction of nitrates to nitrites. (Positive test).
5. If no color appears, add to the tubes a pinch of zinc in powder form (free of nitrates and nitrites).
6. Observe if the red color forms or the culture remains colourless.

Interpretation

- a) If there is no nitrate reduction the zinc will be reduced to nitrite and will form a red color upon reacting with the Gries reagent. The test organism is negative (Absence of nitrates).
- b) If there is no appearance of color, this indicates that the organism reduced the nitrate present in the culture medium to nitrite, possibly carrying the reaction to the gaseous nitrogen. The test organism is positive (Presence of nitrates).

Bibliography

Finegold, S.M., Sutter, V.L.; Ahebery, H.R.; Rosenblatt, J.E.: Isolation of Anaerobic Bacteria. Man. Clin. Micro. Biol. 2nd ed. 1974, 365:375. Finegold, S.M.; Rosenblatt, J.E.: Practical Aspects of Anaerobic Sepsis Medicine. 1973, 52(4), 311:322.

Microbiological Test

Microorganisms	Growth	Colony colour
<i>Enterobacter cloacae</i> ATCC 13047	Acceptable	Orange
<i>Escherichia coli</i> ATCC 25922	Acceptable	Orange
<i>Salmonella enteritidis</i> ATCC 13076	Satisfactory	Blue-greenish
<i>Salmonella thyphimurium</i> ATCC 14028	Satisfactory	Blue-greenish
<i>Shigella flexneri</i> ATCC 12022	Satisfactory	Blue-greenish
<i>Streptococcus faecalis</i> ATCC 11700	----	----

KAA CONFIRMATORY AGAR (CENAN)

Cat. 1027

For the isolation and confirmation of Lancefield Group D streptococci in foods

Formula in grams per liter

<i>Tryptone</i>	20,00	<i>Yeast Extract</i>	5,00
<i>Sodium Chloride</i>	5,00	<i>Disodium Citrate</i>	1,00
<i>Esculin</i>	1,00	<i>Ferric Ammonium Citrate</i>	0,50
<i>Sodium Azide</i>	0,15	<i>Kanamycin Sulfate</i>	0,020
<i>Bacteriological Agar</i>	15,00		

Final pH 7,0 ± 0,2 at 25°C

Preparation

Suspend 48 grams of the medium in one liter of distilled water. Heat with frequent agitation until complete dissolution. Distribute into appropriate containers and sterilize at 121°C (15 lbs. sp.) for 15 minutes. Pour into petri dishes.

Uses

KAA (Kanamycin, Aesculin, Azide) Confirmatory Agar is used to confirm presumptive positives from KAA Broth tubes. Kanamycin and azide have a great inhibition effect on the accompanying bacterial flora and is highly selective for D-streptococci.

Streak to obtain isolated colonies and incubate at 37°C for 48 hours. Lancefield Group D *streptococci* grow forming small, translucent colonies surrounded by a black halo.

Bibliography

M.R. Pascual Anderson. Técnicas para Examen Microbiológico de Alimentos y Bebidas (Centro Nacional de Alimentación y Nutrición) Madrid, 1982.
 Brandl, E. Aspergerger H., Pfleger, F. U-IBEN CH: Zum Vorkommen von D-streptokokken in Käse. 1985.

Microbiological Test

Microorganisms	Growth	Turn
<i>Streptococcus faecalis</i> ATCC 11700	Satisfactory	Olive green-black
<i>Streptococcus faecium</i> ATCC 8043	Satisfactory	Olive green-black
<i>Staphylococcus aureus</i> ATCC 6538	Moderate	---
<i>Escherichia coli</i> ATCC 11775	Inhibited	---
<i>Streptococcus lactis</i> ATCC 19435	Slightly inhibited	olive green-black to colourless

K.A.A. PRESUMPTIVE BROTH

Cat. 1209

For the presumptive detection of Lancefield Group D streptococci from foods

Formula in grams per liter

<i>Tryptone</i>	20,00	<i>Yeast Extract</i>	5,00
<i>Sodium Chloride</i>	5,00	<i>Disodium Citrate</i>	1,00
<i>Esculin</i>	1,00	<i>Ferric Ammonium Citrate</i>	0,50
<i>Sodium Azide</i>	0,150	<i>Kanamycin Sulfate</i>	0,02

Final pH 7,0 ± 0,2 at 25°C

Preparation

Suspend 33 grams of the medium in one liter of distilled water. Mix well. Heat slowly till completely dissolved. Dispense and sterilize at 121°C (15 lbs sp) for 15 minutes. Distribute and sterilize in autoclave at 121°C for 15 minutes.

Uses

Kanamycin and azide have a great inhibition effect on the accompanying bacterial flora and is highly selective for D-streptococci. Inoculation of 1 ml. and 0,2 ml. aliquots of sample are made in tubes of strength medium. 10 ml. sample aliquots or more are made in double strength medium tubes.

Always utilize 5 tubes for the MPN method counts.

Incubate at 37°C for 48 hours. Positive tubes demonstrate a brownish black color. Counts are by the MPN method.

Bibliography

M.R. Pascual Anderson *Técnicas para Examen Microbiológico de Alimentos y Bebidas* (Centro Nacional de Alimentación y Nutrición) Madrid, 1982
Corps pag. 76 Aleman
Brandl, E. Aspergerger H., Pfleger, F. U-IBEN CH: Zum Vorkommen von D-streptokokken in Käse. 1985.

Brandl, E. Aspergerger H., Pfleger, F. U-IBEN CH: Zum Vorkommen von D-streptokokken in Käse. 1985.

Microbiological Test

Microorganisms	Growth
<i>Streptococcus faecalis</i> ATCC 11700	Satisfactory
<i>Streptococcus faecium</i> ATCC 8043	Satisfactory
<i>Staphylococcus aureus</i> ATCC 6538	Moderate
<i>Escherichia coli</i> ATCC 11775	Inhibited
<i>Streptococcus loctis</i> ATCC 19435	Moderate-Inhibited

KF STREPTOCOCCAL AGAR

Cat. 1034

For the isolation and enumeration of fecal streptococci, by direct culture or by membrane filtration.

Formula in grams per liter

Maltose.....	20,00	Peptone Mixture	10,00
Yeast Extract	10,00	Sodium Glycerophosphate	10,00
Sodium Chloride.....	5,00	Lactose	1,00
Sodium Azide.....	0,40	Bacteriological Agar	20,00

Final pH 7,2 ± 0,2 at 25°C

Preparation

Suspend 76,4 grams of the medium in one liter of distilled water. Mix well. Heat with frequent agitation and boil for one minute. Sterilize at 121°C (15 lbs. sp.) for 10 minutes. Cool to 50°C or 60°C and aseptically add 10 ml. of 1% TTC (Trifenil Tetrazolium Chlorure) solution per liter of sterile medium.

Uses

The KF *Streptococcal* Agar is used for the plate counts of *streptococci* in water samples. The plates are incubated for 48 hours at 35°C. At times it is necessary to prolong the incubation for 72 hours.

The addition of 1% TTC allows enterococci to develop a red colour as the result of the reduction of tetrazolium to an acid azodye

The red or rose colonies are counted as streptococci, while colonies with orange, yellow, white or the other

colors are not counted. The number of *streptococci* is calculated per 100 ml. of water.

This medium is used more for determining the presence of *Streptococcus faecalis* in milk and its derivatives, as well as in other foods. Isolation and enumeration of fecal streptococci is made according to APHA.

Bibliography

Ramos Cordova, Mario. "Manual of Methods of Milk and Lactose Analysis". Edition of Author, Mexico, D. F., 1976.
 Kenner, Clark and Kabler, Applied Microbiol. 9:15, 1961.
 Donnelly C.W., R.E. Bracket, D.Doores, W.H. Lee, and J. Lovett. 1992. Compendium of methods for the microbiological examination of foods, 3rd ed. American Public Health Association, Washington, D.C.

Microbiological Test

Microorganisms	Growth	Colony colour
<i>Enterobacter aerogenes</i> ATCC 13048	Inhibited	----
<i>Escherichia coli</i> ATCC 25922	Inhibited	----
<i>Streptococcus faecalis</i> ATCC 19433	Satisfactory	Red
<i>Streptococcus faecalis</i> ATCC 29212	Satisfactory	Red

KING A MEDIUM PSEUDOMONAS P AGAR Cat. 1531

For the identification of *Pseudomonas*, it favours the production of pyocyanin

Formula in grams per liter

<i>Bacteriological Peptone</i>	20,00	<i>Potassium Sulfate</i>	10,00
<i>Magnesium Chloride</i>	1,40	<i>Bacteriological Agar</i>	13,6

Final pH 7,0 ± 0,2 at 25°C

Preparation

Suspend 45 grams of the medium in one liter of distilled water. Add 10 ml. of glycerin. Heat with frequent agitation and boil for 1 minute. Dispense into appropriate containers and sterilize by autoclaving at 121°C (15 lbs sp) for 15 minutes.

Uses

Pseudomonas P Agar promotes the production of pyocyanin and/or pyorubin and inhibits fluorescein production by *Pseudomonas*. Pyocyanin produced by *pseudomonas* gives a blue color diffusing into the medium. A greenish-blue color denotes a small amount of fluorescein production not totally inhibited.

Both *Pseudomonas P* and *Pseudomonas F* Agar should be used in parallel to differentiate the *Pseudomonas* species.

Incubate up to 7 days at 35°C and check for bacteriological growth after 24-48 and 72 hours and then after 6 days. Colonies of *Pseudomonas aeruginosa* will appear surrounded by a blue to green zone.

Bibliography

J. Lab. and Clin. med. 44:3301 USP XIX
King E.O. Ward, M.K. a Raney. Two simple media for the demonstration of pyocyanin and fluorescein. J. Lab. Clin. Med. 44 1954.
U.S. Pharmacopoeia XXIII, 1995.

Microbiological Test

Microorganisms	Growth	Colony colour
<i>Pseudomonas aeruginosa</i> ATCC 9027	Satisfactory	Blue
<i>Pseudomonas aeruginosa</i> ATCC 10145	Satisfactory	----
<i>Pseudomonas aeruginosa</i> ATCC 17934	Satisfactory	----
<i>Pseudomonas aeruginosa</i> ATCC 25619	Satisfactory	Blue-green
<i>Pseudomonas aeruginosa</i> ATCC 27853	Satisfactory	Blue

KING B MEDIUM PSEUDOMONAS F AGAR Cat. 1532

 Medium for the identification of *Pseudomonas*. It favours the production of fluorescein.

Formula in grams per liter

<i>Peptone Mixture</i> 20,00	<i>Dipotassium Phosphate</i> 1,50
<i>Magnesium Sulfate</i> 1,50	<i>Bacteriological Agar</i> 14,00

Final pH 7,0 ± 0,2 at 25°C
Preparation

Suspend 37 grams of the medium in one liter of distilled water. Add 10 ml. of glycerin. Heat with frequent agitation and boil for one minute.

Dispense into appropriate containers and sterilize by autoclaving at 121°C (15 lbs.sp) for 15 minutes.

Uses

Pseudomonas F Agar promotes fluorescein production (while pyocyanin production is inhibited) by *Pseudomonas*. Under UV stimulation fluorescein is demonstrated by a fluorescent yellow color diffused in the medium. When a greenish yellow color appears, it is due to small amounts of pyocyanin not totally suppressed. Cultures of *pseudomonas* exist which produce a pigment or fluorescein or both and, because of this situation, it is

recommended to utilize both *Pseudomonas F* and *Pseudomonas P (Pyocyanin)* Agar to better identify the species.

Colonies of *Pseudomonas aeruginosa* will appear surrounded by a yellow to yellow-green zone resulting from fluorescein production.

Bibliography

J. Lab. and Clin. Med 44:301, 1954 USP XIX
 King E.O. Ward, M.K. a Raney. Two simple media for the demonstration of pyocyanin and fluorescein. J. Lab. Clin. Med. 44 1954.
 U.S. Pharmacopoeia XXIII, 1995.

Microbiological Test

Microorganisms	Growth	Colony colour
<i>Pseudomonas aeruginosa</i> ATCC 9027	Satisfactory	Yellow-green
<i>Pseudomonas aeruginosa</i> ATCC 10145	Satisfactory	Yellow-green
<i>Pseudomonas aeruginosa</i> ATCC 17934	Satisfactory	----
<i>Pseudomonas aeruginosa</i> ATCC 25619	Satisfactory	Yellow-green
<i>Pseudomonas aeruginosa</i> ATCC 27853	Satisfactory	Yellow-green

KING FG AGAR

Cat. 1053

For the recount of psicrotrophic microorganisms.

Formula in grams per liter

<i>Bacteriological Peptone</i>	20,00	<i>Maltose</i>	10,00
<i>Sodium Chloride</i>	5,00	<i>Potassium Phosphate</i>	1,50
<i>Magnesium Sulfate</i>	0,75	<i>Bacteriological Agar</i>	15,00

Final pH 7,0 ± 0,2 at 25°C

Preparation

Suspend 52,25 grams of medium in one liter of deionized or distilled water. Mix well. Heat with frequent agitation and boil for one minute. Sterilize in the autoclave at 121°C (15 lbs psi) for 15 minutes. Cool at 50°C and aseptically add 2 ml. of sterile-filtered 0,05% crystal violet solution. Mix well and dispense into Petri dishes.

Uses

Psychrotrophic organisms are those which can tolerate low temperatures between 4-20°C. Organisms in this group are *Pseudomonas*, *Achromobacter*, *Alcaligenes*, *Flavobacterium*, *Aeromonas* as well as other species of enterobacteria from the genera: *Escherichia*, *Proteus*, *Klebsiella*, *Enterobacter* and *Hafnia*. All are gram-negative microorganisms.

The total count per ml. of food sample is performed using serial dilutions, placing 1.0 ml. of each dilution on the surface of the medium and spreading with a sterile glass rod. Incubation is for 5 days at 17°C. Count only larger (not punctiform) colonies and multiply by the dilution factor to obtain the total count.

Bibliography

Pascual Anderson – Metodología analítica para alimentación y bebidas - Diaz Santos, 199.

Microbiological Test

Microorganisms	Growth
<i>Pseudomonas</i> spp.	Satisfactory
<i>E. Coli</i> ATCC 25922	Satisfactory
<i>Proteus mirabilis</i> ATCC 14273	Satisfactory

KLIGLER IRON AGAR

Cat. 1042

Used for the differentiation of Gram-negative Enterobacteriae.

Formula in grams per liter

Peptone mixture	20,00	Lactose	10,00
Sodium Chloride	5,00	Dextrose	1,00
Ferric Ammonium Citrate	0,50	Sodium Thiosulfate.....	0,50
Phenol Red	0,025	Bacteriological Agar	15,00

Final pH 7,4 ± 0,2 at 25°C

Preparation

Suspend 52 grams of the medium in one liter of distilled water. Mix well and heat with frequent agitation. Boil for one minute. Dispense into tubes and sterilize at 121° C (15lbs. pressure) for 15 minutes. Allow to cool in a slanted position so as to obtain butts of 1'5-2 cm. Depth. For greater accuracy, Kligler Iron Agar should be used on the day of preparation or melted and solidified before use.

Uses

Inoculate the medium with the colony in study by stabbing the butt and streaking the surface of the tube. Lactose fermentating organisms totally acidify the medium,

resulting in a yellow color. The microorganisms which do not ferment lactose acidify only in the bottom of the tube, with the slanted surface remaining the same original cherry red color. Formation of hydrogen sulfide blackens the medium.

The results are interpreted the same as TSI Agar. It is recommended using the medium on the same day of preparation.

Bibliography

J. Bact. 13:183, 1927. J. Bact. Clin. Med. 25:649, 1940.

A.= Acid
() = Occasional reactions
Alk.= Alkaline

ORGANISMS	SLANT	DEPTH	GAS	H ₂ S
<i>Enterobacter & Klebsiella</i>	A.	Alk.	++	-
<i>Hafnia</i>	Alk.	A.	+	-
<i>Escherichia coli</i>	A.(Alk.)	A.	+(-)	-
<i>Shigella</i>	Alk.	A.	-	-
<i>Salmonella typhi</i>	Alk.	A.	-	+(-)
<i>S. paratyphi & S. choleraesuis</i>	Alk.	A.	+	-
<i>Other Salmonella</i>	Alk.	A.	+	+++
<i>Citrobacter</i>	Alk.(A.)	A.	+	+++(-)
<i>Edwardsiella</i>	Alk.	A.	+	+++
<i>Serratia</i>	Alk.(A)	A.	-	-
<i>P. vulgaris</i>	A.(Alk.)	A.	+	+++
<i>P. mirabilis</i>	Alk.(A.)	A.	+	+++
<i>P. morganii & P. rettgeri</i>	Alk.	A.	-	-
<i>Providencia</i>	Alk.	A.	+or-	-

Microbiological Test

Microorganisms	Growth	Slide	Base	H ₂ S	Gas
<i>Escherichia coli</i> ATCC 25922	Good	Yellow	Yellow	-	+
<i>Proteus vulgaris</i> ATCC 6380	Good	Red	Yellow	+	-
<i>Salmonella enteritidis</i> ATCC 13076	Good	Red	Yellow	+	+
<i>Shigella flexneri</i> ATCC 12022	Good	Red	Yellow	-	-
<i>Citrobacter freundii</i> ATCC 8090	Good	Yellow	Yellow	+	+

KOSER CITRATE BROTH

Cat. 1200

For the differentiation of *E.coli* from *Enterobacter* on basis of the citrate utilization.

Formula in grams per liter

<i>Sodium Citrate</i>	3,00	<i>Sodium Ammonium Phosphate</i>	1,50
<i>Monopotassium Phosphate</i>	1,00	<i>Magnesium Sulfate</i>	0,20

Final pH 6,7 ± 0,2 at 25°C

Preparation

Suspend 5,7 grams of the medium in one liter of distilled water. Mix well until completely dissolved. Dispense into screw-capped tubes. Sterilize at 121°C (15 lbs sp) for 15 minutes with loose caps. Tighten the caps after sterilization.

Uses

Koser Citrate Broth is used to differentiate *E. coli* from the *Enterobacter* group in the same way as Simmons Citrate Agar, but its advantage is that it is possible to differentiate between coliforms of fecal origin (majority are citrate-

negative) and organisms from dirt which are 90% positive according to Wilson and Miles. These same authors report only 6,7% of the coliforms isolated from human or animal feces are citrate-positive.

BIBLIOGRAPHY

Koser J. Bact. 8:493, 1973. Wilson G.S. and Miles A.A., "Topley and Wilson's Principles of Bacteriology and Immunology", 4th Ed., Edward Arnold Ltd., London, Vol. 1, page 760.

Microbiological Test

Microorganisms	Growth
<i>Enterobacter aerogenes</i> ATCC 13048	Satisfactory
<i>Enterobacter cloacae</i> ATCC 23355	Satisfactory
<i>Escherichia coli</i> ATCC 25922	Null

LACTOSE BROTH

(EUROPEAN PHARMACOPEIA)

Cat. 1206

Medium used for the study of lactose fermentation.

Formula in grams per liter

Gelatin Pancreatic digest 5,00 *Lactose monohydrate*..... 5,00
Beef Extract 3,00

Final pH 6,9 ± 0,2 at 25°C

Preparation

Suspend 13 grams of the medium in one liter of distilled water. Heat agitating frequently until completely dissolved. Dispense into test tubes with gas collecting tubes. Sterilize in autoclave at 121°C (15 lbs.sp) for 15 minutes. Cool as quickly as possible.

Uses

Check the sterilization of the medium by incubating the tubes at 35°C for 24 hours prior to inoculation

Seed aliquots of 1, 5, or 100 ml. of the sample liquid in containers adequate for the quantity of medium. Incubate at 35°C for 24 to 48 hours and check for the presence of gas, which constitutes a presumptive test.

For details consult the standard methods for water, milk, and food analysis texts, or the European pharmacopoeia.

Bibliography

American Public Health Association. Standard Methods of the Examination of Dairy Products, 12th Edition APHA, New York, 12th, 1967. American Public Health Association. Standard Methods for the Examination of Water and Wastewater Edition APHA, Inc. New York, 1966. European Pharmacopoeia, 4th Edition Microbiological examination of non-sterile products 2.002

Liquid Sample (Inoculum)	Lactose Broth (ml)	Lactose Broth Concentration
1	10	13
10	10	26
10	20	19,5
100	50	39,0
100	35	50,1
100	20	78,0

Microbiological Test

Microorganisms	Growth	Gas production
<i>Escherichia coli</i> ATCC 25922	Satisfactory	+
<i>Klebsiella pneumoniae</i> ATCC 13883	Satisfactory	+
<i>Salmonella typhimurium</i> ATCC 14028	Satisfactory	-
<i>Proteus vulgaris</i> ATCC 13315	Satisfactory	-

LACTOSE SULFITE BROTH BASE (EUROPEAN PHARMACOPEIA)

Cat. 1009

Selective medium recommended for the detection and enumeration of *C. perfringens*

Formula in grams per liter

Lactose monohydrate	10,00	Casein Peptone	5,00
Sodium chloride	2,50	Yeast extract	2,50
Cysteine hydrochloride	0,30		

Final pH 7,1 ± 0,1 at 25°C

Preparation

Suspend 20,3 grams of the medium in one liter of distilled water. Heat with frequent agitation for one minute until completely dissolved. Dispense by 8 ml in tube test with small gas collection durham tubes. Sterilize in autoclave at 121 °C (15 lbs sp) for 15 minutes. Store at 4°C. Before using add to each tube 0.5 ml of a solution of sodium metabisulfite 12 g/liter and 0.5 ml of a solution of 10 gr/liter of ferroammonium citrate, both solutions have to be fresh prepared and sterilized.

any other suitable containers with a small Durham tube. Mix with minimum shaking and incubate at 45,5°C – 46,5°C for 24/48 hour.

The containers showing a blackening due to iron sulfide and abundant formation of gas in the Durham tube (at least 1/10 of the volume) indicate the presence of *C. Perfringens*. Estimate the most probably number use the appropriate table (M.P.N. Table).

Bibliography

European Pharmacopoeia 4th Edition 2002 p. 137-140.

Uses

This is a selective medium used to detect and enumerate *C. perfringens* using the techniques of most probable number of bacteria. The European Pharmacopoeia recommends it and named it Medium R. Use it in tubes or

Microbiological Test

Microorganisms	Growth	Gas production	Blackening
<i>Clostridium perfringens</i> ATCC 12919	Satisfactory	+	+

LAURYL SULFATE AGAR FOR MEMBRANE FILTRATION

Cat. 1309

Selective isolation and count of coliforms.

Formula in grams per liter

<i>Lactose</i>	30,00	<i>Casein Peptone</i>	40,00
<i>Sodium lauryl sulfate</i>	1,00	<i>Yeast extract</i>	6,00
<i>Bacteriological Agar</i>	15,00.		

Final pH 7,4 ± 0,2 at 25°C

Preparation

Suspend 92 grams of medium in one liter of distilled water.
Heat with frequent agitation until completely dissolved.
Sterilize at 121°C (15 lbs. sp.) for 15 minutes .

Uses

Lauryl Sulphate Agar is a selective medium used for the presumptive coliform detection method in milk and food.

Bibliography

APHA 1999 Standard Methods for the examination of water and wastewater, 20th edition.

Microbiological Test

Microorganisms	Growth	Colony colour
<i>Enterobacter aerogenes</i> ATCC 13048	Satisfactory	Pink
<i>Escherichia coli</i> ATCC 25922	Satisfactory	Purple
<i>Salmonella enteritidis</i> ATCC 13076	Satisfactory	Clear
<i>Staphylococcus aureus</i> ATCC 25923	Inhibited	---
<i>Enterococcus faecalis</i> ATCC 19433	Inhibited	---

LAURYL SULFATE BROTH

Cat. 1310

Recommended for use in the detection of coliform organisms in waters. (A.P.H.A)

Formula in grams per liter

<i>Tryptose</i>	20,00	<i>Lactose</i>	5,00
<i>Sodium Chloride</i>	5,00	<i>Dipotassium Phosphate</i>	2,75
<i>Monopotassium Phosphate</i>	2,75	<i>Sodium Lauryl Sulfate</i>	0,10

Final pH 6,8 ± 0,2 at 25°C

Preparation

Suspend 35,6 grams of the medium in one liter of distilled water. Dissolve the medium completely. Dispense in test tubes containing inverted Durham vials. Sterilize by autoclaving at 121°C (15 lbs. sp.) for 15 minutes. Refrigerated broth becomes cloudy, but clears considerably at room or incubator temperatures. Clarity is not required for performance because only gas formation is considered significant.

Uses

Lauryl Sulfate Broth is a selective medium recommended for the enumeration of coliforms in water and dairy products as well as for confirmatory tests of lactose fermentation with gas production by coliforms in foods.

Sporulating aerobic bacteria are completely inhibited.

Another advantage of this medium is the indol test can be performed directly in the tube.

Bibliography

APHA 1999. Standar Methods for the examination a water and wastewater, 20th Edition.

Microbiological Test

Microorganisms	Growth	Gas Production
<i>Enterobacter aerogenes</i> ATCC 13048	Satisfactory	+
<i>Escherichia coli</i> ATCC 25922	Satisfactory	+
<i>Salmonella typhimurium</i> ATCC 14028	Satisfactory	-
<i>Staphylococcus aureus</i> ATCC 25923	Strongly Inhibited	-

LEVINE E.M.B. AGAR (EOSINE METHYLENE BLUE) Cat. 1050

Used for the isolation and differentiation of enteric bacilli and coliform microorganisms.

Formula in grams per liter

<i>Gelatin Peptone</i>	10,00	<i>Lactose</i>	10,00
<i>Dipotassium Phosphate</i>	2,00	<i>Eosin</i>	0,40
<i>Methylene Blue</i>	0,065	<i>Bacteriological Agar</i>	15,00

Final pH 7,1 ± 0,2 at 25°C

Preparation

Suspend 37,5 grams of the medium in one liter of distilled water. Mix well until a uniform suspensions is obtained. Heat with frequent agitation and boil for 1 min. Distribute and sterilize at 121° C (15 lbs. sp) for 15 minutes. Swirl gently the sterile, liquid agar is cooled to 45°C before pouring into Petri dishes.

Uses

Levine E.M.B. Agar is a selective medium for the investigation and differentiation of enteric bacilli and coliform microorganisms. It is also used for the isolation and identification of *Candida albicans* it is:

Characteristics of the colonies

Escherichia coli: 2 to 3 mm. in diameter. Blue-black in the center, with edges clear to transmitted light, often with a metallic green sheen with reflected light.

Enterobacter aerogenes: Large, 4 to 6 mm. in diameter. Elevated and mucoid. Grayish-brown in the center to transmitted light. Generally it does not have a metallic sheen.

Salmonella and *Shigella*: Transparent, amber to colourless.

Proteus: When there is no swarming, similar to *Salmonella* or *Shigella*.

Staphylococcus (coagulase positive): Punctiform, colourless.

Candida albicans: After 24 to 48 hours at 35°C in 10% CO₂, feathery or in the form of a spider web.

Other *Candida*: Flat, round, yeast-like colonies. From time to time *Nocardia* can be isolated.

Rapid Identification of *C. albicans*:

The suspect clinical material such as sputum, expectorations, oral or vaginal secretions and skin and nail scrapings are streaked on the surface of the LEMB Agar which contains added tetracycline. After 24-48 hours of incubation at 35°C in an atmosphere of approximately 10% CO₂, colonies appear feathery or similar to a "spider's web". As the method is not always uniform, check at the same time for the production of chlamydospores in special media (Cornmeal Agar, Czapek Dox Agar, etc.) and conduct rapid tests for sugar fermentations.

The colonies of coagulase positive *staphylococci*, golden colored strains as well as white (*S. aureus* and *epidermidis*), give punctiform and colourless colonies.

Bibliography

Levine, J. Inf. Dis. 22:43, 1981. J. Bact. 45:471, 1943. Vogel, R.A. and Moses, R.M. Weld's Method for the Rapid Identification of *Candida albicans* in Clinical Materials. Am. J. Clin. Path. 28:103-106, 1957.

Microbiological Test

Microorganisms	Growth	Colony colour
<i>Enterobacter aerogenes</i> ATCC 13048	Satisfactory	Pink
<i>Proteus mirabilis</i> ATCC 14273	Satisfactory	Colourless
<i>Salmonella typhimurium</i> ATCC 14028	Satisfactory	Colourless
<i>Escherichia coli</i> ATCC 25922	Satisfactory	Purple-green, sheen
<i>Staphylococcus aureus</i> ATCC 25923	Inhibited	metallic, black centre

LISTERIA OXFORD AGAR BASE Cat. 1133

Selective medium for the detection of *Listeria monocytogenes*.

Formula in grams per liter

<i>Columbia Agar Base</i>	39,00	<i>Lithium Chloride</i>	15,00
<i>Esculine</i>	1,00	<i>Ferric-ammonium Citrate</i>	0,50

Final pH 7,2 ± 0,2 at 25°C

Preparation

Suspend 27,75 grams of medium in 500 ml. of distilled water. Heat with frequent agitation until complete dissolution. Distribute into appropriate containers. Sterilize in autoclave at 121°C (15 lbs. psi) during 15 minutes. Cool to 50°C and aseptically add the reconstituted supplement.

Uses

The selective medium for *Listeria* according to the Oxford formula is recommended for the detection of *Listeria monocytogenes* from clinical samples and food products. The medium uses Lithium chloride as an inhibiting agent as well as other supplements which inhibit the growth of Gram negative bacteria and a large part of Gram positive ones.

The system indicator is esculin and iron for isolation and differentiation of *Listeria*. *Listeria monocytogenes* hydrolyses esculin to esculetin forming black complexes. Apart from that, *Listeria monocytogenes* produces greenish brown colonies with a black zone.

Bibliography

Curtis, G.D.W. , Mitchell, R.G., King, A.F., Griffin E.J.A selective medium for the isolation of *Listeria monocytogenes*. Letters in Appl. Microbiol.8.95-98.

Microbiological Test

Microorganisms	Growth	Colony colour
<i>Listeria monocytogenes</i> ATCC 19117	Good	+
<i>Staphylococcus aureus</i> ATCC 25923	None	-

LISTERIA FRASER ENRICHMENT BROTH BASE

Cat. 1120

Enrichment medium for detection and isolation of *Listeria* in food and environmental samples.

Formula in grams per liter

Sodium Chloride	20,00	Disodium Phosphate	12,00
Tryptone	10,00	Proteose Peptone	5,00
Yeast Extract	5,00	Lithium Chloride	3,00
Monopotassium Phosphate	1,35	Esculine	1,00

Final pH 7,2 ± 0,2 at 25°C

Preparation

Suspend 28,7 grams of medium in 500 ml. of distilled water. Heat with frequent agitation until complete dissolution. Sterilize in autoclave at 121°C (15 lbs. psi) during 15 minutes. Aseptically add the reconstituted supplement. Mix well and distribute. It may present a slight precipitate.

FRASER SUPPLEMENT (For 500 ml of prepared medium) aseptically reconstitute: 1 vial of Acryflavine + Nalidixic Acid in 2,0 ml of distilled water and 1 vial of Ferric Ammonium Citrate in 2,0 ml of distilled water.

Ferric Ammonium Citrate 250 mg / Nalidixic Acid 10,0 mg / Acryflavine 12,5 mg /

Uses

Fraser Broth is an appropriate medium for the detection of *Listeria* spp. in food products and in samples from the environment. All *Listeria* species hydrolyze the esculin to esculetin, this reacts with iron ions producing blackening.

Another advantage is that, the addition of ferric ammonium citrate improves the growth of *L. monocytogenes*. The Lithium chloride inhibits the growth of enterococci which can hydrolyze the esculin.

Inoculate 0,1 ml. of the sample in 10 ml. of Fraser broth. Incubate at 37°C during 26 ± 2 hours in aerobic conditions. Compare each inoculated tube with a non-inoculated control tube with white bottom. The tubes which present blackening should be inoculated again in Oxford medium. The tubes which keep the original colour, are considered as negative.

Bibliography

Fraser J.A. and Sperber W.H (1988) McClain D. and Lee W.H(1988)

Microbiological Test

Microorganisms	Growth
<i>Streptococcus faecalis</i> ATCC 29212	None
<i>Listeria monocytogenes</i> ATCC 19117	Good

LOWENSTEIN JENSEN MEDIUM BASE

Cat. 1116

The addition of whole egg makes it suitable for the cultivation of *M. tuberculosis* and other *Mycobacteria*.

Formula in grams per liter

Potato Flour.....	30,00	Asparagine.....	3,60
Monopotassium Phosphate.....	2,50	Magnesium Citrate.....	0,60
Malachite Green.....	0,40	Magnesium Sulfate.....	0,24

Final pH 7,0 ± 0,2 at 25°C

Preparation

Suspend 37,3 grams of the medium in 600 ml. of distilled water, with 12 ml. of Glycerol (do not add Glycerol if bovine bacilli or other glycerophobic organisms are to be cultivated) Heat with frequent agitation and boil for one minute. Sterilize in autoclave at 121°C (15 lbs sp) for 15 minutes. Cool to 50° C. Meanwhile, prepare one litre of whole egg, aseptically obtained and mixed without introducing air bubbles. Add slowly the egg to the base to obtain an homogeneous mixture without bubbles. Distribute into sterile screw capped tube. Place the tubes in an slanted position. Thyndallise to inspissate at 85-90°C for 45 minutes.

Uses

Lowenstein-Jensen Medium Base can be used, with whole egg, to isolate mycobacteria other than *M. leprae*. With 5% sodium chloride, Lowenstein-Jensen Medium can be used as an aid in the differentiation of mycobacteria on the basis of salt tolerance. *M. fortuitum*, *M. . triviale*, *M.*

chelonei and some strains of *M. flavescens* grow on this medium while most other *mycobacterial* strains are inhibited.

Lowenstein-Jensen Medium in a deep butt tube may be used to aid in the differentiation of *mycobacteria* on the basis of the catalase test.

Lowenstein-Jensen Medium with antibiotics can be used to selectively isolate *mycobacteria* and inhibit contaminating flora. Addition of ribonucleic acid to the Lowenstein-Jensen Medium may increase the number of positive cultures.

Bibliography

Bailey and Scott. Diagnostic Microbiology. The C.V. Mosby Company, Saint Louis, 1978. Diagnostic Procedures and Reagents., APHA. Fifth Ed. 1970, New York. Raiza Nikolajuk of Irurzum and A.J.F., Irurzum. The Laboratory in the Diagnostics of Tuberculosis. Ed. Medical Panamericana, Buenos Aires, 1972.

Microbiological Test

Microorganisms	Growth
<i>Mycobacterium tuberculosis</i> H37RV	Satisfactory
<i>Micobacterium fortuitum</i> ATCC 6841	Satisfactory
<i>Mycobacterium kansasii</i> ATCC 12478	Satisfactory

LYSINE DECARBOXYLASE BROTH

Cat. 1208

Identification of enterobacteria. Lysine Decarboxylase Agar is used in the identification of microorganisms, especially enteric bacilli, based on the decarboxylation of lysine.

Formula in grams per liter

<i>Gelatin Peptone</i>	5,00	<i>L-Lysine</i>	5,00
<i>Yeast Extract</i>	3,00	<i>Dextrose</i>	1,00
<i>Bromcresol Purple</i>	0,02		

Final pH 6,8 ± 0,2 at 25°C

Preparation

Dissolve 14 grams of the medium in one liter of distilled water. Dispense in quantities of 5 ml in screw-capped tubes. Sterilize in autoclave at 121°C (15 lbs sp) for 15 minutes. Let the cap a bit loose to allow a good gas exchange. Close it well after sterilization.

Uses

The tubes are inoculated with the microorganism samples and incubated for 24 hours at 32° to 35°C, or if preferred, at 37°C.

The enteric bacilli produce acid in the initial fermentation of dextrose with a change to a yellow color. The cultures that decarboxylate lysine form cadaverine and the color returns

to the alkaline purple. A yellow color after 24 hours indicates a negative result.

The following chart indicates the typical reactions of the important groups of the *Enterobacteriaceae*:

With the substitution of arginine or ornithine for lysine, this medium (Falkow Broth Base) can be used to study the decarboxylation of these amino acids.

Bibliography

Falkow A. S. Clin. Path. 28:598, 1958.
Ewing Davis and Deaves, Studies in the Serratia Group. U.S. Dept. H.E.W.C.D.C. Atlanta, 1972.
Edwards and Ewing. Identification of Enterobacteriaceae, Burgess Publ. Co. Minneapolis, Minn., 1961.

	Purple
Positive	<i>Escherichia</i> <i>Klebsiella</i> <i>Salmonella</i> , except <i>S. paratyphi A</i> <i>Arizona</i> <i>Alkaescens-Dispar</i> <i>Serratia. Gpo. Hafnia</i>

	Yellow
Negative	<i>Proteus</i> <i>Providencia</i> <i>S. paratyphi A</i> <i>Shigella</i> <i>Aeromonas</i> <i>Citrobacter</i>

Microbiological Test

Microorganisms	Lysine
<i>Salmonella typhi</i> ATCC 6539	+
<i>Salmonella paratyphi A</i>	-
<i>Proteus vulgaris</i> ATCC 13315	-
<i>Salmonella gallinarum</i> NCTC 9240	+
<i>Serratia liquifaciens</i>	(+) slow

LYSINE IRON AGAR

Cat. 1044

Used in studies of decarboxylation of Lysine for rapid differentiation of *Salmonella* and *Arizona*.

Formula in grams per liter

<i>L-Lysine</i>	10,00	<i>Gelatin Peptone</i>	5,00
<i>Yeast Extract</i>	3,00	<i>Dextrose</i>	1,00
<i>Ferric Ammonium Citrate</i>	0,50	<i>Sodium Thiosulfate</i>	0,04
<i>Bromocresol Purple</i>	0,02	<i>Bacteriological Agar</i>	13,50

Final pH 6,7 ± 0,2 at 25°C

Preparation

Suspend 33 grams of the medium in one liter of distilled water. Mix well and dissolve while heating and boil for one minute. Dispense in tubes and sterilize in autoclave at 121° C (15 lbs.sp) for 12 minutes. Cool in a slanted position.

Uses

For the rapid differentiation of enterobacteria, especially *Salmonella* and *Arizona*. Lysine Iron Agar is very useful for the rapid differentiation of *Salmonella* and *Arizona* from *Citrobacter*. It is used to differentiate the *enterobacteria* on the basis of lysine decarboxylation and deamination and H₂S production. Some strains of *Arizona* can rapidly ferment lactose and form colonies that are colourless or pink to red, on media such as MacConkey Agar or Desoxycholate Agar. The strains which rapidly ferment the lactose produce a large quantity of acid, changing the

original purple colour of the medium to yellow. This could cause the *Arizona* strain to be interpreted as a *coliform*.

Lysine Iron Agar is especially formulated to avoid this confusion. *Salmonella* and *Arizona* *alkalinize* the medium by decarboxylating lysine, importing a bluish purple colour to the whole surface.

Proteus and *Providencia* produce a characteristic orange-red colour on the slant while the butt is yellow from the production of acid from the deamination of lysine.

Bibliography

Edwards and Fite Applied Microbiol. 9:478, 1961. Edwards and Ewing. Identification of Enterobacteriaceae. Burgess Publishing Co. Minneapolis, Minn., 1962.

Microbiological Test

Microorganisms	Growth	Slide	Base	H ₂ S
<i>Citrobacter freundii</i> ATCC 8090	Good	Red-purple	Yellow	+
<i>Escherichia coli</i> ATCC 25922	Good	Red-purple	Red-purple	-
<i>Proteus mirabilis</i> ATCC 25933	Good	Red-deep	Yellow	-
<i>Salmonella typhimurium</i> ATCC 14028	Good	Red-purple	Red-purple	+
<i>Shigella flexneri</i> ATCC 12022	Good	Red-purple	Yellow	-
<i>Salmonella arizonae</i>	Good	Red-purple	Red-purple	+

MACCONKEY AGAR (EUR. PHARM) Cat. 1052

Used for the study of Coliform organisms

Formula in grams per liter

<i>Pancreatic Digest of Gelatin</i>	17,00	<i>Lactose monohydrate</i>	10,00
<i>Sodium Chloride</i>	5,00	<i>Peptone Mixture</i>	3,00
<i>Bile Salts n° 3</i>	1,50	<i>Neutral Red</i>	0,03
<i>Crystal Violet</i>	0,001	<i>Bacteriological Agar</i>	13,50

Final pH 7,1 ± 0,2 at 25°C

Preparation

Suspend 50 grams of the medium in one liter of distilled water. Mix well until a uniform suspension is obtained. Heat with frequent gentle agitation and boil for one minute. Sterilize in autoclave at 121° C (15 lbs. sp) for 15 minutes. Cool to 45 °C, and pour into Petri dishes. Allow the plates to solidify and place them upside down to avoid excessive moisture in the surface of the medium.

Uses

For the selective isolation and identification of enterobacteria from feces, urine, wastewater and foods. MacConkey Agar is a selective and differential medium for the isolation of enteric gram negative bacilli.

The specimen can be streaked directly on the medium or inoculated first into an enrichment broth such as Tetrathionate Broth, Selenite Cystine Broth, or GN Broth. Incubate the plates and broth tubes at 35°C for 18 to 24 hours. Subculture the broth tubes onto MacConkey Agar and reincubate.

It is recommended to streak samples onto other selective media such as Eosin Methylene Blue Agar, SS Agar, XLD Agar, Hektoen Enteric Agar, Bismuth Sulfite Agar (especially for *Salmonella typhi*), and/or Brilliant Green Agar, especially for *salmonellas*. See the listings in this manual for these formulations.

Other organisms not belonging to the enterobacteria such as *Pseudomonas* and *Aeromonas* grow on MacConkey Agar. *Enterococci* can also grow as small pinpoint red colonies as well as some strains of *Staphylococci*, whose weak pink colonies are small and opaque.

This medium can also be used for the differentiation of *mycobacteria*.

CHARACTERISTICS OF THE COLONIES:

Escherichia coli: Red to pink. Not mucoid. Can be round with an opaque precipitate of bile salts. ***Klebsiella***: Large, red, mucoid. ***Enterobacter***: Large, red. Not mucoid. ***Serratia***: Red to pink. Not mucoid. ***Arizona and Citrobacter***: Colourless, transparent. Red if lactose is fermented. ***Proteus***: Colourless and transparent. ***Pseudomonas***: Colourless to greenish-brown. Characteristic sweet odor. ***Salmonella***: Colourless, transparent or amber. ***Shigella***: Colourless, transparent or very faintly pink. ***Staphylococcus***: Punctiform, pale pink, opaque and scanty. ***Enterococcus***: Scanty, punctiform, red, opaque with a clear zone about 1 mm in diameter around the colony.

Bibliography

MacConkey J. Hig. 5:33, 1905. Joseph Md. State. Dept. Health. Procedures, 1960.

Microbiological Test

Microorganisms	Growth	Colony colour
<i>Enterobacter aerogenes</i> ATCC 13048	Good	pink-red
<i>Escherichia coli</i> ATCC 25922	Good	pink-red (biliar precipitate)
<i>Proteus vulgaris</i> ATCC 13315	Good	colourless
<i>Salmonella enteritidis</i> ATCC 13076	Good	colourless
<i>Shigella dysenteriae</i> ATCC 13313	Good	colourless
<i>Staphylococcus aureus</i> ATCC 25923	Inhibited	colourless

MACCONKEY AGAR N° 2

Cat. 1035

For the identification of enterococci in the presence of coliforms and non lactose fermenters in water and foods

Formula in grams per liter

<i>Bacteriological peptone</i>	20,00	<i>Lactose</i>	10,00
<i>Sodium Chloride</i>	5,00	<i>Bile Salts no 2</i>	1,50
<i>Neutral Red</i>	0,05	<i>Crystal Violet</i>	0,001
<i>Bacteriological Agar</i>	13,50		

Final pH 7,2 ± 0,2 at 25°C

Preparation

Suspend 50 grams of the medium in one liter of distilled water. Mix well. Heat with frequent agitation and boil until completely dissolved. Dispense into appropriate containers and sterilize at 121° C (15 lbs. sp.) for 15 minutes.

Uses

Fecal streptococci grow as intensely red, small colonies surrounded by a zone of pale red precipitate. These

microorganisms are indicators of fecal contamination. Non-lactose fermenting bacteria form colourless colonies.

Bibliography

Mac Geachie J. and Kennedy A.C. J. Clin. Path. 16, 32-38, 1963

Microbiological Test

Microorganisms	Growth	Colony colour
<i>Escherichia coli</i> ATCC 25922	Satisfactory	Rose-red (biliar precipitate)
<i>Enterococcus faecalis</i> ATCC 29212	Satisfactory	Red
<i>Salmonella enteritidis</i> ATCC 13076	Satisfactory	Colourless
<i>Staphylococcus aureus</i> ATCC 25923	Satisfactory	Colourless

MACCONKEY AGAR WITH SORBITOL

Cat. 1099

Selective and differential medium for the research of *E. coli* 0157:H7

Formula in grams per liter

<i>Gelatin Peptone</i>	20,00	<i>Sorbitol</i>	10,00
<i>Sodium Chloride</i>	5,00	<i>Bile Salts N° 3</i>	1,50
<i>Neutral Red</i>	0,03	<i>Crystal Violet</i>	0,001
<i>Bacteriological Agar</i>	15,00		

Final pH: 7,1 ± 0,2 at 25°C

Preparation

Suspend 51,5 grams of the medium in one liter of distilled water. Heat to boiling with frequent agitation until totally dissolved. Dispense and sterilize at 121° C (15 lbs psi) for 15 minutes. Distribute into sterile Petri dishes. If needed, allow the plate surface to dry.

Uses

Sorbitol MacConkey Agar is based on the formula developed by Rappaport & Henig. This medium is recommended for the research of *E. coli* 0157:H7. The composition is similar to MacConkey Agar but the lactose has been substituted by sorbitol. *E. coli* 0157:H17 does not ferment the sorbitol and therefore produces

colourless colonies. Most of the other *E. coli* do ferment it and therefore their colonies are pink.

E. coli 0157:H7 has been recognized as being responsible for haemorrhagic colitis, characterized by a bleeding diarrhea with intense abdominal ache.

Optimal temperature for *E. coli* 0157:H7 is 35°C -37°C.

Bibliography

Rappaport F. and Henig E. (1952), J.Clin.Path., 5,361. Karmali M.A.(1988), Culture, 9,2. Doyle M.P. and Schoeni S.L (1984), Appl. and Envir. Microbiol., 48, 855-856.

Microbiological Test

Microorganisms	Growth	Colony colour
<i>Escherichia coli</i> ATCC 25922	Good	Pink
<i>Escherichia coli</i> 0157:h7	Good	Colourless

MACCONKEY AGAR WITHOUT CRYSTAL VIOLET

Cat. 1037

Used for the study of coliform organisms

Formula in grams per liter

<i>Gelatin Peptone</i>	17,00	<i>Lactose</i>	10,00
<i>Bile Salts N° 3</i>	5,00	<i>Sodium Chloride</i>	5,00
<i>Peptone Mixture</i>	3,00	<i>Neutral Red</i>	0,03
<i>Bacteriological Agar</i>	12,00		

Final pH 7,4 ± 0,2 at 25°C

Preparation

Suspend 52 grams of the medium in one liter of distilled water. Mix well until a uniform suspension is obtained. Heat with frequent gentle agitation and boil for one minute. Sterilize in autoclave at 121°C (15 lbs. sp.) for 15 minutes. Cool to 45°C and pour into plates.

Uses

For the investigation of enteric microorganisms, especially the enterococci, from water, feces and other material.

MacConkey Agar without Crystal Violet is plated directly with the suspected sample. For suspected pathogens from feces and other material, inoculate also in parallel other selective media such as Desoxycholate Agar or DCLS Agar.

Lacking crystal violet, this medium also supports the growth of enterococci and some *staphylococci*. Plates are incubated at 35°C and examined after 24-48 hours. In general, the characteristics of the colonies are:

Bibliography

Gray, L.D. 1995. *Escherichia*, *Salmonella*, *Shigella* and *Yersinia*, p. 450-456. In P.R. Murray, E.J. Baron, M.A. Pfaller, F.C. Tenover, and R.H. Tenover (ed.), *Manual of clinical microbiology*, 6th ed. American Society for Microbiology, Washington, D.C.
Eaton, A.D., L.S. Clesceri, and A.E. Greenberg (ed.) 1995. *Standard methods for the examination of water and wastewater*, 19th ed. American Public Health Association, Washington, D.C.

ORGANISM	COLOR OF COLONY
<i>E. coli</i>	Red or pink
<i>E. aerogenes</i>	Pink, mucoid
<i>Enterococci</i>	Small, discrete, red
<i>Staphylococci</i>	Red to pink
<i>Salmonella</i> , <i>Shigella</i> & <i>Pseudomonas</i>	Colourless, lactose-negative

Microbiological Test

Microorganisms	Growth	Colony colour
<i>Escherichia coli</i> ATCC 25922	Good	Red-pink
<i>Enterobacter aerogenes</i> ATCC 13048	Good	Colourless
<i>Salmonella enteritidis</i> ATCC 13076	Good	Colourless
<i>Staphylococcus aureus</i> ATCC 25923	Good	Pink
<i>Staphylococcus aureus</i> ATCC 12228	Good	Pink

MACCONKEY AGAR WITHOUT CRYSTAL VIOLET AND WITHOUT SODIUM CHLORIDE

Cat. 1098

Differential medium that inhibits the *Proteus* swarming. Recommended for urine analysis.

Formula in grams per liter

<i>Gelatin Peptone</i>	17,00	<i>Lactose</i>	10,00
<i>Bile Salts N° 3</i>	5,00	<i>Peptone Mixture</i>	3,00
<i>Neutral Red</i>	0,075	<i>Bacteriological Agar</i>	12,00

Final pH 7,4 ± 0,2 at 25°C

Preparation

Suspend 47 grams of the medium in one liter of distilled water. Mix well until a uniform suspension is obtained. Heat with frequent gentle agitation and boil for one minute. Sterilize in autoclave at 121°C (15 lbs. sp.) for 15 minutes. Cool to 45°C and pour into plates.

Uses

It is differential medium used for the detection and isolation of enteric microorganisms. The lack of Sodium Chloride provides an electrolyte deficient medium preventing *Proteus* spp. from spreading (swarming). In addition, this medium does not contain crystal violet

allowing *Staphylococcus*, *Enterococcus* and *Mycobacterium* spp. to grow.

Bibliography

MacConkey, A. 1905 Lactose-fermenting bacteria in feces J. Hyg 5:333-379
Murray, P.R., E.J. Baron, M.A. Pfaller, F.C. Tenover, and R.H. Tenover (eds) Manual of clinical microbiology, 6th ed. American Society for Microbiology, Washington, D.C.
Mazura-Reets, G.T. Neblett, and J.M. Galperin, 1979 MacConkey Agar: Co2 vs. ambient incubation. Abst. Ann. Mtg. American Society for Microbiology. C179.

Microbiological Test

Microorganisms	Growth	Colony colour
<i>Escherichia coli</i> ATCC 25922	Good	Red-pink
<i>Enterobacter aerogenes</i> ATCC 13048	Good	Pink
<i>Proteus vulgaris</i> ATCC 13315	Good	Inhibited swarming
<i>Staphylococcus aureus</i> ATCC 25923	Good	Pale pink
<i>Streptococcus faecalis</i> ATCC 19433	Good	dot like Pink

MACCONKEY BROTH

Cat. 1210

For the detection of coliforms in water, milk and other materials of sanitary importance.

Formula in grams per liter

<i>Gelatin Pancreatic Digest</i>	20,00	<i>Lactose Monohydrate</i>	10,00
<i>Dehydrated Oxbile</i>	5,00	<i>Bromcresol Purple</i>	0,01

Final pH 7,3 ± 0,2 at 25°C

Preparation

Suspend 35 grams of medium in one liter of distilled water. Heat with frequent agitation until completely dissolved. To analyze 10 ml samples, prepare a double concentration medium. Dispense in test tubes with a gas collecting tube (Durham) 10 ml amount for samples of 1 ml or less. Sterilize in an autoclave at 121°C (15 lbs. of pressure) for 15 minutes.

Uses

MacConkey broth is used for cultivating gram-negative, lactose, fermenting bacilli in water and foods, as a presumptive test medium for the presence of *coliform* organisms in water and other materials of sanitary importance. Formation of gas and acid confirm the

presence of coliforms, as demonstrated by the change of medium color from purple to yellow.

Bibliography

MacConkey, A. 1905. Lactose-fermenting bacteria in faeces. J. Hyg 5:333-379.
MacConkey, A. 1908. Bile salt media and their advantage in some bacteriological examinations. J. Hyg. 8:322-334.
Chils, E., and L. A. Allen. 1953. Improved methods for determining the most probable number of *Bacterium coli* and of *Streptococcus faecalis*. J. Hyg. Camb. 51:468-477.

Microbiological Test

Microorganisms	Growth	Acid	Gas
<i>Enterobacter aerogenes</i> ATCC 13048	Satisfactory	+	+
<i>Escherichia coli</i> ATCC 25922	Satisfactory	+	+
<i>Salmonella choleraesuis</i> ATCC 12011	Acceptable	-	-
<i>Staphylococcus aureus</i> ATCC 25923	Null	-	-

MALT EXTRACT AGAR

Cat. 1038

For the cultivation of fungi and yeasts

Formula in grams per liter

Malt Extract.....	12,75	Dextrin.....	2,75
Glycerin.....	2,35	Gelatin Peptone.....	0,78
Bacteriological Agar.....	15,0		

Final pH 4,7 ± 0,2 at 25°C

Preparation

Suspend 33,6 grams of the medium in one liter of distilled water. Homogenize and heat with frequent agitation. Boil for one minute. Sterilize in autoclave at 118°C (12 lbs. sp.) for 10 minutes.

NOTE: If the medium is overheated the agar loses its capacity to solidify.

Uses

Malt Extract Agar has been used for years to cultivate fungi and yeast cultures in the sugar industry, in the manufacturing of syrups, soft drinks, and other drinks.

It is also recommended in conjunction with other specific media which are included in this manual.

Bibliography

Thom and Raper, Manual of the Aspergilli 39:1945.

Microbiological Test

Microorganisms	Growth
<i>Saccharomyces cerevisiae</i> ATCC 9763	Satisfactory
<i>Saccharomyces uvarum</i> ATCC 9080	Satisfactory
<i>Candida Albicans</i> ATCC 10231	Satisfactory
<i>Aspergillus niger</i> ATCC 16404	Satisfactory

MALT EXTRACT BROTH

Cat. 1245

For the isolation and count of yeast and moulds, as well as for sterility tests

Formula in grams per liter

Malt Extract	6,00	Maltose Certified	6,00
Dextrose	6,00	Yeast Extract	1,20

Final pH 4,7 ± 0,2 at 25°C

Preparation

Suspend 19 grams of the medium in one liter of distilled water. Mix well. Heat with frequent agitation to completely dissolve the medium. Dispense and sterilize at 115°C – 118°C (10-12 lbs sp) for 15 minutes. **DO NOT OVERHEAT.**

Uses

Malt Extract Broth contains a malt extract purified and clarified for microbiological use.

It is used to cultivate yeasts and molds within a short time period from foods and beverages.

Bibliography

Galloway L.D. and Burgess R. "Applied Mycology and Bacteriology" 3rd Ed. Leonard Hill London, 54-57, 1952.
Recommended methods for the Microbiological Examination of Foods APHA Inc. New York, 1958.

Microbiological Test

Microorganisms	Growth
<i>Saccharomyces cerevisiae</i> ATCC 9763	Satisfactory
<i>Saccharomyces uvarum</i> ATCC 9080	Satisfactory
<i>Candida albicans</i> ATCC 10231	Satisfactory
<i>Aspergillus niger</i> ATCC 16404	Satisfactory

MANNITOL NITRATE MOTILITY MEDIUM

Cat. 1509

For the rapid differentiation of enterobacteria

Formula in grams per liter

Casein Peptone	10,00	Mannitol	7,50
Potassium Nitrate	1,00	Phenol Red	0,04
Bacteriological Agar	3,50		

Final pH 7,6 ± 0,2 at 25°C

Preparation

Suspend 22 grams of the medium in one litre of distilled water. Mix well. Heat with frequent agitation to boiling and completely dissolved. Dispense into tubes to obtain a butt depth of 6-7 cm.
Sterilize at 121°C (15 lbs. sp.) for 15 minutes.

Uses

This semi-solid medium permits the rapid identification of *enterobacteria* on the basis of motility, mannitol utilization and nitrate reduction to nitrite.

The medium is inoculated by stabbing the center of the tube to its base and incubating at 37°C for 18-24 hours.

Motile bacteria show a diffuse turbidity away from the inoculation line while non-motile organisms only grow

along the stab line. If mannitol is fermented, the medium changes color from red to yellow

Adding Gries Reagent (sulfanilic acid-alpha-naphthylamine) to the surface of the medium can demonstrate the reduction of nitrate to nitrite.

Bibliography

Titters R.R. and L.A. Sancholzer 1936. The use of semi-solid agar for the detection of bacterial motility, J. Bacteriol 31: 575-580.
Snell and Wright; 1941, J. Biolog. Chem. 13: 675.
Compendiu of methods for the microbiological examination of foods. Am. Public. Health Association.

Microbiological Test

Microorganisms	Motility	Mannitol	Nitrates
<i>Escherichia coli</i> ATCC 25922	+	+	+
<i>Klebsiella pneumoniae</i> ATCC 13883	-	+	+
<i>Proteus mirabilis</i> ATCC 25933	+	-	+
<i>Acinetobacter anitratum</i> ATCC 17924	-	-	-

MANNITOL SALT AGAR

Cat. 1062

Used for the isolation of pathogenic *Staphylococci*.

Formula in grams per liter

Sodium Chloride.....	75,00	Peptone Mixture.....	10,00
D-Mannitol.....	10,00	Beef Extract.....	1,00
Phenol Red.....	0,025	Bacteriological Agar.....	15,00

Final pH 7,4 ± 0,2 at 25°C

Preparation

Suspend 111 grams of the medium in one litre of distilled water. Mix well and heat with frequent agitation until complete dissolution. Boil for one minute.

Sterilize in autoclave at 121°C (15 lbs. of steam pressure) for 15 minutes. Pour into Petri dishes.

Uses

This is a selective medium prepared according to the recommendations of Chapman for the isolation of presumptive pathogenic *staphylococci*. Most of the other bacteria are inhibited by the high concentration of salt.

The degradation of mannitol with the production of acid changes the color of the medium from rose to yellow. Due to its high content of sodium chloride, a heavy inoculum of the material in study can be used.

Generally the plates are incubated for 36 hours, colonies of non-pathogenic *staphylococci* appearing as small colonies surrounded by a red or purple zone. The mannitol fermenting pathogenic *staphylococci* are larger and are surrounded by a yellow zone. The addition of 5% Pronadisa's Egg Yolk Emulsion allows to detect the lipase activity of *staphylococci*, as well as mannitol fermentation. The high concentration of salt in the medium clears the egg yolk emulsion and lipase production is detected as a yellow opaque zone around the colonies of *staphylococci* that produce this enzyme. This phenomenon, together with a positive coagulase test confirms the organism as a pathogenic *staphylococci*.

Bibliography

McColloch Am. J. Vet. Research, 8:173, 1947. Velilla, Faber, and Pelczar Am. J. Vet. Research, 8:275, 1947.
Chapman, G.H. 1945 J. Bact. 50:201-203

Microbiological Test

Microorganisms	Growth	Colony colour
<i>Escherichia coli</i> ATCC 25922	Inhibited	----
<i>Enterobacter aerogenes</i> ATCC 13048	Inhibited	----
<i>Staphylococcus aureus</i> ATCC 25923	Satisfactory	Yellow
<i>Staphylococcus epidermidis</i> ATCC 12228	Acceptable	Red
<i>Staphylococcus epidermidis</i> ATCC 14990	Satisfactory	Red

MARINE AGAR

Cat. 1059

Used for the recount and isolation of the heterotropic marine bacteria

Formula in grams per liter

<i>Sodium Chloride</i>	19,40	<i>Bacteriologic Peptone</i>	5,00
<i>Magnesium Chloride</i>	8,80	<i>Sodium Sulfate</i>	3,24
<i>Calcium Chloride</i>	1,80	<i>Yeast Extract</i>	1,00
<i>Potassium Chloride</i>	0,55	<i>Sodium Bicarbonate</i>	0,16
<i>Ferric Citrate</i>	0,10	<i>Potassium Bromide</i>	0,08
<i>Strontium Chloride</i>	0,034	<i>Boric Acid</i>	0,022
<i>Disodium Phosphate</i>	0,008	<i>Sodium Silicate</i>	0,004
<i>Sodium Fluoride</i>	0,0024	<i>Ammonium Nitrate</i>	0,0016
<i>Bacteriological Agar</i>	15,00		

Final pH 7,6 ± 0,2 at 25°C

Preparation

Suspend 55,1 grams of the medium in one liter of distilled water. Heat to boiling agitating frequently until completely dissolved. Dispense into appropriate containers. Sterilize by autoclaving at 121°C (15 lbs sp) for 15 minutes. The colour of the prepared medium is clear transparent amber or slightly opalescent colour, may present a light precipitation. It is recommended to homogenize the medium in its container before pouring into plates.

Uses

This medium contains all the nutrients necessary to cultivate the majority of marine bacteria.

Using the conventional plate count technique or the streaking the surface of the plate, results are good. However, precaution must be taken in the pour plate method to cool the medium to 45°C before pouring as the majority of marine organisms are heat-sensitive.

Bibliography

J. Marine Research N:42, 1941. Limnology and Oceanography 5:78, 1960.

Microbiological Test

Microorganisms	Growth
<i>Vibrio fischeri</i>	Good
<i>Vibrio harveyi</i>	Good

MARINE BROTH

Cat. 1217

Used for the recount and isolation of heterotropic marine bacteria.

Formula in grams per liter

<i>Sodium Chloride</i>	19,40	<i>Magnesium Chloride</i>	8,80
<i>Bacteriological Peptone</i>	5,00	<i>Sodium Sulfate</i>	3,24
<i>Calcium Chloride</i>	1,80	<i>Yeast Extract</i>	1,00
<i>Potassium Chloride</i>	0,55	<i>Sodium Bicarbonate</i>	0,16
<i>Ferric Citrate</i>	0,10	<i>Potassium Bromide</i>	0,08
<i>Strontium Chloride</i>	0,034	<i>Boric Acid</i>	0,022
<i>Sodium Silicate</i>	0,004	<i>Sodium Fluoride</i>	0,0024
<i>Ammonium Nitrate</i>	0,0016	<i>Disodium Phosphate</i>	0,008

Final pH 7,6 ± 0,2 at 25°C

Preparation

Suspend 40,0 grams of the medium in one liter of distilled water. Heat until boiling to dissolve completely. Dispense into appropriate containers. Sterilize by autoclaving at 121°C (15 lbs pressure) for 15 minutes. The colour of the prepared medium is clear transparent amber or slightly opalescent colour, may present a light precipitation.

It contains all the nutrients necessary for the cultivation of most marine bacteria.

Bibliography

ZoBell, C.E. 1941. Studies on marine bacteria. I. The cultural requirements of heterotrophic aerobes. J. Mar. Res. 4:42-75.
Buck, J.D., and R.C. Cleverdon. 1960. The spread plate as a method for the enumeration of marine bacteria. Limnol. Oceanogr.
Weiner, R.M., A.M. Segall, and R.R. Colwell. 1985.

Uses

Marine Broth is similar to the formula for Marine Agar, lacking the agar.

Microbiological Test

Microorganisms	Growth
<i>Vibrio fischeri</i>	Good
<i>Vibrio harveyi</i>	Good

MIO MEDIUM

Cat. 1510

For enterobacteria identification

Formula in grams per liter

<i>Gelatin Peptone</i>	10,00	<i>Casein Peptone</i>	10,00
<i>L-Ornithine</i>	5,00	<i>Yeast Extract</i>	3,00
<i>Dextrose</i>	1,00	<i>Bromcresol Purple</i>	0,02
<i>Bacteriological Agar</i>	2,00		

Final pH 6,5 ± 0,2 at 25°C

Preparation

Suspend 31 grams of the medium in one liter of distilled water. Heat agitating frequently and boil for one minute until completely dissolved. Distribute in screw-capped tubes and sterilize at 121°C (15 lbs sp) for 15 minutes.

Uses

The cultures are inoculated by stabbing the MIO medium and incubating for 18 to 24 hours at 35° C. Read the reactions of motility and ornithine decarboxylase before adding the Kovacs Reagent for the indol test.

The motility is indicated by cloudiness in the media or growth extending away from the line of inoculation. Ornithine decarboxylation is indicated by a purple color in the medium. A negative ornithine reaction produces a

yellow color in the bottom of the tube. For the indol test, add 3 to 4 drops of Kovacs reagent, and shake the tube gently. The appearance of a red or rose color in the reagent layer is a positive indication of indol. Compare the results with an uninoculated test tube.

Bibliography

Ederer, G.M., and M. Clark. 1970. Motility-Indole-Ornithine medium. Appl. Microbiol. 2:849.
 Oberhofer, T.R., and R. Hajkowski. 1970. Evaluation of non-lactose-fermenting members of the Klebsiella-Enterobacter-Serratia Division. I. Biochemical characteristics. Am. J. Clin. Pathol. 54:720.

Microbiological Test

Microorganisms	Growth	Motility	Indol	Ornithine(dexc.)
<i>Escherichia coli</i> ATCC 25922	Satisfactory	+	+	+
<i>Enterobacter aerogenes</i> ATCC 13048	Satisfactory	+	-	+
<i>Klebsiella pneumoniae</i> ATCC 13883	Satisfactory	-	-	-
<i>Proteus mirabilis</i> ATCC 25933	Satisfactory	±	-	+

MOELLER KCN BROTH BASE

Cat. 1112

Used for the differentiation of enteric bacilli

Formula in grams per liter

Sodium Phosphate.....	5,64	Sodium Chloride	5,00
Peptone Mixture.....	3,00	Potassium Phosphate.....	0,225

Final pH 7,6 ± 0,2 at 25°C

Preparation

Dissolve 14 grams of the medium in one liter of distilled water. Dispense and sterilize in autoclave at 121°C (15 lbs.sp) for 15 minutes. Allow to cool to room temperature. Add 15 ml of a 0,5% solution of potassium cyanide (0,5 g per 100 ml distilled water) and close containers tightly. 5 ml of a 1% 2,3,5 Triphenyltetrazol solution per liter of base may be added if desired.

Caution: Do not inhale the cyanide with the pipette.

Uses

Inoculate the medium lightly so that the inoculum cannot be misinterpreted as growth when cultures are examined. This may be accomplished by using a 3 mm. loopful of an overnight (24 hours) broth culture or transferring a light inoculum from an agar slant culture with a straight wire. KCN Broth facilitates the recognition and identification of microorganisms similar to *Citrobacter freundii*, especially

those that ferment lactose slowly but develop rapidly in the presence of cyanide. Also, it is very useful in differentiating *Salmonella* and *Arizona* from the Bethesda-Ballerup group.

GROWTH

Enterobacter/Klebsiella/Bethesda-Ballerup/Proteus/Citrobacter/Providencia/Hafnia/Serratia.

NO GROWTH

Escherichia/Arizona/Salmonella/Shigella.

Bibliography

Moeller. Acta Path. and Microbiol. Scand., 134:115, 1954.
Gershamand Cn. J. Microbiol, 1, 1960
Edwards and Ewing, Identification of Enterobacteriaceae. Burgess Publ. Co., Minneapolis, Minn., 1972.

Microbiological Test

Microorganisms	Growth
<i>Enterobacter spp.</i>	+
<i>Citrobacter freundii</i> ATCC 8090	+
<i>Proteus vulgaris</i> ATCC 6380	+
<i>Escherichia coli</i> ATCC 25922	-
<i>Salmonella enteritidis</i> ATCC 13076	-
<i>Shigella flexneri</i> ATCC 12022	-

MOSSEL EE BROTH (EUROPEAN PHARMACOPEIA) Cat. 1202

For the selective enrichment of enterobacteria in foods specially Salmonellas and Coliforms

Formula in grams per liter

<i>Dehydrated Oxbile</i>	20,00	<i>Gelatin Pancreatic Digest</i>	10,00
<i>Disodium Phosphate</i>	8,00	<i>Glucose Monohydrate</i>	5,00
<i>Monopotassium Phosphate</i>	2,00	<i>Brilliant Green</i>	0,015

Final pH 7,2 ± 0,2 at 25°C
Preparation

Suspend 45 g of the medium in a liter of distilled water. Heat with frequent agitation until completely dissolved. Heat at 100°C for 30 minutes. Cool immediately.
DO NOT STERILIZE IN AUTOCLAVE.

Uses

Enterobacteriaceae which contaminate foods grow well in this medium while undesirable gram-positive organisms are inhibited. *E. coli*, even though it is present in small numbers as a contaminant in foods, grows easily in this medium.

Inoculate 10 grams of the food sample in 100 ml. of EE Broth (Mossel) and agitate vigorously to form a homogeneous suspension. Incubate at 35°C. After 3 hours resuspend the sample. At the end of the incubation time of 8-24 hours, observe for turbidity. Subculture to selective solid media such as Violet Red Bile Agar. Proceed with normal isolation and identification with these media.

Bibliography

Mossel D.A.A., Visser M. and Cornelissen A.M.R.J. App. Bact. 24:444, 1963.
 Mossel D.A.A. et al. J. Bact. 84:381, 1982.

Microbiological Test

Microorganisms	Growth	Ac. prod. yellow
<i>Enterobacter aerogenes</i> ATCC 13048	Satisfactory	+
<i>Escherichia coli</i> ATCC 25922	Satisfactory	+
<i>Salmonella enteritidis</i> ATCC 13076	Satisfactory	± (could be slow)
<i>Staphylococcus aureus</i> ATCC 25923	Inhibited	-

M.R.S. AGAR

Cat. 1043

Medium recommended to favor the growth of lactobacilli in general.

Formula in grams per liter

Dextrose	20,00	Bacteriological Peptone	10,00
Beef extract	8,00	Sodium acetate	5,00
Yeast extract	4,00	Dipotassium phosphate	2,00
Ammonium citrate	2,00	Tween 80	1,00
Magnesium sulfate	0,20	Manganese sulfate	0,05
Bacteriological agar	10,00		

Final pH 6,2 ± 0,2 at 25°C

Preparation

Suspend 62 grams of the medium in one liter of distilled water. Heat with frequent agitation until boiling. Dispense it in adequate containers and sterilize in autoclave at 121°C (15 lbs sp) for 12 minutes.

Uses

The MRS formulation was developed by de Man, Rogosa and Sharpe to replace a variable product (tomato juice) at the same time to provide a medium with would support good growth of *Lactobacilli* in general, those strains which showed pour growth in existing media.

Lactobacilli are microaerophilic and generally require layer plates for aerobic cultivation on solid media. Submerged or surface colonies may be compact or feathery, and are small, opaque and while.

The pour plate method deposits 1 ml. of the previously diluted sample into a sterile Petri dish and the cooled (45°C-50°C) medium is added. After solidification, a second layer is poured. The plates are incubated at 37°C for 3 days or better, at 30°C for 5 days. It is important to maintain a humid atmosphere because the plates should not dry out during incubation which is in 5% CO₂.

Bibliography

Briggs M (1.953) "An Improved Medium for Lactobacilli" J. Dairy Res. 20, 36-40.
Man, J.C. de Rogosa M., Sharpe, M. Elisabeth (1960) "A Medium for the Cultivation of Lactobacilli". J. Appl. Bact. 23, 130-135.

Microbiological Test

Microorganisms	Growth
<i>Lactobacillus acidophilo</i> ATCC 4356	Good
<i>Lactobacillus casei</i> ATCC 393	Good
<i>Escherichia coli</i> ATCC 25922	Moderate-Good
<i>Staphylococcus aureus</i> ATCC 25923	Inhibited

MRS BROTH

Cat. 1215

Formula developed by Man, Rogosa and Sharpe to facilitate the growth of lactobacilli in general.

Formula in grams per liter

<i>Dextrose</i>	20,00	<i>Bacteriological Peptone</i>	10,00
<i>Beef Extract</i>	8,00	<i>Sodium Acetate</i>	5,00
<i>Yeast Extract</i>	4,00	<i>Dipotassium Phosphate</i>	2,00
<i>Ammonium Citrate</i>	2,00	<i>Polysorbate 80 (Tween 80)</i>	1,00
<i>Magnesium Sulfate</i>	0,20	<i>Manganese Sulfate</i>	0,05

Final pH 6,2 ± 0,2 at 25°C

Preparation

Suspend 52 grams of the medium in one liter of distilled water. Mix well and heat agitating frequently until complete dissolution of the medium. Dispense in adequate containers and sterilize in autoclave at 121°C (15 lbs.sp) for 12 minutes.

Uses

This medium is selective for *Lactobacilli*. Times and temperatures of incubation are the same as MRS Agar (37°C for 3 days or better, 30°C for 5 days). Tubes showing growth are subcultured to MRS Agar to confirm the presence of lactobacilli. MRS Broth may be used for test in the identification of *Lactobacilli*, such as

temperature dependence, growth in 4% NaCl, growth in 0,4% Teepol, etc. as recommended by Sharpe, Fryer and Smith.

Bibliography

Sharpe M. Elisabeth, fryer T.F. and Smith D.G. (1966) "Identification of the Lactic Acid Bacteria in Identification Method for Microbiologist Part A" (Gibbs B.M. and Skinner F.A. eds.) London and New York, Academic Press.
 Briggs M. (1953) J. dairy Res., 20: 36-40
 Reuter G. (1985) Intern. J. Food Microbiol 2: 55-68.

Microbiological Test

Microorganisms	Growth
<i>Lactobacillus acidophilus</i> ATCC 4356	Good
<i>Lactobacillus casei</i> ATCC 393	Good
<i>Lactobacillus fermentum</i> ATCC 9338	Moderate-Good
<i>Escherichia coli</i> ATCC 25922	Moderate-Good
<i>Pseudomonas aeruginosa</i> ATCC 27853	Inhibited

MR VP MEDIUM

Cat. 1512

Used for the differentiation of group *Escherichia-Enterobacter*
(Methyl Red and Voges-Proskauer reactions)

Formula in grams per liter

Peptone mixture 7,00 Dextrose 5,00
Potassium Phosphate 5,00

Final pH 6,9 ± 0,2 at 25°C

Preparation

Suspend 17 grams of the medium in one liter of distilled water. Mix well. If needed, heat slightly to dissolve completely. Dispense in tubes and sterilize at 121°C (15 lbs sp) for 15 minutes.

Uses

For the differentiation of the enteric gram negative bacilli, especially the *Escherichia-Enterobacter* group. MR-VP Medium is used as an aid in the differentiation of enteric gram negative bacilli on the basis of methyl red and acetylmethylcarbinol (Voges Proskauer) reactions of the *Escherichia-Enterobacter* group.

In 1915 Clark and Lubs used methyl red as an indicator of acidity in the cultures of the *Coli-Enterobacter* group. This test is now known as the methyl red test and serves to distinguish between those microorganisms that produce and maintain a high concentration of acid from those that initially produce a small amount of acid and are capable of later attacking those same acids, turning the medium to neutral or alkaline, such as *Enterobacter*.

Voges and Proskauer described in 1898 a fluorescent red coloration that appeared in certain cultures upon adding drops of KOH solution. Later it was supposed that this reaction was due to oxidation of acetylmethylcarbinol to diacetyl which reacted with the peptone of the medium to give a red color. *Enterobacter* oxidizes the acetylmethylcarbinol and gives the red coloration, in contrast to *Escherichia coli* which does not.

Method

Methyl red test:

Add 5 drops of a 0.4% solution of methyl red to 5 ml. of a culture incubated for 3 to 5 days. A positive reaction will give a red color, and a negative a yellow color. The reaction is immediate.

Voges-Proskauer test:

To 5 ml. of medium inoculated and incubated up to 5 days, add 0.6 ml. of 5% alpha-naphthol in absolute ethanol and 0.2 ml. of 40% sodium hydroxide and shake from time to time over a 15 minute period. The tube may be held at room temperature or incubated at 35-37° C. It is important that the reagents be added in sequence. A positive test is indicated by development of a faint pink to red color. The test should not be read after one hour because negative VP cultures may develop a copper color after that time.

Bibliography

Clark and Lubs. J.: Inf. Dis. 17:160, 1955.
Ewing. Enterobacteriaceae. USPHS.
Edwards and Ewing. Identification of Enterobacteriaceae Burgess Publ. Co. Minneapolis, Minn., 1962.
Voges, O., and B. Proskauer. 1898. Z. Hyg. 28: 20-22.
Association of Official Analytical Chemists. 1995. Bacteriological analytical manual, 8th ed. AOAC International, Gaithersburg, MD.

Microbiological Test

Microorganisms	Growth	MR	VP
<i>Enterobacter aerogenes</i> ATCC 13048	Good	- (yellow)	+ (red)
<i>Escherichia coli</i> ATCC 25922	Good	+ (red)	- (without change)
<i>Klebsiella pneumoniae</i> ATCC 23357	Good	+	-

MUELLER HINTON AGAR

Cat. 1058

Recommended for sensitivity tests on antibiotics and sulfamides and for the primary isolation of *Neisseria*.

Formula in grams per liter

<i>Beef Infusion</i>	2,00	<i>Casein Peptone H</i>	17,50
<i>Starch</i>	1,50	<i>Bacteriological Agar</i>	17,00

Final pH 7,4 ± 0,2 at 25°C

Preparation

Suspend 38 grams of medium in one liter of distilled water. Mix well. Heat agitating frequently and boil for about one minute. Dispense and sterilize in autoclave at 116 - 121°C (15 lbs.sp) for 15 minutes. Cool to 45° or 50° C and add defibrinated blood if desired. The blood mixture should be chocolate by heating to 80° C for 10 minutes if *Neisseria* development is desired. **DO NOT OVERHEAT**. To remelt the cold medium, heat as briefly as possible.

Uses

This Medium is specified in the FDA Bacteriological Analytical Manual for Food Testing. Is also recommended for testing most commonly encountered aerobic and facultative anaerobic bacteria. Mueller Hinton Agar can be used to cultivate *Neseria* specimens. It is recommended to incubate the plates at 35°C in a CO₂ atmosphere.

The mayor use of Mueller Hinton Agar is for antimicrobial susceptibility testing. It has become the standard medium for the Bauer Kirby method and its performance is specified by the NCCLS.

In the light of such criticisms the NCCLS called interested manufactures together to discuss the standardization and stabilization of Mueller Hinton Agar. Control methods were established whereby critical antimicrobial organism combination had to yield consistent zones of inhibition within 2 mm of the specified diameters in the standards.

Bibliography

Mueller and Hinton A. Protein-Free Medium for Primary Isolation of the *Gonococcus* and *Meningococcus*. Proc. Soc. Exp. Biol. and Med. 48:330, 1941. Harris and Coleman Diagnostic. Procedures and Reagents. 4th Edition APH, Inc. New York, 1963. National Committee for Clinical Laboratory Standards. 1993. Atlas, R.M. 1993 Handbook of microbiological media. CRC Press, Boca Raton. Fl.

Microbiological Test

Microorganisms	Diameter inhibition halo in mm according to NCCLS				
	Ampicillin 10 µg	Tetracycline 30 µg	Gentamicyne 10 µg	Polimixyn B300 UI	Sulfametoxazole 1,25 µg Trimethoprim 23,75 µg
<i>Escherichia coli</i> ATCC 25922	15-20	18-25	19-26	12-16	24-32
<i>Staphylococcus aureus</i> ATCC 25923	24-35	19-27	19-27	7-13	24-32
<i>Streptococcus faecalis</i> ATCC 33186	-	-	-	-	16-23
<i>Pseudomonas aeruginosa</i> ATCC 27853	-	-	16-21	-	-

MUELLER HINTON II AGAR

Cat. 1055

Recommended for antibiotics sensitivity tests and for the primary isolation of gonococci and meningococci.

Formula in grams per liter

Beef Infusion	2,00	Casein Peptone H.....	17,50
Starch	1,50	Bacteriological Agar.....	17,00

Final pH 7,4 ± 0,2 at 25°C

Preparation

Suspend 38 grams of the medium in one liter of distilled water. Mix well. Heat agitating frequently and boil for about one minute. Dispense and sterilize in autoclave at 116 - 121°C (12 -15 pounds steam pressure) for 15 minutes. Cool to 45° or 50° C and add defibrinated blood if desired. If Neisseria development is desired the blood mixture should be chocolate by heating to 80° C for 10 minutes . **DO NOT OVERHEAT.** To remelt the cold medium, heat as briefly as possible.

Uses

Mueller Hinton Agar II can be used to cultivate Neisseria specimens. It is recommended to incubate the plates at 35°C. in a CO₂ atmosphere. The major use of Mueller Hinton Agar is for antimicrobial susceptibility testing. It has become the standard medium for the Bauer-Kirby method and its performance is

specified by NCCLS (National Committee for Clinical Laboratory Standards).

The medium complies with the requirement of NCCLS and is manufactured to contain low concentrations of tyamine and thymidine as well as appropriate levels of calcium and magnesium ions.

Bibliography

Bauer, A.L., W.M. Kirby, J.C. Sherris, and M. Turck. 1966. Antibiotic susceptibility testing by a standardized single disc method. Am. J. Clin. Pathol 45: 493-496.
Wood, G.L. and J.A. Washington, 1995 Antibacterial susceptibility tests, dilution and disk diffusion methods, p. 1327-1341. In Murray, P.R., E.J. Baron, M.A. Tenover, F.C. Tenover.

Microbiological Test

Microorganisms	Growth
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Response to the sensibility tests against the different antibiotics, using type cultures and observed after 24 hours.

TYPE CULTURE	Diameter halo in mm according to NCCLS				
	ESSAY DISKS				
	Ampicillin 10 µg	Tetracycline 30 µg	Gentamicin 10 µg	Polymixin B300 UI	Sulfamethoxazole 1,25 µg Trimethoprim 23,75 µg
<i>Escherichia coli</i> ATCC 25922	15-20	18-25	19-26	12-16	24-32
<i>Staphylococcus aureus</i> ATCC 25923	24-35	19-27	19-27	7-13	24-32
<i>Streptococcus faecalis</i> ATCC 33186	/	/	/	/	16-23
<i>Pseudomonas aeruginosa</i> ATCC 27853	/	/	16-21	/	/

MUELLER HINTON BROTH

Cat. 1214

Used for the development of gonococci and meningococci as well as for sensitivity testing in liquid medium to different antibiotics.

Formula in grams per liter

Beef infusion	2,00	Acid casein peptone (H).....	17,50
Corn Starch.....	1,50		

Final pH 7,4 ± 0,2 at 25°C

Preparation

Dissolve 21 grams of medium in one liter of distilled water. Mix well. Heat with frequent agitation and boil for one minute. Sterilize in autoclave at 121°C (15 lbs.sp) for 15 minutes. **Do not overheat at any time during the process.**

Uses

Mueller Hinton media was developed for the cultivation of pathogenic *neisserias* and other fastidious microorganisms. The starch performs as a growth factor, probably functions like a colloid protector and neutralizes toxic products that are able to form during the development of the organisms. Mueller Hinton Broth can be used with complete confidence because it is a rich medium able to grow fastidious organisms. Also it is used simultaneously together with the Agar of the same name, to carry out sensitivity testing of a great number of

antimicrobial agents, for determination dilution MIC studies.

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 Olsen A.M. and Scott, W.J. Nature, 557; 337, 1946.
 Bauer, A.L., W.M. Kirby, J.C. Sherris, and M. Turck. 1966. Antibiotic susceptibility testing by a standardized single disc method. Am. J. Clin. Pathol 45: 493-496.

Wood, G.L. and J.A. Washington, 1995 Antibacterial susceptibility tests, dilution and disk diffusion methods, p. 1327-1341. In Murray, P.R., E.J. Baron, M.A. Pfaller, F.C. Tenover.

Microbiological Test

Microorganisms	Growth
<i>Staphylococcus aureus</i> ATCC 25923	Good
<i>Escherichia coli</i> ATCC 25922	Good
<i>Streptococcus faecalis</i> ATCC 33186	Good
<i>Pseudomonas aeruginosa</i> ATCC 27853	Good
<i>Streptococcus pyogenes</i> ATCC 19615	Good
<i>Listeria monocytogenes</i> ATCC 19113	Good

MUELLER KAUFMAN BROTH BASE

Cat. 1130

For the selective enrichment of Salmonella from meats and other foods

Formula in grams per liter

Sodium Tiosulphate	40,70	Calcium Carbonate	25,00
Ox Bile	4,75	Sodium Chloride	4,50
Meat Extract	4,50	Yeast Extract	1,80
Beef Extract	0,90		

Final pH 7,6 ± 0,2 at 25°C

Preparation

Dissolve 82 grams of the medium in one liter of distilled water. If necessary heat briefly and cool quickly. A sediment of calcium carbonate will remain. Do the autoclave. Before use add 20 ml/liter of iodine and potassium iodide solution and 10 ml/liter of brilliant green 0,1% solution. Distribute in tubes after homogenizing the possible precipitate. Once added these substances, do not heat again. **Preparation of the iodine and potassium iodide: 5 gr. of potassium iodide, 4 gr. of iodine, 20 ml. of distilled water.**

Uses

Mueller recommended Tetrathionate Broth as a selective medium for the isolation of Salmonella Kauffman modified the formula to include oxbile and brilliant green as selective agents to suppress bacteria such as Proteus spp.

The British Standard Specification specifies Brilliant Green Tetrathionate Broth for isolating Salmonella from meat and meat products and from poultry and poultry products. It is also a recommended selective broth for isolating Salmonella from animal feces and sewage polluted water.

Using more than one selective broth increases the isolation of Salmonella from samples with multiple sero types.

Use the medium in the day that it's produced. Sodium Tiosulphate plus iodine added produce Tetrathionate formation and in this way *Coliforms* and Intestinal Bacteria are inhibits.

Salmonella and Proteus they are not inhibit because they reduce Tetrathionate.

Salmonella growing its stimulated by bile but inhibit other germs.

Brilliant Green Inhibits Gram (+)

Bibliography

Kauffman, F. 1935. Weitere erfahrungen mit dem kombinierten anreicherungsverfahren fur Salmonella bazillen. Ztschr. F. Hyg. 117: 26-32.

A manual for recommended methods for the microbiological examination of poultry and poultry products. 1982.

Microbiological Test

Microorganisms	Concentration Inoculum	Growth	
		6 hours	24 hours
<i>Escherichia coli</i> ATCC 25928	approx. 99%	< 30%	< 5%
<i>Salmonella typhimurium</i> ATCC 14028	approx. 1%	> 70%	> 95%

MYCOBIOTIC AGAR (FUNGAL SELECTIVE AGAR) Cat. 1072

For the isolation of moulds in highly contaminated samples.

Formula in grams per liter

Soy Peptone	10,00	Dextrose	10,00
Cycloheximide (actidione)	0,40	Chloramphenicol	0,05
Bacteriological Agar	15,50		

Final pH 6,9 ± 0,2 at 25°C

Preparation

Suspend 36 grams of the medium in one liter of distilled water. Mix well until a uniform suspension is obtained. Soak for 10-15 minutes. Heat with frequent agitation and boil for one minute. Distribute and sterilize at 118°C (15 lbs. sp) for 15 minutes. Cool and use immediately. Once cold, remelt just one time with the minimum heat. **DO NOT OVERHEAT.**

Uses

For the cultivation and selective isolation of pathogenic fungi. Mycobiotic Agar is a medium for the selective cultivation of fungal pathogens from diverse clinical samples and other materials contaminated with a mixed associated flora. Basically this medium is Mycology Agar to which has been added chloramphenicol which inhibits bacterial development and cycloheximide which inhibits the growth of saprophytic fungi. Mycobiotic Agar is very useful to isolate pathogenic fungi from diverse types of highly samples highly contaminated with different types of accompanying flora, such as those of the head, skin scrapings, nails, bronchial lavages, gastric juices, soil, etc.

It is recommended to inoculate several plates or tubes with the same sample in study and incubate them at ambient temperature (22-25°C) and at 35°C. The toxic effect of the antimicrobial mixture is greater in the ambient temperature, for which reason the number of positive

isolates is higher at temperatures below 35°C incubation than at 25°C.

It is recommended to inoculate at the same time other culture media like Littman Bile Agar, Biggy Agar, etc., with the object to obtain a greater number of isolates. The dermatophytes and other numerous groups of pathogenic fungi grow quickly in the Mycobiotic Agar which inhibits most of the bacteria and the *fungal saprophytes* or commensal contaminants.

Nevertheless, it should be noted that *Allescheria boydii*, *Aspergillus fumigatus*, *Cryptococcus neoformans*, *Actinomyces bovis*, and *Nocardia asteroides*, are inhibited by the antibiotics present in the medium. The first three can be isolated on Littman Bile Agar with the addition of streptomycin, and *Nocardia asteroides* on Mycological Agar or in Trypticasein Soy Agar with added cycloheximide. *Actinomyces bovis* grow well on the plates of Anaerobic Agar and in Thioglycollate Medium without Indicator.

Bibliography

Dean and Halley, Public Health Reports, 77:61, 1972. Hupper and Walker, A.J. Clin. Path. 29:291, 1958.
McDonough Ajello, Georg, and Brinkman J. Lab. and Clin. Med. 55:116, 1960.

Microbiological Test

Microorganisms	Growth
<i>Escherichia coli</i> ATCC 25922	Inhibited
<i>Staphylococcus aureus</i> ATCC 25923	Inhibited
<i>Trichophyton mentagrophytes</i>	Satisfactory
<i>Trichophyton rubrum</i>	Satisfactory
<i>Candida albicans</i> ATCC 2091	Satisfactory
<i>Aspergillus niger</i>	Inhibited/light
<i>Penicillium</i> spp.	Inhibited/light

NITRATE MOTILITY BASE MEDIUM

Cat. 1565

Recommended medium for the confirmation of *Clostridium perfringens*

Formula in grams per liter

<i>Casein Peptone</i>	5,00	<i>Galactose</i>	5,00
<i>Beef Extract</i>	3,00	<i>Disodium Phosphate</i>	2,50
<i>Potassium Nitrate</i>	1,00	<i>Bacteriological Agar</i>	3,50

Final pH 7,3 ± 0,2 at 25°C

Preparation

Suspend 20 grams of the medium in one liter of distilled water. Mix well . Heat with frequent agitation to boiling and completely dissolved. Dispense into tubes to obtain a butt depth of 6-7 cm. Sterilize at 121 (15 lbs .sp) for 15 minutes.

Uses

Nitrate reduction is a valuable criteria for differentiating and identifying various types of bacteria. Certain bacteria reduce nitrates to nitrites only, while others are capable of further reducing nitrite to free nitrogen or ammonia. Nitrites are colourless, however, in an acid environment, they will react to produce a pink or red colour. When specific reagents are added and the nitrate positive organisms reduces nitrates to nitrites, a pink colour

develops in the broth medium. Nitrate negative organisms are unable to reduce nitrates and they yield no colour after adding the reagents.

Bibliography

Titters R.R. and L.A. Sancholzer 1936. The use of semi-solid agar for the detection of bacterial motility, J. Bacteriol 31: 575-580.
Snell and Wright; 1941, J. Biolog. Chem. 13: 675.
Compendiu of methods for the microbiological examination of foods. Am. Public. Health Association.

Microbiological Test

Microorganisms	Mobility	Nitrate
<i>Clostridium perfringens</i>	-	+
<i>Clostridium bifermentans</i>	+	-

NUTRIENT AGAR

Cat. 1060

Used for the enumeration of organisms in water, faeces and other materials

Formula in grams per liter

Gelatin Peptone..... 5,00	Beef Extract.....3,00
Bacteriological Agar..... 15,00	

Final pH 6,8 ± 0,2 at 25°C

Preparation

Suspend 23 grams of the medium in one liter of distilled water. Mix well and leave to stand until the mixture is uniform. Heat with gentle agitation and boil for one or two minutes, or until completely dissolved. Dispense and sterilize at 121°C (15 lbs.sp) for 15 minutes.

Uses

For the cultivation of non fastidious microorganisms. Nutrient Agar is a general purpose medium, not selective but suitable for the cultivation of non fastidious microorganisms. It can be used as a colony count medium in sanitation, medical, and industrial bacteriology. There

are many uses for Nutrient Agar in the bacteriological analysis of drinking water, waste water, milk and other foods. It is also used in the multiplication of microorganisms to produce vaccines and antigens in general; in the tests of sensitivity and resistance, and as a base to prepare an enriched medium by adding ascitic fluid, etc. It is used in biochemical test, for example indol decarboxylase and lysine decarboxylase.

Bibliography

Greenberg and Cooper Can. Med. Assn. J. 83:143, 1960.
Wetmore and Gochenour J. Bact. 72:79, 1956.

Microbiological Test

Microorganisms	Growth
<i>Staphylococcus aureus</i> ATCC 25923	Good
<i>Escherichia coli</i> ATCC 25922	Good
<i>Salmonella typhimurium</i> ATCC 14028	Good
<i>Streptococcus pyogenes</i> ATCC 12344	Good
<i>Streptococcus pneumoniae</i> ATCC 6301	Good

NUTRIENT AGAR (D.E.V. REGULATIONS) Cat. 1314

To enumerate organisms in water, faeces and other materials

Formula in grams per liter

<i>Meat Peptone</i> 10,00	<i>Beef Extract</i> 10,00
<i>Sodium Chloride</i> 5,00	<i>Bacteriological Agar</i> 18,00

Final pH 7,3 ± 0,2 at 25°C

Preparation

Suspend 43,0 grams of the medium in one liter of distilled water. Mix well. Heat agitating frequently and boil for one or two minutes, or until completely dissolved. Dispense and sterilize at 121°C (15 lbs.sp) for 15 minutes. Cool to 45°C and pour into Petri dishes.

Uses

Nutrient Agar is used for cultivating a wide variety of microorganisms.

The American Public Health Association (APHA) suggested this standard culture medium for use in bacterial processing for water analysis.

In Standard Methods of Water Analysis and Standard Methods of Milk Analysis, the APHA advocated the use of dehydrated media for bacterial examination of water and milk.

Bibliography

American Public Health Association. 1923. Standard methods of milk analysis, 4 Th. Ed. American Public Health Association, Washington, D.C.
Association of Official Analytical Chemists. 1995. Official methods of analysis of AOAC International, 16th ed. AOAC International, Arlington, VA.

Microbiological Test

Microorganisms	Growth
<i>Escherichia coli</i> ATCC 25922	Satisfactory
<i>Salmonella typhimurium</i> ATCC 14028	Satisfactory
<i>Proteus vulgaris</i> ATCC 13315	Satisfactory
<i>Streptococcus faecalis</i> ATCC 11700	Satisfactory
<i>Klebsiella pneumoniae</i> ATCC 13883	Satisfactory

NUTRIENT BROTH

Cat. 1216

Used for the enumeration of organisms in water, faeces, and other materials.

Formula in grams per liter

Gelatin Peptone 5,00 *Beef Extract* 3,00

Final pH 6,8 ± 0,2 at 25°C

Preparation

Suspend 8 grams of the medium in one liter of distilled water. Mix well and leave to stand until the mixture is uniform. Heat with gentle agitation and boil for one or two minutes, or until complete dissolution. Dispense and sterilize at 121° C (15 lbs.sp) for 15 minutes.

Uses

For the general cultivation of non fastidious microorganisms. Nutrient Broth is a liquid medium, produced according to the formula from APHA and AOAC and support the growth of a great variety of microorganisms that are not very particular in nutritional needs.

Nutrient Broth is used in many laboratory procedures as is, or with added indicators, carbohydrates, organic liquids, salts, etc. This medium is used in accordance with official

recommended procedures for the bacteriological analyses of water, milk and dairy products, in foods of important sanitation, tests for sensitivity and resistance, and as a base to prepare media supplemented with other nutrients.

Bibliography

Walsbren, Carr, and Dunnette A. J. Clin. Path. 21:884, 1951.
American Public Health Association. 1923. Standar methods of water analysis, 5th ed. American Public Health Association, Washington, D.C.
Marshall, R.T. (ed) 1993 Standard methods for the microbiological examination of dairy products, 16th ed. American Public Health Association, Washington, D.C.

Microbiological Test

Microorganisms	Growth
<i>Enterobacter aerogenes</i> ATCC 13048	Satisfactory
<i>Escherichia coli</i> ATCC 25922	Satisfactory
<i>Salmonella typhi</i> ATCC 6539	Satisfactory
<i>Staphylococcus epidermis</i> ATCC 14990	Satisfactory
<i>Streptococcus pyogenes</i> ATCC 12344	Moderate

NUTRIENT GELATIN

Cat. 1300

Used for tests of microorganisms which liquefy gelatin..

Formula in grams per liter

Gelatin 120,00	Gelatin Peptone 5,00
Beef Extract 3,00	

Final pH 6,8 ± 0,2 at 25°C

Preparation

Suspend 128 grams of the medium in one liter of distilled water. Heat gently agitating frequently until completely dissolved. Sterilize at 121° C (15 lbs. sp.) for 15 minutes.

Uses

For the detection of proteolytic bacteria. Nutrient Gelatin was one of the first solidifying agents used in the beginning of bacteriology. It is used to investigate the presence of proteolytic microorganisms, especially in the bacteriological analysis of water. For the plate count of organisms in water, this medium is being replaced by solid media with agar.

Nutrient Gelatin was originally used in the standard method for water and wastewater as a direct plate count technique, replacing the dilution method. However, this method required incubation at approximately 20°C, not ideal for most organisms, and the medium is now principally used for the detection of proteolysis as evidenced by the liquefaction of gelatin.

The tubes are inoculated by stabbing with a needle (straight wire) and incubated at 20-23° C for up to 30 days. Refrigerate the test cultures together with an uninoculated

Nutrient Gelatin control tube and read the reactions as soon as the control tube has hardened.

This is determined by inverting the tube. A strong positive remains liquid.

If plates of Nutrient Gelatin are utilized, they can be streaked or seeded with aliquots of the sample in a pour-plate technique. Check for hydrolysis of gelatin on the streaked plate by adding a drop of saturated ammonium sulfate or 20% sulfasalicylic acid to an isolated colony. Look for a zone of clearing around the colony (Stone reaction) in 10 minutes. The Stone reaction is also used on *Staphylococcus* Medium N° 110.

Bibliography

Ewing Enterobacteriaceae USPHS Publication 734 Washington, 1960.
Edwards and Ewing. Identification of Enterobacteriae, Burgess Publ. Co. Minneapolis, Minn., 1962.
Standard Methods for the Examination of Water and Sewage, Nineth Edition APHA Inc. New York, 1960.

Microbiological Test

Microorganisms	Growth	Gelatinase
<i>Bacillus subtilis</i> ATCC 6633	Satisfactory	+
<i>Clostridium perfringens</i> ATCC 12924	Satisfactory	+
<i>Escherichia coli</i> ATCC 25922	Satisfactory	-
<i>Staphylococcus aureus</i> ATCC 25923	Satisfactory	+

OF BASAL MEDIUM (HUGH AND LEIFSON) Cat. 1500

For the identification of non fermenting bacilli of medical and sanitary importance.

Formula in grams per liter

Casein Peptone	2,00	Sodium Chloride.....	5,00
Dipotassium Phosphate	0,30	Bacteriological Agar	2,50
Bromthymol Blue	0,03		

Final pH 7,1 ± 0,2 at 25°C

Preparation

Suspend 9,8 grams of medium in one litre of distilled water. Heat with frequent agitation until dissolved. Sterilize in an autoclave at 121° C (15 lbs. sp.) for 15 minutes. Add 10 ml. of 10% glucose (or any suitable sugar) solution sterilized by filtration to 100 ml. of liquid medium. Mix and dispense aseptically 5 ml. per tube. If preferred, add 1,0 grams of carbohydrate directly to 100 ml. of medium and sterilize in an autoclave at 118° C (12 lbs.) for 10 minutes to avoid the degradation of the sugar. The color of the prepared medium is green.

Uses

Inoculate 2 fresh tubes by stabbing with a fresh culture of the organism in study. If the medium has been prepared and stored, remelt in a water bath to expel the dissolved gases. After inoculation add to one of the tubes a layer of 4 to 5 mm. of paraffin oil. It is not recommended to use mineral oil. Incubate both tubes at 35° C for 48 hours or more, up to 7 days with the caps loose. To facilitate the identification of Gram-negative non-fermenting *bacilli*, use also Indol Nitrate Medium

Results

Fermentation: Yellow color in both tubes with or without formation of gas. **Oxidation:** Yellow color only in the tube that does not contain the oil. **No oxidation/fermentation:** No change in the color of the tubes. The carbohydrates have not been fermented or oxidized. Inert microorganisms, e.g. *Alcaligenes faecalis*.

Moraxellas are Gram-negative, oxidase +, non-fermenting coccobacilli which rarely cause any pathogenic condition. *M. osloensis* (formerly *Mima polymorpha* var *oxidans*) can easily be confused with *gonococci* when only microscopic analysis of the urigenital specimen is performed. This organism can also be isolated rarely from other products such as blood and cerebrospinal fluid and be confused with *Neisseria meningitidis*. However differentiation from pathogenic neisserias is relatively easy and simple; biochemical tests utilizing Indol Nitrate Medium, OF Basal Medium and growth in Nutrient Agar are extensively used for this purpose.

Bibliography

Hugh, R. and Leifson, E.J. Bact. 66:24-26, 1953. Lisenko J. Gen. Microbiol., 35:379, 1961. Edwards y Ewing Identification of Enterobacteriaceae. Burgess Publ. Co. Minneapolis, Minn., 1972.

ORGANISM	TUBE W/O OIL	TUBE W/O OIL	MOTILITY
<i>Alcaligenes</i>	-	-	+
<i>Mimapolymorpha</i> (<i>Acinetobacter</i>)	-	-	-
<i>Pseudomonas</i>	Acid (ox)	-	+
<i>Herellea</i> ** (<i>Acinetobacter</i>)	Acid (ox)	-	-
<i>Shigella</i>	Acid	Acid (Ferm.)	-
<i>Salmonella</i>	Acid and gas	Acid and gas	+

* *Acinetobacter calcoaceticus* var *lwoffii*.

** *Acinetobacter calcoaceticus* var *anitratus*.

Microbiological Test

Microorganisms	Without sugar		With Glucose		With Lactose		With sucrose	
	○	●	○	●	○	●	○	●
<i>Alcaligenes faecalis</i> ATCC 8750	K	K	K	K	K	K	K	K
<i>Escherichia coli</i> ATCC 25922	K	K	AG	AG	AG	AG	K	K
<i>Pseudomonas aeruginosa</i> ATCC 27853	K	K	A	K	K	K	K	K
<i>Salmonella enteritidis</i> ATCC 13076	K	K	AG	AG	K	K	K	K
<i>Shigella flexneri</i> ATCC 12022	K	K	A	A	K	K	K	K

○ = Opened ● = Closed K = Alkaline, green (without change) A = Acid, yellow G = Gas, sometimes perceptible

O.G.A. MEDIUM (OXITETRACYCLINE AGAR BASE) Cat. 1527

For recount and selection of yeast and moulds in food samples.

Formula in grams per liter

Dextrose 10,00	Yeast Extract..... 5,00
Bacteriological Agar 15,00	

Final pH 6,5 ± 0,2 at 25°C

Preparation

Suspend 30 grams of the medium in one liter of distilled water. Mix well. Heat with frequent agitation and boil until completely dissolved. Distribute into appropriate containers and sterilize in autoclave at 121° C (15 lbs.sp) for 10 minutes. Allow to cool to 45-50°C and aseptically add 100 mg of oxytetracycline per liter of medium. Mix well and pour into petri dishes.

Uses

The pour plate method is recommended to count up incubation at 20°C-25°C and exam daily from the second day to the 6th.

In Neutral pH the oxytetracycline produce best results than when you use low pH medium to inhibit bacterial forms.

These mediums inhibit the acidophilus (Lactobacillus included) that produce no desired growing in acid pH mediums.

Bibliography

American Public Health Association. Standard Methods for the Examination of Dairy Products, 13th Ed. APHA, Inc. New York, 1960.
Thom and Raper, Manual of the Aspergilli 39:1945.

Microbiological Test

Microorganisms	Growth
<i>Escherichia coli</i> ATCC 25922	Inhibited
<i>Pseudomonas aeruginosa</i> ATCC 27853	Inhibited
<i>Candida Albicans</i> ATCC 10231	Satisfactory
<i>Penicillium spp.</i> ATCC 12022	Satisfactory
<i>Aspergillus niger</i>	Satisfactory

ORANGE SERUM AGAR

Cat. 1307

Medium used for isolation and detection of different acid tolerant pathogen germs in fruit juices

Formula in grams per liter

Casein Peptone	10,00	Orange Extract	5,00
Glucose	4,00	Monopotassium Phosphate	3,00
Yeast Extract	3,00	Bacteriological Agar	15,00

Final pH 5,5 ± 0,2 at 25°C

Preparation

Suspend 40 grams of the medium in one liter of distilled water. Heat with frequent agitation to boiling, and keep boiling for one minute. Dispense into appropriate containers. Sterilize in autoclave at 118° C (15 lbs.psi) for 15 minutes. **DO NOT OVERHEAT**

Uses

This culture medium as contains orange serum, is specially indicated for the existing micro flora in citric juices, as for example Bacillus, Lactobacillus, moulds, etc. It's a medium

specially indicated for production control in Fruit Juice Industry.

Bibliography

Hays G.L.(1951), Proc. Florida State Hort. Soc. , 94th Ann.
 Murdock D.I. and Brokaw C.H.(1958), Food Tech. , 12, 573-576.
 American Public Health Association (1976), Compendium of Methods for the Microbiological Examination of Foods, APHA Inc. Washington DC.

Microbiological Test

Microorganisms	Growth
<i>Aspergillus niger</i> ATCC16404	Satisfactory
<i>Lactobacillus fermentum</i> ATCC 9338	Satisfactory
<i>Saccharomyces cerevisiae</i> ATCC 9763	Satisfactory

OSMOPHILIC AGAR

Cat. 1057

For the research of osmophilic yeasts in foods

Formula in grams per liter

Fructose..... 60,00 Yeast Extract..... 5,00
Bacteriological Agar 15,00

Final pH 7,0 ± 0,2 at 25°C

Preparation

Suspend 80 grams of the medium in one liter of distilled water. Heat with frequent agitation until boiling and completely dissolved. Distribute into appropriate containers. Sterilize in autoclave at 121° C (15 lbs.sp) for 15 minutes. The high concentration of fructose makes this medium selective and it is recommended to count yeasts that develop in media with a high osmophilic pressure.

Uses

This medium is selective because of the high concentration of sugar and supports the growth of osmophilic yeasts, capable of growing on media with an elevated osmotic pressure. These yeasts can change or affect, therefore, fruit concentrates, syrups and honey, etc.

From 1 grams of food sample, make dilutions and place 1 ml. aliquots in Petri dishes and add the medium cooled to 45-50° C. Swirl gently and allow to solidify. Incubate at 22° C for 72 hours.

This medium is formulated according to the standards of the National Center for Foods and Nutrition (CeNAN) for total counts of osmophilic yeasts.

Bibliography

Pascual Anderson. "Técnicas para el Análisis Microbiológico de Alimentos y Bebidas" (Centro Nacional de Alimentación y Nutrición (Madrid 1982).

Microbiological Test

Microorganisms	Growth
<i>S. rouxii</i>	Satisfactory
<i>S. mellis</i>	Satisfactory
<i>Zygosaccharomyces spp.</i>	Satisfactory

PALCAM LISTERIA AGAR BASE

Cat. 1141

Selective and differential medium for the diagnose and detection of *Listeria monocytogenes*.

Formula in grams per liter

<i>Columbia Agar Base</i>	39,00	<i>Lithium chloride</i>	15,00
<i>Mannitol</i>	10,00	<i>Yeast Extract</i>	3,00
<i>Esculin</i>	0,80	<i>Glucose</i>	0,50
<i>Ferric Ammonium Citrate</i>	0,50	<i>Phenol Red</i>	0,08

Final pH 7,2 ± 0,2 at 25°C

Preparation

Suspend 34,5 grams of medium in 500 ml. of distilled water. Heat with frequent agitation until complete dissolution. Distribute into appropriate containers. Sterilize in autoclave at 121°C (15 lbs. psi) during 15 minutes. Cool to 50°C and aseptically add the reconstituted supplement.

Uses

Palcam medium is recommended for isolation of *Listeria monocytogenes* in food products. It is highly selective due to the presence of lithium chloride, Ceftazidime, Polymixin B and Acryflavine. This allows the easy differential diagnose of *Listeria monocytogenes* using a double system indicator: Esculin and iron and Mannitol and phenol red.

Listeria monocytogenes hydrolyses the Esculin which brings about the formation of a black Zone around the colony. *Listeria monocytogenes* does not ferment the

mannitol; differentiation of contaminants is easy as enterococci and estafilococci ferment same and produce a change from red to yellow due to the pH indicator of phenol red.

The addition of egg yolk emulsion favors the recuperation of harmed *Listeria* strains.

Bibliography

Van Netten, P., I. Perales A. Van de Moosalijk G.D.W. Curtis and DAA Mossel 1989 Liquid and solid selective differential media for the detection and enumeration of *L. Monocytogenes* and other *Listeria* spp. Int. J. of Food Microbiol 8: 299-317.
Farber JMDW Warburton and T. Babiuk, 1994 Isolation of *Listeria monocytogenes* from all food and environmental samples.

Microbiological Test

Microorganisms	Growth	<u>BLACK ZONE</u>
<i>Listeria monocytogenes</i> ATCC 19117	Good	+
<i>Staphylococcus aureus</i> ATCC 25923	Good	-

PEPTONE WATER (CENAN)

Cat. 1403

Liquid medium used to cultivate and for carbohydrate fermentation studies as well as to perform the Indol test.

Formula in grams per liter

Bacteriological peptone..... 10,00

Sodium Chloride 5,00

Final pH 7,2 ± 0,2 at 25°C

Preparation

Suspend 15 grams of the medium in one liter of distilled water. Dissolve the medium completely. Distribute into appropriate containers and sterilize in autoclave at 121°C (15 lbs sp) for 15 minutes.

Uses

Used for cultivation, fermentation studies of carbohydrates and to perform the indol test.

This formula, according to the CeNAN (National Center for Food and Nutrition), is recommended for the investigation of indol production in coliforms. A loopful from each tube of presumptive broth should be inoculated into Peptone

Water and incubated at 44°C for 48 hours. Add Kovacs Reagent to determine indol production.

Bibliography

M.R. Pascual Anderson (1982) Tecnicas para Analisis Microbiologico de Alimentos y Bebidas, CeNAN.
MacFaddin, J.F. 1985. Media for isolation-cultivation-identification-maintenance of medical bacteria, vol. 1. p. 610-612. Williams & Wilkins, Baltimore, M.D.
Finegold, S.M., and W. martin, 1982. Bailey and Scott's diagnostic microbiology, 6th ed. St. Louis.

Microbiological Test

Microorganisms	Growth
<i>Escherichia coli</i> ATCC 25922	Satisfactory
<i>Salmonella typhimurium</i> ATCC 14028	Satisfactory
<i>Staphylococcus aureus</i> ATCC 25923	Satisfactory

PHENOL RED BROTH BASE

Cat. 1115

For the study of carbohydrate fermentations

Formula in grams per liter

Casein Peptone 10,00 Sodium Chloride 5,00
 Phenol Red 0,018

Final pH 7,4 ± 0,2 at 25°C

Preparation

Dissolve 15 grams of medium in one liter of distilled water. Add 5-10 g/l of the desired carbohydrate. (you may add 0,5-1,0 g/l of agar if the medium is going to be utilized for anaerobes). Heat with frequent agitation until complete dissolution. Dispense into tubes and add gas collecting tubes Durham for gas detection. Sterilize at 116-118°C (10-12 lbs. psi.) for 15 minutes.

Uses

A basal medium for determining the fermentation reactions of microorganisms must be capable of supporting growth of test organisms and be free from fermentable carbohydrates. Vera used a fermentation test medium employing the pH indicator phenol red and obtained highly accurate results.

Phenol Red Broth Base is used for carbohydrate fermentation studies of many microorganisms. Control tubes of uninoculated medium should be run in parallel with inoculated tubes. Tubes should be examined frequently because different carbohydrates are utilized at variable speeds. The appearance of a yellow color is the indication of fermentation, with or without gas formation.

Phenol Red Broth Base is an excellent substrate for streptococci, as well for other less fastidious bacteria, the growth promotion on the medium can be greatly improved for fastidious, and microaerophilic.

For anaerobes the medium should be used on the day of preparation or the medium must be heated and cooled before use.

Bibliography

Ewing, W.H. 1986. Edwards and Ewing's identification of Enterobacteriaceae, 4th edition. Elsevier Science Publishing Co., Inc. New York.
 Vera H.D. 1950 Relation of peptones and other culture media ingredients to accuracy of fermentation tests. Am. J. Public Health 40:1267.
 MaFaddin, J.F. 1985. Media for isolation-cultivation-identification-maintenance of medical bacteria. Williams & Wilkins, Baltimore, MD.

Microbiological Test

Microorganisms	Glucose		Lactose	
	acid	gas	acid	gas
<i>Escherichia coli</i> ATCC 25922	+	+	+	+
<i>Proteus vulgaris</i> ATCC 6380	+	+	-	-
<i>Salmonella typhimurium</i> ATCC 14028	+	+	-	-

PHENOL RED DEXTROSE AGAR

Cat. 1023

Medium similar to the Dextrose Agar, with Phenol Red as pH indicator

Formula in grams per liter

Peptone Mixture.....	10,00	Dextrose.....	10,00
Sodium Chloride.....	5,00	Phenol Red	0,025
Bacteriological Agar	15,0		

Final pH 7,4 ± 0,2 at 25°C

Preparation

Suspend 40 grams of the dehydrated medium in one liter of distilled water. Soak 10 to 15 minutes. Heat with frequent agitation and boil for one minute. Sterilize at 121°C (15 lbs sp.) for 15 minutes. Once sterilized, cool to 40°C-45°C and pour into petri dishes.

Uses

Phenol Red Broth Base is recommended for use to determine the ability of organisms to ferment various carbohydrates.

Phenol Red Broth Base is an excellent substrate for streptococci, as well as for other less fastidious bacteria, the growth promotion of the medium can be greatly improved for fastidious.

Phenol Red Dextrose Agar is similar to Dextrose Agar with the addition of phenol red as a pH indicator. It is used to

study fermentation reactions of all types of microorganisms.

Bibliography

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Microbiological Test

Microorganisms	Growth	Acid	Gas production
<i>Alcaligenes faecalis</i> ATCC 8750	Satisfactory	-	-
<i>Escherichia coli</i> ATCC 25922	Satisfactory	+	+
<i>Klebsiella pneumoniae</i> ATCC 13883	Satisfactory	+	+
<i>Proteus vulgaris</i> ATCC 6380	Satisfactory	+	+
<i>Salmonella typhimurium</i> ATCC 14028	Satisfactory	+	+
<i>Shigella flexneri</i> ATCC 12022	Satisfactory	+	-

PHENOL RED DEXTROSE BROTH

Cat. 1235

For sucrose fermentation studies

Formula in grams per liter

Casein Peptone	10,00	Dextrose	5,00
Sodium Chloride	5,00	Phenol Red	0,018

Final pH 7,4 ± 0,2 at 25°C

Preparation

Dissolve 20 grams of the medium in one liter of distilled water. If the medium is for the cultivation of anaerobes, add 0,5-1 grams of agar. Mix well. Heat with frequent agitation to dissolve the medium completely. Dispense in 5 ml amounts into test tubes with gas collecting tube (Durham). Sterilize at 116-118°C (12 lbs sp) for 15 minutes. **DO NOT OVERHEAT.**

Uses

Phenol Red Dextrose Broth contains casein peptone which is rich in nutrients and is obtained by the enzymatic digestion of casein. It allows for abundant growth of a wide variety of fastidious microorganisms. Being free of carbohydrates it is useful in fermentation studies.

The complete medium functions very well in rapid bacterial susceptibility tests for antimicrobial agents. With pure cultures results can be obtained in approximately 3 hours. Some cases require up to 8 hours incubation.

Phenol red indicator changes to yellow in acid conditions as a result of bacterial fermentation. Durham tubes trap any gases produced during fermentation. Additional tubes of Phenol Red Broth Base without carbohydrates should be inoculated at the same time to avoid false positive results caused by fermentable material present in one or more of the components.

Bibliography

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 Association of Official Analytical Chemists. 1995 official methods of analysis of AOAC Arlington, VA:
 Baron EJ LR Peterson and S.M. Finegold 1994. Bailey & Scott's diagnostic microbiology, 9th edition. Mosby-Year Book, Inc. St. Louis, MO.
 Murray, PR., E.J. Baron M.A. Pfaller F.C. Tenover and R.H. Tenover (ed) 1995. Manual of clinical microbiology, 6th edition. American Society for Microbiology, Washington DC.

Microbiological Test

Microorganisms	Glucose	
	acid	gas
<i>Escherichia coli</i> ATCC 25922	+	+
<i>Proteus vulgaris</i> ATCC 6380	+	+
<i>Salmonella typhimurium</i> ATCC 14028	+	+

PHENOL RED SUCROSE BROTH

Cat. 1239

For sucrose fermentation studies

Formula in grams per liter

Casein Peptone.....	10,00	Sucrose	5,00
Sodium Chloride.....	5,00	Phenol Red	0,018

Final pH 7,4 ± 0,2 at 25°C

Preparation

Dissolve 20 grams of the medium in one liter of distilled water. If the medium is for the cultivation of anaerobes, add 0,5-1 grams of agar. Mix well. Heat with frequent agitation to dissolve the medium completely. Dispense in 5 ml amounts into test tubes with gas collecting tube (Durham). Sterilize at 116-118°C (12 lbs sp) for 15 minutes. **DO NOT OVERHEAT.**

Uses

This medium is the same to Phenol Red Broth Base (Cat. 1115) having added Sucrose for fermentation studies.

A basal medium for determining the fermentation reactions of microorganisms must be capable of supporting growth of test organisms and be free from fermentable carbohydrates. Vera used a fermentation test medium employing the pH indicator phenol red and obtained highly accurate results.

Phenol Red Broth Base is used for carbohydrate fermentation studies of many microorganisms. Control tubes of uninoculated medium should be run in parallel with inoculated tubes. Tubes should be examined frequently because different carbohydrates are utilized at

variable speeds. The appearance of a yellow color is the indication of fermentation, with or without gas formation.

A positive test is indicated by a color change from red to yellow, with or without gas production.

For anaerobes the medium should be used on the day of preparation or the medium must be heated and cooled before use.

Bibliography

Rogers, Ryan and Severans. Antibiotic and Chemother 5:382, 1955
Association of Official Analytical Chemists. 1995 official methods of analysis of AOAC Arlington, VA:
Baron EJ LR Peterson and S.M. Finegold 1994. Bailey & Scott's diagnostic microbiology, 9th edition. Mosby-Year Book, Inc. St. Louis, MO.
Murray, PR., E.J. Baron M.A. Pfaller F.C. Tenover and R.H. Tenover (ed) 1995. Manual of clinical microbiology, 6th edition. American Society for Microbiology, Washington DC.

Microbiological Test

Microorganisms	Sucrose	
	acid	gas
<i>Escherichia coli</i> ATCC 25922	-	-
<i>Proteus vulgaris</i> ATCC 6380	+	+
<i>Salmonella typhimurium</i> ATCC 14028	-	-

PHENYLALANINE AGAR

Cat. 1040

Used for the differentiation of enteric bacilli which deaminate phenylalanine to phenyl pyruvic acid

Formula in grams per liter

<i>D-L Phenylalanine</i>	2,00	<i>Yeast Extract</i>	3,00
<i>Sodium Chloride</i>	5,00	<i>Sodium Phosphate</i>	1,00
<i>Bacteriological Agar</i>	12,00		

Final pH 7,3 ± 0,2 at 25°C

Preparation

Suspend 23 grams of the medium in one liter of distilled water. Mix well. Heat with frequent agitation and boil for one minute. Dispense and sterilize in autoclave at 121°C (15 lbs. sp.) for 10 minutes. Allow the tubes to solidify in a slanted position.

Uses

Phenylalanine Agar is used for differentiating *Proteus* and *Providencia* species from other Enterobacteriaceae, based on deamination of phenylalanine Battiaux, Osteaux, Fresnoy and Meriametz, developed a method to differentiate members of the *Proteus* and *Providencia* groups from other Enterobacteriaceae, based on the ability of *Proteus* and *Providencia* to determinate phenylalanine to phenylpyruvic acid by enzymatic activity.

Proteus and *Providencia* are the only enterobacteria which have a positive reaction, the others are negative.

To differentiate *Proteus* and *Providencia* seed heavily the suspicious organisms in Urea Agar Base (Christensen), or

Urea Broth. *Proteus* hydrolyzes the urea. The *Providencia* is negative for urease production.

Inoculate heavily with the sample organism. Incubate for 18 to 24 hours at 35°C. Add 4 to 5 drops of 10% ferric chloride. The immediate appearance of an intense green color (1-5 minutes) indicates the presence of phenylpyruvic acid.

Bibliography

Bailey and Scott. Diagnostic Microbiology. The C.V. Mosby Company. Saint Louis, 1978. Edwards and Ewing. Identification of Enterobacteriaceae. Burgess Publ. Co. Minneapolis, Minn., 1972. Ewing. Enterobacteriaceae. USPH. Publication 734. Washington, 1969. Lennette E.H., Spaulding and S.P. Truant. Manual of Clinical Microbiology, A.S.M. MaFaddin, J.F. 1985. Media for isolation-cultivation-identification-maintenance of medical bacteria, vol. 1, p. 634-636. Williams & Wilkins, Baltimore, MD.

Microbiological Test

Microorganisms	Growth	Phenyl piruvic Ac.(deam.)
<i>Escherichia coli</i> ATCC 25922	Satisfactory	-
<i>Enterobacter aerogenes</i> ATCC 13048	Satisfactory	-
<i>Proteus vulgaris</i> ATCC 13315	Satisfactory	+
<i>Providencia</i> spp.	Satisfactory	+

POTATO DEXTROSE AGAR

Cat. 1022

Used for the identification, cultivation and enumeration of yeasts and moulds.

Formula in grams per liter

Potato Infusion (solids) 4,00 Dextrose 20,00
Bacteriological Agar 15,00

Final pH 5,6 ± 0,2 at 25°C

Preparation

Suspend 39 grams of the medium in one liter of distilled water. Mix well and heat agitating frequently. Boil for one minute and sterilize at 121°C (15 lbs. sp.) for 15 minutes.

Uses

Potato Dextrose Agar can be used in the analysis of dairy products, bottled drinks, frozen food, and other types of food. It can also be used in the identification of *fungi* and yeasts in parallel with their cellular morphology or in methods of micro cultivation in slides.

When the medium is to be used for enumeration of molds and yeasts, add to the medium, sterilized and cooled to 45-50°C, approximately 14 ml. of a sterilized 10% solution

of tartaric acid to obtain a pH of 3,5. Do not heat the medium after adding the acid, because the agar may hydrolyze and not solidify.

Bibliography

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Association of Official Analytical Chemists. 1995. Bacteriological analytical manual, 8th ed. AOAC International. Gaithersburg, MD.

Microbiological Test

Microorganisms	Growth
<i>Aspergillus niger</i> ATCC 16404	Satisfactory
<i>Candida albicans</i> ATCC 10231	Satisfactory
<i>Saccharomyces cerevisiae</i> ATCC 9763	Satisfactory

POTATO DEXTROSE BROTH

Cat. 1261

Used for the identification, cultivation and enumeration of yeasts and moulds

Formula in grams per liter

Dextrose..... 20,00 Infusion from potato (Solids).....6,50

Final pH 5,6 ± 0,2 at 25°C

Preparation

Suspend 26,5 grams of the medium in one liter of distilled water. Boil for one minute and sterilize at 121° C (15 lbs. steam pressure) for 15 minutes.

Uses

Potato Dextrose Broth is used for cultivating yeast and moulds, the nutritionally rich base (potato infusion) encourages mould sporulation and pigment production in some demartophytes, but it also encourages luxuriant fungal growth.

Bibliography

Association of Official Analytical Chemists. 1995. Bacteriological analytical manual, 8th ed. AOAC International, Gaithersburg, MD.
MacFaddin, J.F. 1985. Media for isolation-cultivation-identification-maintenance of medical bacteria, vol. 1 Williams & Wilkins, Baltimore, MD.
Frank, J.F. G.L. Christen, and L.B. Bullerman (G.H. Richardson, Tech. Comm.) 1993. Tests for groups of microorganisms. P. 271-286, In Marshall, R.T. (ed.). Standard methods for the microbiological examination of dairy products, 16th ed. American Public Health Association, Washington, D.C.

Microbiological Test

Microorganisms	Growth
<i>Aspergillus niger</i> ATCC 16404	Satisfactory
<i>Candida albicans</i> ATCC 10231	Satisfactory
<i>Saccharomyces cerevisiae</i> ATCC 9763	Satisfactory

PPLO AGAR BASE W/O CRYSTAL VIOLET

Cat. 1140

For the isolation and culture of Mycoplasma in clinical specimens and mixed cultures

Formula in grams per liter

Peptone 10,00	Beef Heart Infusion 6,00
Sodium Chloride 5,00	Bacteriological Agar 14,00

Final pH 7,8 ± 0,2 at 25°C

Preparation

Suspend 35 grams of the medium in one liter of distilled water. Mix well. Heat agitating frequently until completely dissolved. Sterilize in autoclave at 121°C (15 pounds sp.) for 15 minutes. Let it cool under 50°C and if desired aseptically add 1% of serum fraction for PPLO or 25% of ascitic fluid, mixing well.

Uses

PPLO Agar was described by Morton, Smith and Leberman. PPLO Agar was used in study of the growth requirements of Mycoplasma, along with the identification and cultivation of this organism.

Store the dehydrated medium below 30°C. The dehydrated medium is very hygroscopic. Keep container tightly closed.

PPLO colonies are round with a dense center and a less dense periphery, giving a - fried egg -appearance on PPLO Agar.

Bibliography

Adler, H.E. and AJ Da Massa. 1967 Use of formalized Mycoplasma gallisepticum antigens and chicken erythrocytes in hemagglutination and hemagglutination-inhibition studies. Appl. Microbiol 15:245-248.
Morton HE and JG Lecce. 1953. Selective action of thallium acetate and crystal violet for pleuropneumonia like organisms of human origin. J. Bacteriol 66:646-649.

Microbiological Test

Microorganisms	Growth
<i>Mycoplasma bovis</i> ATCC 25523	Satisfactory
<i>Mycoplasma pneumoniae</i> ATCC 15531	Satisfactory

PPLO BROTH BASE W/O CRYSTAL VIOLET

Cat. 1262

Basal medium recommended for the enrichment of microorganisms PPLO: Mycoplasma

Formula in grams per liter

Beef Heart Infusion 6,00 *Peptone* 10,00
Sodium Chloride 5,00

Final pH 7,8 ± 0,2 at 25°C

Preparation

Suspend 21 grams of the medium in one liter of distilled water. Dissolve completely and sterilize in autoclave at 121°C (15 pounds sp.) for 15 minutes. Let it cool under 50°C and aseptically add the desired supplements and selective agents.

Uses

PPLO Agar was described by Morton, Smith and Leberman. PPLO Agar was used in study of the growth requirements of Mycoplasma, along with the identification and cultivation of this organism.

PPLO Broth w/o is prepared according to the formula described by Morton and Lecci. Crystal Violet is omitted from this formula due to its inhibitory action on some Mycoplasma. PPLO Broth w/op has been used for the cultivation of Mycoplasma for research studies.

Store the dehydrated medium below 30°C. The dehydrated medium is very hygroscopic. Keep container tightly closed.

Bibliography

Leland DS, MA Lapworth, RB Jones and MLV French 1982. Comparative evaluation of media for isolation of Ureaplasma urealyticum and genital Mycoplasmas species. J. Clin. Microbiol. 16:709-714.

Kenny GE 1985 Mycoplasmas, p. 407-411 In EH Lennette, A Balows Manual of clinical microbiology, 4th ed. American Society for Microbiology, Washington DC.

Microbiological Test

Microorganisms	Growth
<i>Mycoplasma bovis</i> ATCC 25523	Satisfactory
<i>Mycoplasma pneumoniae</i> ATCC 15531	Satisfactory
<i>Mycoplasma gallinarum</i> ATCC 19708	Satisfactory
<i>Streptococcus pneumoniae</i> ATCC 6303	Null to Satisfactory (*)

(*) Depending of the selective agents

PSEUDOMONAS F AGAR KING B MEDIUM Cat. 1532

Medium for the identification of *Pseudomonas*. It favors the production of fluorescein

Formula in grams per liter

Peptone Mixture..... 20,00	Dipotassium Phosphate..... 1,50
Magnesium Sulfate..... 1,50	Bacteriological Agar..... 14,00

Final pH 7,0 ± 0,2 at 25°C

Preparation

Suspend 37 grams of the medium in one liter of distilled water. Add 10 ml. of glycerin. Heat with frequent agitation and boil for one minute.
Dispense into appropriate containers and sterilize by autoclaving at 121°C (15 lbs.sp) for 15 minutes.

Uses

Pseudomonas F Agar is used for detecting and differentiating *Pseudomonas aeruginosa* from other *Pseudomonas* based on fluorescein production.

Incubation times and temperatures are similar to King A Medium.

This medium promotes the production of pyoverdin, a green fluorescing pigment which, unlike pyocyanin, is not soluble in chloroform. The pigment diffuses throughout the medium and is observed by use of a Wood's UV lamp. Positive organisms are *P. fluorescens*, *P. putida*.

Bibliography

King E.O. Ward M.K. Raney 1954. Two simple media for the demonstration of pyocyanin and fluorescein. J. lab Clin. Med. 44:301.
The United States Pharmacopoeia 1995. 23rd ed. United States Pharmacopoeia Convention, Rockville MD.

Microbiological Test

Microorganisms	Growth	Colony colour
<i>Pseudomonas aeruginosa</i> ATCC 9027	Satisfactory	Yellow-green
<i>Pseudomonas aeruginosa</i> ATCC 10145	Satisfactory	Yellow-green
<i>Pseudomonas aeruginosa</i> ATCC 17934	Satisfactory	----
<i>Pseudomonas aeruginosa</i> ATCC 25619	Satisfactory	Yellow-green
<i>Pseudomonas aeruginosa</i> ATCC 27853	Satisfactory	Yellow-green

PSEUDOMONAS P AGAR

KING A MEDIUM

Cat. 1531

 For the identification of *Pseudomonas*. It favors the production of pyocyanin

Formula in grams per liter

<i>Bacteriological Peptone</i> 20,00	<i>Potassium Sulfate</i> 10,00
<i>Magnesium Chloride</i> 1,40	<i>Bacteriological Agar</i> 13,60

Final pH 7,0 ± 0,2 at 25°C
Preparation

Suspend 45 grams of the medium in one liter of distilled water. Add 10 ml. of glycerin. Heat with frequent agitation and boil for 1 minute. Dispense into appropriate containers and sterilize by autoclaving at 121°C (15 lbs sp) for 15 minutes.

Uses

This medium is designed for the presumptive identification of *Pseudomonas aeruginosa* and promotes pyocyanin production.

Pseudomonas Agar P, patterned after the formulations described by King Ward and Raney, are modified to USP specifications. *Pseudomonas agar P* enhances the production of pyocyanin and inhibits the formation of fluorescein. Both pigments diffuse from *Pseudomonas* colonies into the medium in which they grow. Pyocyanin elaborated is a blue color.

The cultures are incubated at 30°C and observed regularly at 24-48 hours up to 6-7 days. Typical cultures are various shades of green and, at times, red if there is production of pyocyanin.

Pyocyanin can be removed by chloroform extraction. Adding 2 ml. of chloroform to a tube of medium and gently shaking will remove the pigment.

Bibliography

King E.O. Ward M.K. Raney D.E.-J. Lab. and Clin Med, 1954, 44, 301-307
 Bacteriological Analytical Manual, 8th edition. 1995. AOAC International, Gaithersburg, MD.
 The United States Pharmacopoeia. 1995. The United States pharmacopoeia, 23rd ed. United States Pharmacopoeial Convention, Rockville, MD.

Microbiological Test

Microorganisms	Growth	Colony colour
<i>Pseudomonas aeruginosa</i> ATCC 9027	Satisfactory	Blue
<i>Pseudomonas aeruginosa</i> ATCC 10145	Satisfactory	---
<i>Pseudomonas aeruginosa</i> ATCC 17934	Satisfactory	---
<i>Pseudomonas aeruginosa</i> ATCC 25619	Satisfactory	Blue-green
<i>Pseudomonas aeruginosa</i> ATCC 27853	Satisfactory	Blue

RAKA-RAY AGAR BASE

Cat. 1061

The addition of phenylethanol and sorbitan monooleate makes a selective medium to isolate lactic-acid bacteria in beer and fermentation processes of beer.

Formula in grams per liter

<i>Tryptone</i>	20,00	<i>Maltose</i>	10,00
<i>Cycloheximide</i>	0,007	<i>Yeast Extract</i>	5,00
<i>Fructose</i>	5,00	<i>Glucose</i>	5,00
<i>Potassium Glutamate</i>	2,50	<i>Betaine HCL</i>	2,00
<i>Diammonium Hydrogen Citrate</i>	2,00	<i>Magnesium Sulfate</i>	2,00
<i>Potassium Phosphate</i>	2,00	<i>Liver Extract</i>	1,00
<i>Manganese Sulfate</i>	0,66	<i>N-Acetylglucosamine</i>	0,50
<i>Bacteriological Agar</i>	17,00	<i>Potassium Aspartate</i>	2,50

Final pH 5,4 ± 0,2 at 25°C

Preparation

Suspend 77,2 grams of the medium in one liter of deionized or distilled water. To which have been previously added 10 ml. of sorbitan monooleate. Heat with frequent agitation to dissolve the medium completely. Do not overheat. Sterilize at 121°C (15 lbs.sp) for 15 minutes. Cool to 45°C-50°C and aseptically add 3 g. of phenylethanol.

Uses

RAKA-RAY Agar yields very good results in the detection of *lactobacilli* in the fermentation processes of beer. These organisms can change the organoleptic characteristics of the beer by their metabolites. The detection is complicated because of the nutritional and environmental requirements of these organisms. For these reasons, several formulations have been described to optimize the medium and obtain good growth. Higher counts of *lactobacilli* in comparative tests have been obtained with this medium because it contains growth stimulants such as liver extract, yeast extract, N-acetylglucosamine and sorbitan

monooleate. Maltose is added as a source of carbohydrate when certain *lactobacilli* cannot utilize glucose. Selectivity is obtained by adding 3 g/l of 2-phenylethanol, which inhibits Gram-negative bacteria, and cycloheximide, which inhibit yeasts.

The inoculation can be made by direct streaking of the agar surface or by the double layer pour plate method. Incubation is carried out at 25-30°C in anaerobic conditions for 4 days. Some organisms grow slower and may require 7 or more days.

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Methods of Analysis of the ASBC (1976) 7th Edition, The Society, St. Paul, Mn. USA.
European Brewing Convention, EBC Analytica Microbiologica: Part II J. Inst. Brewing (1981) 87,303-321.

Microbiological Test

Microorganisms	Growth
<i>Lactobacillus fermentans</i> ATCC 9338	Good
<i>Escherichia coli</i> ATCC 25922	Inhibited

RAPPAPORT SOY BROTH (VASSILIADIS)

Cat. 1240

Enrichment medium for *Salmonella*

Formula in grams per liter

<i>Magnesium Chloride</i>	13,58	<i>Sodium chloride</i>	7,20
<i>Soy Peptone</i>	4,50	<i>Monopotassium Phosphate</i>	1,26
<i>Disodium Phosphate</i>	0,18	<i>Malachite Green</i>	0,036

Final pH 5,2 ± 0,2 at 25°C

Preparation

Suspend 26,75 grams of the medium in one liter of distilled water. Heat with frequent agitation until complete dissolution. Dispense and sterilize at 115°C (12 lbs.sp) for 15 minutes.

DO NOT OVERHEAT.

Uses

A medium recommended for the selective isolation of *Salmonella* in food or in environmental samples, as well as in feces without preenrichment. It constitutes a modification of the Rappaport Vassiliadis medium with the advantage that offers a better stability of the pH of the prepared medium and optimizes the concentration of Magnesium Chloride. These two modifications improve notably the performance of the medium. It must not be used if there is any suspect of the presence of *Salmonella typhi*. The best recuperation are obtained by incubating at 42°C.

Proceed a usual for the sampling of foods:

- Transfer 0,1 ml. of Preenrichment Broth (25 g. sample in 225 ml. of Buffered Peptonized Water incubating at 37°C for 20 hours) to 10 ml. of Rappaport Soy Broth Vassiliaded.
- Incubate for 24 hours at 42°C.
- Confirm in suitable plates and verify the biochemical and serological characteristics of the suspicious colonies.

Bibliography

Rappaport F., Konforti N. and Navon B. (1956) J. Clin Pathol., 9,261.
 Peterz M. Wiberg C. and Norberg P. (1989) J. Appl. Bact. 66: 523-528.

Microbiological Test

Microorganisms	Growth
<i>Escherichia coli</i> ATCC 25922	< 5% (conc. 99%)
<i>Salmonella typhimurium</i> ATCC 14028	> 95% (conc. 1%)

REINFORCED CLOSTRIDIAL AGAR

Cat. 1087

For the culture and recount of *Clostridium* and other anaerobic microorganisms.

Formula in grams per liter

<i>Beef extract</i>	10,00	<i>Peptone</i>	10,00
<i>Dextrose</i>	5,00	<i>Sodium chloride</i>	5,00
<i>Yeast extract</i>	3,00	<i>Sodium acetate</i>	3,00
<i>Soluble Starch</i>	1,00	<i>L-Cysteine Hydrochloride</i>	0,50
<i>Bacteriological Agar</i>	12,50		

Final pH 6,8 ± 0,2 at 25°C

Preparation

Suspend 50 grams of the medium in one liter of distilled water. Heat with frequent agitation until completely dissolved. Dispense into tubes and sterilize in the autoclave at 121°C (15 lbs sp) for 15 minutes. Cool to 45-50°C and add if wanted 0,02 gr./liter of Polymixin B in sterile filtered solution form

Uses

The Reinforced *Clostridium* Agar, lacks inhibitory substances. If you want to inhibit the Gram-negative bacteria Polymixin can be added as previously indicated. Reinforced Clostridial Medium is a semisolid medium formulated by Hirsch and Grinstead. Their work demonstrated that the medium outperformed other media

in supporting growth of clostridia from small inocula and produced higher viable cell counts.

Bibliography

Barnes, EMJE Despaul and M. Ingram 1963. The behavior of a food poisoning strain of *Clostridium welchii* in beef. J. Appl. Bacteriol 26:415.
MacFaddin JF. 1985 Media for isolation-cultivation-identification-maintenance of medical bacteria, vol. 1, p. 660-668. Williams & Wilkins, Baltimore MD.

Microbiological Test

Microorganisms	Growth
<i>Clostridium bifermentans</i> ATCC 19299	Good
<i>Clostridium difficile</i>	Good
<i>Clostridium perfringens</i> ATCC 13124	Good
<i>Clostridium perfringens</i> ATCC 10543	Good
<i>Escherichia coli</i> ATCC 25922	Good
<i>Bacillus cereus</i> ATCC 11778	Good

REINFORCED CLOSTRIDIAL MEDIUM (EUROPEAN PHARMACOPEIA)

Cat. 1007

For cultivating and enumerating Clostridia, other anaerobes, and other species bacteria from foods and clinical specimens.

Formula in grams per liter

Beef extract.....	10,00	Casein peptone.....	10,00
Dextrose.....	5,00	Sodium chloride.....	5,00
Yeast extract.....	3,00	Sodium acetate.....	3,00
Starch.....	1,00	L-Cysteine chloride.....	0,50
Bacteriological Agar.....	0,50		

Final pH 6,8 ± 0,2 at 25°C

Preparation

Suspend 38 grams of the medium in one liter of distilled water. Heat with frequent agitation until completely dissolved. Dispense into tubes and sterilize in the autoclave at 121°C (15 lbs sp) for 15 minutes. Cool to 45-50°C and add if wanted 0,02 gr./liter of Polymixin B in sterile filtered solution form.

Uses

Reinforced *Clostridium* Medium, is a semisolid medium formulated by Hirsch and Grinstead. Their work demonstrated that the medium outperformed other media in supporting growth of clostridia from small inocula and produced higher viable cell counts.

The medium in a non selective enrichment medium and grows various anaerobic and facultative bacteria when incubated anaerobically.

Bibliography

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Andrews, W.H. (ed) 1995. Microbial methods p. 1-119. In Official methods of analysis of AOAC International. 16th ed.

Microbiological Test

Microorganisms	Growth
<i>Clostridium bifermentans</i> ATCC 19299	Good
<i>Clostridium difficile</i>	Good
<i>Clostridium perfringens</i> ATCC 13124	Good
<i>Clostridium perfringens</i> ATCC 10543	Good
<i>Escherichia coli</i> ATCC 25922	Good
<i>Bacillus cereus</i> ATCC 11778	Good

ROGOSA SL AGAR

Cat. 1096

Selective medium for the cultivation of lactobacilli in medical and food microbiology

Formula in grams per liter

<i>Sodium Acetate</i>	15,00	<i>Tryptose</i>	10,00
<i>Dextrose</i>	10,00	<i>Monopotassium Phosphate</i>	6,00
<i>Yeast Extract</i>	5,00	<i>Sucrose</i>	5,00
<i>Arabinose</i>	5,00	<i>Ammonium Citrate</i>	2,00
<i>Sorbitan Monoleate</i>	1,00	<i>Magnesium Sulfate</i>	0,57
<i>Manganese Sulfate</i>	0,12	<i>Ferrous Sulfate</i>	0,03
<i>Bacteriological Agar</i>	15,00		

Final pH 5,4 ± 0,2 at 25°C

Preparation

Suspend 75 grams of the medium in one liter of distilled water. Heat with frequent agitation and boil to dissolve it completely. Add 1,32 ml. of Acetic Acid Glacial, mix well. Heat again at 90° -100° C for two minutes. **DO NOT AUTOCLAVE.** Dispense into sterilized appropriate containers. Cool the medium at 40° -45° C to obtain plate counts.

Uses

This medium is used for isolation, enumeration and identification of lactobacilli in oral bacteriology, feces, vaginal specimens and foodstuffs. The low pH and high acetate concentrations effectively suppress other bacterial flora allowing lactobacilli to flourish.

Rogosa SL Agar is a selective medium, modified by Rogosa to contain high levels of sodium acetate at a low pH which inhibits most microorganisms but allows for the growth of lactobacilli. Direct inoculation or plate count methodologies can be used.

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Microbiological Test

Microorganisms	Growth
<i>Lactobacillus casei</i> ATCC 9595	Satisfactory
<i>Lactobacillus fermentum</i> ATCC 9338	Satisfactory
<i>Lactobacillus plantarum</i> ATCC 8014	Satisfactory
<i>Lactobacillus leichmannii</i> ATCC 4797	Satisfactory
<i>Staphylococcus aureus</i> ATCC 25923	Inhibited

ROGOSA SL BROTH

Cat. 1234

Selective medium to cultivate lactobacilli in medical and food microbiology

Formula in grams per liter

<i>Sodium Acetate</i>	15,00	<i>Tryptose</i>	10,00
<i>Dextrose</i>	10,00	<i>Monopotassium Phosphate</i>	6,00
<i>Yeast Extract</i>	5,00	<i>Sucrose</i>	5,00
<i>Arabinose</i>	5,00	<i>Ammonium Citrate</i>	2,00
<i>Sorbitan Monooleate</i>	1,00	<i>Magnesium Sulfate</i>	0,57
<i>Manganese Sulfate</i>	0,12	<i>Ferrous Sulfate</i>	0,03

Final pH 5,4 ± 0,2 at 25°C

Preparation

Suspend 60 grams of the medium in one liter of distilled water and heat till boiling for one minute. Add 1,32 ml. of Glacial Acetic Acid and mix well. Distribute in tubes and heat again to 90-100°C for 2-3 minutes. **DO NOT AUTOCLAVE.**

Uses

Rogosa SL Broth is a modification of media described by Rogosa, Mitchell and Wiseman.

This medium is used for isolation, enumeration and identification of lactobacilli in oral bacteriology, feces, vaginal specimens and foodstuffs. The low pH and high acetate concentrations effectively suppress other bacterial flora allowing lactobacilli to flourish.

Rogosa SL Broth is similar to Rogosa SL Agar, lacking agar and is very selective by means of its high sodium acetate concentration and its low pH, very advantageous for the cultivation of *lactobacilli*.

Bibliography

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 MacFaddin, J. D. 1985. Media for isolation-cultivation-identification-maintenance of medical bacteria, vol. 1, p. 678-680. Williams & Wilkins, Baltimore, M.D.

Microbiological Test

Microorganisms	Growth
<i>Lactobacillus casei</i> ATCC 9595	Satisfactory
<i>Lactobacillus fermentum</i> ATCC 9338	Satisfactory
<i>Lactobacillus plantarum</i> ATCC 8014	Satisfactory
<i>Lactobacillus leichmannii</i> ATCC 4797	Satisfactory
<i>Staphylococcus aureus</i> ATCC 25923	Inhibited

ROSE BENGAL AGAR

Cat. 1081

For the culture and selective isolation of yeast and moulds

Formula in grams per liter

Dextrose	10,00	Bacteriological peptone	5,00
Potassium phosphate	1,00	Magnesium sulfate.....	0,50
Chloramphenicol	0,50	Bengal rose	0,05
Bacteriological Agar	15,00		

Final pH 7,2 ± 0,2 at 25°C

Preparation

Suspend 32 grams of the medium in one liter of distilled water. Mix well and heat with frequent agitation until boiling. Boil for one minute. Distribute into appropriate containers and sterilize in autoclave at 121°C (15 lbs. sp.) for 15 minutes.

Uses

This is a selective medium for fungi and yeasts in foods. The Bengal Rose inhibits the massive growth of fast-growing so that the development of other slow growths can be detected on addition. The yeasts appear rose colored, being stained by this product. On the other hand, the chloramphenicol inhibits the *bacterial* growth.

The inoculation can be carried out from a diluted source whether by extension of 0.1 ml. of each dilution into the prepared plates, or by the pouring method, depositing 1

ml. of each dilution into the empty plate, pouring the medium immediately afterward (once it has been cooled at 45°C). Incubate for 5 days at 22°C.

Bibliography

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Koburger J.A. 1972. Fungi in foods. Effect of plating medium pH on counts. J. Milk Food Technol. 35:659-660.
Papvizas, G.C., and C.B. Davey. 1959. Evaluation of various media and antimicrobial agents for isolation of soil fungi.
Marshall, R.T. (ed) 1993. Standard methods for the examination of dairy products, 16th ed. American Public Health assoc., Washington, DC.

Microbiological Test

Microorganisms	Growth	Colony appearance
<i>Candida albicans</i> ATCC 10231	Good	Rose, plane, bulky
<i>Aspergillus niger</i> ATCC 1015	Good	White, filamentose, wien become black
<i>Escherichia coli</i> ATCC 25922	Inhibited	----

ROTHE BROTH (GLUCOSE BROTH WITH AZIDE) Cat. 1238

For the quantitative determination of faecal streptococci

Formula in grams per liter

<i>Peptone Mixture</i> 15,00 <i>Sodium Chloride</i> 7,50 <i>Sodium Azide</i> 0,20	<i>Glucose</i> 7,50 <i>Beef Extract</i> 4,50
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Final pH 7,2 ± 0,2 at 25°C

Preparation

Dissolve 34,7 grams in one liter of distilled water. To prepare a double strength broth use 69.4 grams per liter. Mix well. Dispense into appropriate containers and sterilize at 118 °C (12 lbs sp) for 15 minutes

Uses

Rothe Broth is a selective medium incorporating sodium azide to inhibit Gram-negative flora. Rothe Broth is ideal for enumeration of streptococci from residual waters, foods and products susceptible to contamination by residual water, by the serial dilution method. The presence of fecal streptococci indicates fecal contamination. They are the best indicators of contamination on chloride water as *E. Coli* has a better resistance to chloride. Malmann and Seligmann recommended Rothe broth for the quantification of Streptococci in water, food and other materials suspect of being contaminated by waste waters

Inoculate 10 ml. of the sample in 10 ml. tubes of double strength Rothe Broth (or 1 ml. of the sample in 10 ml. of single strength medium). Utilize 5 tubes for each dilution (according to Mallmann and Seligmann).

Incubate all tubes at 37°C for 48 hours. Confirmation of fecal streps is obtained by subsequent inoculation of positive tubes into EVA Broth.

Bibliography

Mallmann W.L. Seligmann E.B. AJPH, 1950, 40 286-289
 Standard Methods for the Examination of Water and Wastewater. Eleventh Edition APHA Inc. New-York 1960
 Edwards S.J. (1933) J. Comp. Path Therap., 46,211.

Microbiological Test

Microorganisms	Growth
<i>Escherichia coli</i> ATCC 25922	Inhibited
<i>Staphylococcus aureus</i> ATCC 25923	Inhibited
<i>Streptococcus faecalis</i> ATCC 19433	Good

R2A AGAR (EUROPEA PHARMACOPOEIA) Cat. 1071

Recommended by the European Pharmacopoeia for total aerobe count in waters.

Formula in grams per liter

<i>Peptone</i>	0,75	<i>Yeast extract</i>	0,50
<i>Dextrose</i>	0,50	<i>Sodium pyruvate</i>	0,30
<i>Dipotassium phosphate</i>	0,30	<i>Tryptone</i>	0,25
<i>Starch</i>	0,50	<i>Magnesium sulphate</i>	0,024
<i>Bacteriological agar</i>	15,00		

Final pH 7,2 ± 0,2 at 25°C

Preparation

Suspend 18,1 grams of the dehydrated medium in one liter of distilled water. Mix well. Heat with frequent agitation and boil for one minute until completely dissolved. Sterilize in autoclave at 121° C (15 lbs. sp) for 15 minutes. Cool to 45 °C and pour into Petri dishes.

Uses

R2A Agar was developed by Reasoner and Geldreich for bacteriological plate counts of treated potable water. A low nutrient medium, such as R2A Agar, in combination with a lower incubation temperature and longer incubation time simulates the growth of stressed and chlorine-tolerant bacteria.

R2A Agar is recommended in Standard Methods for the Examination of water and wastewater for pour plate, spread plate and membrane filter methods for heterotrophic plate counts.

Bibliography

American Public health Association (1985) Standard Methods for the enumeration of Water and Wastewater. European Pharmacopoeia fourth Edition published 20 September 2001.

Microbiological Test

Microorganisms	Growth
<i>Escherichia coli</i> ATCC 25922	Satisfactory
<i>Escherichia coli</i> ATCC 11775	Satisfactory
<i>Staphylococcus aureus</i> ATCC 25923	Satisfactory
<i>Staphylococcus epidermis</i> ATCC 12228	Satisfactory

SABOURAUD DEXTROSE AGAR

Cat. 1024

Used for the cultivation of fungi

Formula in grams per liter

Dextrose..... 40,00	Peptone Mixture 10,00
Bacteriological Agar..... 15,00	

Final pH 5,6 ± 0,2 at 25°C

Preparation

Suspend 65 grams of the medium in one liter of distilled water. Mix well until a uniform suspension is obtained. Heat with frequent agitation and boil for one minute. Distribute and sterilize at 118°C-121°C (no more than 15 lbs. sp.) for 15 minutes. Avoid overheating, as it, facilitates the hydrolysis of the components, and the medium remains soft.

Uses

Sabouraud Dextrose Agar can be used for the isolation, identification and maintenance of pathogenic and saprophytic fungi.

When the materials in study are highly contaminated, the isolation can be improved by adding a selective antimicrobial package. Georg and collaborators recommended aseptically adding 0,5 mg. of cycloheximide, 20 units of penicillin, and 40 mg. of streptomycin per ml. of medium, minutes before using, for the inhibition of contaminating flora which can obstruct the growth of fungal cultures.

To diminish the growth of other microorganisms several inhibitors such as tellurite, bile salts, and dyes can be incorporated into the medium.

The incubation of the plates should be at 25°C to 35°C. The addition of 0,1 grams of triphenyl tetrazolium chloride (TTC) for each 100 ml. of medium greatly facilitates the identification of different species of the genus *Candida* because these yeasts yield colonies of different colors such as whites, roses, reds, and violets. One can obtain a very rich Sabouraud medium by dissolving the medium in one litre of Brain Heart Infusion. If desired, antimicrobial agents can be added to this enriched combination of media.

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 Beuchat, L.R., J.E. Corry, A.D. King, Jr. and J.I. Pitt (ed) 1986. Methods for the mycological examination of food. Plenum Press, New York.

Microbiological Test

Microorganisms	Growth
<i>Aspergillus niger</i> ATCC 16404	Good
<i>Candida albicans</i> ATCC 26790	Good
<i>Escherichia coli</i> ATCC 25922	Moderate-Satisfactory
<i>Lactobacillus casei</i> ATCC 9595	Good
<i>Saccharomyces cerevisiae</i> ATCC 9763	Good

SABOURAUD DEXTROSE AGAR+CHLORAMPHENICOL (EUROPEAN PHARMACOPOEIA)

Cat. 1134

For the selective cultivation and isolation of fungi

Formula in grams per liter

Dextrose	40,00	Peptone Mixture	10,00
Chloramphenicol	0,05	Bacteriological Agar	15,00

Final pH 5,6 ± 0,2 at 25°C

Preparation

Suspend 65 grams of the medium in one liter of distilled water. Mix well until a uniform suspension is obtained. Heat with frequent agitation till boiling. Distribute and sterilize at 118 °C (no more than 12 lbs. pressure) for 15 minutes. Avoid excessive heating, as it will facilitate the hydrolysis of the components, and the medium will remain soft.

Uses

Sabouraud Dextrose Agar is used for culturing yeast, molds and aciduric microorganisms. This medium is a modification of the Dextrose Agar described by Sabouraud. It is used for cultivating pathogenic fungi, particularly those associated with skin infections. The high

dextrose concentration and acidic pH make this medium selective for fungi.

Sabouraud Dextrose Agar is also used for determining the microbial content of cosmetics and for the mycological evaluation of food.

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Microbiological Test

Microorganisms	Growth
<i>Candida albicans</i> ATCC 2091	Satisfactory
<i>Candida tropicalis</i> ATCC 750	Satisfactory
<i>Escherichia coli</i> ATCC 25922	Inhibited
<i>Staphylococcus aureus</i> ATCC 25923	Inhibited

SABOURAUD DEXTROSE AGAR WITH CHLORAMPHENICOL

Cat. 1090

Used for the selective cultivation and isolation of fungi

Formula in grams per liter

Dextrose.....	40,00	Peptone Mixture	10,00
Chloramphenicol.....	0,50	Bacteriological Agar	15,00

Final pH 5,6 ± 0,2 at 25°C

Preparation

Suspend 65,5 grams of the medium in one liter of distilled water. Mix well until a uniform suspension is obtained. Heat with frequent agitation and boil for one minute. Distribute and sterilize at 118°-121° C (not more than 15 lbs. pressure) for 15 minutes. Avoid undue exposure to heat, which facilitates the hydrolysis of the components, and the medium remains soft.

Uses

Sabouraud Dextrose Agar is used for culturing yeast, molds and aciduric microorganisms. This medium is a modification of the Dextrose Agar described by Sabouraud. It is used for cultivating pathogenic fungi, particularly those associated with skin infections. The high dextrose concentration and acidic pH make this medium selective for fungi.

Sabouraud Dextrose Agar is also used for determining the microbial content of cosmetics and for the mycological evaluation of food.

This selective medium is recommended for the isolation of yeasts and dermatophytes from contaminated samples. The presence of chloramphenicol inhibits a great majority of bacterial contaminants.

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 Jarett, L., and A.C. Sonnenwirth (ed) 1980. Gradwohl's clinical laboratory methods and diagnosis, 8th ed. CV Mosby.
 Curry, A. S., J. G. Graf, and G. N. McEwen, Jr. (ed) 1993. CTFA Microbiology Guidelines. The Cosmetic, Toiletry, and Fragrance Association, Washington, D.C.

Microbiological Test

Microorganisms	Growth
<i>Candida albicans</i> ATCC 2091	Satisfactory
<i>Candida tropicalis</i> ATCC 750	Satisfactory
<i>Escherichia coli</i> ATCC 25922	Inhibited
<i>Staphylococcus aureus</i> ATCC 25923	Inhibited

SABOURAUD DEXTROSE AGAR WITH CHLOR.+CYCLO HEXIMIDE Cat. 1089

Used for the selective cultivation of pathogenic fungi

Formula in grams per liter

Dextrose	40,00	Peptone mixture.....	10,00
Chloramphenicol	0,05	Cycloheximide.....	0,40
Bacteriological Agar	15,00		

Final pH 5,6 ± 0,2 at 25°C

Preparation

Suspend 65,5 grams of the medium in one liter of distilled water. Mix well to obtain a uniform suspension. Heat with frequent agitation and boil for one minute. Dispense and sterilize at 118°C for 15 minutes. which could hydrolyze the medium to a weak gel.

DO NOT OVERHEAT

Uses

Sabouraud Dextrose Agar is used for culturing yeast, molds and aciduric microorganisms. This medium is a modification of the Dextrose Agar described by Sabouraud. It is used for cultivating pathogenic fungi, particularly those associated with skin infections. The high dextrose concentration and acidic pH make this medium selective for fungi.

Sabouraud Dextrose Agar is also used for determining the microbial content of cosmetics and for the mycological evaluation of food.

It is the right selective medium for the growth of pathogenic fungi. The chloramphenicol inhibits a great majority of bacterial contaminants. The cycloheximide (actidione) inhibits the growth of saprophytic fungi.

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Microbiological Test

Microorganisms	Growth
<i>Candida albicans</i> ATCC 10231	Satisfactory
<i>Candida tropicalis</i> ATCC 750	Partially inhibited
<i>Escherichia coli</i> ATCC 25922	Inhibited
<i>Trichophyton mentagrofites</i>	Satisfactory
<i>Penicillium</i> spp.	Partially inhibited

SABOURAUD DEXTROSE AGAR WITH CYCLOHEXIMIDE (ACTIDIONE)* Cat. 1088

Used for the selective culture of fungi

Formula in grams per liter

Dextrose..... 40,00	Peptone Mixture 10,00
Cycloheximide (Actidione)..... 0,40	Bacteriological Agar 15,00

Final pH 5,6 ± 0,2 at 25°C

Preparation

Suspend 65,4 grams of the medium in one liter of distilled water. Mix well until a uniform suspension is obtained. Heat with frequent agitation and boil for one minute. Distribute and sterilize at 118 °C - 121 °C (not more than 15 pounds pressure). Avoid undue exposure to heat, which facilitates hydrolysis of the components, and the medium remains soft. The cycloheximide inhibits the growth of saprophytes fungi.

Uses

Sabouraud Dextrose Agar is used for culturing yeast, molds and aciduric microorganisms. This medium is a modification of the Dextrose Agar described by Sabouraud. It is used for cultivating pathogenic fungi, particularly those associated with skin infections. The high dextrose concentration and acidic pH make this medium selective for fungi. Sabouraud Dextrose Agar is also used for determining the microbial content of cosmetics and for the mycological evaluation of food.

This medium contains added cycloheximide to inhibit the saprophytic fungi but allow for growth of the pathogenic fungi. *Cryptococcus neoformans*, *Aspergillus fumigatus* and some species of *Candida* (*tropicalis*, *krusei*) are inhibited by cycloheximide while other species of *Candida* and all the dermatophytes, in general, grow without problems.

(*) Actidione, Trademark Upjohn Pharmaceutical Co.

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Microbiological Test

Microorganisms	Growth
<i>Candida albicans</i> ATCC 2091	Good
<i>Escherichia coli</i> ATCC 25922	Good/Moderate
<i>Aspergillus niger</i> ATCC 16404	Inhibited/Light
<i>Penicillium</i> spp.	Inhibited/Light
<i>Trichophyton mentagrophytes</i>	Good

SABOURAUD DEXTROSE BROTH

Cat. 1205

Used for the culture of fungi

Formula in grams per liter

Dextrose 20,00 Peptone Mixture 10,00

Final pH 5,7 ± 0,2 at 25°C

Preparation

Suspend 30 grams of the medium in one liter of distilled water. Mix well until a uniform suspension is obtained. Heat with frequent agitation and boil for one minute. Distribute and sterilize at 118-121°C (no more than 15 lbs sp) for 15 minutes. Do not overheat.

Uses

Sabouraud Dextrose Broth, is used for culturing yeast, molds and aciduric microorganisms. This medium is modification of the Dextrose Agar described by Sabouraud. It is used for cultivating pathogenic fungi, particularly these associated with skin infections The high dextrose

concentration and acidic pH make this medium selective for fungi.

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Microbiological Test

Microorganisms	Growth
<i>Aspergillus niger</i> ATCC 16404	Satisfactory
<i>Candida albicans</i> ATCC 26790	Satisfactory
<i>Escherichia coli</i> ATCC 25922	Partially inhibited
<i>Lactobacillus casei</i> ATCC 9595	Satisfactory
<i>Saccharomyces cerevisiae</i> ATCC 9763	Satisfactory

SABOURAUD FLUID MEDIUM

Cat. 1506

For cultivation of yeast and molds

Formula in grams per liter

Dextrose..... 20,00 Casein Peptone..... 5,00
 Meat Peptone 5,00

Final pH 5,7 ± 0,2 at 25°C

Preparation

Suspend 30 grams of the medium in one liter of distilled or deionized water. Heat agitating frequently until completely dissolved. Dispense and sterilize at 121°C(15 lbs.sp) for 15 minutes. **Do not overheat**, as the medium contains high levels of carbohydrates which can darken (caramelize) and lose effectiveness.

Uses

Sabouraud fluid Medium is employed in sterility test procedures for determining the presence of molds, yeasts and aciduric microorganisms. The acid reaction of the final medium is inhibitor to a large number of bacteria

and makes the medium particularly well suited for cultivating fungi and acidophilic microorganisms.

Sabouraud Liquid Medium is used in the tests of sterility of pharmaceutical products, in special parenterals, such as antisera, antibiotic preparations, venipuncture equipment, and saline and glucose solutions. The formula meets the standards of the U.S.P

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 United States Pharmacopeial Convention. 1995. The United States pharmacopoeia, 23rd ed. The United States Pharmacopeial Convention, Rockville, M.D.

Microbiological Test

Microorganisms	Growth
<i>Aspergillus niger</i> ATCC 16404	Satisfactory
<i>Candida albicans</i> ATCC 26790	Satisfactory
<i>Escherichia coli</i> ATCC 25922	Inhibited partially
<i>Lactobacillus casei</i> ATCC 9595	Satisfactory
<i>Saccharomyces cerevisiae</i> ATCC 9763	Satisfactory

SABOURAUD MALTOSÉ AGAR

Cat. 1054

Used for the cultivation of fungi and yeasts

Formula in grams per liter

Maltose 40,00 Peptone mixture 10,00
Bacteriological Agar 15,00

Final pH 5,6 ± 0,2 at 25°C

Preparation

Suspend 65 grams of the medium in one liter of distilled water. Mix well until a uniform suspension is obtained. Heat with frequent agitation and boil for one minute. Distribute and sterilize at 118°C-121°C (not more than 15 lbs. sp.) for 15 minutes. Avoid overheating as it facilitates the hydrolysis of the components, and the medium remains soft.

Uses

Sabouraud Maltose Agar is a modification of Sabouraud Dextrose Agar with maltose substituted for dextrose. It is a selective medium due to its acid pH. Davidson, Dawding and Buller reported that Sabouraud Maltose Agar was satisfactory medium in isolating *T. gypseum* from a case of tinea barbae and in their studies of the infections caused by *Microsporon audonini*, *M. lanosum* and *Trichophyton gypseum*.

The medium can be used as the original formula or can be modified if the sample material is contaminated. To prepare a selective culture medium add to every litre of sterilized medium one of the following antimicrobials: 20,000 u of penicillin + 40 mg of streptomycin or 200 mg of chloramphenicol or 250 mg of neomycin.

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Microbiological Test

Microorganisms	Growth
<i>Aspergillus niger</i> ATCC 16404	Satisfactory
<i>Candida albicans</i> ATCC 26790	Satisfactory
<i>Escherichia coli</i> ATCC 25922	Partially inhibited
<i>Lactobacillus casei</i> ATCC 9595	Satisfactory
<i>Saccharomyces cerevisiae</i> ATCC 9763	Satisfactory

SABOURAUD MALTOSÉ BROTH

Cat. 1213

For the cultivation of yeast, moulds and acidophilic bacteria, as well as for sterility tests

Formula in grams per liter

Maltose..... 40,00 Peptone Mixture 10,00

Final pH 5,6 ± 0,2 at 25°C

Preparation

Suspend 50 grams of the medium in one liter of distilled water. Mix well until completely dissolved. Dispense and sterilize at 121°C (15 lbs sp) for 15 minutes.

Uses

Sabouraud Maltose Broth is a modification of Sabouraud Dextrose Broth in which Maltose is substituted for Dextrose. It is a selective broth because of its acid pH. Sabouraud Maltose Broth is used for the cultivation of yeasts, molds, acidophilic bacteria as well as for sterility tests for yeasts and molds.

The growth of moulds appears as cotton balls in the medium. Initially they form a membrane at the top of the liquid/air surface.

The growth of yeasts and bacteria are manifested by a homogeneous turbidity which can be then stained and viewed microscopically.

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 Association of Official Analytical Chemists. 1995. Bacteriological analytical manual, 8th ed. AOAC International, Gaithersburg, MD.

Microbiological Test

Microorganisms	Growth
<i>Aspergillus niger</i> ATCC 16404	Satisfactory
<i>Candida albicans</i> ATCC 26790	Satisfactory
<i>Escherichia coli</i> ATCC 25922	Partially inhibited
<i>Lactobacillus casei</i> ATCC 9595	Satisfactory
<i>Saccharomyces cerevisiae</i> ATCC 9763	Satisfactory

SALINE PEPTONE WATER

Cat. 1405

Recommended as diluent and for the homogenization of microbiological samples.

Formula in grams per liter

Sodium chloride 8,50

Bacteriological Peptone..... 1,00

Final pH 7,0 ± 0,2 at 25°C

Preparation

Dissolve 9,5 grams of the medium in one liter of distilled water. Mix well until obtaining a uniform suspension. Heat, agitating frequently and boil for one minute or until completely dissolved. Distribute into appropriate containers and sterilize at 121°C (15 lbs sp) for 15 minutes.

Uses

This medium is used for the growth of bacterial cultures, such as marine bacteria (2). It is also used as a diluent for

microbiological samples (1) in many food studies, environment studies, etc.

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- (2) Destoumieux – garzon D. Saulnier, D. Garnier 3. et al. J. Biol Chem. Vol. 276, Issue 50 -47070-47077 (14 December-2001).

Microbiological Test

Microorganisms	Growth
<i>Escherichia coli</i> ATCC 25922	Good
<i>Salmonella typhimurium</i> ATCC 14028	Good
<i>Staphylococcus aureus</i> ATCC 25923	Good

SALMONELLA CHROMOGENIC AGAR

Cat. 1122

Medium used for the isolation of Salmonella in clinical samples and foods

Formula in grams per liter

<i>Sodium citrate</i>	8,50	<i>Sodium Desoxicholate</i>	5,00
<i>Casein Peptone</i>	5,00	<i>Beef extract</i>	5,00
<i>Ferroammonium citrate</i>	0,50	<i>Chromogenic mixture</i>	0,31
<i>Bacteriological Agar</i>	12,00		

Final pH 7,2 ± 0,2 at 25°C

Preparation

Suspend 36,3 grams of the medium in one liter of distilled water. Dissolve by heating agitating frequently. Boil for one minute. **DO NOT OVERHEAT. DO NOT AUTOCLAVE.** Pour into Petri dishes. Keep plates refrigerated at 6-8°C protecting them from light (may present a slight precipitate). It is recommended to prepare the plates on the same day to be used.

Uses

This new selective chromogenic medium is used for detecting and presumptive identification of Salmonella Sp. from stool samples.

This medium contains two chromogenic substrates (Magenta and X-Gal) in a chromogenic mixture. These both substrates will differentiate non-Salmonella organisms (that appear blue or are not stained by any of

the chromogens of the medium from Salmonella organisms in magenta colonies.

On the basis of its good sensitivity and specificity, and also for the celerity of its results, this medium is recommended for primary plating when human stool samples are screened for Salmonella spp.

Bibliography

Journal Clinical Microbiology, Vol. 41 n° 7 p. 3229-3232, July 2003 Robert Cassar and Paul Cuschieri.
 J.D. Perry, Michael Furs, Jeffrey Taylor, Et. Al. Journal Clinical Microbiology, March 1999, pag. 766-768 Vol. 37, n° 3
 Gallioto di camillo, p. Et. Al. (J. Clinil Microbiol. March 1999.

Microbiological Test

Microorganisms	Growth	Medium colour
<i>Escherichia coli</i> ATCC 25922	Good	blue-green
<i>Salmonella enteritidis</i> ATCC 13076	Good	Magenta
<i>Salmonella typhi</i> ATCC 19430	Good	Magenta
<i>Salmonella typhimurium</i> ATCC 14028	Good	Magenta
<i>Proteus vulgaris</i> ATCC 13315	Inhibited	colourless

SALMONELLA SHIGELLA AGAR

Cat. 1064

Selective medium for the isolation of Salmonella and Shigella

Formula in grams per liter

Lactose	10,00	Bile salts mixture	8,50
Sodium Citrate	8,50	Sodium Thiosulphate	8,50
Beef Extract	5,00	Peptone Mixture	5,00
Ferric Citrate	1,00	Neutral Red	0,025
Brilliant Green	0,330mg	Bacteriological Agar	13,50

Final pH 7,0 ± 0,2 at 25°C

Preparation

Suspend 60 grams of the medium in one liter of distilled water. Mix well until a homogeneous suspension is obtained. Heat with frequent agitation and boil for one minute. **DO NOT STERILIZE IN AUTOCLAVE.** Cool to 45°C and 50° C and distribute in Petri plates. Allow the medium to solidify partially uncovered.

Uses

SS agar is a selective and differential medium widely used in sanitary bacteriology to isolate *Salmonella* and *Shigella* from feces, urine, and fresh and canned foods. Inhibition of Gram-positive microorganisms is obtained by the bile salts mixture. Due to its strong inhibitory power, SS Agar can be streaked with a heavy inoculum but other less inhibitory media such as Desoxicholate Agar, MacConkey Agar, Eosin Methylene Blue (EMB) Agar, XLD Agar, and Hektoen Enteric Agar should be streaked in parallel.

Non-lactose fermenting bacteria (supposed pathogens) produce clear colonies, transparent or colourless, while coliforms are sufficiently inhibited, and form small colonies that vary from rose to red in color.

The H₂S producing bacteria produce colonies with black centers and a clear halo such as *Proteus* and other species of *Salmonella*.

The plates of the medium can be kept for at least a week in refrigeration.

Bibliography

Pub. Health Reports. 65:1075, 1950. Paper Read at Microbiological Congress, 1950.
Proc. 22nd Ann. Meet. Northeastern Conf. Lab. Workers in Pullorum Disease Control Burlington, Vermont, June 20-21, 1950.

BACTERIA	COLONIES
<i>Shigella</i> and the major part of <i>salmonellas</i>	Clear, colourless, transparent.
<i>Escherichia coli</i>	Small, rose to red.
<i>Enterobacter, Klebsiella</i>	Large than <i>E. coli</i> , mucoid, pale opaque cream to rose.
<i>Proteus</i> and some <i>salmonellas</i>	Colourless, transparent, with a black center if H ₂ S is produced.

Microbiological Test

Microorganisms	Growth	Colony colour
<i>Proteus mirabilis</i> ATCC 25933	Partially inhibited	Colourless
<i>Enterobacter aerogenes</i> ATCC 13048	Partially inhibited	Cream-rose
<i>Salmonella enteritidis</i> ATCC 13076	Satisfactory	Colourless
<i>Salmonella typhi</i> ATCC 6539	Satisfactory	Colourless
<i>Salmonella typhimurium</i> ATCC 14028	Satisfactory	Colourless
<i>Shigella flexneri</i> ATCC 12022	Satisfactory	Colourless
<i>Streptococcus faecalis</i> ATCC 19433	Inhibited	-----
<i>Escherichia coli</i> ATCC 25922	Inhibited	-----

SCHAEDLER AGAR

Cat. 1066

For the cultivation of anaerobic microorganisms from contaminated specimens

Formula in grams per liter

<i>Trypticasein soy broth</i>	10,00	<i>Peptone mixture</i>	5,00
<i>Dextrose</i>	5,00	<i>Yeast extract</i>	5,00
<i>Tris(Hydroxymethyl Aminomethane)</i>	3,00	<i>Hemin</i>	0,01
<i>L-Cystine</i>	0,40	<i>Bacteriological agar</i>	13,50

Final pH 7,6 ± 0,2 at 25°C

Preparation

Suspend 41,9 grams of the medium in one liter of distilled water. Mix and heat with frequent agitation and boil for one minute. Sterilize in autoclave at 121°C (15 lbs. sp.) for 15 minutes. Once sterilized cool to 45°C -50°C and add, if desired, 5% of defibrinated blood.

Uses

Because of its superior nutritive properties and its low oxidation-reduction potential, Schaedler Agar can easily support the growth of anaerobes from the intestinal and digestive tracts and other organ sites without the interference of the accompanying aerobic flora. In normal conditions, the multiplication of anaerobes is diminished by the rapid increase of *enterococci*, *E. coli*, *Enterobacter*, and other facultative bacteria of the intestine.

Methodology

It is recommended to consult methods for the cultivation of anaerobic organisms in food analysis.

Suspend a determined amount of the sample in a known volume of physiological saline. Take a small aliquot and make serial dilutions. With a calibrated loop inoculate duplicate plates previously dried and incubate at the appropriate time and temperature. Select for enumeration those plates, which contain 30 to 100 colonies.

For enumeration of *Streptococcus fecalis*, the aerobe and facultative anaerobe which is an indicator of contamination (with feces), in dehydrated and frozen food and for the detection of *Clostridium welchii*, Schaedler Agar can be used in the following manner:

Inoculate the food sample (frozen, precooked) in suspension, by streaking. Incubate aerobically at 25°C and at 35°C for 24 to 48 hours, and count *S. fecalis*.

If testing precooked meat, inoculate also the base medium (with added neomycin) to investigate the presence and number of *Clostridium welchii*. Incubate anaerobically.

Although thioglycollate is widely used to lower the oxidation-reduction potential favoring the development of anaerobes, it has been proven that it is an inhibitor of other organisms. In this case the medium should include cystine, which together with glucose, acts as a reducing agent, with the additional advantage that cystine inhibits the development of *Escherichia coli* in vitro.

Schaedler used the basic medium adding to it selective substances for the isolation and recovery of *lactobacilli*, *streptococci*, *clostridia*, *Bacteroides*, and *Flavobacterium* from feces and contents of the intestinal tract.

For each litre of the base add the following:

1. For anaerobic lactobacilli and streptococci.
Sodium Chloride 10,0 g.
Neomycin 0,002 g.
Incubate anaerobically at 35°C for a minimum of 48 hours.
2. For Bacteroides and clostridia.
Placenta powder (Nutritional Biochemical Corp. Cleveland, OH) 2,0 g.
Neomycin 0,002 g.
Incubate anaerobically at 35°C.
3. For Flavobacterium.
Tyrotricine in ethanol at 0.5%..... 7,0 ml

Bibliography

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Microbiological Test

Microorganisms	Growth
<i>Bacteroides fragilis</i> ATCC 25285	Good
<i>Clostridium butyrium</i> ATCC 9690	Good
<i>Clostridium perfringens</i> ATCC 13124	Good
<i>Streptococcus pyogenes</i> ATCC 19615	Good

SCHAEDLER BROTH

Cat. 1218

For the cultivation of anaerobes present in clinical samples and food

Formula in grams per liter

<i>Trypticasein Soy Broth</i>	10,00	<i>Dextrose</i>	5,00
<i>Yeast Extract</i>	5,00	<i>Tris (Hydroxymethyl aminomethane)</i>	3,00
<i>asein Peptone</i>	2,50	<i>Meat Peptone</i>	2,50
<i>L-Cystine</i>	0,40	<i>Hemin</i>	0,01

Final pH 7,6 ± 0,2 at 25°C

Preparation

Dissolve 28,4 grams of the medium in one liter of distilled water. Mix well. Allow to stand for 10-15 minutes. Heat with frequent gentle agitation and boil for one minute. Sterilize at 121°C (15 lbs sp) for 15 minutes.

Uses

Schaedler Broth is a liquid medium rich in nutrients, like that of Schaedler Agar but lacking agar. A large number of pathogenic anaerobic organisms involved in diverse diseases affecting humans as well as animals grow abundantly in this medium.

Schaedler Broth can be used advantageously over other liquid media for primary isolation of anaerobes, for blood cultures and other clinical materials. It is useful for the determination of minimum inhibitory concentration (MIC) of antimicrobials used in sensitivity tests. The solid medium is not used to perform sensitivity tests because there is no effective agreement between the concentration of the drug and the diameters of the zones of inhibition that are observed when the solid medium is used.

To determine the MIC in this medium, Fass and collaborators described a simple method that does not require an anaerobic atmosphere. They recommend placing in the bottom of the test tube a glass bead of 6 mm. in diameter before sterilizing in an autoclave. The bacterial growth is observed after 18 to 24 hours of incubation at 35°C. In order to know if the Schaedler Broth that has been stored has deteriorated or oxidized, add 0.01 grams of resazurin for each 100 ml. of the medium. To cultivate anaerobic *cocci*, the authors recommend adding 1 ml. of inactivated horse serum for every 100 ml. of broth.

Bibliography

- Fass R.J. Prior R.B. and Rotille C. A. 1975
Antimicrobial Agents Chemother. 8, 444-452.
Rotille C.A. and Col. 1075 Antimicrob. Agents Chemother. 7, 311-315.
Isenberg HD. (ed) 1992. Clinical microbiology procedures handbook. American Society for Microbiology, Washington, DC.
Atlas RM. 1993 Handbook of microbiological media, p. 794-795
CRC Press, Boca Raton, FL.

Microbiological Test

Microorganisms	Growth
<i>Bacteroides fragilis</i> ATCC 25285	Satisfactory
<i>Clostridium butyrium</i> ATCC 9690	Satisfactory
<i>Clostridium perfringens</i> ATCC 13124	Satisfactory
<i>Streptococcus pyogenes</i> ATCC 19615	Satisfactory

SELENITE CYSTINE BROTH

Cat. 1220

For the selective enrichment of *Salmonella* and some other *Shigella* strains

Formula in grams per liter

<i>Sodium phosphate</i>	10,00	<i>Peptone mixture</i>	5,00
<i>Lactose</i>	4,00	<i>Sodium Selenite</i>	4,00
<i>L-Cystine</i>	0,01		

Final pH 7,0 ± 0,2 at 25°C

Preparation

Suspend 23 grams of the medium in one liter of distilled water. Mix well and heat slowly until the medium is dissolved. Dispense in screw-capped test tubes sterilize under flowing steam for 5 minutes. **Do not autoclave.** The color of medium should be beige to pale pink.

Uses

In 1953, North and Bartram modified an enriched medium prepared by Leifson in 1936 by adding the amino acid cystine. This amino acid establishes a redox potential that seems to be very good for enrichment and recovery of *Salmonella* and some strains of *Shigella*, present in limited numbers in feces, diverse foods, and other products of sanitary concern. Selenite Cystine Broth is used particularly to limit the loss of sensitivity that affects other enrichment media especially in food products with a high content of organic material, for example, foods of egg and egg powder.

Selenite Cystine Broth inhibits the early multiplication of bacteria such as coliforms, but allows the *salmonellas* to grow with ease. Nevertheless, after 18 hours of incubation, the commensal microorganisms rapidly increase and begin to impede the isolation of *salmonellas*, so that it is necessary to restreak or subculture before the elapse of this critical time. These inoculations to differential solid media should be performed at the end of 8-12 hours of incubation.

Follow the usual methods used in the microbiological analysis of food.

Bibliography

Leifson E. (1936) Am. J. Hyg 24: 423-432
 American Public Health Association (1976) Compendium of Methods for the Microbiological Examination of Foods.
 Fricker CR. (1987) J. Appl. Bact. 63: 99-116

Microbiological Test

Microorganisms	Growth
<i>Escherichia coli</i> ATCC 25922	Increase no
<i>Salmonella pullorum</i> ATCC 9120	Satisfactory
<i>Salmonella choleraesuis</i> ATCC 12011	Satisfactory
<i>Salmonella typhi</i> ATCC 6539	Satisfactory

SELLERS AGAR

Cat. 1065

Differential medium used in studies of Gram negative non-fermenting bacilli

Formula in grams per liter

Gelatin Peptone	20,00	Sodium Chloride	2,00
Dipotassium Phosphate.....	1,00	Magnesium Sulfate	1,50
Sodium Nitrate.....	1,00	Yeast Extract.....	1,00
L-Arginine	1,00	Sodium Nitrite.....	0,35
Bromthymol Blue.....	0,04	Phenol Red	0,008
Bacteriological Agar	13,50	D-mannitol.....	2,00

Final pH 6,7 ± 0,2 at 25°C

Preparation

Suspend 43,4 grams of the medium in one liter of water. Mix well. Heat with frequent agitation and boil for one minute. Dispense into test tubes and sterilize at 121° C (15 lbs psi) for 10 minutes. Cool the tubes in a slanted position with a slant length of 7-7,5 cm and a butt depth of 3,5-cm. **Important: Immediately before inoculation, add 0,15 ml or 2 drops of 50% aqueous solution of dextrose, allowing it to run down the side of the tube opposite to the slant.**

Uses

Sellers Agar is inoculated by stabbing with a needle to the base of the tube and streaking the slant. Incubate at 35°C for 24 hours. It is a very useful medium to identify and differentiate *Pseudomonas aeruginosa*, *Herellea vaginicola*, *Mima polymorpha* and *Alcaligenes fecalis*. To aid in the identification of the non-fermenters, other media such as OF Basal Medium, Indol Nitrate Medium, etc. should be used. *Mima* and *Herellea* (*Acinetobacter*

calcoaceticus) morphologically resemble *Neisseria* and frequently are erroneously reported as causes of gonococcal urethritis and meningococcal (resistant to penicillin) meningitis.

The differentiation is based on the detection of fluorescence, glucose oxidation, production of nitrogen gas and pH changes. Under UV light only the *pseudomonas* exhibit fluorescence, which is stimulated by magnesium and mannitol in the medium. At times it is necessary to hold the tubes 2 days for *Pseudomonas* to produce a typical alkaline (blue color) reaction in the medium. After incubation, check for oxidation of glucose by the appearance of a yellow band, which can disappear after 24 hours.

Bibliography

Sellers J. Bact. 87: 46, 1964 Lennette E.H., Spaulding H.E. and Truant P.J. Manual of Clinical Microbiology, 2nd Ed. 1974.

Typical Reactions				
MICROORGANISM	<i>PSEUDOMONAS</i>	<i>MIMA</i> *	<i>HERELLEA</i> **	<i>A. FECALIS AND VIBRIO</i>
Colour of Slant	Green	Blue	Blue	Blue
Colour of Butt	Blue or no change	No change	No change	Blue or no change
Colour of Band	Blue at times	Absent	Yellow	Absent
Fluorescence on Slant	Yellow Green	No	No	No
Nitrogen gas	Yes	No	No	No

* *A. calcoaceticus* var. *Lwoffii*

** *A. calcoaceticus* var. *Anitratius*

Microbiological Test

Microorganisms	Growth	Slide	Base	Strip	Fluorescence
<i>Acinetobacter calcoaceticus</i> ATCC 19606	Good	Blue	Green	Yellow	-
<i>Acinetobacter lwoffii</i> ATCC 9957	Good	Blue	Blue	-	-
<i>Alcaligenes faecalis</i> ATCC 8750	Good	Blue-green	Blue-green	-	-
<i>Pseudomonas aeruginosa</i> ATCC 27853	Good	Blue-green	Blue-green	Blue	+

SIM MEDIUM

Cat. 1514

Used for the identification and differentiation of enterobacteria

Formula in grams per liter

Casein Peptone	20,00	Meat Peptone	6,10
Ferric Ammonium Sulfate.....	0,20	Sodium Thiosulfate.....	0,20
Bacteriological Agar.....	3,50		

Final pH 7,3 ± 0,2 at 25°C

Preparation

Suspend 30 grams of the medium in one liter of distilled water. Leave to soak for 5 to 10 minutes. Mix well until a uniform suspension is obtained, heat agitating constantly and boil for one minute until completely dissolved. Dispense and sterilize by autoclaving at 121°C (15 lbs sp) for 15 minutes.

Uses

Inoculate the pure culture by stabbing to a depth of 3/4 of the tube. Incubate at 35° C for 18 to 24 hours and read the results. Darkening indicates the production of H₂S. Growth only along the inoculation line indicates non-motility. The mobility is indicated by a diffuse turbidity away from the line of inoculation. Production of indol by adding Ehrlich or

Kovacs reagents gives a purple-red coloration to the reagents. Alternatively, a strip of filter paper impregnated with an oxalic acid solution placed in the top of the tube (above the medium) can be used for the detection of indol (red color).

Bibliography

S.A.B. Manual of Microbiological Mc. Graw-Hill, Book Co. New York, 1957.
Greene, Bilum de Cora, Fairchail, Kaplan, Landau and Sharp. J. Bact. 63:347, 1951.
Harrigan WF. And MacCarice ME (1966) Laboratory Methods in Microbiology Academic Press., 53.

ORGANISM	H ₂ S	INDOL	MOTILITY
<i>Salmonella typhi</i>	+ or -	-	+
<i>Salmonella</i>	+ or -	-	+
<i>Shigella</i>	-	+ or -	-
<i>E. coli</i>	-	+	+ or -
<i>Klebsiella</i>	-	+ or -	-
<i>Enterobacter</i>	-	-	+
<i>Citrobacter</i>	+	-	+

Microbiological Test

Microorganisms	Growth	H ₂ S	Mobility	Indol
<i>Escherichia coli</i> ATCC 25922	Good	-	+	+
<i>Salmonella typhimurium</i> ATCC 14028	Good	+	+	-
<i>Shigella flexneri</i> ATCC 12022	Good	-	-	-

SIMMONS CITRATE AGAR

Cat. 1014

For the determination of citrate utilization by enterobacteria.

Formula in grams per liter

<i>Ammonium Dihydrogen Phosphate</i>	1,00	<i>Dipotassium Phosphate</i>	1,00
<i>Sodium Chloride</i>	5,00	<i>Sodium Citrate</i>	2,00
<i>Magnesium Sulfate</i>	0,20	<i>Bacteriological Agar</i>	15,00
<i>Bromthymol Blue</i>	0,08		

Final pH 6,9 ± 0,2 at 25°C

Preparation

Suspend 24,3 grams of the medium in one liter of distilled water. Mix well and heat with frequent agitation until completely dissolved. Dispense in tubes and sterilize in the autoclave at 121°C (15 lbs sp.) for 15 minutes. Cool the tubes in a slanted position so that the base is short (1-1,5 cm. deep). Alternatively, the media can be poured into petri plates.

Uses

Simmons Citrate Agar is used to differentiate enteric Gram-negative bacilli on the basis of sodium citrate utilization as a source of carbon and inorganic ammonium salt utilization as a source of nitrogen. It is recommended for the differentiation of coliforms isolated from water. It is used in the same manner as Koser Citrate Broth for the utilization of citrate as one of the IMVIC reactions. It can be poured into plates or dispensed in tubes with long

slants. The surface of the slant is inoculated and the base stabbed. The tubes are incubated at 35-37°C for 4 days. If good results are not obtained, as in the case of some *Providencia* strains, incubate for 7 days. Only those organisms capable of utilizing citrate as a source of carbon grow on the slant and produce a color change from green to blue (alkaline).

This medium can be used especially for the differentiation of enteric organisms as follows:

Bibliography

Simmons. J. Inf. Dis. 39:209, 1926. Standard Methods for the Examination of Water and Wastewater. Eleventh Edition. APHA Inc. New York, 1960. Edwards & Ewing. Enterobacteriaceae. USPHS. Publications 743, Washington, 1972. Torregrosa and Ortiz, Pediatrics 59:35, 1961.

NEGATIVE	POSITIVE	
<i>Escherichia</i>	<i>Arizona</i>	<i>Enterobacter</i>
<i>Shigella</i>	<i>Citrobacter</i>	<i>Klebsiella</i>
<i>S. typhi</i>	<i>Salmonella paratyphi B</i>	<i>Serratia</i>
<i>S. paratyphi A</i>	<i>S. Typhimurium</i>	

Microbiological Test

Microorganisms	Growth	Medium colour
<i>Enterobacter aerogenes</i> ATCC 13048	Good	Blue
<i>Escherichia coli</i> ATCC 25922	Inhibited	Green
<i>Salmonella enteritidis</i> ATCC 13076	Good	Blue
<i>Shigella dysenteriae</i> ATCC 13313	Inhibited	Green
<i>Salmonella typhimurium</i> ATCC 14028	Good	Blue
<i>Salmonella typhi</i> ATCC 19430	Good	Green

SLANETZ - BARTLEY MEDIUM

(ISO 7899-2)

Cat. 1109

For the detection and count of intestinal Enterococcus by the membrane filtration technique.

Formula in grams per liter

Tryptose	20,00	Yeast Extract	5,00
Disodium Phosphate	4,00	Glucose Anhydrous	2,00
Sodium Azide	0,40	Tetrazolium Chloride	0,10
Bacteriological Agar	10,00		

Final pH 7,2 ± 0,1 at 25°C

Preparation

Suspend 42 grams of the medium in one litre of distilled water, dissolve with frequent agitation until boiling and completely dissolved **DO NOT OVERHEAT. DO NOT AUTOCLAVE.** Dispense into Petri plates and leave it to solidify

Uses

This medium is very selective for *streptococci*. When incubated at elevated temperatures (44-45°C), all red or brown colonies are confirmed as fecal streps (Taylor and Burman, 1964 and Mead, 1966).

Burkwall and Hartman demonstrated that the addition of 0,5 ml. of Tween 80 and 20 ml. of a 10% solution of sodium carbonate or bicarbonate to each litre of medium was valuable when investigating *streptococci* in frozen foods.

The British Ministry of Health (1969) in its "Report 71" recommended this medium for the enumeration of fecal streptococci in water systems. Water is filtered through a membrane which is then placed on the surface of a plate of Slanetz and Bartley Medium. The plate is incubated at 37°C for 4 hours and then at 44-45°C for 44 hours. The

membrane is examined with a magnifying lens under good light and all red or brown colonies are counted as fecal streps.

Food samples can be examined as suggested by the Nordic Committee of Food Analysis (1968). The samples are homogenized and diluted in a physiological saline solution and inoculated to yield 15-150 colonies per plate. The inoculum is spread uniformly on the surface of the plate by a sterile glass rod. The plates are inverted and incubated at 47°C for 48 hours. After incubation typical colonies (pink to dark red, with a thin white edge) are counted.

Bibliography

Slanetz L.W. and Bartley C.H. 1957. J. Bact. 74; 591-595.
 Nordic Committee of Food analysis 1968 Leaflet 68.
 Department of Health and Social Security report 71 1982.
 The Bacteriological examination of drinking water supplies, Hmbo, London.

Microbiological Test

Microorganisms	Growth	Red colonies
<i>Streptococcus pyogenes</i> ATCC 12344	Moderate	-
<i>Streptococcus agalactiae</i> ATCC 13813	Null/light	-
<i>Streptococcus faecalis</i> ATCC 11700	Satisfactory	+
<i>Streptococcus faecalis</i> ATCC 19433	Satisfactory	+
<i>Staphylococcus aureus</i> ATCC 25923	Null	
<i>Escherichia coli</i> ATCC 25922	Null	

SODIUM SELENITE BROTH

Cat. 1222

For the selective isolation of *Salmonella*.

Formula in grams per liter

Peptone Mixture.....	5,00	Lactose.....	4,00
Sodium Phosphate.....	10,00	Sodium Selenite.....	4,00

Final pH 7,0 ± 0,2 at 25°C

Preparation

Suspend 23 grams of the medium in one liter of distilled water. Mix well and heat gently until dissolved. Dispense and sterilize by exposing the medium to flowing steam for 5 minutes. Excessive heating is detrimental. **DO NOT STERILIZE IN AUTOCLAVE.** If the broth is to be used immediately, sterilization is unnecessary. Broth which has been tubed and steamed may be kept for months under refrigeration, with precautions to prevent evaporation.

Uses

Sodium Selenite Broth can be made more selective for the isolation of *Salmonella* in meat products when it is incubated for 16 to 18 hours at 43°C instead of 37°C. It is recommended for the transport of specimens of *Vibrio cholera* because these organisms can survive 2 to 5 days

in the Sodium Selenite Broth. If the pH is adjusted to 7,8 by means of the addition of sodium carbonate, the vibrio survive 8 to 10 days at temperatures between 22°C and 25°C.

Bibliography

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Harvey and Phillips J. Hyg. 59:93, 1961.
Felsenfeld, Waters, and Ishihara. Illinois Branch Meeting. Soc. Exper. Biol. and Med., 1950.

Microbiological Test

Microorganisms	Growth
<i>Escherichia coli</i> ATCC 25922	Partially inhibited
<i>Salmonella choleraesuis</i> ATCC 12011	Satisfactory
<i>Salmonella typhi</i> ATCC 6539	Satisfactory
<i>Salmonella typhimurium</i> ATCC 14028	Satisfactory

SPS AGAR

Cat. 1082

For the isolation of *Clostridium perfringens* from foods

Formula in grams per liter

Casein Peptone	15,50	Yeast Extract	10,00
Sodium Sulfite.....	0,30	Sulfadiazine	0,12
Ferric Citrate	0,50	Polymixin B.....	0,01
Bacteriological Agar.....	13,00		

Final pH 7,0 ± 0,2 at 25°C

Preparation

Suspend 40 grams of the medium in one litre of distilled water. Mix well. Heat with frequent agitation and boil for one minute. Dispense and sterilize at 118°C (12 lbs. sp.) for 15 minutes. Cool to 45-50°C.

Uses

SPS Agar is a moderately selective medium containing antimicrobial agents to inhibit undesirable species. *Clostridium perfringens* reduces the sulfite in the formula and produces black colonies.

The medium is a modification of the Wilson-Blair and the more recent Mossel formula for recovery of clostridia with Miller-Prickett tubes. SPS Agar eliminates the need for these tubes by the incorporation of sulfadiazine.

The authors isolated *C. perfringens* from dried meats and frozen pastries. Very few microorganisms other than *C. perfringens* grow on SPS Agar but can form small black colonies.

Material samples are prepared in an homogenizer or other equipment and serial dilutions are plated in SPS Agar

previously cooled to 45-50°C. Incubate anaerobically (The authors used a mixture of 90% nitrogen and 10% CO₂).

Serial dilutions in tubed media can also be made and incubated aerobically at 35-37°C for 24 hours.

Because other organisms can grow on this medium, perform a Gram stain and look for spores. Many common microorganisms are totally or partially inhibited, but if they develop, generally do not form black colonies, do not form spores, do not reduce nitrate and are non-motile Gram-positive vegetative bacilli.

The lack of motility and the capacity to reduce nitrate can be determined on Indol Nitrite Medium with 2 g/l. of added agar.

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Mossel. J.SCI. Agr. 10: 662. 1959.
Mossel de Bruin Van Diepen, Vendrig y Zoutwelle J. Applied Bact, 19: 142. 1956.

Microbiological Test

Microorganisms	Growth	Colony colour
<i>Clostridium perfringens</i> ATCC 12919	Satisfactory	Black
<i>Clostridium sporogenes</i> ATCC 11437	Moderate	Black
<i>Escherichia coli</i> ATCC 25922	Inhibited	---
<i>Staphylococcus aureus</i> ATCC 6538	Moderate-Inhibited	White

STANDARD METHODS AGAR (PLATE COUNT AGAR)

Cat. 1056

For total microbial plate count in milk and other materials of sanitary significance. (APHA* Formula)

Formula in grams per liter

Casein Peptone.....	5,00	Yeast Extract.....	2,50
Dextrose	1,00	Bacteriological Agar.....	15,00

Final pH 7,0 ± 0,2 at 25°C

Preparation

Suspend 23,5 grams of the medium in one liter of distilled water. Heat agitating frequently until boiling and completely dissolved. Dispense into appropriate containers and sterilize at 121 °C (15 lbs. sp) for 15 minutes.

Uses

Standard Methods Agar is recommended by APHA when enumerating bacteria of sanitary interest, which are indicators of contamination or microbial load in foods. In general, 1 ml. of the appropriate dilution is added to the sterile agar at a temperature of 44-45°C, mixed gently and poured into sterile Petri dishes.

Incubate the Petri dishes at a specified temperature and time period and count the developed colonies. Consult the specific texts of APHA for the particular sample applications.

Bibliography

Standard Methods for the Examination of Dairy Products, 13th Ed. APHA, 1972. American Public Health Association. Recommended Methods for the Microbiological Examination of Foods, APHA Inc. New York, 1958. Standard Methods for the Examination of Water and Wastewater, APHA Inc. New York, 1960.

Microbiological Test

Microorganisms	Growth
<i>Escherichia coli</i> ATCC 25922	Satisfactory
<i>Escherichia coli</i> ATCC 13762	Satisfactory
<i>Staphylococcus aureus</i> ATCC 25923	Satisfactory
<i>Staphylococcus epidermidis</i> ATCC 12228	Satisfactory

STANDARDS METHODS AGAR WITH POWDERED MILK

Cat. 1033

For use in bacterial plate counts of microorganisms from milk and dairy derivatives (Formula APHA*)

Formula in grams per liter

<i>Casein Peptone</i>	5,00	<i>Yeast Extract</i>	2,50
<i>Dextrose</i>	1,00	<i>Skimmed milk powder</i>	1,00
<i>Bacteriological Agar</i>	15,00		

Final pH 7,0 ± 0,2 at 25°C

Preparation

Suspend 24,5 grams of the medium in one liter of distilled water. Boil until it is completely dissolved and sterilize at 121°C (15 lbs. sp.) for 15 minutes. Cool to 45°C-50°C.

* APHA: American Public Health Association Inc.

Uses

This medium is used with the same techniques as Standard Method Agar.

Bibliography

R.C. MARSHALL (1.993) Standard Methods for the Microbiological examination of dairy products, 16th Ed. (American Public Health Association, Washington, D.C.). England and Wales. The Dairy Products (Hygiene) Regulations 1995 Statutory Instrument No. 1086. London: HMSO, 1995. British Standards Institution. BS 4285 Microbiological examination for dairy purposes. Section 2.1 Enumeration of microorganisms by poured plate technique for colony count. London: BSI, 1984.

Microbiological Test

Microorganisms	Growth
<i>Escherichia coli</i> ATCC 25922	Satisfactory
<i>Escherichia coli</i> ATCC 13762	Satisfactory
<i>Staphylococcus aureus</i> ATCC 25923	Satisfactory
<i>Staphylococcus epidermidis</i> ATCC 12228	Satisfactory

STAPHYLOCOCCUS AGAR N° 110

Cat. 1032

Used for the isolation of *Staphylococcus*

Formula in grams per liter

Sodium Chloride.....	75,00	Gelatin.....	30,00
Casein Peptone.....	10,00	D-Mannitol.....	10,00
Dipotassium Phosphate.....	5,00	Yeast Extract.....	2,50
Lactose.....	2,00	Bacteriological Agar.....	15,00

Final pH 7,0 ± 0,2 at 25°C

Preparation

Suspend 149 grams of the medium in one liter of distilled water. Mix well. Heat with frequent agitation and boil for one minute. Dispense and sterilize in autoclave at 121°C (15 lbs. sp.) for 15 minutes.

Uses

Staphylococcus Agar No. 110 is used to isolate *staphylococci* from purulent processes, cases of pneumonia, meningitis, furunculosis, urethritis, vaginitis, etc. This medium is also used for isolating *staphylococci* which contaminate a wide variety of foods and produce food poisoning.

It is possible to enrich the media by adding 5% blood, which also produces good hemolytic reactions and formation of golden yellow colonial pigments. Mannitol fermentation is detected by adding a few drops of Bromothymol blue and looking for a yellow halo around

the colony. The plates can be flooded with 5 ml. of a saturated solution of ammonium sulfate, or better yet, with a drop of 20% sulfasalicylic acid and incubated for 12 minutes to observe the hydrolysis of the gelatin: clearing around the colony constitutes a positive hydrolysis (Stone Reaction).

Bibliography

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Mac Faddin, J.F. 1985 Media for isolation cultivation identification maintenance of medical bacteria, vol. 1 p. 722-726. Williams & Wilkins, Baltimore, MD.
Association of Official Analytical Chemists 1995. Bacteriological analytical manual, 8th ed. AOAC International, Gaithersburg, MD.

Microbiological Test

Microorganisms	Growth	Pigment production
<i>Bacillus subtilis</i> ATCC 6633	Satisfactory	-
<i>Escherichia coli</i> ATCC 25922	Inhibited	-
<i>Staphylococcus aureus</i> ATCC 25923	Satisfactory	+
<i>Staphylococcus aureus</i> ATCC 6538	Satisfactory	+
<i>Staphylococcus epidermidis</i> ATCC 12228	Satisfactory	-

STREPTOCOCCUS SELECTIVE AGAR (STREPTOSEL AGAR)

Cat. 1070

For the enrichment and isolation of *Streptococcus* from diverse clinical materials and of highly contaminated products of sanitary importance.

Formula in grams per liter

Casein peptone	15,00	Soy peptone.....	5,00
Sodium chloride.....	4,00	Sodium citrate.....	1,00
L-Cystine.....	0,20	Sodium sulfite.....	0,20
Dextrose.....	5,00	Sodium azide.....	0,20
Crystal violet	0,0002	Bacteriological agar.....	12,00

Final pH 7,4 ± 0,2 at 25°C

Preparation

Suspend 42,6 grams of the medium in one litre of distilled water. Mix well and leave to soak 10-15 minutes to allow the agar particles to hydrate properly. Heat agitating frequently and boil for 1 minute. Sterilize in an autoclave at (12 lbs. of pressure) 118°C for 15 minutes. Avoid overheating. Cool to 45-50°C and pour into Petri dishes. Invert the solidified agar plates to avoid excess water condensation.

Uses

Basically this medium is the same as *Streptococcus* Selective Broth (Streptosel Broth) to which has been added 1,5% agar.

It has the same use as the broth previously mentioned. Adding 0,5% of sterile defibrinated sheep or rabbit blood notably increases its nutritional power and hemolytic studies can be conducted. These conditions yield good results in the isolation and identification of different groups of *Streptococcus* such as the alpha and beta-hemolytic, and the non-hemolytic.

Bibliography

Washington, D.C. 2nd Ed., 1974.

Microbiological Test

Microorganisms	Growth
<i>Escherichia coli</i> ATCC 25922	Inhibited
<i>Streptococcus faecalis</i> ATCC 19433	Satisfactory
<i>Streptococcus faecium</i> ATCC 27270	Satisfactory

STREPTOCOCCUS SELECTIVE BROTH (STREPTOSEL BROTH)

Cat. 1204

For the selective growth of streptococci from clinical samples.

Formula in grams per liter

Casein Peptone.....	15,00	Soy Peptone.....	5,00
Sodium Chloride.....	4,00	Sodium Citrate.....	1,00
L-Cystine.....	0,20	Sodium Sulfite.....	0,20
Dextrose.....	5,00	Sodium Azide.....	0,20
Crystal Violet.....	0,0002		

Final pH 7,4 ± 0,2 at 25°C

Preparation

Suspend 30,6 grams of the medium in a litre of distilled water. Heat with frequent agitation and boil for one minute. Dispense in 10 ml. amounts into screw-capped tubes and sterilize in the autoclave at 118°C (12 lbs. sp.) for 15 minutes. DO NOT OVERHEAT, or the medium will become too inhibitory.

Uses

Clinical material, obtained by a swab of the nasal passage or pharynx, is inoculated into this selective medium and the tubes are incubated at 35°C for 18-24 hours in a normal atmosphere. If one wants to streak Blood Agar and/or *Streptococcus* Selective Agar with 5% sheep or rabbit blood, incubate these plates in a 5-10% CO₂ atmosphere. CDC (Center for Disease Control, Atlanta, GA.) does not recommend the use of candle jars to generate CO₂. It is recommended to inoculate the Blood Agar plates by the pour plate method (in thick plates) or to inoculate the plates with a streak and make several stabs with the loop and incubate in a normal atmosphere.

Many organisms such as saprophytic *Neisseria*, *Staphylococcus*, *Haemophilus*, non-hemolytic *streptococci*, and a certain number of *enterobacteria* will not grow or only scarcely, in this medium. The growth of streptococci can be determined by the formation of a granular precipitate in the bottom of the tube, with the liquid above clear or slightly turbid. At this point, perform a Gram stain and restreak on Blood Agar to purify the strain.

It is convenient to place bacitracin and optochin discs in the area of heavy inoculum on the Blood Agar plate and

incubate for 18-24 hours at 35°C under the recommended conditions. It is important to remember that the discs are used for differentiation of *streptococci* and *pneumococci* and are not to be confused with antibiotic sensitivity discs of higher concentration.

Subculture the organism growing in the zone of inhibition from 10-18 mm. in diameter around the bacitracin disc into 2,5 ml. of the *Streptococcus* Selective Broth and incubate under the normal conditions. Perform a Gram stain and observe for formation of *coccal* chains. Perform the catalase and bile solubility tests on characteristic colonies taken from the Blood Agar Plate or from the growth obtained from the broth.

The presence of variable length chains of Gram-positive cocci inhibited by bacitracin in low concentration, catalase negative and insoluble in bile or bile salts, constitute a valid presumptive identification of Group A beta-hemolytic *streptococci*. The definitive identification of the *streptococcal* groups can be made by performing other biochemical tests such as esculin hydrolysis, pyruvate hydrolysis, etc. Also, serological typing, using Lancefield antisera methods, or easier or more conveniently, the techniques of coagglutination of Edwards and Larson can be performed.

Bibliography

Washington, D.C. 2nd Ed., 1974.

Microbiological Test

Microorganisms	Growth
<i>Escherichia coli</i> ATCC 25922	Inhibited
<i>Streptococcus faecalis</i> ATCC 19433	Satisfactory
<i>Streptococcus faecium</i> ATCC 27270	Satisfactory

STUART TRANSPORT MEDIUM

Cat. 1518

For transport and maintenance of all kind of samples.

Formula in grams per liter

Agar N° 2.....	3,00	Sodium Thioglycollate.....	1,00
Sodium Glycerophosphate.....	10,00	Methylene Blue.....	0,002
CaCl ₂	0,10		

Final pH 7,4 ± 0,2 at 25°C

Preparation

Suspend 14,1 grams of the medium in one litre of distilled water. Heat with frequent agitation and boil for one minute. Dispense in screw-capped tubes and sterilize in an autoclave at 121°C (15 lbs. sp.) for 15 minutes.

Uses

For the transport of all types of specimens. Stuart Transport Medium is a semisolid medium used in the transport and preservation of specimens for the cultivation of diverse organisms such as *gonococci*, *streptococci*, *enterobacteria*, etc. It is essentially non-nutritive and contains sodium thioglycollate to retard oxidation.

The original formula was developed for the preservation and transport of *Neisseria gonorrhoeae* and *Trichomonas vaginalis*. Later it was demonstrated that the medium could be used in the handling and cultivation of *Haemophilus influenzae*, alpha and beta hemolytic

streptococci, *pneumococci*, and *enterobacteria* which can survive at an ambient temperature for 6 to 8 weeks. However, it is recommended to send the sample to the laboratory as soon as possible. For the transport of delicate microorganisms it is advised to use cotton swabs impregnated with charcoal which are commercially available.

Bibliography

Beakley, J. W. 1975. The toxicity of wooden applicator sticks for *Neisseria gonorrhoeae*. Pub. Hlth. Lab. 15 (1), 11:16.
 Stuart, R.D. Toshach, Sh. R., and Patsula, M. T.: 1954. The problem of transport of specimen for cultura of gonococci. Canad. J. Publ. Hlth. 45(2), 13:83.
 Stuart, R. D. 1954. Transport medium for specimens in Public Health Bacteriology. Pub. Hlth. Rep. Wash. 74(5), 431:438.

Microbiological Test

Microorganisms	Recovery
<i>Bordetella pertusis</i> ATCC 9340	Good
<i>Haemophilus influenzae</i> ATCC 19418	Good
<i>Neisseria gonorrhoeae</i> ATCC 19424	Good
<i>Neisseria meningitidis</i> ATCC 13090	Good
<i>Shigella flexneri</i> ATCC 12022	Good
<i>Streptococcus pneumoniae</i> ATCC 6301	Good

TCBS AGAR

Cat. 1074

For the selective isolation of *vibrium*.

Formula in grams per liter

Yeast Extract.....	5,00	Casein Peptone	5,00
Meat Peptone.....	5,00	Sodium Citrate	10,00
Sodium Thiosulfate	10,00	Ox Bile	5,00
Sodium Chololate.....	3,00	Sucrose	20,00
Sodium Chloride.....	10,00	Ferric Citrate	1,00
Thymol Blue	0,04	Bromthymol Blue.....	0,04
Bacteriological Agar	14,00		

Final pH 8,6 ± 0,2 at 25°C

Preparation

Suspend 88 grams of the medium in one litre of distilled water. Mix from 10 to 15 minute. Heat with frequent agitation and boil for 1 minute until completely dissolved. Cool to 45-50°C and pour in Petri dishes. Do not sterilize in an autoclave.

Uses

TCBS Agar is widely used to isolate and cultivate diverse species of the genus *Vibrio* that can cause cholera, cholera diarrhea or food poisoning from contaminated foods. The last 2 conditions especially can be caused by ingesting raw or partially processed fish or seafood containing *Vibrio parahaemolyticus*.

TCBS Agar is highly inhibitory to the *enterobacteria*, including *coliforms* and *Proteus*. *Enterococci* are also

greatly inhibited and allows the proliferation of *vibrio*, such as *V. cholerae* and *V. alginolyticus*.

The suspect material (feces, vomit, rectal swabs, fish, and other food), is heavily inoculated on the surface of the plate, incubated at 35°C for 18 to 24 hours.

Almost all *vibrios* ferment sucrose and yield yellowish colonies from the production of acid. Some types of *Proteus* (fermenters of sucrose) can form yellowish colonies similar to those of *vibrios*.

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Felsemfeld, Bull World Othg. 34:161, 1966. Kobayashi. T. Enomoto S. Sakasaki, R. Y.
Kwajaras, S., Jap. J. Bact. 18 387 291, 1963.

MICROORGANISMS	CHARACTERISTICS OF THE COLONIES
<i>Vibrio cholerae</i> and its biotype Tor	Large, smooth, elevated, yellow or pale yellowish brown. 2 to 3 mm. in diameter. Yellow agar.
<i>V. parahaemolyticus</i> (GROUP I)	Colourless with a green center. 3 to 4 mm. in diameter. No color change in agar.
<i>V. parahaemolyticus</i> (GROUP II)	Yellow or pale yellowish brown. 3 to 4 mm. in diameter. Yellow agar.
<i>V. alginolyticus</i>	Yellow, large.
<i>Enterobacteriaceae</i>	Scanty growth, punctiform, transparent. No color change in agar.
<i>Pseudomonas</i> , <i>Aeromonas</i>	Blue, small, punctiform.
<i>Enterococcus</i>	Scanty growth, punctiform. Yellow agar.

Microbiological Test

Microorganisms	Growth	Medium colour
<i>Vibrio cholerae</i> Inaba	Satisfactory	Yellow
<i>Vibrio cholerae</i> Ogawa	Satisfactory	Yellow
<i>Vibrio alginolyticus</i>	Moderate	Yellow
<i>Vibrio parahaemolyticus</i> ATCC 17802	Satisfactory	Blue
<i>Enterobacter cloacae</i> ATCC 13047	Light	Yellow
<i>Proteus mirabilis</i> ATCC 14273	Moderate	Light blue
<i>Escherichia coli</i> ATCC 25922	Negative	---
<i>Pseudomonas aeruginosa</i> ATCC 27853	Negative-light	Blue

TETRATHIONATE BROTH BASE

Cat. 1114

Used as a selective enrichment medium to isolate *Salmonella* from feces, urine and other materials.

Formula in grams per liter

Peptone mixture	5,00	Bile Salts	1,00
Calcium Carbonate	10,00	Sodium Thiosulfate	30,00

Final pH 8,4 ± 0,2 at 25°C

Preparation

Suspend 46 grams of the medium in one litre of distilled water. Mix well and heat to boiling. Cool and dispense by 10 ml in tubes continually swirling the flask to maintain homogeneity. Add 20 ml per litre of a iodine solution to the amount of medium to be used on the same day. Prepare the solution by dissolving 6 gr of iode and 5 gr of potassium iodiure in 20 ml of distilled water. Once the medium is prepared, store refrigerated.

Uses

Tetrathionate Broth Base is used as a selective enrichment for the cultivation of *Salmonella* species that may be present in small numbers and compete with intestinal flora. It is also used in processing fecal cultures for bacteria.

Inoculate each 10 ml. tube with 1-2 grams of the sample (feces, waste water, etc.) and incubate for 12-24 hours. Using this culture, streak onto selective plated media such as MacConkey Agar, Bismuth Sulfite Agar, Desoxycholate Agar, Brilliant Green Agar, XLD Agar or Hektoen Enteric Agar. The organisms which reduce the tetrathionate, such as *Salmonella*, proliferate in this medium. *Proteus* can also reduce tetrathionate and thus diminish the effectiveness of the medium. This negative situation can greatly minimized by adding 4 mg/l. novobiocin before adding the iodine solution.

Bibliography

American Public Health Association Recommended Methods for the Microbiological Examination of Foods, APHA, Inc. New York, 1958. American Public Health Association Standard Methods for the Examination of Dairy products. Eleventh Edition, APHA, Inc. New York, 1960.

Microbiological Test

Microorganisms	Growth
<i>Escherichia coli</i> ATCC 25922	Scarce-null
<i>Salmonella choleraesuis</i> ATCC 12011	Satisfactory
<i>Salmonella typhi</i> ATCC 6539	Satisfactory
<i>Salmonella typhimurium</i> ATCC 14028	Satisfactory

THIOGLYCOLLATE BROTH (NIH)

Cat. 1241

For sterility assays of biological and pharmaceutical products

Formula in grams per liter

<i>Casein Peptone</i>	15,00	<i>Yeast Extract</i>	5,00
<i>Dextrose</i>	5,00	<i>Sodium Chloride</i>	2,50
<i>Sodium Thioglycollate</i>	0,50	<i>L-Cystine</i>	0,50

Final pH 7,1 ± 0,2 at 25°C

Preparation

Suspend 28,5 grams of the medium in one liter of distilled water. Mix well. Heat with frequent agitation and boil until complete dissolution. Dispense in fermentation tubes or in adequate containers and sterilize in autoclave at 121°C (15 lbs. sp.) for 18 minutes.

Uses

This medium is used in detecting microorganisms in normally sterile materials.
Thioglycollate Broth is prepared according to the formula of the National Institute of Health (NIH) and the United States Pharmacopoeia (USP.).

Better results are obtained if the broth is used within a few days of preparation as the medium oxidizes rapidly. If kept longer, heat in a water bath to remove dissolved oxygen.

Bibliography

U.S. Pharmacopoeia XVI, 1960

Microbiological Test

Microorganisms	Growth
<i>Bacillus subtilis</i> ATCC 6633	Satisfactory
<i>Candida albicans</i> ATCC 10231	Satisfactory
<i>Clostridium sporogenes</i> ATCC 19404	Satisfactory
<i>Streptococcus pyogenes</i> ATCC 19615	Satisfactory
<i>Bacteroides fragilis</i> ATCC 25285	Satisfactory
<i>Escherichia coli</i> ATCC 25922	Satisfactory

THIOGLYCOLLATE FLUID MEDIUM (FTM)

Cat. 1508

Used as a culture medium in sterility tests.

Formula in grams per liter

Casein Peptone.....	15,00	L-Cystine.....	0,50
Dextrose.....	5,50	Yeast Extract.....	5,00
Sodium Chloride.....	2,50	Sodium Thioglycollate.....	0,50
Resazurin.....	0,001	Bacteriological Agar.....	0,75

Final pH 7,1 ± 0,2 at 25°C

Preparation

Suspend 29,5 grams of the medium in one liter of distilled water. Mix well until obtaining a uniform suspension. Heat with frequent agitation. Boil for 1-2 minutes or until completely dissolved. Dispense in 15x2 cm test tubes (15 ml in each tube. Sterilize for 15 to 18 minutes at 121°C (15 lbs. sp.). Cool before using, and store in the dark. Once prepared it can be used some time after preparation until it is 30% oxidized, which is indicated by a pink colour on the resazurine surface. If the oxidation is greater, reheat the medium only once, with steam or boiling water, cool and use.

Uses

This medium is used for detecting microorganisms in normally sterile materials, and also is accepted by the European Pharmacopoeia for sterility testing of pharmaceutical biologic products and devices. Sodium thioglycollate neutralizes the bacteriostatic effect of the compounds used as preservatives in pharmaceutical preparations, especially injectables.

When this medium oxidizes, indicated by the appearance of a rose color throughout the medium, do not use. The medium is satisfactory for use if the oxidized zone does

not exceed 30% of the liquid volume. In this case heat to a boil until the color disappears to expel the dissolved oxygen. Do not heat the medium more than one time.

With this medium it is not necessary to use a cap of sterile paraffin oil or incubate in special containers for anaerobes. The anaerobic organisms develop in the bottom of the tube; the microaerophiles in the middle of the medium and the aerobes in the top oxidized layer. It is recommended to incubate up to 8 days and check for growth at different intervals.

When the material in study contains other preservatives, use a sufficient amount of thioglycollate to dilute the inoculum beyond its bacteriostatic strength level.

Bibliography

Brewer. JAMA, 115, 1940. Uera. J. Bact. 47:59, 1944. Pittman. J. Bact. 51:19, 1946.
Kurtin A. J. Clin. Path. 30:229, 1958.
Baron, E.J. C.R. Peterson, S.M. Finegold 1994. Bailey and Scott's diagnostic Microbiology, 9th ed. MosBy-Year Book, Inc; St. Louis, M.O. The United States Pharmacopoeial Convention, 1995, 23th ed. P. 1686-1690.

Microbiological Test

Microorganisms	Growth
<i>Bacillus subtilis</i> ATCC 6633	Good
<i>Candida albicans</i> ATCC 10231	Good
<i>Neisseria meningitidis</i> ATCC 13090	Good
<i>Staphylococcus aureus</i> ATCC 6538P	Good
<i>Clostridium sporogenes</i> ATCC 11437	Good
<i>Streptococcus pyogenes</i> ATCC 19615	Good

THIOGLYCOLLATE MEDIUM WITHOUT INDICATOR Cat. 1516

For an abundant development of aerobic, anaerobic and facultative microorganisms.

Formula in grams per liter

Casein Peptone.....	17,00	Soy Peptone.....	3,00
Dextrose.....	6,00	Sodium Thioglycollate.....	0,50
Sodium Chloride.....	2,50	Bacteriological Agar.....	0,70
L-Cystine.....	0,25	Sodium Sulfite.....	0,10

Final pH 7,0 ± 0,2 at 25°C

Preparation

Suspend 30 grams of the medium in one litre of distilled water. Mix until a uniform suspension is obtained. Heat with frequent agitation and boil for one minute. Dispense in 20 x 150 mm. test tubes, filled half way, using 15 to 18 ml. Sterilize at 121° C (15 lbs. sp.) for 15 minutes.

The tightly capped test tubes should be stored in refrigeration. For optimal performance the tubes should be boiled and cooled at ambient temperature before use. The boiling restores the uniformly hazy appearance of the medium.

Uses

Thioglycollate Medium without Indicator is characterized by its ability to support growth from a minimal inoculum of a great variety of aerobes and anaerobes. Strict aerobes develop in the upper part, whereas anaerobes develop in the bottom of the medium tube.

Incorporating casein and soy peptones allows for the growth of aerobic microorganisms such as members of the genus *Brucella*. This medium supports the growth of strict anaerobes such as *S. acetobutyricum*, *Clostridium novyi*,

Actinomyces bovis, *Bacteroides*, *Lactobacillus*, and other bacteria. Pathogenic fungi frequently grow well in this medium.

The medium can be used with the addition of 10% serum for the cultivation of *Trichomonas vaginalis* and other microorganisms that utilize serum for added growth.

TM w/o Indicator is satisfactorily used as an enriched culture medium for several types of pathogenic specimens and as a transport medium.

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Kurtin A. J. Clin. Path. 30:229, 1958.
Baron, E.J. C.R. Peterson, S.M. Finegold 1994. Bailey and Scott's diagnostic Microbiology, 9th ed. MosBy-Year Book, Inc; St. Louis, M.O. The United States Pharmacopoeial Convention, 1995, 23th ed. P. 1686-1690.

Microbiological Test

Microorganisms	Growth
<i>Bacillus subtilis</i> ATCC 6633	Satisfactory
<i>Candida albicans</i> ATCC 10231	Satisfactory
<i>Streptococcus pyogenes</i> ATCC 19615	Satisfactory
<i>Bacteroides vulgatus</i> ATCC 8482	Moderate
<i>Neisseria meningitidis</i> ATCC 13090	Satisfactory

THIOGLYCOLLATE USP MEDIUM

Cat. 1533

For the cultivation of aerobic and anaerobic microorganisms and for sensitivity testing

Formula in grams per liter

<i>Yeast Extract</i>	5,00	<i>Bacteriological Peptone</i>	15,00
<i>Dextrose</i>	5,50	<i>Sodium Thioglycollate</i>	0,50
<i>Sodium Chloride</i>	2,50	<i>L-Cystine</i>	0,50
<i>Resazurin</i>	0,001	<i>Bacteriological Agar</i>	0,50

Final pH 7,1 ± 0,2 at 25°C

Preparation

Suspend 29,5 grams of the medium in one litre of distilled water. Mix well. Heat to boiling to completely dissolve the medium. Dispense and sterilize at 121°C (15 lbs. sp.) for 15 minutes. Cool to room temperature (25°C).

If the stored medium exhibits greater than 20% pink color (due to oxidation), the tubes should be reheated in a water bath to expel the oxygen. Do not reheat more than one time.

Uses

This medium is excellent for the cultivation of aerobic and anaerobic microorganisms without the need for an anaerobic system.

Sodium thioglycollate in the medium neutralizes the bacteriostatic effect produced by mercurial compounds used as preservatives in pharmaceutical solutions. It is necessary to establish the bacteriostatic activity of the product by the method described in the USP (1970) in

order to avoid false negative results. Thioglycollate USP medium is also recommended for the cultivation of *Clostridium*.

Prepared according to the U.S.A. Pharmacopoeia to perform sterility test.

Procedure

The medium is used in liquid form in test tubes or as a slanted solid with added agar (1,5%). The medium or slant agar tube can be inoculated directly and incubated at 35 to 37°C. The presence or absence of growth distinguishes the diverse groups like those indicated in the preceding chart.

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Vera. J. Bact. 47:59, 1944. King. Annals. N.Y. Acad. Sci. 98:615, 1962. Alvarez, A.J.: Med. Tech. 21:249, 1955. Vera and Petran. Bull. Natl. Assn. Clin. Lab. 5:90, 1964. Tarshis J. Lab. and Clin. Med., 54:630, 1959.

Microbiological Test

Microorganisms	Growth
<i>Bacillus subtilis</i> ATCC 6633	Good
<i>Candida albicans</i> ATCC 10231	Good
<i>Clostridium sporogenes</i> ATCC 11437	Good
<i>Streptococcus pyogenes</i> ATCC 19165	Good
<i>Bacteroides fragilis</i> ATCC 25285	Good
<i>Escherichia coli</i> ATCC 25922	Good

TODD-HEWITT BROTH

Cat. 1236

For the cultivation of beta-hemolytic streptococci for serological typing

Formula in grams per liter

<i>Beef Infusion</i>	3,10	<i>Bacteriological Peptone</i>	20,00
<i>Dextrose</i>	2,00	<i>Sodium Chloride</i>	2,00
<i>Disodium Phosphate</i>	0,40	<i>Sodium Carbonate</i>	2,50

Final pH 7,8 ± 0,2 at 25°C

Preparation

Suspend 30 grams of the medium in one liter of distilled water. Mix well. Heat with frequent agitation until completely dissolved. Dispense into appropriate containers and sterilize at 121°C (12 lbs. sp.) for 15 minutes.

Uses

Todd Hewitt Broth was originally developed for the production of *streptococcal* hemolysin. The broth was modified by Updyke and Nickle and is used preferentially for the cultivation of beta-hemolytic strains, especially for serological typing, from clinical specimens and for epidemiological studies.

Todd-Hewitt Broth is recommended as an enrichment medium for the growth of streptococcal cells in the identification of groups A and B by if staining this medium was used as an enrichment broth for group a streptococci in a comparison study of a rapid antigen test.

To prepare Todd Hewitt Agar, add 13-15 g/l. to the broth and sterilize as above.

Bibliography

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Isenberg H.D. (ed) 1992. Clinical Microbiology procedures handbook, American Society for Microbiology, Washington, D.C.
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Microbiological Test

Microorganisms	Growth
<i>Neisseria meningitidis</i> ATCC 13090	Satisfactory
<i>Streptococcus mitis</i> ATCC 9895	Satisfactory
<i>Streptococcus pneumoniae</i> ATCC 6303	Satisfactory
<i>Streptococcus pyogenes</i> ATCC 19615	Satisfactory

TOMATO JUICE AGAR

Cat. 1073

For the cultivation and enumeration of lactobacilli

Formula in grams per liter l

<i>Tomato Juice (dried)</i> 20,00	<i>Bacteriological Peptone</i> 10,00
<i>Peptone Milk</i> 10,00	<i>Bacteriological Agar</i> 15,00

Final pH 6,4 ± 0,2 at 25°C

Preparation

Suspend 55 grams of the medium in one litre of distilled water. Mix well. Heat with frequent agitation and boil for one minute. Dispense and sterilize at 121°C (15 lbs. sp.) for 15 minutes. Do not overheat.

Uses

On ordinary media lactobacilli show little or no growth. With tomato juice added the medium greatly improves the recovery of these organisms. The colonies are larger and more characteristic than on other media.

Tomato Juice Agar is recommended for enumeration in direct plating of lactobacilli in the saliva which can be an indication of the predisposition for caries. Adjusting the pH to 5.0 makes the medium more selective and increases colony size while inhibiting a large part of the accompanying bacteria in the saliva.

Bibliography

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 Davis GHG (1959) Lab. Prac., 8 161-167

Microbiological Test

Microorganisms	Growth
<i>Lactobacillus casei</i> ATCC 9595	Good
<i>Lactobacillus leich mannii</i> ATCC 4797	Good
<i>Lactobacillus</i> spp.	Good

TRIPLE SUGAR IRON AGAR (EUROPEAN PHARMACOPOEIA) Cat. 1046

For identification and the differentiation of enteric bacteria.

Formula in grams per liter

Peptone Mixture.....	20,00	Lactose.....	10,00
Sucrose.....	10,00	Sodium Chloride.....	5,00
Beef Extract.....	3,00	□ Yeast Extract.....	3,00
Dextrose.....	1,00	Ferrous Ammonium Citrate.....	0,30
Sodium Thiosulphate.....	0,30	□ Phenol Red.....	0,025
Bacteriological Agar.....	12,00		

Final pH 7,4 ± 0,2 at 25°C

Preparation

Suspend 64,6 grams of the medium in one liter of distilled water. Dissolve by heating and agitating frequently for one minute. Distribute in tubes and sterilize at 121° C (15 lbs.sp) for 15 minutes and cool in a slanted position, as to obtain butts of 1.5 – 2 cm depth.

Uses

Triple Sugar Iron Agar (TSI) may be used to differentiate enteric gram-negative bacteria on the basis of carbohydrate fermentation and H₂S production. It is used as an aid in the identification of pathogenic and saprophytic enterobacteria isolated from routine bacteriological analysis of material samples such as feces. This medium is used as a key to initiate the identification of enterobacteria in some FDA schemas.

The mode of action is similar to Kligler Iron Agar which contains two sugars. The addition of 1% sucrose in the TSI Agar allows for the recognition and exclusion of *Proteus*, *Hafnia* and *Providencia* do not ferment the lactose or only slowly but do ferment sucrose rapidly which excludes them from the *Salmonella-Shigella* group.

Bibliography

Standard Methods for the Examination of Dairy Products. APHA, 1972.
Food and Drug Administration. Bacteriological Analytical Manual, 1976.
Vanderzant, C. and D.F. Splitt stresser (ed) 1992. Compendium of methods for the microbiological examination of foods, 3rd ed. American Public Health Association, Washington D.C.

Microbiological Test

Microorganisms	Growth	Slide	Bottom	H ₂ S	Gas
<i>Escherichia coli</i> ATCC 25922	Good	Yellow	Yellow	-	+
<i>Proteus vulgaris</i> ATCC 13315	Good	Yellow	Yellow	+	+
<i>Salmonella enteritidis</i> ATCC 13076	Good	Red	Yellow	+	+
<i>Shigella flexneri</i> ATCC 12022	Good	Red	Yellow	-	-

TRYPTICASEIN DEXTROSE MEDIUM

Cat. 1003

For the general use in microbiology and for the differentiation of aerobic and anaerobic microorganisms, for dextrose fermentations and detection of motility

Formula in grams per liter

Casein Peptone 20,00	Dextrose 5,00
Bromothymol Blue 0,01	Bacteriological Agar 3,50

Final pH 7,3 ± 0,2 at 25°C

Preparation

Suspend 28,5 grams of the medium in one litre of distilled water. Mix well. Heat with frequent agitation and boil for one minute. Dispense into tubes, filling to half capacity. Sterilize at 116-118°C (10-12 lbs. sp.) for 15 minutes. Cool and tighten caps to prevent dehydration. Stored at ambient temperature, the medium can be used several weeks after preparation.

Uses

Trypticasein Dextrose Medium is inoculated by stabbing to half the depth of the medium. Reactions are generally complete after 18-24 hours incubation at 35-37°C.

The fermentation of dextrose is demonstrated by a color change from purple to yellow. The presence of gas is

observed by formation of bubbles in the agar or foam on the surface of the tube. Motility is seen by diffusion away from the line of inoculation (positive) and the medium becomes cloudy. Nonmotile organisms only grow in the inoculation line.

Bibliography

Recommended Methods for the Microbiological Examination of Foods APHA Inc., New York.
 COMPENDIUM OF METHODS FOR THE MICROBIOLOGICAL EXAMINATION OF FOOD. 3RD edition APHA 1992.
 Standard Methods for the Examination of Dairy Products. 11th Edition. APHA., Inc. New York, 1960.
 Greenberg and Cooper Can. Med. Assn. J. 83:143, 1960.
 Wetmore and Gochenour J. Bact. 72:79, 1956.

Microbiological Test

Microorganisms	Growth
<i>Aspergillus niger</i> ATCC 16404	Satisfactory
<i>Candida albicans</i> ATCC 26790	Satisfactory
<i>Escherichia coli</i> ATCC 25922	Partially inhibited
<i>Lactobacillus casei</i> ATCC 9595	Satisfactory
<i>Sacharomyces cerevisiae</i> ATCC 9763	Satisfactory

TRYPTICASEIN GLUCOSE EXTRACT AGAR

Cat. 1041

For the plate count enumeration of bacteria in potable and waste water

Formula in grams per liter

Casein Peptone.....	5,00	Beef Extract.....	3,00
Dextrose.....	1,00	Bacteriological Agar.....	15,00

Final pH 7,0 ± 0,2 at 25°C

Preparation

Suspend 24 grams of the medium in one litre of distilled water. Soak 10-15 minutes. Mix well. Heat with frequent agitation and boil for one minute. Sterilize at 121° C (15 lbs. sp.) for 15 minutes. Cool to 45° -50° C and pour into Petri dishes.

Uses

Trypticasein Glucose Extract Agar is used for the enumeration of bacteria from potable and waste water by the plate count method. Follow the procedures in the current Standard Methods for the Examination of Water and Wastewater. The Casein Peptone and the Beef Extract provide the carbon and nitrogen sources, required

for growth of a wide variety of organisms dextrose is a source of fermentable carbohydrate (energy source).

Bibliography

Standard Methods for the Examination of Water and Wastewater. 11th Edition APHA Inc. New York, 1960.
Standard Methods for the examination of dairy products, 16th ed. American Public Health Association; Washington D.C. Marshall, R.T. (1993).
Standard Methods for the examination of water and wastewater 18th ed. American Public Health Association, Washington D.C. 1992.

Microbiological Test

Microorganisms	Growth
<i>Staphylococcus aureus</i> ATCC 25923	Good
<i>Streptococcus faecalis</i> ATCC 11700	Good
<i>Escherichia coli</i> ATCC 25922	Good
<i>Salmonella typhimurium</i> ATCC 14028	Good
<i>Pseudomonas aeruginosa</i> ATCC 27853	Good (production of pigment)
<i>Bacillus cereus</i> ATCC 11778	Good

TRYPTICASEIN SOY AGAR **(ACC. EUROPEAN PHARMACOPOEIA)** **Cat. 1068**

It is very useful for the determination of hemolytic reactions.

Formula in grams per liter

<i>Casein Pancreatic Digest</i>	15,00	<i>Soy Peptone</i>	5,00
<i>Sodium Chloride</i>	5,00	<i>Bacteriological Agar</i>	15,00

Final pH 7,3 ± 0,2 at 25°C

Preparation

Suspend 40 grams of the medium in one liter of distilled water. Mix well. Heat with frequent agitation and boil for one minute, until the medium is completely dissolved. Dispense and sterilize in autoclave 121°C (15 lbs. pressure) for 15 minutes. If large quantities are to be prepared, sterilization time in autoclave, may be increased. but not temperature. To prepare blood plates for hemolysis studies, add 5 to 10% of defibrinated sterile blood, rabbit or sheep, to the sterile medium, cooled to about 45°C.

Uses

Trypticasein Soy Agar is a medium very rich in nutrients for "general use" in microbiological laboratories. It supports the abundant growth of fastidious organisms such as *pneumococci*, *streptococci*, *neisserias*, etc. Containing two peptones obtained by enzymatic hydrolysis of casein and soy protein, this medium supports the growth of a great variety of microorganisms, including fastidious aerobes and anaerobes.

Since it lacks carbohydrates it is very useful in the study of hemolytic reactions and also in the preparation of chocolate agar.

If desired, antibiotics can easily be incorporated as well as other supplements or inhibitory agents.

A short list of microorganisms that grow on this medium are the following: *Streptococcus*, *Neisseria*, *Brucella*, *Corynebacterium*, *Listeria*, *Pasteurella*, *Vibrio*, *Haemophilus vaginalis*, *Candida*, etc.

Bibliography

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 Standard Methods for the Examination of Dairy Products. 11th Edition. APHA., Inc. New York, 1960.
 Hentges, A. J. Clin. Path, 38:304, 1962. Kereluk and Gunderson. Applied Microbiol. 22:299, 1959.
 Curry, A.S., G. Joyce and G.N. Mcerven, Jr. 1993 CTFA Microbiology guideline. The Cosmetic Toiletry and Fragrance Association, Inc. Washington D.C.

Microbiological Test

Microorganisms	Growth	Growth with 5% sheep's blood	Hemolysis
<i>Neisseria meningitidis</i> ATCC 13090	Good	Good	---
<i>Staphylococcus aureus</i> ATCC 25923	Good	Good	beta
<i>Staphylococcus epidermidis</i> ATCC 12228	Good	Good	---
<i>Streptococcus pneumoniae</i> ATCC 6303	Good	Good	alpha
<i>Streptococcus pyogenes</i> ATCC 19615	Good	Good	beta

TRYPTICASEIN SOY BROTH (EUROPEAN PHARMACOPOEIA) Cat. 1224

Recommended for general laboratory use and to cultivate fastidious microorganisms

Formula in grams per liter

Casein Peptone.....	17,00	Soy Peptone.....	3,00
Sodium Chloride.....	5,00	Dipotassium Phosphate.....	2,50
Dextrose	2,50		

Final pH 7,3 ± 0,2 at 25°C

Preparation

Suspend 30 grams of the medium in one litre of distilled water. Mix well. Heat slightly until complete dissolution of the medium, if necessary. Dispense in tubes and sterilize in autoclave at 121°C (but not more than 15 lbs. steam pressure) for 15 minutes. Larger quantities may require longer sterilization time, but the temperature should not be increased.

Uses

Trypticasein Soy Broth is used frequently in many procedures of diagnostic research or microbiology. For example, it is used for the isolation and sensitivity testing of all types of pathogens, and for the production of antigens for agglutination and serological tests. Other uses include:

1. Urine cultures.
2. Blood cultures.
3. Cultivation of cerebrospinal fluid.
4. Antibiotic sensitivity testing.

5. Cultivation of anaerobic microorganisms, vibrios, and Bacteroides.
6. Preparation of bacterial antigens.
7. Examination of solid foods.
8. Tests for bile solubility.
9. Qualitative examination of yeasts and molds.
10. With blood the medium becomes richer so that it can cultivate a wider variety of microorganisms.

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MacFaddin, J.D. 1985. Media for isolation-cultivation-identification-maintenance of medical bacteria, p. 797, vol. 1. Williams & Wilkins, Baltimore, MD.

Microbiological Test

Microorganisms	Growth
<i>Brucella abortus</i> ATCC 4315	Good
<i>Staphylococcus aureus</i> ATCC 25923	Good
<i>Escherichia coli</i> ATCC 25922	Good
<i>Enterobacter aerogenes</i> ATCC 13048	Good
<i>Candida albicans</i> ATCC 10231	Good
<i>Streptococcus pyogenes</i> ATCC 19615	Good
<i>Streptococcus pneumoniae</i> ATCC 6303	Good

TRYPTONE BILE SALTS AGAR (TBA) (ISO 9308-1) Cat. 1013

Detection and enumeration of *E. coli* and coliform bacterias in waters

Formula in grams per liter

Tryptone 20,00 *Bile Salts* 1,50
Bacteriological Agar 15,00

Final pH 7,2 ± 0,2 at 25°C

Preparation

Suspend 36,5 g. of the dehydrated medium in one liter of distilled water. Heat agitating frequently until boiling. Distribute into appropriate containers and sterilize at (121 ± 3)°C for 15 minutes. Allow the medium to cool to 50 ± 5°C and distribute into double layer plates, making a layer of no less of 5 mm depth.

Uses

Medium used for the quick test for the detection and enumeration of coliform bacteria and *E. Coli* by the

membrane filtration technique as per the Standard ISO 9308-1.

Bibliography

Sahidi S.A. and Ferguson A.R. (1971) Appl. Microbiol., 21 500-506.
 Harmon S.M., Kauttar D.A. and Peeler J.T. (1971) Appl. Microbiol. 21. 922-927.
 Hauschild A.H.W and Hilsheimer R. (1973) Appl. Microbiol. 27. 78-82.

Microbiological Test

Microorganisms	Growth
<i>Escherichia coli</i> ATCC 25922	+
<i>Klebsiella pneumoniae</i> ATCC 13833	+

TRYPTONE SOY AGAR (ISO 9308-1) Cat. 1138

For the detection and enumeration of *Escherichia coli* and coliform bacteria.

Formula in grams per liter

<i>Casein peptone</i>	15,00	<i>Soy Peptone</i>	5,00
<i>Sodium Chloride</i>	5,00	<i>Bacteriological Agar</i>	15,00

Final pH 7,2 ± 0,1 at 25°C

Preparation

Suspend 40,0 grams of the medium in one liter of distilled water. Mix well and heat agitating frequently till boiling. Distribute into appropriate containers and sterilize at 121 °C (15 lbs. sp.) for 15 minutes. Allow the medium to cool to 50°C and distribute in double layer plates, making the layers no less than 5 mm depth.

Uses

This medium is used for the quick and standard test for the detection and count of coliform bacteria and *E. coli* by the membrane filtration method as directed in the ISO 9308-1:2.000 Regulation.

It has a general use, the two different peptones it contains allows to cultivate a great variety of microbes., even anaerobic bacteria when seeded in anaerobic conditions. It also serves as blood agar base as it does not contain any sugars, hemolytic reactions can be studied when blood is added

Bibliography

ISO 9308-1:2.000 Regulation water quality-Detection and count of *Escherichia coli* and coliform bacteria.
Anon. 1987 J. Food Microbiol., 5: 291-296.

Microbiological Test

Microorganisms	Growth
<i>Klebsiella pneumoniae</i> ATCC 13833	+
<i>Escherichia coli</i> ATCC 25922	+

TRYPTOPHAN CULTURE BROTH

(ISO 9308-1)

Cat. 1237

For the detection and enumeration of *Escherichia coli* and coliform bacteria.

Formula in grams per liter

Casein Peptone..... 10,00	Sodium chloride.....5,00
L-Tryptophan 1,00	

Final pH 7,5 ± 0,1 at 25°C

Preparation

Suspend 16,0 grams of medium in one liter of distilled water. Heat to boiling agitating frequently. Distribute in test tubes, 3 ml each. Close the tubes with cotton or with a plastic or metallic cap. Sterilize at 121° C (15 lbs. sp.) for 15 minutes.

Uses

This medium is used for the quick and standard test for the detection and count of coliform bacteria and *E. coli*

by the membrane filtration method as directed in the ISO 9308-1:2.000 Regulation.

Bibliography

ISO 9308-1:2.000 Regulation water quality-Detection and count of *Escherichia coli* and coliform bacteria.

Microbiological Test

Microorganisms	Growth
<i>Escherichia coli</i> ATCC 25922	+
<i>Klebsiella pneumoniae</i> ATCC 13833	-

TSN AGAR

Cat. 1075

For the selective isolation of *Clostridium perfringens* from foods and other material

Formula in grams per liter

Casein Peptone.....	15,00	Yeast Extract.....	10,00
Sodium Sulfite.....	1,00	Ferric Citrate.....	0,50
Neomycin Sulfate.....	0,02	Polymixin Sulfate.....	0,05
Bacteriological Agar.....	13,50		

Final pH 7,0 ± 0,2 at 25°C

Preparation

Suspend 40 grams of the medium in one litre of distilled water mix .Mix well. Heat with frequent agitation and boil for one minute. Dispense and sterilize at 118°C (12 lbs. sp.) for 10 minutes. DO NOT OVERHEAT. Cool to 45-50°C.

Uses

TSN Agar can be used in tubes or plates for the identification and enumeration of *C. perfringens* in foods and other materials, especially from mixed contaminating flora.

Incubation at 46°C makes the medium very selective while neomycin inhibits the growth of the majority of

enterobacteria and *C. bifermentens* (partially). Use an anaerobic jar for incubation if possible.

Read within half an hour after taking plates out of the jars and observe for black colonies which can lose their color by oxidation in air after this time period.

Bibliography

Angelotti, Nall, Foter y Lewis. Applied Microbiol. 10: 193. 1962.
Mossel. J.SCI. Agr. 10: 662. 1959.
Mossel de Bruin Van Diepen, Vendrig y Zoutwelle J. Applied Bact, 19: 142. 1956.

Microbiological Test

Microorganisms	Growth	Colony colour
<i>Clostridium perfringens</i> ATCC 10543	Satisfactory	Black
<i>Clostridium perfringens</i> ATCC 13124	Satisfactory	Black
<i>Escherichia coli</i> ATCC 25922	Inhibited	----
<i>Pseudomonas aeruginosa</i> ATCC 27853	Inhibited	----

T. S. C. AGAR BASE (TRYPTOSE SULFITE CYCLOSERINE AGAR BASE) Cat. 1029

Base media used for detection and enumeration of *Clostridium perfringens*

Formula in grams per liter

Tryptose 15,00	Soy Peptone 5,00
Yeast extract 5,00	Sodium Bisulfite 1,00
Ferroamonium 1,00	Bacteriological Agar 15,00

Final pH 7,6 ± 0,2 at 25°C

Preparation

Suspend 42 grams. of the medium in one liter of distilled water. Mix well . Heat agitating frequently and boil for one minute until completely dissolved. Distribute into appropriate containers and sterilize at 121°C (15 lbs. sp.) for 15 minutes.

Uses

The T.S.C. Base Agar, is a nutritive media, that is supplemented with egg yolk, due to the capacity of certain *Clostridium perfringens* strains to produce an opaque area in the colony surroundings. This is not recognized as a universal character for all *C. perfringens*. After 24 hour

incubation all black colonies lecithinase positive as well as the lecithinase negative ones, have to be considered presumptive *C. perfringens* positive.

Bibliography

Sahidi S.A. and Ferguson A.R. (1971) Appl. Microbiol, 21 500-506. Harmon S.M., Kauttar D.A. and Peeler J.T. (1971) Appl. Microbiol. 21 922-927. Hauschild A.H.W. and Hilsheimer R. (1973) Appl. Microbiol. 27. 78-82.

Microbiological Test

Microorganisms	Growth	Colony Colour
<i>Clostridium perfringens</i> spp.	Satisfactory	Black

TTC CHAPMAN AGAR

Cat. 1076

Recommended for the recount of coliforms in drinking waters by filtration technique

Formula in grams per liter

Meat peptone	10,00	Beef extract	5,00
Lactose	20,00	Yeast extract	6,00
Heptadecil Sodium sulfate	0,10	Bromothymol blue	0,05
Bacteriological Agar	15,00		

Final pH 7,2 ± 0,2 at 25°C

Preparation

Suspend 56,2 grams of the dehydrated medium in one litre of distilled water. Mix well. Heat with frequent agitation to boiling. Dispense in adequate containers and sterilize at 121°C for 15 minutes. Leave it cool at 45°C and add 3 ml. of Triphenyltetrazolium Chloride (TTC) sterile solution at 1% to each litre of medium. Homogenize and pour into Petri dishes. Do not heat again the medium.

Uses

This medium is adapted to the presumptive control of *coliforms* in waters by the filtration technique according to spanish legislation.

Two samples of water must be taken on two membranes and incubate them on TTC CHAPMAN at 35°C and 44°C respectively. After 48 hours of incubation:

- *E. coli* and *Citrobacter spp.* present yellow colonies with orange-coloured center.
- *Enterobacter spp.* brick red coloured colonies and dark yellow with orange-coloured center. The medium is yellow.
- *Klebsiella spp.* brick red coloured or yellow, but without center. The medium is yellow.

- *Bacteria* not fermentative of lactose, the colonies are violaceous and the medium blue.

The results will always refer to recounts of 100 ml. of sample (considering if it has been necessary to dilutions). The colonies grown at 35°C will be considered as fecal *coliforms* and the colonies grown at 44°C considered as *E. coli*.

It must be realized a confirmation of the colonies in EMB, Kligler, etc. for the verification of the Biochemical characteristics.

Bibliography

Chapman G.H. 1946, A single culture medium for selective isolation of plasma coagulating staphylococci and for improved testing of chromogenesis (J. Bacteriol. 51: 409-410).

Tittsler R.P. and L.A. Sandholzer. 1936. The Use of Semi-Solid Agar for the detection of bacteria motility. (J. Bacteriol 31: 575-580)

Microbiological Test

Microorganisms	Growth	Colony Colour
<i>Escherichia coli</i> ATCC 25922	Satisfactory	Yellow with orange center
<i>Citrobacter spp.</i>	Satisfactory	Yellow with orange center
<i>Klebsiella spp.</i>	Satisfactory	Red to yellow
<i>Enterobacter</i> ATCC 13048	Satisfactory	Red to dark yellow with orange center
<i>Not fermenting species</i>	Satisfactory	Light violet

UREA AGAR BASE (CHRISTENSEN) Cat. 1110

For the differentiation of enteric bacilli on the basis of urease production

Formula in grams per liter

<i>Gelatin Peptone</i>	1,00	<i>Dextrose</i>	1,00
<i>Sodium Chloride</i>	5,00	<i>Monopotassium Phosphate</i>	2,00
<i>Urea</i>	20,00	<i>Phenol Red</i>	0,012

Final pH 6,8 ± 0,2 at 25°C

Preparation

Dissolve 29 grams of the medium in 100 ml. of distilled water. Sterilize by filtration. Separately dissolve 15 grams of agar in 900 ml. of distilled water by boiling. Sterilize in autoclave at 121°C (15 lbs.sp) for 15 minutes. Cool to 50°C and add to the 100 ml. of the sterile Urea Agar Base. Mix well and dispense aseptically in sterile tubes. Leave the medium to set in a slanted position so as to obtain deep butts. At a pH of 6.8 to 7.0 the solidified medium should have a light pinkish yellow colour. Do not remelt the slanted agar.

Uses

Urea Agar Base may be used as an aid in the differentiation of microorganisms, particularly enteric gram-negative *bacilli*, on the basis of urea hydrolysis.

The solid medium is used to differentiate enteric bacilli on the basis of urea decomposition. *Proteus*, some

paracolons, and a few other organisms give a positive (purple) reaction.

To obtain good results, inoculate heavily over the slant as the speed of the reaction depends on the relation of organism amount and medium surface. Do not inoculate the butt of this medium as it is used as a negative color control. A positive test is denoted by a change in color, due to ammonia production, from pinkish yellow to a deep purple or bluish red on the slant surface. Observations of the tubes should be made at 2-4 hours. Re-incubate all negative cultures daily for up to 7 days for positives such as *Brucella*.

Bibliography

Christensen J. Bact. 52:641, 1946. Thal and Chen J. Bact. 69:10, 1955. Ewing Enterobacteriaceae. USPHS, Publication 734.

Microbiological Test

Microorganisms	Growth	Urease
<i>Enterobacter aerogenes</i> ATCC 13048	Satisfactory	-
<i>Escherichia coli</i> ATCC 25922	Satisfactory	-
<i>Klebsiella pneumoniae</i> ATCC 13883	Satisfactory	+
<i>Proteus vulgaris</i> ATCC 13315	Satisfactory	+
<i>Salmonella typhimurium</i> ATCC 14028	Satisfactory	-

UREA BROTH

Cat. 1226

For the differentiation of enterobacteria particularly *Proteus* from *Salmonella* and *Shigella*.

Formula in grams per liter

Urea	20,00	Monopotassium Phosphate.....	9,10
Sodium Phosphate.....	9,50	Yeast Extract.....	0,10
Phenol Red.....	0,01		

Final pH 6,8 ± 0,2 at 25°C

Preparation

Suspend 38,7 grams of the medium in 100 ml. of distilled water without heating. When the powder is dissolved, sterilize by filtration.

Dispense in small sterile tubes in quantities of 0,5 to 2 ml. Larger volumes can be used but the reactions will be slower.

When there is no filter available the medium can be sterilized in an autoclave at 5 to 8 lbs. of pressure for 15 minutes. If the medium is prepared and inoculated immediately it provides good results without sterilizing.

Uses

Urea Broth can be used for the determination of the urea activity in *enterobacteria* as well as microorganisms of the

general *Brucella*, *Bacillus*, *Micrococcus*, and *Mycobacterium*.

Developed by Rustigian and Stuart, this highly buffered medium usually reacts only to the high outputs of ammonia by *Proteus*, *Morganella* and *Providencia rettgeri* in the first 24 hours of incubation. An alkaline reaction produces a purple color in the presence of the phenol indicator.

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Rustigian and Stuart. Proc. Soc. Exp. Biol. and Med. 47:109, 1941. Stuart, Van Stratum and Rustigian. J. Bact. 48:437, 1945. McKay, Edwards and Leonar A. J. Clin. Path. 17:479, 1947. Gordon and Mihn. J. Gen. Microbiol., 21:736, 1959. Goldsmith and Latlief. Applied Microbiol., 3:195, 1955.

Microbiological Test

Microorganisms	Urease
<i>Escherichia coli</i> ATCC 25922	-
<i>Klebsiella pneumoniae</i> ATCC 13833	+
<i>Salmonella typhimurium</i> ATCC 14028	-
<i>Proteus vulgaris</i> ATCC 13315	+

UREA INDOL BROTH

Cat. 1227

For the identification of enterobacteria on the basis of urease and indol production and the transdeamination of tryptophan (TDA)

Formula in grams per liter

Monopotassium Phosphate	1,00	Dipotassium Phosphate	1,00
Sodium Chloride	5,00	Urea	20,00
L-Tryptophan	3,00	Phenol Red	0,025

Final pH 6,8 ± 0,2 at 25°C

Preparation

Suspend 30 grams of the medium in one litre of distilled water. Mix well. Add 10 ml. of ethanol.95°. Dispense in 1-5 ml. amounts into sterile tubes.

Uses

Prepare a heavy suspension of the organism isolated from plated media and inoculate the Urea Indol Broth tubes. Incubate at 37°C for 18-24 hours. Observe at 3-4 hours for any positive urease tubes which turn the indicator to a deep violet red color (alkalinization), typical of *Proteus* or *Yersinia*. *Klebsiella* and some *Citrobacter* develop positive tubes after 18 hours.

Indol production is determined by adding a few drops of Kovacs Reagent. A positive test is indicated by the development of a red color in the reagent layer. Tryptophan deaminase (TDA) is demonstrated by adding to a 24 hour culture a few drops of a 30% solution, diluted 1:3, of iron perchloride. The appearance of a maroon or reddish maroon color indicates a positive TDA.

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	UREA	INDOL	TDA
<i>Escherichia coli</i>	-	+	-
<i>Shigella dysenteriae, boydii, flexneri</i>	-	d	-
<i>Shigella sonnei</i>	-	-	-
<i>Salmonella</i>	-	-	-
<i>S. arizonae SG III</i>	-	-	-
<i>Citrobacter</i>	-	-	-
<i>Edwardsiella</i>	-	+	-
<i>Proteus vulgaris</i>	+	+	+
<i>Proteus rettgeri</i>	+	+	+
<i>Proteus morgani</i>	+	+	+
<i>Proteus mirabilis</i>	+	-	+
<i>Providencia</i>	-	+	+
<i>Yersinia enterocolitica</i>	+	d	-
<i>Y. pseudotuberculosis</i>	+	-	-
<i>Klebsiella pneumoniae</i>	+(slow)	-	-
<i>K. oxytoca</i>	+(slow)	+	-
<i>Enterobacter aerogenes</i>	-	-	-
<i>E. cloacae, E. hafniae</i>	-	-	-
<i>E. agglomerans</i>	-	d	-
<i>Serratia marcescens, liquefaciens</i>	-	-	-

d = variable according to different biochemical types

Microbiological Test

Microorganisms	Urease	Indol
<i>Escherichia coli</i> ATCC 25922	-	+
<i>Klebsiella pneumoniae</i> ATCC 13883	+	-
<i>Proteus vulgaris</i> ATCC 13315	+	+
<i>Salmonella typhimurium</i> ATCC 14028	-	-

VIOLET RED BILE AGAR WITH GLUCOSE

Cat. 1092

For the cultivation and enumeration of enterobacteria in water, foods and other materials

Formula in grams per liter

Yeast Extract.....	3,00	Bacteriological Peptone.....	7,00
Glucose.....	10,00	Bile Salts n° 3.....	1,50
Sodium Chloride.....	5,00	Crystal Violet.....	0,002
Neutral Red.....	0,03	Bacteriological Agar.....	15,00

Final pH 7,4 ± 0,2 at 25°C

Preparation

Suspend 41,5 grams of the medium in one litre of distilled water. Mix well. Heat with frequent agitation and boil for one minute. Cool to 45°C and dispense immediately. Alternatively, sterilize the medium at 118°C (12 lbs. sp.) for 15 minutes. Do not overheat or remelt the medium.

Uses

This medium is used to detect coliform bacteria as indicators of fecal contamination in water or food. *Coliforms* will ferment the glucose and produce acid with or without gas. Lactose-negative *Salmonella* and *Shigella* types and enteropathogenic *E. coli* grow on this medium as well as *Klebsiella* and *Citrobacter* which are more heat-resistant than coliforms and can indicate a production process defect (insufficient heating).

It is convenient to use the pour plate method by placing 1 ml. of the desired dilution in a sterile Petri dish, adding 15

ml. of medium, cooled to 45° to 50°C, and rotating gently before allowing to solidify. The pour plate method suppresses the growth of gram-negative non-fermenting bacteria by its anaerobic conditions. The fermentation of glucose is likewise stimulated and results in the formation of purplish-red colonies, clearly visible, surrounded by a zone of the same color.

Bibliography

D.A. Mossel, M. Koopmans, F. Van Rossem (1979) Influence of carbon source, bile salts and incubation temperature on recovery of enterobacteriaceae from foods using MacConkey types agars. (J. Food Protect 42: 470-475).
D.A. Mossel, (1985) Media for Enterobacteriaceae (Inst. J. Food Microbiol 2:27).

Microbiological Test

Microorganisms	Growth	Colony colour
<i>Escherichia coli</i> ATCC 11775	Satisfactory	Red
<i>Salmonella gallinarum</i> NCTC 9240	Satisfactory	Red
<i>Staphylococcus aureus</i> ATCC 6538	Inhibited	----
<i>Shigella flexneri</i> ATCC 29903	Satisfactory	Red
<i>Streptococcus lactis</i> ATCC 19435	Inhibited	----

VIOLET RED BILE AGAR WITH GLUCOSE, LACTOSE (V.R.B.G.L.) (EUR. PHARM.) Cat. 1144

Recommended for the detection and enumeration of enterobacteria

Formula in grams per liter

<i>Glucose Monohydrate</i>	10,00	<i>Lactose Monohydrate</i>	10,00
<i>Gelatin Pancreatic Digest</i>	7,00	<i>Sodium Chloride</i>	5,00
<i>Yeast Extract</i>	3,00	<i>Bile Salts N° 3</i>	1,50
<i>Neutral Red</i>	0,03	<i>Crystal Violet</i>	0,002
<i>Bacteriological Agar</i>	15,00		

Final pH 7,4 ± 0,1 at 25°C
Preparation

Suspend 51,5 grams of the medium in one liter of distilled water. Mix well. Heat with frequent agitation until complete dissolution. Boil for one minute. Cool to 45 °C. and use immediately. It can also be dispensed and sterilized in autoclave at 118 °C (12 lbs. sp.) for 15 minutes.

Uses

Medium recommended by the European Pharmacopoeia for the selective isolation of Gram-negative bacteria.

Microbiological examination of non-sterile products, test for specified micro-organisms.

Subculture on plates of agar this medium. Incubate at 35°C to 37°C for 18 h to 24 h. The product passes the test if there is no growth of colonies of gram-negative bacteria on any plate.

Bibliography

Hitchins, A.D., P.A. Hartman, and E.C.D. Todd. 1992. Coliforms – *Escherichia coli* and its toxins, p. 325-369. In Vanderzant, C., and D.F. Splittstoesser (ed.) Compendium of methods for the microbiological examination of foods, 3rd ed. American Public Health Association, Washington, DC.

Microbiological Test

Microorganisms	Growth	Colony colour
<i>Escherichia coli</i> ATCC 11775	Satisfactory	Red
<i>Salmonella gallinarum</i> NCTC 9240	Satisfactory	Red
<i>Staphylococcus aureus</i> ATCC 6538	Inhibited	----
<i>Shigella flexneri</i> ATCC 29903	Satisfactory	Red
<i>Streptococcus lactis</i> ATCC 19435	Inhibited	----

VIOLET RED BILE AGAR WITH LACTOSE

Cat. 1093

Selective and differential medium for the detection and enumeration of coliforms in milk and dairy products.

Formula in grams per liter

Yeast Extract.....	3,00	Gelatin Peptone	7,00
Bile Salts n° 3.....	1,50	Lactose.....	10,00
Sodium Chloride.....	5,00	Bacteriological Agar.....	15,00
Neutral Red.....	0,03	Crystal Violet.....	0,002

Final pH 7,4 ± 0,2 at 25°C

Preparation

Suspend 41,5 grams of the medium in one liter of distilled water. Mix well. Heat with frequent agitation and boil for one minute. Cool to 45 °C, and use immediately. It can also be dispensed and sterilized in autoclave at 118° (12 lbs. sp.) for 15 minutes.

Uses

For the detection and enumeration of coliforms in milk, food and other materials. Violet Red Bile Agar (VRBA) is a differential and mildly selective medium for the detection of coliforms in water as well as milk and other food materials. Gram-positive organisms are markedly inhibited by the bile salts and the crystal violet. The colonies of lactose fermenting bacteria are red in color whose size depends on the number of colonies on the plate. Occasionally the cocci of the intestinal tract can develop as small, punctiform red colonies.

Violet Red Bile Agar can be utilized for the presumptive identification of coliforms in milk and other food materials according to the APHA (Standard Methods for the Examination of Milk Products).

The material sample is seeded in small aliquots immediately onto VRBA. If desired, after the plates have solidified and been stored, but before the sample is seeded, another thin layer can be poured on top. Some laboratories are accustomed to this method and dismiss any growth on the lower layer as contamination.

In the studies of Hartman, he demonstrated that media prepared only by boiling gave the same results as media sterilized by autoclaving.

Bibliography

Collins, J. Milk and Food Tech 18:169, 1955. Hartman, J. Milk and Food Tech 23:43, 1960
Speck, M.L. (ed) 1976. Compendium of Methods for the Microbiological Examination of Foods (APHA).

Microbiological Test

Microorganisms	Growth	Colony colour
<i>Escherichia coli</i> ATCC 25922	Good	Purple
<i>Enterobacter aerogenes</i> ATCC 13048	Good	Purple
<i>Salmonella enteritidis</i> ATCC 13076	Good	Colourless
<i>Staphylococcus aureus</i> ATCC 25923	Inhibited	----
<i>Enterococcus faecalis</i> ATCC 19433	Inhibited	----

VOGEL JOHNSON AGAR

Cat. 1079

For isolation of *S. aureus* mannitol fermenters, coagulase positive, in clinical samples and foods

Formula in grams per liter

<i>Tryptone</i>	10,00	<i>Yeast Extract</i>	5,00
<i>Mannitol</i>	10,00	<i>Dipotassium Phosphate</i>	5,00
<i>Lithium Chloride</i>	5,00	<i>Glycine</i>	10,00
<i>Phenol Red</i>	0,025	<i>Bacteriological Agar</i>	15,00

Final pH 7,2 ± 0,2 at 25°C

Preparation

Suspend 60 grams of the medium in one litre of distilled water. Mix well, and heat with frequent agitation. Boil for one minute or until the medium is completely dissolved. Sterilize in the autoclave at 121°C (15 lbs. sp.) for 15 minutes. Cool to 45-50°C and add 20 ml of an sterile solution of potassium tellurite 1%. Mix well and dispense. To prepare a less selective medium add only 10 ml of the potassium tellurite solution.

Uses

Vogel Johnson Agar plates can be streaked heavily with a swab and incubated at 35-37°C for 24-48 hours, looking for black colonies surrounded by a yellow zone. During the first 24 hours the majority of microorganisms other than coagulase-positive staphylococci are totally or markedly inhibited. At 48 hours many coagulase-negative staphs, mannitol-positive and mannitol-negative, begin to appear.

S. epidermidis, almost always inhibited early, forms small grayish-black colonies without yellow zones.

Coagulase-positive staphs form black colonies on the red medium. If they ferment mannitol, the colonies are surrounded by a yellow zone. Mannitol-negative organisms do not change the red color of the medium.

The medium is excellent for the detection of staph carriers as well as studies of sanitary concern.

Bibliography

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Zebrovitz E. Evans, J.B. add Niven C.P. (1955) J. Bact. 70, 687.

Microbiological Test

Microorganisms	Growth	Colony colour
<i>Escherichia coli</i> ATCC 25922	Inhibited	---
<i>Proteus mirabilis</i> ATCC 25933	Negative to poor	Black
<i>Staphylococcus aureus</i> ATCC 25923	Good	Black with yellow haes
<i>Staphylococcus epidermis</i> ATCC 12228	Moderate	Translucid to black

WILKINS CHALGREN MEDIUM

Cat. 1503

Used for susceptibility testing as well as for the isolation and culture of anaerobic bacteria in general.

Formula in grams per liter

<i>Tryptone</i>	10,00	<i>Yeast Extract</i>	5,00
<i>Bacteriological Peptone</i>	10,00	<i>Dextrose</i>	1,00
<i>Sodium Chloride</i>	5,00	<i>Sodium Pyruvate</i>	1,00
<i>L-Arginine</i>	1,00	<i>Vitamin K1</i>	0,0005
<i>Hemin</i>	0,0005	<i>Bacteriological Agar</i>	15,00

Final pH 7,1 ± 0,2 at 25°C

Preparation

Suspend 48 grams of the medium in one litre of distilled water. Mix well. Heat by boiling until the medium is completely dissolved. Dispense, if desired and sterilize at 121°C (15 lbs. sp.) for 15 minutes. Cool 45°C before adding antibiotics. Mix gently and pour into Petri dishes.

Uses

Wilkins and Chalgren designed this medium for use in the determination of minimum inhibitory concentrations (MIC) of antibiotics for anaerobic bacteria by the agar dilution method. It has the advantage over other media in that it does not need the addition of blood to obtain satisfactory growth of clinically important anaerobic bacteria, as it

includes Yeast Extract that provides the most needed growing factors to cultivate bacteroides melaninogenicus.

It has the same performance in petri dishes as in tubes.

Bibliography

Wilkins T.D. and Chalgren S. (1976) Antimicrob. Agents. Chemother., 10, 926-928.
Sutter V.L., Barry A.L., Wilkins T.D. and Zabransky R.J. (1979) and Microb. Agents Chemother, 16, 495-502.
Brown W.J. and Waatti P.E. (1980) Antimicrob. Agents Chemother., 17, 629-635.

Microbiological Test

Microorganisms	Growth
<i>Bacteroides fragilis</i> ATCC 25285	Good
<i>Bacteroides melanogenicus</i> ATCC 25611	Good
<i>Clostridium perfringens</i> ATCC 13123	Good

WL DIFFERENTIAL AGAR

Cat. 1026

Employed to control Industrial fermentation processes especially in brewery

Formula in grams per liter

<i>Yeast Extract</i>	4.00	<i>Casein Peptone</i>	5.00
<i>Dextrose</i>	50.00	<i>Monopotassium Phosphate</i>	0.55
<i>Potassium Chloride</i>	0.425	<i>Calcium Chloride</i>	0.125
<i>Magnesium Sulfate</i>	0.125	<i>Ferric Chloride</i>	0.0025
<i>Manganese Sulfate</i>	0.0025	<i>Bromocresol Green</i>	0.022
<i>Cycloheximide</i>	0.004	<i>Bacteriological Agar</i>	20.00

Final pH 5,5 ± 0,2 at 25°C

Preparation

Suspend 80 grams of the medium in one litre of distilled water. Soak 10-15 minutes. Heat evenly while stirring frequently and boil the medium for a minute. Dispense in test tubes or flasks and sterilize in an autoclave at 121°C (15 lbs. sp.) for 15 minutes.

Uses

For the control of industrial fermentations by yeasts. WL Differential Agar (Wallerstein Laboratories) is used together with the WL Nutrient Agar for the control of the manufacture of beer and other fermentation processes.

The medium allows for the selective multiplication of yeast cells in fermentation liquids, which contain a microflora mix consisting of fungi and bacteria. When the number of yeast cells present is relatively small, certain bacteria can also be detected.

The addition of 0,004 grams of cycloheximide (actidione) converts the nutrient agar formula into a differential medium, which inhibits the development of yeasts and molds while permitting the notable proliferation of the bacteria present in the fermentation liquids and subsequent identification and enumeration.

The quantity and composition of the microflora present in beer and in other industrial fermentations, are very important factors which must be controlled during different manufacturing processes. Green and Grey found the

microscopic counts of organisms did not give sufficient information to control those processes.

Both media are widely used in the industries of vinegar, bread yeasts, grape and wine growing, and distilled spirits. In the production of yeasts for the bakery and distillery industries, the pH of the media is adjusted to 6,5.

Time and temperature of incubation are important factors according to the type of yeast. In general, temperatures of 25°C with the beer yeasts and 30°C with the bread and other alcoholic yeast fermentations are appropriate. The time of incubation varies from 2 to 7 days depending on the flora found, which can extend to 14 days if necessary.

Likewise, the atmosphere chosen for incubating the culture must be appropriate. The bread yeasts are incubated aerobically while the alcoholic fermentation yeasts are incubated anaerobically and in the presence of CO₂.

Bibliography

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Microbiological Test

Microorganisms	Growth
<i>Escherichia coli</i> ATCC 25922	Good
<i>Lactobacillus fermentum</i> ATCC 9338	Good
<i>Saccharomyces cerevisiae</i> ATCC 9763	Inhibited
<i>Saccharomyces uvarum</i> ATCC 9080	Inhibited
<i>Proteus mirabilis</i> ATCC 25933	Good

WL NUTRIENT AGAR

Cat. 1086

For the determination of microbial flora in beer fermentation processes and manufacturing

Formula in grams per liter

Yeast Extract.....	4,00	Tryptone.....	5,00
Dextrose.....	50,00	Monopotassium Phosphate.....	0,55
Potassium Chloride.....	0,425	Calcium Chloride.....	0,125
Magnesium Sulfate.....	0,125	Ferric Chloride.....	0,0025
Manganese Sulfate.....	0,0025	Bromocresol Green.....	0,022
Bacteriological Agar.....	15,00		

Final pH 5,5 ± 0,2 at 25°C

Preparation

Suspend 75 grams of the medium in one litre of distilled water. Heat with frequent agitation and boil for one minute. Sterilize at 121°C (15 lbs. sp.) for 15 minutes.

Uses

WL Nutrient Agar, based on the Green and Grey formulation, is recommended for the control of industrial fermentations, particular the manufacturing of beer. With a pH of 5,5, true counts of beer yeasts can be made. With a pH of 6,5, the medium is ideal for bakery and distilled spirit yeasts.

The medium can be made selective and differential by adding cycloheximide (actidione), suppressing the yeast growth but allowing for proliferation of undesirable of bacterial contaminants.

Both the WL Nutrient and Differential Agar formulas are used in conjunction: 1 plate of WL Nutrient Agar and 2 plates of WL Differential Agar.

The WL Nutrient Agar plate is incubated aerobically for total plate count of yeasts. One of the WL Differential Agar

plates is incubated aerobically for acetic acid bacteria-*Flavobacterium*, *Proteus*, *thermophilic bacteria* and others- whereas the second plate is incubated anaerobically for investigation of lactic-acid bacteria and species of *Pediococcus*.

All plates are incubated, in general, at 25°C as in the case of beer, and at 30°C for bakery and malt alcoholic yeasts. Plates are incubated for 2-10 days up to 2 weeks, according to the flora present. Counts are made at regular intervals during this period.

Bibliography

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Green, S.R. and P.P. Gray 1950. A differential procedure applicable to bacteriological investigation in brewing. Wallerstein Lab. Commun 13:357.
MacFaddin J.D. 1985. media for isolation cultivation-identification-maintenance of medical bacteria, vol. 1, p. 854-856 Williams Wilkins, Baltimore, MD.

Microbiological Test

Microorganisms	Growth
<i>Escherichia coli</i> ATCC 25922	Moderate
<i>Lactobacillus fermentum</i> ATCC 9338	Moderate
<i>Proteus mirabilis</i> ATCC 25933	Moderate
<i>Saccharomyces cerevisiae</i> ATCC 9763	Good
<i>Saccharomyces uvarum</i> ATCC 9080	Good

XLD AGAR (EUR. PHARM.)

XYLOSE LYSINE DESOXYCHOLATE AGAR

Cat. 1080

For the isolation of enteropathogenic bacteria, especially from the genera of Shigella, Salmonella, and Arizona

Formula in grams per liter

Xylose	3,50	L-Lysine	5,00
Lactose Monohydrate	7,50	Sucrose	7,50
Sodium Chloride	5,00	Yeast Extract	3,00
Phenol Red	0,08	Bacteriological Agar	13,50
Sodium Desoxycholate	2,50	Sodium Thiosulfate	6,80
Ferric Ammonium Citrate	0,80		

Final pH 7,4 ± 0,2 at 25°C

Preparation

Suspend 55 grams of the medium in one liter of distilled water. Heat with frequent agitation until a temperature of approximately 90°C. Do not boil. Transfer immediately into a water bath at about 50°C. Pour into Petri plates as soon as it has cooled. The medium should have a reddish color and be clear, or almost clear. Excessive heating or a prolonged stay in the water bath produces precipitation. When this occurs, reactions are satisfactory, but colonies may be slightly smaller. This precipitation can be eliminated by paper filtration.

Uses

In XLD Agar it is possible to obtain the following differential this medium was developed principally for isolating Shigella and Providencia. It has been shown to be more effective than other enteric differential media, reactions: the degradation of xylose, lactose and sucrose, with the production of acid, manifested in the color change from red to yellow. Sodium thiosulfate serves as a reactive substance with the iron salt as an indicator of the formation of hydrogen sulfide. The bacteria that decarboxylate the lysine to cadaverine are identified by the presence of a purple-red color around the colonies due to the elevation of pH.

The characteristics of the colonies are:

Arizona: Red and transparent with a black center.
Citrobacter: Yellow and opaque. Can present a black center and clear edges.
Edwardsiella: Red with a black center and clear edges.
E. coli, Enterobacter, Serratia: Yellow and opaque. Zone of yellow precipitation around the colonies.
Klebsiella: Large, yellow, pale, mucoid and opaque. Zone of yellow precipitation around the colonies.
Proteus mirabilis and P. vulgaris: Yellow, transparent, with clear edges. Black center especially P. Mirabilis.
Proteus morganii and P. rettgeri: Red and transparent.
Providencia and Shigella: Red and transparent.
Salmonella: Red, transparent, yellow edges with black centers only if H₂S is produced.

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 Rollender, W. U. Beckford; R.D. Belsky, B. Krostoff (1969) Comparison of Xylose Lysine desoxycholate agar and MacConkey agar for the isolation of Salmonella and Shigella from clinical specimens (tech. Bull. Reg. Med. Tech, 39 (1):8-p)

Microbiological Test

Microorganisms	Growth	Colony colour
<i>Proteus mirabilis</i> ATCC 14273	Good	Yellow(may have black center)
<i>Escherichia coli</i> ATCC 25922	Moderate	Yellow (precipitated)
<i>Salmonella arizonae</i> ATCC 13314	Good	Transparent red (black center)
<i>Salmonella typhimurium</i> ATCC 14028	Good	Transparent red (black center)
<i>Shigella sonnei</i> ATCC 25931	Good	Red
<i>Staphylococcus aureus</i> ATCC 25923	Inhibited	---

YEAST EXTRACT AGAR (ISO 6222:1999) Cat. 1049

Nutrient medium for the recount of germs in water

Formula in grams per liter

Yeast extract 3,00 Bacteriological Agar..... 15,00
Tryptone 6,00

Final pH 7,2 ± 0,2 at 25°C

Preparation

Suspend 23 grams of the medium in one litre of distilled water. Heat with frequent agitation and boil for one minute. Do not overheat. Sterilize in an autoclave at 121°C for 15 minutes.

Uses

Yeast Extract, is a medium rich in nutrients which permits the recovery of a wide spectrum of bacteria, yeast and Prepare decimal dilutions and make recount for pouring in

plate. Incubate two series of plates, one at 37°C for 24 hours and the other at 20-22°C for 3 days.

Bibliography

International Organization for Standardization: Water Quality. Enumeration of cultural micro-organisms. Colony count by inoculation in a nutrient agar culture medium, International Standard ISO 6222 (1999).

Microbiological Test

Microorganisms	Growth
<i>Escherichia coli</i> ATCC 25922	Satisfactory
<i>Candida albicans</i> ATCC 10131	Satisfactory
<i>Staphylococcus aureus</i> ATCC 25923	Satisfactory
<i>Aspergillus niger</i>	Satisfactory
<i>Penicillium spp.</i>	Satisfactory

YEAST EXTRACT AGAR (FOR MOULDS) Cat. 1312

For the cultivation of moulds and yeast from diverse materials, specially milk and dairy products

Formula in grams per liter

Dextrose..... 10,00 Yeast Extract..... 5,00
Bacteriological Agar..... 20,00

Final pH 6,5 ± 0,2 at 25°C

Preparation

Suspend 35 grams of the dehydrated medium in one liter of distilled water. Heat agitating frequently until completely dissolved. Sterilize in autoclave at 121°C (15 lbs. sp.) for 15 minutes.

Uses

Medium suitable to cultivate moulds and yeast from milk and dairy products. The inoculation method can be either by flooding or in surface, depending on the purpose for with the medium is intended to be used for. Incubation time will be of 7 days at a temperature of 28°C and in aerobic condition.

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Overcase, W.W. and D.J. Weakley. 1969. An aureomycin-rose Bengal agar for enumeration of yeast and mold in cottage cheese.
International Dairy Federation. Standard Method ISO/DIS 6611.
Koburger, J.A.. 1970. Fungi in foods: 1. Effect of inhibitor and incubation temperature on enumeration. J. Milk Food Technol. 33:433-434.

Microbiological Test

Microorganisms	Growth
<i>Escherichia coli</i> ATCC 25922	Good
<i>Staphylococcus aureus</i> ATCC 25923	Good
<i>Candida albicans</i> ATCC 1023	Good
<i>Aspergillus niger</i>	Good
<i>Penicillium</i> spp.	Good

YEAST EXTRACT SOY AGAR

Cat. 1097

Medium used for selective isolation of dermatophytes and other pathogen fungus in clinic samples

Formula in grams per liter

Dextrose	40,00	Soy peptone	10,00
Yeast extract	5,00	Chloramphenicol	0,05
Streptomycine	0,03	Bacteriological agar	17,00

Final pH 6,6 ± 0,2 at 25°C

Preparation

Suspend 72 grams of the dehydrated medium in one liter of distilled water. Heat agitating frequently until completely dissolved. Sterilize in autoclave at 118°C (12 lbs.sp) for 15 minutes.

Uses

Yeast Extract Soy Agar is a modification of the Sabouraud Medium and was formulated by Carmichael and Claus for the selective isolation of Trychophyton verrucosum as well as other fungi associated with contagious diseases. Yeast Extract Soy Agar contains streptomycine and chloramphenicol, antibiotics that inhibit the bacterial grow but allow to detect pathogenic fungi.

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Beuchat, L.R. 1979. Comparison of acidified and antibiotic-supplemented potato dextrose agar from three manufactures for its capacity to recover fungi from foods. J. Food Prot. 42: 427-428.

Microbiological Test

Microorganisms	Growth
<i>Candida albicans</i> ATCC 10231	Satisfactory
<i>Escherichia coli</i> ATCC 25922	Inhibited
<i>Trychophyton mentagrophytes</i> ATCC 9533	Satisfactory

AGAR, PEPTONES AND OTHER INGREDIENTS

Agar-Agar

Etymologically the word "Agar" comes from the Malayan language which describes the red algae from the *genus Eucheuma*.

Agar is a dried colloidal substance extracted from one of several species of red seaweeds, particularly of the genera *Gelidium*, *Gracilaria*, *Pterocladia*, and *Anthopeltis*. Because of different quality and use requirements, agar is divided into two groups: industrial and bacteriological types.

The increase in use of agar for industrial applications such as foods (tinned meats and vegetables, sweets, pastries, ice cream, etc.) has been enormous because of its properties as a dispersing agent, stabilizer, thickener, and gelling agent. Because of its many advantages, agar replaces pectin. Since it is a vegetal gelatin of marine origin in definition, it is the perfect substitute for the gelatin of animal origin because it has approximately eight times (8x) the gellification power of animal gelatin.

BACTERIOLOGICAL AGAR

EUROPEAN TYPE

Cat. 1800

AMERICAN TYPE

Cat. 1802

The use of agar in bacteriology is known to all. It was the school of bacteriology of Robert Koch that introduced agar which until then had been a curious oriental food. Today, agar is utilized around the world in bacteriological culture media as the only gelling agent of choice.

Bacteriological agar is incorporated into culture media for the isolation of bacterial and fungal microorganisms as well as the differentiation of strains and the study of their susceptibility to chemotherapeutic agents.

This high quality agar has been an indispensable tool in shaping the development of bacteriology in its present form.

Agar is a unique colloid, which remains liquid down to its melting point (approx. 36°C). This allows for mixing of blood with culture media for determination of hemolytic reactions. Likewise, once solidified, agar will remain solid until its melting point temperature (approx. 85°C) is reached allowing for studies of thermophilic bacteria incubated at 60°C or higher.

Owing to differences in bacteriological techniques around the world, our R&D department has developed two types of agar to address the specifications for the U.S. and European market; European type and American type.

EUROPEAN TYPE: The European approach of bacteriology is to use as little agar as possible in order not to introduce unknown substances to the culture media. For this reason, the gel strength is higher (800-1100 g/cm²), and can be used at lower concentrations. The ash content is low (< 4.5%).

AMERICAN TYPE: In the American concept of agar it is considered not only as a gelling agent but as a source of indefinable but indispensable trace elements crucial to the growth of many bacteria and fungi. The gel strength is

lower (550-850 g/cm²) and should be used in a higher concentration. The ash is also slightly higher (< 6.5%).

INDUSTRIAL AGAR

Cat. 1804

The experimented increase in the use of Agar-Agar for industrial applications such as foods (tinned meats and vegetables, sweets, pastries, ice creams, etc) has been enormous because of its properties as a dispersing agent, stabilizer, thickener and gelling agent. Because of its many advantages, it replaces pectin and since it is a vegetal gelatin of marine origin in definition, it is the perfect substitute for the gelatin of animal origin, being so that it has eight times more the gellification power of animal gelatin.

PURIFIED AGAR

Cat. 1806

This agar is highly purified with a very low ash content for use in microbiology and biochemistry. It is subjected to rigid tests which guarantee its excellent performance in biochemical, bacteriological and mycological applications. It can be used in special studies such as yeast assimilation and vitamin assays.

VITRO AGAR

Cat. 1808

This agar was developed especially for "in vitro" cell culture. Because of its physical-chemical characteristics, color, transparency, degree of purity and, above all, its high gel strength (approximately 1000 g/cm²) which allows usage levels as low as 0.4-0.5%, this agar is recommended for micropropagation techniques (initiation, propagation, radiation, etc.).

This product is strictly controlled and is designed to give high yields in large industrial operations for growing tissue culture plants (ornamentals, horticulture, woody plants, etc.).

Carbohydrates and Glucosides

Carbohydrates constitute more than half the organic material in the world. In culture media, carbohydrates and glucosides are used as a source of energy by bacteria and to differentiate genera and identify species.

The ability of a microorganism to attack a particular carbohydrate is a defining characteristic in bacterial species which under strict physico-chemical controls remains constant through generations of growth on artificial culture media.

DEXTROSE

Cat. 1900

Dextrose is offered at a very high grade purity. It is used as a source of energy to cultivate microorganisms and for fermentation studies. It is free of all other sugars and starches, proteins, alcohols and heavy metals. It is a

white, crystalline powder in appearance. Its specific rotation is between +52.5°C and +52.76°C.

Dextrose (D-Glucose) is widely used in the study of fermentations carried out by microorganisms. In liquid culture media it is generally used in a 0.5% concentration while in solid media formulations it can be used in higher concentrations.

This hexose sugar has a beneficial effect on old cultures of many types of microorganisms because it is easily assimilated. Adding 0.05% dextrose to a culture medium free of carbohydrates can increase the speed and recovery of many organisms.

Dextrose is incorporated into many culture media formulas, such as those employed in the selective isolation of enterobacteria.

LACTOSE

Cat. 1902

This disaccharide, along with dextrose, constitute the most commonly used carbohydrates used in biology today. It is comprised of a molecule of d-glucose and a molecule of d-galactose. It is free of dextrose, casein and other proteins, starches and alcohol. It does not contain traces of heavy metals and so can be used with great confidence in biological applications.

Lactose is not fermented by Salmonella or Shigella which would indicate that it is free of dextrose.

It can be incorporated into media formulas alone or in combination with other fermentable substances, such as the differential and selective media for the detection of coliforms in products of sanitary interest (water, milk, and other foods). It is also one of the components of culture media used to detect the presence of enteropathogenic bacteria.

MALTOSE CERTIFIED

Cat. 1904

Maltose Certified, is a pure carbohydrate prepared especially for use in bacterial culture media. It is used in media such as Trypticasein Agar Base and Phenol Red Broth Base at concentrations from 0,5% to 1,0%.

It is used also in culture media for the isolation of yeasts and molds. It meets USP specifications.

SUCROSE

Cat. 1906

Sucrose is a disaccharide composed of a molecule of glucose and a molecule of fructose. Its specific rotation is +65,9°C and is free of other substances. It is a popular addition to culture media formulations.

Peptones

The term "peptone" is used to define a product soluble in water which is obtained by hydrolysis of particular protein or proteins. This material contains a mixture of free amino acids, peptides and proteases which remain in solution after heating to 100°C. The presence of alkaline metals or phosphates can cause the precipitation of the peptones at a neutral pH. For this reason peptones produced at a pH of neutrality should be utilized in media formulas. All the peptones bearing the mark *PRONADISA* are manufactured under strict conditions of quality control. A great variety of peptones exist because of the different growth requirements of the organisms for certain amino acids and peptides. In general, the proteins used in the production of peptones are of two types, animal proteins (casein, gelatin, meat) and vegetal proteins (soy).

Peptones are obtained by various types of digestion such as acid, alkaline or enzymatic processes.

Acid hydrolysis ruptures all the proteins and peptides and produces only free amino acids; at the same time, it destroys some important amino acids such as tryptophan.

Peptones can be used by bacteria as a source of energy and enhances the production of proteins, H₂S, indol, amines, etc.

In the preparation of culture media one should use the type peptone which provides the characteristics appropriate for the test. For instance, in the test for indol one should use a peptone rich in tryptophan (casein peptone).

It is also important to realize that apart from the amino acids present, peptones contain other constituents which can stimulate growth such as nucleic acids, minerals, vitamins, and occasionally carbohydrates as in the case of soy peptone.

ACID CASEIN PEPTONE

Cat. 1604

Acid Casein Peptone is an acid hydrolysate of casein low in cystine and tryptophan. It is used for vitamin determinations by microbiological methods because it is free of vitamins destroyed by the acid treatment.

BACTERIOLOGICAL GELATIN

Cat. 1704

Gelatin Bacteriological is a refined product approved for use in bacteriology and has no fermentable carbohydrates. It is used for the identification of proteolytic organisms and is generally incorporated in media at 3-5%.

On occasion it is used as a support for culture media. It is used in tests for the liquefaction of gelatin by microorganisms at a concentration of 12% in water.

BACTERIOLOGICAL PEPTONE

Cat. 1616

This peptone is standardized for the preparation of many bacteriological culture media. It is an excellent source of

nitrogen for bacterial growth. It is completely soluble giving a clean solution in the concentrations utilized in culture media.

BEEF EXTRACT

Cat. 1700

Beef Extract is prepared from fresh meat and can be used in general bacteriology and in various media formulas for the growth of streptococci and staphylococci and media for febrile antigen production.

Normally, Beef Extract is utilized at concentrations from 0,5-0,8% and has the same properties as beef extract paste with the advantages that is much easier to handle and goes into solution without difficulty.

BILE SALTS N° 3

Cat. 1706

Bile Salts N° 3 is a mixture of bile extracts especially prepared for use in selective media such as MacConkey Agar and *Salmonella Shigella* Agar. It is an excellent inhibitor of gram-positive bacteria such as *streptococci* and *staphylococci*.

BIOTRYPTASE CL PEPTONE

Cat. 1605

This ingredient is a mixture of peptones high in nutrient value. It is recommended for the recovery of fastidious microorganisms such as *Brucella*, *Pasteurella* and particularly in the production of febrile antigens as well as in blood culture bottle formulas.

CASEIN CC PEPTONE

Cat. 1603

This peptone is a pancreatic digest of casein especially designed for use in the production of tetanus toxin by *Clostridium tetani*. It can also be used for fastidious microorganisms and some fermentation processes.

CASEIN PEPTONE

Cat. 1602

Casein peptone is a pancreatic digest of casein designed for incorporation into a wide range of culture media formulations for growth of all types of fastidious and non-fastidious microorganisms. The enzymatic treatment of casein is gentle and produces a source rich in vitamins and amino acids such as tryptophan which encourages the growth of difficult-to-grow organisms.

Casein peptone is recommended for the enrichment of culture media for both pathogenic microorganisms and food-borne bacteria. It is used to demonstrate production of indol because of the high content of tryptophan, and in other media for the identification tests of bacteria such as carbohydrate fermentation and nitrate reduction. This peptone can be used in media for sterility testing

according to the USP and for potency tests of antibiotics and other antimicrobial agents.

GELATIN PEPTONE

Cat. 1606

Gelatin Peptone is a pancreatic digest of gelatin characterized by a low content of cystine, tryptophan and the absence of carbohydrates. It is used to promote the growth of various organisms under controlled conditions and for culture media for fermentation studies.

HEMOGLOBIN

Cat. 7004

Hemoglobin is a dried preparation of bovine erythrocytes. It forms a stable solution at 2% after sterilization.

It is used as an enrichment substance in certain culture media such as Trypticasein and phosphate broth, to isolate by hemoculture fastidious germs as hemophilus, streptococcus, etc... and specially to prepare the Chocolate Agar Media, widely used on the isolation of pathogenic *Neisserias*, *gonococcus* and *meningococcus*.

Generally, the basic media are prepared separately at double concentration, just like the hemoglobin suspension.

It is sterilized and mixed at equal volumes, being the concentration of the complete medium reduced to a normal level.

MALT EXTRACT

Cat. 1708

Malt Extract is widely used in culture media for growing fungi. It is prepared by extracting the soluble fraction of malted barley at low temperatures to preserve the maximum levels of nitrogenous and carbohydrate components.

MEAT PEPTONE

Cat. 1600

Meat peptone is a peptic digest of animal tissue. Because of its high sulfur content, it is used extensively in H₂S production studies. Meat peptone is an excellent promoter of growth over a wide range of microorganisms.

POLYPEPTONE

Cat. 1610

Polypeptone is a combination of casein peptone and meat peptone designed for incorporation into several formulas where abundant growth is desired. It is recommended for enterobacteria and can be used in both liquid and solid media.

PROTEOSE PEPTONE

Cat. 1609

This mixed peptone is used for difficult to grow microorganisms because of its high nutritional value. It can be used with excellent results in the production of bacterial toxins. It contains a peptic digest of animal tissue.

PROTEOSE PEPTONE N° 3

Cat. 1607

This is a mixed peptone of animal tissue digested to an optimum degree to produce a source rich in proteases and peptides for growing fastidious microorganisms such as *gonococci*.

This peptone is also used for the production of toxins, especially diphtheria toxin.

SOY PEPTONE

Cat. 1608

Soy Peptone is a papaic digest of soy which is utilized to grow a wide range of bacteria. It is rich in carbohydrates and is generally incorporated into culture media at between 0,3-0,5% concentration.

TRYPTONE

Cat. 1612

This pancreatic digest of casein is utilized as a source of nitrogen in many culture media for growing bacteria as well as fungi.

The lack of detectable carbohydrates makes this peptone an excellent choice for bacterial studies based on

fermentation reactions. Its high level of tryptophan makes it useful in the production of indol.

It is recommended for use in all types of culture media including sanitary bacteriology of foods, water (treated and untreated), sterility tests (USP) and susceptibility tests according to official publications.

TRYPTOSE

Cat. 1614

Tryptose is an enzymatic digest of protein which can be an excellent sole source of nitrogen, demonstrating a superiority over meat peptone in this regard. It is used to grow many fastidious microorganisms such as *Brucella*, *Streptococcus*, and *Neisseria*.

YEAST EXTRACT

Cat. 1702

Yeast Extract is produced from autolyzed yeast cells and is very soluble in water. It is used as an enrichment in a large number of culture media for general bacteriology and in media for sterility according to the USP.

Because of its high content of carbohydrates, Yeast Extract should not be used in fermentation studies.

GENERAL SUGGESTION FOR THE USE
AND MAINTENANCE OF
DEHYDRATED MEDIA

Rehydration

The dissolution of the media frequently determines the clarity and yield of the final product. It is essential to obtain a homogeneous solution with minimal exposure to heat.

You must use only purified water.

The required quantity of powder material should be added to half the volume of water. After total mixing, add the rest of the water, taking caution to rinse the sides of the container and stir the contents carefully.

Previous heating of the water to a temperature of 45 to 50°C favors dispersion and rapid dissolution of the powder. Allowing the mixture to stand for 5 minutes helps to obtain a uniform suspension. Many formulas that do not contain gelatin, agar or cystine, dissolve without heat, but others require direct heat for complete dissolution. Apply heat evenly, boil it as briefly as possible (normally a minute or two is sufficient).

Sterilization

Follow the instructions that appear on the label. In general, these instructions are for smaller volumes of media. For larger volumes increase the time of sterilization to 30 minutes, but the temperature or steam pressure should not exceed the indication on the label. The media that contain carbohydrates should not be autoclaved at a temperature that exceeds 116°C to 118°C. Always avoid overheating.

Storage of dehydrated media

When the bottle of powdered medium has been opened for use, it should be closed immediately to avoid rehydration. Store it in a cool dry place out of direct light. If the medium hydrates (cakes), it will become contaminated and be difficult to sterilize in which case, the bottle should be discarded.

It is important that the inventory powdered of media be large enough to address all the necessary applications, but sufficiently small to assure constant rotation. Although many media can be kept at ambient conditions for long periods of time, not all, however, are stable indefinitely.

Presentations

All our Dehydrated Culture Media, Peptones and Agars, can be supplied in the following presentations:

5 Kgs. Drum	10 Kgs. Drum
25 Kgs. Drum	50 Kgs. Drum

Cautions

You can find below our Dehydrated Culture Media that require the following cautions:



R:22 Toxic when swallowed.
S:45 In case of accident or uneasiness, go to the doctor immediately (show the label if possible). In case of accident or uneasiness, go to the doctor immediately (show the label if possible).

- AZIDE BLOOD AGAR BASE
- BILE ESCULIN AZIDE AGAR
- CONFIRMATORY K.A.A. AGAR
- E.V.A. BROTH
- KF STREPTOCOCCAL AGAR
- PRESUMPTIVE K.A.A. BROTH
- ROTHE BROTH
- SABOURAUD DEXTROSE AGAR WITH CHLOR.+CYCLOHEXIMIDE
- SLANETZ BARTLEY MEDIUM
- STREPTOCOCCUS SELECTIVE AGAR
- STREPTOCOCCUS SELECTIVE BROTH

R:22/23 Toxic by inhalation and swallowing. Danger of accumulative effects.

S:23/45 Do not inhale vapors. In case of accident or uneasiness, go to the doctor immediately (show the label if possible). In case of accident or uneasiness, go to the doctor immediately (show the label if possible).

- BRILLIANT GREEN SELENITE BROTH
- SELENITE CYSTINE BROTH
- SODIUM SELENITE BROTH

R:40 Possibility of irreversible effects.
S:36/37 Use appropriate clotting and protecting gloves.

- ACETAMIDE BROTH

GUIDE FOR THE USE OF DEHYDRATED CULTURE MEDIA
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Cat.	DESCRIPTION
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MEDIA FOR GENERAL USE

1108	BLOOD AGAR BASE
1048	BRAIN HEART INFUSION AGAR
1400	BRAIN HEART INFUSION BROTH
1402	BUFFERED PEPTONE WATER
1104	COLUMBIA AGAR BASE
1021	DEXTROSE AGAR
1029	EMERSON AGAR
1036	EUGON AGAR
1203	GLUCOSE BROTH (DEXTROSE BROTH)
1214	MUELLER HINTON BROTH
1060	NUTRIENT AGAR
1216	NUTRIENT BROTH
1403	PEPTONE WATER (CeNAN)
1023	PHENOL RED DEXTROSE AGAR
1056	STANDARD METHODS AGAR
1033	STANDARD METHODS AGAR WITH POWDERED MILK
1003	TRYPTICASEIN DEXTROSE MEDIUM
1041	TRYPTICASEIN GLUCOSE EXTRACT AGAR
1068	TRYPTICASEIN SOY AGAR
1224	TRYPTICASEIN SOY BROTH

ISOLATION AND IDENTIFICATION MEDIA

Enterobacteriaceae

1045	DCLS AGAR
1067	DESOXYCHOLATE CITRATE AGAR
1025	DESOXYCHOLATE LACTOSE AGAR
1039	EOSIN METHYLENE BLUE AGAR
1030	HEKTOEN ENTERIC AGAR
1042	KLIGLER IRON AGAR
1050	LEVINE EOSIN METHYLENE BLUE AGAR
1052	MACCONKEY AGAR
1035	MACCONKEY AGAR Nº 2
1037	MACCONKEY AGAR WITHOUT CRYSTAL VIOLET
1403	PEPTONE WATER (CeNAN)
1040	PHENYLALANINE AGAR
1014	SIMMONS CITRATE AGAR
1046	TRIPLE SUGAR IRON AGAR
1092	VIOLET RED BILE AGAR WITH GLUCOSE
1080	XLD AGAR
1212	EWING MALONATE BROTH
1504	INDOLE NITRATE MEDIUM
1208	LYSINE DECARBOXYLASE BROTH
1509	MANNITOL NITRATE MOTILITY MEDIUM
1510	MIO MEDIUM
1112	MOELLER KCN BROTH BASE
1202	MOSSEL EE BROTH
1514	SIM MEDIUM
1110	UREA AGAR BASE (CHRISTENSEN)
1226	UREA BROTH
1227	UREA INDOL BROTH

Cat.	DESCRIPTION
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Coliforms

1051	B.C.P. AGAR
1010	BRILLIANT GREEN BILE AGAR
1228	BRILLIANT GREEN BILE BROTH 2%

1020	DESOXYCHOLATE AGAR
1025	DESOXYCHOLATE LACTOSE AGAR
1522	E.C. MEDIUM
1118	ENDO AGAR BASE
1039	EOSIN METHYLENE BLUE AGAR
1042	KLIGLER IRON AGAR
1200	KOSER CITRATE BROTH
1206	LACTOSE BROTH
1310	LAURYL SULFATE BROTH
1052	MACCONKEY AGAR
1210	MACCONKEY BROTH
1512	MR-VP MEDIUM
1403	PEPTONE WATER (CeNAN)
1014	SIMMONS CITRATE AGAR
1227	UREA INDOL BROTH
1093	VIOLET RED BILE AGAR WITH LACTOSE

Salmonella sp.

1011	BISMUTH SULFITE AGAR
1030	HEKTOEN ENTERIC AGAR
1042	KLIGLER IRON AGAR
1044	LYSINE IRON AGAR
1052	MACCONKEY AGAR
1014	SIMMONS CITRATE AGAR
1078	BRILLIANT GREEN AGAR
1221	BRILLIANT GREEN SELENITE BROTH
1402	BUFFERED PEPTONE WATER
1212	EWING MALONATE BROTH
1403	PEPTONE WATER (CeNAN)
1240	RAPPAPORT BROTH
1064	SALMONELLA SHIGELLA AGAR
1220	SELENITE CYSTINE BROTH
1222	SODIUM SELENITE BROTH
1114	TETRATHIONATE BROTH BASE
1227	UREA INDOL BROTH
1080	XLD AGAR

Streptococci sp.

1113	AZIDE BLOOD AGAR BASE
1031	BILE ESCULIN AGAR
1005	BILE ESCULIN AZIDE AGAR
1539	ELLIKER MEDIUM
1018	ENTEROCOCCUS CONFIRMATORY AGAR
1230	EVA BROTH (ETHYL VIOLET AZIDE BROTH)
1027	KAA CONFIRMATORY AGAR (CeNAN)
1209	KAA PRESUMPTIVE BROTH (CeNAN)
1034	KF STREPTOCOCCAL AGAR
1035	MACCONKEY AGAR Nº 2

Cat.	DESCRIPTION
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Streptococci sp.

1037	MACCONKEY AGAR WITHOUT CRYSTAL VIOLET
1238	ROTHE BROTH
1109	SLANETZ AND BARTLEY MEDIUM
1070	STREPTOCOCCUS SELECTIVE AGAR (STREPTOSEL AGAR)
1204	STREPTOCOCCUS SELECTION BROTH
1236	TODD-HEWITT BROTH

Staphylococcus sp.

1113	AZIDE BLOOD AGAR BASE
1100	BAIRD PARKER AGAR BASE

Staphylococcus sp.

1017	CHAPMAN STONE AGAR
1028	DNASE TEST AGAR
1232	GIOLITTI-CANTONI BROTH
1062	MANNITOL SALT AGAR
1032	STAPHYLOCOCCUS AGAR 110
1079	VOGEL JOHNSON AGAR

Fungi and Yeast

1038	MALT EXTRACT AGAR
1245	MALT EXTRACT BROTH
1072	MYCOBIOLOGIC AGAR (FUNGAL SELECTIVE AGAR)
1022	POTATO DEXTROSE AGAR
1024	SABOURAUD DEXTROSE AGAR
1090	SABOURAUD DEXTROSE AGAR WITH CHLORAMPHENICOL
1088	SABOURAUD DEXTROSE AGAR WITH CYCLOHEXIMIDE
1506	SABOURAUD FLUID MEDIUM
1054	SABOURAUD MALTOSE AGAR
1213	SABOURAUD MALTOSE BROTH

Osmophilic Yeast

1057	OSMOPHILIC AGAR
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Pseudomonas

1211	ACETAMIDE BROTH
1207	ASPARAGINE BROTH
1102	CETRIMIDE AGAR BASE
1531	KING A MEDIUM
1532	KING B MEDIUM
1532	PSEUDOMONAS F AGAR (KING B)
1531	PSEUDOMONAS P AGAR (KING A)

Cat.	DESCRIPTION
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Lactic Bacteria

1539	ELLIKER MEDIUM
1043	M.R.S. AGAR
1215	M.R.S. BROTH

1096	ROGOSA SL AGAR
1234	ROGOSA SL BROTH
1073	TOMATO JUICE AGAR

Marine Heterotrophic Bacteria

1059	MARINE AGAR
1217	MARINE BROTH

Anaerobic Bacteria

1000	ANAEROBIC AGAR
1066	SCHAEDLER AGAR
1218	SCHAEDLER BROTH
1503	WILKINS CHALGREN MEDIUM

Clostridium Perfringens

1082	SPS AGAR
1075	TSN AGAR

Bacillus

1500	OF BASAL MEDIUM
1065	SELLERS AGAR

Brucella

1012	BRUCELLA AGAR
1223	BRUCELLA BROTH

Bordetella

1107	BORDET-GENGOU AGAR BASE
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Candida

1006	BIGGY AGAR
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Neisseria and Haemophilus

1106	GC AGAR BASE
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Cat.	DESCRIPTION
Mycobacterium	
1116	LOWENSTEIN-JENSEN MEDIUM BASE
Vibrio	
1074	TCBS AGAR
ANTIBIOTIC ASSAY MEDIA	
1520	ANTIBIOTIC MEDIUM No. 1 (SEED AGAR)
1002	ANTIBIOTIC MEDIUM No. 2 (BASE AGAR)
1534	ANTIBIOTIC MEDIUM No. 3
1524	ANTIBIOTIC MEDIUM No. 5 (STREPTOMYCIN ASSAY AGAR)
1004	ANTIBIOTIC MEDIUM No. 8 (BASE AGAR WITH LOW pH)
1528	ANTIBIOTIC MEDIUM No. 11 (NEOMYCIN ASSAY AGAR)
STERILITY TEST MEDIA	
1241	THYOGLYCOLLATE BROTH (NIH)
1508	THYOGLYCOLLATE FLUID MEDIUM (FTM)
1516	THYOGLYCOLLATE MEDIUM WITHOUT INDICATOR
1533	THYOGLYCOLLATE USP MEDIUM
RESISTANCE TEST MEDIA	
1058	MUELLER HINTON AGAR
1214	MUELLER HINTON BROTH
MICROBIAL COUNTS MEDIA	
1056	STANDARD METHODS AGAR
1033	STANDARD METHODS AGAR WITH POWDERED MILK
Urine Microbial Counts	
1016	CLED AGAR

Cat.	DESCRIPTION
Investigation and Recount of Microorganisms (proteolytic)	
1069	CALCIUM CASEINATE AGAR
1300	NUTRIENT GELATIN
Investigation and Recount of Microorganisms (psicrotrofic)	
1053	KING FG AGAR
MAINTENANCE AND MOTILITY MEDIA	
1502	C.T.A. MEDIUM
1509	MANNITOL NITRATE MOTILITY MEDIUM
Transport Medium	
1535	AMIES TRANSPORT MEDIUM
1530	AMIES TRANSPORT MEDIUM WITHOUT CHARCOAL
1529	CARY BLAIR MEDIUM
1518	STUART TRANSPORT MEDIUM
Media for fungi and bacteriae in which nitrate is the only source of nitrogen supplied	
1015	CZAPEK DOX MODIFIED AGAR
1250	CZAPEK DOX MODIFIED BROTH
Media for beer fermentation processes	
1061	RAKA-RAY AGAR BASE
1026	WL DIFERENTIAL AGAR
1086	WL NUTRIENT AGAR
Media for carbohydrates fermentation	
1203	GLUCOSE BROTH (DEXTROSE BROTH)
1115	PHENOL RED BROTH BASE
1235	PHENOL RED DEXTROSE BROTH
1239	PHENOL RED SUCROSE BROTH